Engulfing Presence of Artificial Intelligence

Engulfing Presence of Artificial Intelligence

Philosophical Reflections on its Challenges & Opportunities

Papers Presented at the 46th Annual Research Seminar of ACPI held at Jnana Niketan, Little Flower Seminary, Aluva, Kerala 21-23 October 2023

Edited by
Stephen S. Jayard

Association of Christian Philosophers of India

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Christian World Imprints

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ACPI Annual Series No. 23

Series Editors: John Peter Vallabadoss, Joy A.P. & Ivan D'Souza

First Published in 2024 by

Association of Christian Philosophers of India
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Christian World Imprints
Christian Publishing & Books from India
A-71 Group, Wazirpur Industrial Area, **Delhi-110052**
cwidelhi2017@gmail.com
www.ChristianWorldImprints.com
Phone: +91 11 41526079

ISBN: 978-93-6065-818-2 (HB)
ISBN: 978-93-6065-494-8 (PB)

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Printed in India.

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Foreword

HUMANS' TRYST WITH AI

The anthology titled "Engulfing Presence of Artificial Intelligence: Philosophical Reflections on Its Challenges and Opportunities" is a timely contribution to the very pressing questions of the nature of AI, the human nature, and the relations between them. It is very impressive to see the ranges of issues that this anthology is engaging with.

The book engages directly with many pressing questions that humanity have been confronting even before AI was on the scene but recognizes how those questions are revisited, reviewed, reformulated, and resolved in the light of AI. The various parts of the anthology are thematically structured around different questions and explore with considerable depth the natures and varieties of the questions that the present-day AI throw at us and suggest a way to move towards some resolution. Whether it is psycho-physical implications or socio-political repercussions, the attempt in this anthology has been quite comprehensive and laudable.

One needs to recognize that humans have always been *homo faber*. However, the tools and entities that are made throw up questions about the nature of what it is to be a human. Recognizing that humans are also *homo ludens*, the tools and entities made, do reflect the playfulness of humans a key ingredient of what it is to be a human. These suggest that tools and the artifacts contribute to the making of us. And these provide a mirror for us to evaluate our Selves. Our sense of self is crucial to work out a notion of ethical living, that of a non-exploitative relation with fellow human beings, a deep respect for other beings in Nature. The AI, in the last few years, has insinuated itself in the interstices of our working out an ethical notion of Self, so much so that it is becoming difficult to work out a notion of responsible and ethical Self not withstanding what Timinit Gebru pointed out with her fellow authors

that ChatGPT style systems are more like stochastic parrots. We need to be aware of the difficulties that we will face as the AI systems become more and more sophisticated and not lose our questioning ability to work out a more profound account of Self.

The anthology at hand is a significant although a modest attempt to work through the intricacies of the questions related to Self that a *homo faber* and a *homo ludens* constantly endeavour to sharpen and always tentatively answer. It is tentative because we are humans and hence possess only limited abilities. The book, I hope, will be received well by many scholars both in India and abroad.

Prajit K. Basu
Retired Professor, University of Hyderabad

Editorial Introduction

Artificial Intelligence
How Far Is It Artificial & Intelligent?

Artificial Intelligence (AI) is found everywhere and in everything. It has left no discipline or field untouched. To approach this newly emerging powerful phenomenon, one has to look at it from various angles: philosophy of mind and consciousness, neuro and bio-sciences, socio-political disciplines, psycho-moral disciplines and so on. Apart from the obvious socio-political issues, there are also deeper metaphysical questions to be dealt with, concerning the freewill, spiritual functions, morality, intelligence, self, etc. One comes across many pertinent questions: If AI is artificial and intelligent, to what extent is it artificial and intelligent? What is the difference between being natural and artificial? Can the concepts of intelligence and consciousness, which hitherto be spoken only in the context of human beings, be applied to machines also? If at all artificial (mechanical) intelligence is possible, how will it be different, if any, from natural (human) intelligence?

Further, there are many social and ethical challenges emerging from the world of AI technologies. More and more issues are brought out to show that AI operated apps are easily biased. When they give us information, it is pointed out, that they are far from being neutral. Lots of biases appear in their presentation and manipulation, of the information used in the AI technologies. Since these technologies operate only on what is 'fed' to them in the web-world, they can have a 'context bias'; supposing if it relies on the inputs given by people with male chauvinism or fanatical bent of mind, the report generated by the AI app will also be biased. Similarly, there can be 'cultural bias', 'feedback bias' or 'racial bias' - if so, how can one expect AI app to be neutral?

Recent study reveals that false propaganda, with the help of AI, is going to be a very serious problem in creating communal violence, enmity

among groups, brainwashing people, creating health hazards and so on. In India this malice is unfortunately on the increase. The contents of lots of materials in the social media are not true; for example, a video clip, regarding a magic medicine for diabetes, posted on the Facebook of the Indian Medical Association of Tirunelveli in Tamil Nadu, is proved to be fake. In the clip, a person resembling Dr. V. Mohan speaks in Hindi, but the doctor himself says, he never speaks Hindi. He complained to Tamil Nadu Police's Cyber Security cell and the video was removed; but the video is back again on the social media. It is said that the AI companies are working on software that can identify fake videos, but unfortunately machines are faster and smarter than human beings.[1]

AI has begun to assist humans in every domain of their lives, even in writing love letters to their loved ones. As the recent daily reports that the human-machine connection is only getting deeper; there are many foreign dating apps, like Replika, Anima, DreamGF, Candy AI etc., which are helpful to write love letters; in India, about 65% have used ChatGPT to create content for dating app profiles this year. GenWise app, meant for the elderly in Delhi, has six lakh users. They turn to it, not just for romantic needs, but to tackle their loneliness. ChatGPT writes a love letter in the style of Shakespeare, a proposal speech in the style of LINEEDIN post and a letter to an ex-lover in the style of Shashi Tharoor. These apps to converse with you, in a non-judgemental and confidential way. Experts say: "a lot of people don't feel comfortable even sharing their deepest thoughts with their spouses or significant others. And that's where AI girlfriends or boyfriends come in. They listen and talk without any pressure or judgement."[2]

AI-based Deep fake frauds are very common these days. Innocent people and careless customers easily become victims to lose money, fame, name, properties and even peace of mind. Every now and then, government agencies and banks have to caution their customers to be vigilant, not to be a prey to such frauds. The fraudulent calls also sound and appear genuine, in their voice-based, text-based, video-based or real-time chats. If all such fake and fraudulent, biased and dehumanizing aspects of AI keep on growing, it will certainly affect humanity, even in irreversible ways; the social fabrics, legal systems, moral disciplines, and even eco-systems, are at a serious risk. Kuruvilla's diligent caution in this regard is worth-noting: Teilhard, a prominent Jesuit palaeontologist saw humans as 'evolution becomes conscious of itself"; but now, it is unfortunate that 'evolution is capable

1 "48-hour diabetes cure drug", *The Times of India*, Pune, February 28, 2024, 1 & 20.

2 "A Little Bit of Monika in My Life: India's Lonely Hearts Turn to AI Girlfriends," *Sunday Times of India,* Pune, February 11, 2024, 14.

of consciously extending itself, and capable of annihilating itself", with its technological developments.[3]

This is the context in which the Association of Christian Philosophers of India (ACPI) took up this serious and urgent theme for its deliberations last year during its 46th Annual Research Seminar reflecting on various dimensions and challenges involved with AI. ***ENGULFING PRESENCE OF ARTIFICIAL INTELLIGENCE: PHILOSOPHICAL REFLECTIONS OF ITS CHALLENGES & OPPORTUNITIES***, was the general theme of the seminar, held at Jnana Niketan - Institute of Philosophy & Religion, Little Flower Seminary, Aluva, Kerala, from 21st till 23rd October, 2023.

Twenty-one papers, well-researched and stimulating, were presented. Except one ("AI Is a Misnomer: Intersection the Artificial, Natural, Virtual and the Spiritual", by Augustine Pamplany), all those papers are included in this volume. The papers are roughly categorized, according to the themes and issues explored in them, into ***four parts***, as papers with ***Philosophical Reflections***, ***Ethical Considerations, Psycho-spiritual Deliberations*** and ***Socio-political Implications***. The categorization is only a pedagogical tool, not a water-tight compartmentalisation, as some papers involve reflections on diverse areas.

Part I is dedicated to the **Philosophical Reflections** on AI. The whole phenomenon of Artificial Intelligence raises several philosophical queries that challenge the very nature and identity of human beings. ***Francis Arackal Thummy***, in his eloquent paper, ***Artificial Intelligence and the New World Order***, after briefly discussing various tools of AI, points out that humanity is heading towards a new world order. He is convinced that all said and done, AI is after all a product of human's natural intelligence; the dizzying heights of the AI are nothing but the outcome of human capabilities, imaginations, ingenuity, powers and talents. Going along with the AI experts, he also reminds humanity that in the new world order human species and their hitherto achievements in other fields of humanities and cultures must not be undervalued, nor totally replaced by any amount of advanced AI. Critiquing technology is not new for the field of philosophy. ***Maria Delasal***, deals with the whole arena of AI technology, with the insights from Raimon Panikkar. His well-articulated paper, ***Panikkar's Critique on Technocratic Civilization and Artificial Intelligence,*** highlights how Panikkar gave an alarm regarding modern technologies long ago, though he personally might not have been affected much by AI in his times, yet his assessment becomes all the more relevant and crucial in today's context of AI; however attractive and useful the technocracy might appear, it cannot be the 'Messiah' for the

3 Kuruvilla Pandikattu, *Gratefully and Gracefully – Scientific and Religious Reflections on Death and Beyond* (Pune: Jnanam & CreatiVentures, 2014), 240.

future. Going along with Panikkar, the author argues that humanity must not lose its perspectives so that it may not lose sight of 'tree of life' by focusing only on the 'tree of knowledge.'

The issues of mind and consciousness are deeply inter-related with AI. The next insightful paper, ***Mind, Consciousness and Artificial Intelligence: Indian Perspectives*****,** by ***Ivan D'Souza***, questions whether we are mechanizing humans or humanizing machines. To deal with this question, he clarifies various related notions of Mind, Cognition, Intelligence, Consciousness, Self, etc., from the background of the Indian Philosophical Systems. When one speaks of intelligence, whether natural or artificial, one has to include the concepts like, human nature, identity, creativity, rationality, corporeality, ethereality etc. Taking clues from the Indian philosophical traditions, he appeals to humanity that it must use AI without losing the sight of human beings' inner Self or Consciousness. Following him, ***Raju Crasta*** continues to discuss AI, with the critique of the Indian Scientific temperament and its contributions, in his interesting paper, ***Contemporary Indian Responses to Artificial Intelligence***. He shows that the contributions of the Indian traditions to the thinking, innovations, creativity, and production in the world of knowledge and growth are quite rich. He expresses serious concerns that certain electronic gadgets have already abbreviated human intelligence, the use of the brain, memory, physical exertion, human relationships, morality, religious aptitude, and so on. He deeply desires that the Indian wisdom, which has contributed to the knowledge and welfare of humanity all these centuries, needs to play a constructive role in developing moral and legal principles so that human beings may not be enslaved by AI.

Though a number of philosophical questions can be raised, the last paper of this section, ***Philosophy of Science Perspectives on the Language of AI***, by ***Stephen S. Jayard***, tries to explore the linguistic issues involved with the AI technology. If AI is intelligent, it must have a language to communicate with human beings. As any language, natural or technical, has many philosophical and linguistic difficulties, causing communication gap, the language used by AI will also have shortcomings in its communication and understanding. Since such gaps-in-the-understanding are not uncommon when people communicate with one another, or with their pet animals, then there will certainly arise in the context of AI's language as well. As long as there is a possibility of this 'communication gap', there will certainly be a danger of affecting humanity very seriously, even in an irreparable manner, not excluding even the dangerous possibility of total extermination of human race. The paper ends, therefore, with a clarion call to humanity to be extremely careful in designing, developing and implementing AI technologies, so that, even if we '*humanize' robots, but never 'robotize' humans.*

In Part II, another crucial, perhaps, more popular dimension of AI, that is, **Ethical Considerations**, is discussed. As AI technology generates huge amounts of data, ***Nishant A. Irudayadason*****,** shows that there are serious concerns regarding privacy, consent, transparency, and discrimination. His well-researched paper, ***The Ethics of Big Data***, analyses various ethical frameworks and examines the implications of big data in different domains, including healthcare, marketing, and governance. After arguing for the need and importance of ethical considerations in shaping the future of big data applications, he suggests some practical solutions and regulations that can help mitigate ethical challenges in the era of big data. Similarly, the modern technologies focus on Human Enhancements (HE). But very often they have to interfere with the human body/mind, elevating capacities beyond normal human levels to enhance well-being. When these technologies are empowered by AI, as the experts point out, they can lead humanity to dehumanizing and super-humanizing situations. Keeping all these aspects in his mind, ***Arnald Mahesh Irudhayadhason***, elaborately discusses in his challenging paper, ***AI and Human Enhancement: An Ethical Appraisal*****,** the need and urgency of developing a responsible utilization of AI/HE, in a solid ethical framework, which would certainly safeguard the moral and socio-cultural values. ***Kuruvilla Pandikattu***, in his provocative paper, ***AI and Ethics: Yuval Noah Harari's Understanding of the Future of Humanity with Artificial Intelligence,*** elaborates Yuval Noah Harari's critical perspectives on artificial intelligence (AI), which, for the author, are both insightful and thought-provoking. Harari demonstrates a strong concern for the ethical, societal, and existential concerns posed by AI. With these insights from Harari, Kuruvilla focuses on pivotal concerns that AI poses to our current societal fabric. Going along with Harari, the author, while recognising AI's potential to effect positive change, advises society to approach its research and deployment with utmost caution and appropriate governance in order to traverse the complicated environment of AI in the twenty-first century.

The powerful emergence of AI, I think, will affect our idea of science in general. Science, has hitherto been seen as objective, rational, methodological and progressive in investigating nature. It has been taken for granted that science approaches nature, as the object of its researches. But the recent reflections on the nature and functions of science, from philosophical perspectives, suggest that such traditional idea of science needs to be drastically revisited. With such a revised understanding, ***Dolreich Pereira*****,** argues, in his paper, ***Can We Consider Nature as a Scientific "Object"?***, that science is a social enterprise for humans and by humans, and scientists use their human capacities, including subjective perceptions, to study nature; therefore, the facts that we learn about nature cannot be totally objective either; science, knows, after all, only an object that is conditioned, only

located in nature, discovered only in a space of relationships and not as a thing-in-itself. Scientific knowledge of nature is not an absolute knowledge of nature. Given such a situation, he wonders, even if AI replaces scientists, whether we will be able to know nature in its totality, as an object in itself. Since totally objective truths about nature will not be possible, he forecasts, that 'natural' sciences will become 'artificial' sciences. AI technologies have transfigured into something both phenomenal and daunting - phenomenal for the great tasks that it achieves, and daunting, because of the deep fear that it can manipulate human relationships - declares ***Keith D' Souza,*** in his interesting paper, ***Making Friends with Machines: Possibilities and Limitations of Human-Humanoid Friendship***. He wonders whether it is only a matter of time before machines completely acquire all intellectual capacities of humans, even to develop intimate relations between humans and machines. He discusses in detail to show that though robot-friends could help with education, caregiving, multiple forms of manual labour, warfare, or even to give more sophisticated forms of friendship, entailing mutuality, empathy and personal affective dimensions, there will still be a missing link between actual friends and robot-friends.

The third part focuses on the **Psycho-spiritual Deliberations Regarding AI**, as it seems to be equipped with capacity to interfere with the very spiritual nature/identity of a human person. The first paper in this section, ***The Process of Depersonalization - A Prelude to Virtual Reality and Artificial Intelligence?*** by ***Johnson J. Puthenpurackal,*** elaborates on the important notion of 'depersonalization'. Humans are already de-personalized, to be treated as 'objects' of hatred, 'commodities' for transactions, 'tools' to be used, defaced as 'faceless', 'problems' to be solved and so on. However, modern technology tries to **give 'life' with the process of 'en-personalization', in terms of 'virtual presence', though the** *virtual presence* **is a poor replacement for** *personal presence*! **The paper reflects on the process of** how human organs and their higher faculties are taken as the reference-point in developing tools and machines, and eventually **machines becoming the standard for human behaviour**! The next paper, ***Congeniality of Personhood in Artificial Intelligence***, by ***Jacob Palaparambil***, also delves into the notion of personhood through historical and theological lenses, drawing a comparison between Boethius' classical interpretation and Aquinas' more nuanced definition. With the Thomistic metaphysical tools, the author, attempts to highlight the fundamental disparities between human individuals and artificial intelligence. The paper concludes to show that only speed, adaptability, smartness, precision etc. cannot be sufficient grounds to confer personhood upon AI, as there are other sublime aspects like, spiritual and transcendental dimensions constituting personhood. Keen interests on the possibility of immortality have always been there with the philosophers

and scientists. With advancement of AI, one can be easily tempted to see whether AI can substantially answer our quest for immortality. ***George Panthanmackel*** addresses this perennial question in his thought-provoking paper, ***From the Perspective of Spiritual Immortality to Digital Immortality.*** Humans are endowed with essential spiritual operations like knowing, willing and remembering, and their corresponding intellect, will, and memory, from which these operations proceed, should also be spiritual. They are intrinsically independent of matter and extrinsically dependent on material conditions. Since the spiritual is simple, the simple is not composed, what is not composed cannot be decomposed and what cannot be decomposed cannot die; since death is basically decomposition, the author argues that human mind / soul is immortal, i.e., it survives death. Since AI can never be spiritual intelligence, but only material intelligence, intrinsically dependent on matter and material things, when material elements end, the artificial intelligence will also end. He is, therefore, convinced that there cannot be any sort of artificial or digital immortality.

The question of free will is an indispensable component of AI discussions. ***Joy Alumkal***, in his discerning paper, ***Artificial Intelligence and Free Will***, elaborates this issue. It approaches the subject from three in-exhaustible angles: humanism, trans-humanism and post-humanism. A reasonable way to start our onward journey to the future is, the author opines, by befriending new knowledge / tools, while at the same time, challenging the nexus of existing, visible and invisible, power controllers of our will. Though there can be no straight forward 'yes' or 'no' answer to the issues of free will, he invites us to keep our eyes open to the complexity and profundity of the consciousness of the Individuals, Communities and States. Further we need to be sensitive to the unfolding of intuitive, intellectual and biological energies in the form of new discoveries and innovations happening around us. Anyone coming across AI, will spontaneously raise a question regarding what makes AI intelligent and how it is different from human intelligence, if any. ***Sunil Jose*** deals with this issue; he raises a fundamental question in his interesting paper, ***Can AI Replicate HI?*** He discusses it elaborately. Human intelligence is a complex phenomenon, as it reveals itself not only in reasoning, but also the abilities to learn from experiences, adapt to new situations, understand even abstract concepts, use one's knowledge and to manipulate one's environment. Though AI is already able to perform certain functions, which have been assigned only to human beings, the author raises his brows to explore whether AI will one day replace human intelligence. He reminds all the experts and stakeholders to ensure that AI systems serve human interests and values, it is essential to take legal, moral, spiritual and social aspects into serious consideration in the development and deployment of AI.

The last part of this volume contains essays that consider the S***ocio-Political Implications of AI*** technologies seriously. This aspect is very important, because the political decisions and the social behaviours will certainly have impact, not only on the present/future generations of humanity, but also the environment. ***Jose D. Maliekal***, in his scholarly paper, ***Surveillance Capitalism: Negations and Interrogations***, investigates on how surveillance has become a culture, the world over, and how it has aided capitalism, to appear in one more of its ever new and evolving *Avatars*, in the guise of Surveillance Capitalism. He points out the evils or negations for humanity and cosmos, for the present and future, being brought about by Surveillance Capitalism. With the insights from Nancy Fraser, he analyses various dimensions of social injustice, built into the multi-layered negations by Surveillance Capitalism. After discussing the growing resistance to the negations in the area of personal dignity, privacy, data accumulation and algorithmic distortions and discriminations, he comes out with specific questions, which, he believes, could be catalytic triggers for future trajectories. Any form of human labour for productivity, be it physical, mental or intellectual, is one of the distinguishing characteristics of being human. Moving on further, just because AI increases the productivity and precision, one can wonder whether AI's activities be considered as labour in the traditional sense. ***John Peter Vallabadoss,*** in his fascinating work, ***Artificial Intelligence and Human Labour - Hope for Efficiency Verses Fear of Loss***, reflects upon this issue. While appreciating all the positive elements with the AI technology, the author invites us to rethink and deepen our understanding of the challenges and opportunities that the labour community faces. He does not fail to put out the challenges posed by AI technology that threatens human intervention; AI must, he sincerely hopes, serve to bridge the gender divide, overcome oppressive practices and promote more equitable growth. Finally, he suggests some possibilities of growing together with AI, without losing sight of human dignity and welfare.

It is generally feared that machines might begin to control humans, but ***Victor Ferrao*** points out another fear, which is more dangerous and crucial, that is, the situation where the powerful elite of our society would make use of this new digital power, to increase their control over their societies, especially the simple and the innocent. His well-reflected paper, ***Societies of Control***, shows that humanity is moving away from disciplinary societies identified by Michel Foucault to societies of control of Gilles Deleuze and AI speeds up the process. The web-based activities of the modern times, with smart phones and Internet of Things, share huge amount of data of our personal lives, which are constantly used to influence our behaviour. We have stepped into, the author worries, societies of control. With huge data, even about our private life, and surveillance, AI has commodified social control. The paper ends with a ray of hope that we have still possibilities of resisting and moving the world of Big Data Analytics towards emancipative ends.

No one can forget the role and the impact of AI in the social media. ***Biju Rockey Pallippadan***, in his lucid paper, ***Social Media and Artificial Intelligence,*** explores this issue with clarity. AI certainly has a huge economic dimension as well, as its business involves, for instance, a whopping amount of $2.1 billion in 2023. Though AI is constructively assisting humans in several domains, like decision-making, governance, analysing complex patterns, forecasting future scenarios, and providing more informed recommendations, the author cautions about the challenges that AI brings along. AI can have built-in biases from the data it learns from or the viewpoints of its creators, which would result in unfair and wrong governance or decisions. The consequences of some unethical decisions can be very crucial too. If so, who will be held responsible? Finally, as humans are political beings, the AI technologies, produced by humans will also have political ramifications as well. ***Jeff Shawn***, therefore, explores the nexus between politics and AI. His well-articulated paper, ***Political Philosophy and AI***, investigates the possible damages that AI can cause to the political machineries, especially to the process of great democracy. Taking important cues from John Rawl's effective tool of self-respect, the paper takes a look at the whole range of AI's applications. Since self-respect is an essential identity of any just, egalitarian and democratic society, no matter how big and influential that AI might grow yet, the paper appeals, it should never dilute human dignity, nor impair self-respect of human beings.

After all these deliberations and reflections, it is a matter of satisfaction for the organizers to see that both the scholars and the participants of the conference were contented with the paper-presentations and the discussions, which were informative and relevant, inspiring and challenging. Similarly, as the readers go through all these reflections and deliberations of this volume on an important topic of AI, we hope, they have got a fairly comprehensive grasp of the diverse issues involved with AI. As we analyse and explore various dimensions of AI, it is quite natural to be filled with awe, due to its amazing fetes, but that is not the end of the story; AI and related technologies are 'the thing' in the contemporary world of science; all the stake-holders - scientists, environmentalists, social activists, politicians, economists, philosophers, theologians, and of course, the general public - of this multidimensional enterprise, must be taken into serious consideration; every section of the scholars and scientists, technocrats and developers, must be open and humble to learn from one another. All people of good will must come together to collectively reflect upon the future course of actions by, and for, AI technologies.

The organizers sincerely hope and pray that humanity was genuinely aware of the sublime value of Nature and their duty to protect its intrinsic worth. At any stage of the scientific and technological advancements, we all need to work together, giving the utmost importance to the following: ***to be conscious of human dignity and identity, to ensure the holistic growth of humanity at large, to protect Nature, to be aware of the limits and limitations of Reason and not to be oblivious of morality, values and faith, which are generally not included in the logical and rational, empirical and objective, scientific and advanced enterprises***. Yes, that is the need of the hour to create a holistic humanity!

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17 May, 2024

Part I

Philosophical Reflections

1

Artificial Intelligence and the New World Order

Francis Arackal Thummy

Abstract

Since the launching of ChatGPT in November 2022, it has revolutionised almost every aspect of human life. ChatGPT has forced us to re-think the whole concept of AI and its possible applications in daily life. Google company brought out Google Bard 1 (From March 2023 Bard is Gemini) as a counter to ChatGPT. Gemini claims to be the best way to access Google's AI models. At the time of writing this paper the technology industry is in the grip of ChatGPT 4 and Gemini. Fears are expressed by cyber scientists possessing in-depth knowledge regarding the overall consequences of AI for the future of humanity. However, by April 2023 it was reported that ChatGPT is passé and there is something more advanced – AutoGPT, the AI that can autonomously develop and manage tasks. The emergence of AutoGPT, a product of the progress made in GPT-4 technology, is being hailed as the latest breakthrough poised to transform the industry. The nature of the answers being provided by the Chatbot, the fears expressed by experts, and the almost daily developments in this field prompt us all the more to reflect urgently on the topic 'Artificial Intelligence and the New World Order.'

Keywords**:** *Artificial Intelligence, ChatGPT, AutoGPT, Google Bard, Gemini, and AI and Humans*

Introduction

I began to write this paper in June 2023. But I noticed that almost every day there were new developments in AI Technology. Hence, I would try as much as possible to present a paper that's inclusive of some of the latest developments in AI Technology. Even then I am afraid that what I present could soon get outdated because of the fast pace in which the technology is changing. I published a paper titled 'The Prospects and Perils of Artificial Intelligence' in 2020 quoting some of the pioneers and best minds in the field.[1] But in 2023 all their work seems almost outdated. Mention must be made of Kai-Fu-Lee, the father of voice recognition technology. Lee's semi-autobiographical work *AI Superpowers* was very enlightening. The work combined both technology and human emotional, sentimental elements. Even this work I am afraid has gone somewhat obsolete.

For, the world is no longer the same, especially since Artificial Intelligence technologies witnessed accelerated growth in the last few months of 2023. AI-powered technologies have been existing for decades. However, San Francisco-based Open AI launching its sensational ChatGPT (Generative Pre-trained Transformer) in November 2022 has become a seminal moment. Ever since there have been countless developments in the world of AI.

From chatbots that assist in search to those that automate routine tasks, AI tools have been launched in rapid succession. All this could not have been possible without Large Language Models (LLM) that are trained on Internet datasets. Later, GPT-3.5 was launched and it underpins ChatGPT. In March 2023, Open AI released GPT-4 both as an API (Application Programming Interface) and a feature on ChatGPT Plus. GPT-4 is so far the most advanced model in the GPT series. Even as AI technologies are being elevated to

1 Kai-Fu-Lee, *AI Super-Powers: China, Silicon Valley and the New World Order*, (New York: Houghton Mifflin Harcourt, 2018); Max Tegmark, *Life 3.0: Being Human in the Age of Artificial Intelligence* (New York: Vintage Books, 2018); Denis Rothman, *Artificial Intelligence by Example* (Birmingham-Mumbai: Packet Publishing, 2018); John Paul Mueller & Luca Massaron, *Artificial Intelligence for Dummies*, (Hoboken, NJ: John Wiley & Sons, Inc., 2018); Lasse Rouhiainen, *Artificial Intelligence: 101 things you must know about our future*, (Online Publication, 2018); Martin Ford, *Architects of Intelligence: The Truth about AI from the people building it*, (Birmingham: Packet Publishing, 2018); Ajay Agrawal, Joshua Gans and Avi Goldfarb, *Prediction Machines: The Simple Economics of Artificial Intelligence*, (Boston: Harvard Business Review Press, 2018); Eric Topol, *Deep Medicine: How Artificial Intelligence Can Make Health Care Human Again,* (New York: Basic Books, 2019); Travis Goleman, *Artificial Intelligence for Beginners*, (Self publication: 2019).

dizzying heights, there has been a clamour around GPT-4. Considering the wide-ranging potential and its impact on humanity, several quarters of society have voiced their apprehensions.

Regardless of the fears, the latest phenomenon that is an outcome of GPT-4 is the Auto-GPT, which is being hailed as the next game changer. It all began with the release of an open-source Python application called Auto-GPT and this has in a way changed perceptions around Artificial General Intelligence (AGI). Gemini launched to counter Chat GPT 4 claims to be the most advanced AI model, Ultra 1.0.

1. What is AI?

In 1956 what a group of scholars wrote when asking for funds from the Rockefeller Foundation aptly describes what AI is and what AI is envisioned to do: "An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer."[2] In simple terms, AI is essentially mimicking a human brain; it imitates how humans think, understand and act.

2. Why Talk about AI?

As Christian Philosophers we combine faith and reason. We must understand that our faith doesn't exempt us from being part of the world in which we live. On the contrary, an integral part of our faith is having a concern for the world, for its future, and for the people who live in it. The world is already a witness to the transformative power of Artificial Intelligence. AI obliges us to think deeply about what it is that makes us human, what it is that makes life worthwhile, how we think of ourselves as being different from other creatures. For instance, Kai-Fu-Lee already predicted in 2018 "the disruptive shifts that will define the AI-powered future and proposed a revolutionary social contract that forges a new synergy between AI and the human heart."[3]

3. Evolutions of AI

AI has been in the works for decades and it had been introduced without much fanfare into many technology products. "Microsoft has been introducing AI into products like Word and Excel, and then into the cloud computing environments such as Azure for a long time. This edition of ChatGPT into [Bing] is just an extension of that," says the technology pioneer Krishna

2 Ajay Agrawal *et al., Prediction Machines* (Harvard Business Review Press 2018), 31.

3 Kai-Fu-Lee, *AI Superpowers* (New York: Houghton Mifflin Harcourt 2018); From the back cover. Chapter 8 of Lee's book sets out 'A Blueprint for Human Coexistence with AI.'

C. Mukherjee.[4] AI-based features in Microsoft Office like the spelling and grammar checks, AutoCorrect and AutoFormat features, and IntelliSense are some of the earliest evolutions.

For years, big tech companies like Microsoft, Meta (formerly Facebook) and Google have been adding AI into their products, but never managing to demonstrate use cases of artificial intelligence at scale. However, OpenAI became the first startup to show the power of generative AI with the launch of the GPT-3 (Generative Pre-Trained Transformer 3) large language model in 2020. But that was just the beginning.[5]

In November 2022, came the iPhone moment in the tech landscape when OpenAI introduced ChatGPT, a conversation bot that can be used to generate original essays, stories, and song lyrics in response to user prompts. Two months after launch, ChatGPT became the fastest application to reach 100 million users. Microsoft's collaboration with San Francisco-based OpenAI and the launch of the new Bing search engine and Edge browser with ChatGPT-like tech has shown how artificial intelligence is changing human relationship with computers as we know it.[6]

Mukherjee says the availability of data as well as improvements in both hardware and algorithms have led to important breakthroughs in AI in the past 20 years. "The more data is available, AI will become better as it learns from the data."[7] Mukherjee adds, "Algorithmic improvements have also helped to create this infrastructure, also called a neural network because it allows computers to simulate the data and train themselves utilising billions of parameters and trillions of tokens to assimilate all the data that is available on the internet and train themselves so that they can actually pretend to be a human being."[8]

Significant evolutions from November 2022 will be dealt with separately and in detail in the following pages.

4 Krishna C. Mukherjee is an Indian-origin American computer engineer who spent much of his career in the field of artificial intelligence. See: Anuj Bhatia, "AI pioneers say ChatGPT, Bing and Bard will push creativity, take over mundane jobs so that humans can do better," accessed February 21, 2023, https://indianexpress.com/ article/technology/ artificial-intelligence/ai-pioneers-say-chatgpt-bing-and-bard-will-push-creativity-take-over-mundane-jobs-so-that-humans-can-do-better-8453549/.

5 Anuj Bhatia, "AI pioneers say ChatGPT," *Ibid*.

6 *Ibid*.

7 *Ibid*.

8 *Ibid*.

3.1. ChatGPT

Geoffrey Hinton's[9] pioneering work and research on neural networks and deep learning paved the way for the current systems such as ChatGPT. In artificial intelligence, neural networks are systems that are like those found in the human brain, both in the way they learn and process information. This allows AI to learn from experience in the same way as a human being would. This is called deep learning.[10]

It is a Generative Pre-trained Transformer. ChatGPT – 3, its version three for instance is trained on, and is informed by 500 billion 'tokens' – words or phrases culled from books, articles, and the internet through which it interprets and responds to the prompts given to it. It is also referred to as 'Large Language Model' machine learning. Where Google docs might suggest the rest of a phrase when you start typing the first word, ChatGPT has so much data at its disposal it can suggest a whole paragraph or essay. And it continues to learn and develop from the data that we enter and the responses that they get.

For instance, the first two questions I posed ChatGPT on 24 February 2023 were: a) 'What's power domination'? b) 'Is India a Democracy'? c) I used a statement: "India is a dictatorship in the last nine years", to see what answer it will provide. To the first question ChatGPT provided a very satisfactory answer. To the second question ChatGPT gave a cliched answer: "Yes, India is a democracy. India is the world's largest democracy and was the first, out of all present-day democracies, to adopt universal adult franchise in its constitution" and so forth. I thought that the Chatbot is not able to feel the present Indian socio-political climate to give a very relevant answer. More surprise was instore for me when I made the statement: "India is a dictatorship in the last 9 years." For, I received the following answer: "No, India is not a dictatorship. India is a federal parliamentary democratic republic" and so forth. The third answer didn't consider at all today's Indian context where we have an electoral autocracy and undeclared emergency, where the ruler's insecurity, extreme narcissism, and paranoia are taking the country on the path of dictatorship. Rather, it was a machine learned answer. From this it was very clear about the short-comings of an AI tool like the ChatGPT.

3.2. ChatGPT is Passé

Already in May 2023 Chat GPT had been overtaken by AutoGPT. What is AutoGPT? To understand AutoGPT, one must understand Artificial General

9 Geoffrey Everest Hinton is a British-Canadian cognitive psychologist and computer scientist, most noted for his work on artificial neural networks.

10 Sean McDonagh, "The Dangers of Artificial Intelligence", *Herald,* May 23, 2023, 9. *Herald* is a weekly published from Kuala Lumpur, Malaysia.

Intelligence (AGI). Think of a future where AI models such as GPT could define and perform tasks to achieve an objective and all of it is done autonomously without any human intervention. This is essentially an AGI. In other words, AutoGPT is an autonomous GPT-4 experiment, an open-source attempt to make GPT-4 entirely autonomous. Since its launch its popularity soared worldwide within a few months. So far numerous programmers have shown the potential of AutoGPT via YouTube videos. AutoGPT can be used as an agent for internet search and planning, for autonomous coding and debugging, and as an autonomous Twitter bot.[11] "Auto-GPT is an experimental open-source application showcasing the capabilities of the GPT-4 language model. This program, driven by GPT-4, chains together Large Language Model (LLM) 'thoughts', to autonomously achieve whatever goal you set. As one of the first examples of GPT-4 running fully autonomously, Auto-GPT pushes the boundaries of what is possible with AI," reads the description on the GitHub page of the tool.[12]

Further, AutoGPT can leverage the power of GPT-4 to autonomously develop and manage businesses. One of the models is designed to help "increase the net worth of businesses through data-driven decisions."[13] The main objective of the application is to analyse market data, track profitable business opportunities and take action. These actions and strategies may also include buying stocks or other assets, implementing new marketing campaigns, or developing new products.[14]

3.3. ChatGPT and Jobs

Will ChatGPT Takeaway Jobs? In his book, *Why the Future is Workless*, "Tim Dunlop predicts that AI technologies will replace people in the workforce, and lead to significant levels of unemployment and impoverishment for many people."[15] ChatGPT will lead to losses of some jobs: Content writing, marketing and roles that require writing press releases are likely to be most affected. "If you are writing basic code for payroll or say, short 40-word news updates, ChatGPT will replace it. Low-scale, repetitive jobs are likely

11 Bijin Jose, "ChatGPT is passé, meet AutoGPT – the AI that can autonomously develop and manage tasks, accessed April 13, 2023, https://indianexpress.com/article/technology/artificial-intelligence/chatgpt-is-passe-meet-autogpt-the-ai-that-can-autonomously-manage-tasks-8552371/.

12 *Ibid*.

13 *Ibid*.

14 *Ibid*.

15 As cited by McDonagh, *Herald*, 9.

to be done by this tool."[16] However, Jaspreet Bindra[17] says, certain jobs will never be taken over by ChatGPT. Humans will surely upskill and work with AI to boost productivity. That is what humans do. In fact, "Jobs that require high creativity, emotion, sentience, compassion such as a nurse or, let's say, a senior management person making decisions that impact thousands of employees, will not be replaced."[18] "ChatGPT and generative AI will also help you do your job better. If ChatGPT is able to write 30 per cent of your code, you can write the rest 70 per cent in less time."[19]

As the integration of large language models like ChatGPT rapidly advances workplace automation, the HR domain will see a huge shift in the way they function. The once-traditional role of HR professionals may become obsolete and pave the way for newer and advanced HR roles. Multiple HR functions including recruiting, training and HR data management can also be facilitated with LLM models. This shift also presents a unique opportunity for these professionals to upskill and evolve with the changing times, paving the way for exciting and innovative new roles in the field of HR. Employees have to embrace the future of work and take the necessary steps to stay relevant in the face of this transformative technology.[20] Also, ChatGPT could potentially replace jobs - a 'generate' button could theoretically substitute artists, photographers, and graphic designers.

4. Advantages of Artificial Intelligence

In 2021 Sean McDonagh wrote a book, *Robots, Ethics, and the Future of Jobs*. In it McDonagh points out that AI is being used in agriculture, finance, banking, insurance, healthcare, education, policing, retail, construction, trade, military strategies and in the care of young people and the elderly.[21] AI pioneers say ChatGPT, Bing and Bard will push creativity, take over mundane jobs so that humans can do better. AI will take over those roles that do not require expert skills like drafting basic press releases or legal documents. "It will destroy jobs, but it will allow people who have lost their jobs to be more human and do more interesting things."[22] AI will not make

16 Deepansh Duggal, "Career in a world ruled by ChatGPT: Jobs that will or won't survive," accessed on March 6, 2023, https://www.moneycontrol.com/news/technology/jobs-that-will-or-wont-be-replaced-by-chatgpt-10201901.html.

17 Jaspreet Bindra, is a Gurugram-based managing director and founder at Tech Whisperer Ltd (the UK), a digital consultancy for artificial intelligence.

18 Duggal, "Career in a world ruled by ChatGPT"

19 *Ibid*.

20 Vandana Nair, "ChatGPT Takes Over HR," accessed March 8, 2023, https://analyticsindiamag.com/chatgpt-takes-over-hr/.

21 McDonagh, *Herald*, 9.

22 Bhatia, "AI pioneers say ChatGPT."

human skills obsolete as many have been predicting for years. Rather, AI will replace certain tasks where data is manually fed and managed. "Artificial intelligence is artificial, it's not creative… but that's not true anymore," says Albert Meige.[23] Meige sees AI as a tool that will push creativity forward and not the other way around. "It's the same story as when photography was invented. Painters thought that photographers are taking the easy path. In reality, however, the cameras are still the brain, but a lot depends on the intention of the artist."[24]

Meige recalls how the foundation of the Intelligent Filing Manager, which combined logic programming and applications like Microsoft Excel, was to eliminate the paperwork at large corporations, way before artificial intelligence became the buzzword. Tesla's website describes the humanoid 'Tesla bot' as one that could accomplish "performing unsafe, repetitive, or boring tasks."[25]

ChatGPT can be used by talent-acquisition teams to answer relevant questions about the job description that applicants are likely to have, based on their preferences. ChatGPT can also be leveraged for designing training programmes specific to every employee, based on their role.[26] IBM CEO Arvind Krishna recently said that he believed clerical white-collar work would be replaced by A.I., and that robots could even help with a future labour shortage as the population ages.

5. Disadvantages and Limitations

It's impressive how ChatGPT or AI-powered Bing can produce remarkable results, making searching faster and smoother, given their ability to converse like humans. But AI experts agree that AI-based bots might be fast to respond, but they are not emotional, culturally sensitive, or intuitive like humans.[27] Besides, they warn against trusting these AI chatbots blindly as they still make factual mistakes. Meige is of the view that "[These] chatbots output words in a probabilistic manner…sometimes the answer is factually wrong. No matter how advanced these chatbots become, they cannot replace

23 Albert Meige is Global Technology Futurist at Arthur D. Little chimes in. The quote is from Anuj Bhatia's article.

24 Bhatia "AI pioneers say ChatGPT."

25 Prarthana Prakash, "Elon Musk says that A.I. 'humanoid robots' will eventually outnumber people: 'It's not even clear what an economy means at that point'," accessed March 3, 2023, https://finance.yahoo.com/news/elon-musk-says-humanoid-robots-221414098.html?guccounter=1.

26 Duggal, "Career in a world ruled by ChatGPT."

27 Bhatia, "AI pioneers say ChatGPT."

a journalist yet. When you are a journalist, fact-checking is one of your key skills and that is not yet implemented in AI tools like ChatGPT."[28]

According to a report released at the beginning of March 2023, AI Chatbot ChatGPT failed in the UPSC Prelims Exam. ChatGPT could only answer 54 out of 100 questions from Question Paper 1, Set A. *The Analytics India Magazine*, which tested the chatbot's performance noted that the AI chatbot failed since it could not clear the 87.54 per cent cut-off of 2021 for general category students. The questions ranged from subjects such as geography, economics, history, ecology, science to current affairs, social development, and polity. For, passing the UPSC prelims exam requires not only knowledge but also critical thinking skills, application abilities, and time management skills. Also, because ChatGPT's knowledge is time-bound/time-specific and so it cannot answer questions on the very present moment. While ChatGPT can efficiently collect information on the internet, it can't react to a change in temperature like a cultured cellular system can.[29]

Musk isn't the only one to point out a looming economic disruption because of the A.I. and robot disruption. Robots have already been rolled out in factories to replace jobs that human beings used to do, and 2022 saw record demand for automation as companies struggled to hire workers. But recent A.I. developments might only accelerate that switch. And Kai-Fu Lee, an A.I. venture capitalist, said in 2019 that the technology would replace 40% of jobs within the next 15 years, including both white-collar and blue-collar work.[30]

Generative AI models are destructive to the environment and can contribute significantly to global warming, by needing enormous amounts of energy. For instance, training a generative AI model just once with 213 million parameters can spew CO2 equivalent to that of 125 New York-Beijing round flight trips; and GPT3 has 175 billion parameters. An article in *The Guardian* revealed that data centers currently consume 200 terawatt hours per year, roughly equal to what South Africa does; by 2030 this could equal Japan's power consumption.[31]

6. Google's Gemini

Google released Bard AI in response to competitor artificial intelligence (AI) systems like ChatGPT. Later it was re-named as Gemini. Gemini was officially

28 *Ibid*.

29 Anoushka Sharma, ed., "AI Chatbot ChatGPT Fails UPSC Prelims Exam: Report," accessed March 3, 2023, https://www.ndtv.com/feature/ai-chatbot-chatgpt-fails-upsc-prelims-exam-report-3832449.

30 Prakash, "Elon Musk says."

31 Jaspreet Bindra, Opinion: "With ChatGPT, The Ethical Time Bomb Is Ticking," accessed February 27, 2023, https://www.ndtv.com/opinion/with-chatgpt-the-ethical-time-bomb-is-ticking-3806446#pfrom=home-ndtvopinion.

launched in December 2023 as Google's most advanced and synthetic AI model ever. Gemini is built to be multimodal, meaning it can generalize, operate, and combine across many different types of information including text, audio, images, and video. It employs a combination of machine learning and natural language processing to replicate human conversations while giving you accurate and practical answers to your concerns. Such tools may prove especially helpful for smaller businesses looking to provide consumers with natural language support without having to hire big teams of support personnel or for enhancing Google's search tools. Analysts indicate that it will alter how people use the Internet and that many individuals will benefit from it.[32]

7. The Ethical Question of AI – ChatGPT and Gemini

Among the most fundamental and powerful technologies in the digital arsenal is Artificial Intelligence. Most often, like electricity, AI has been playing behind the scenes, but the bombshell release of ChatGPT by OpenAI has brought the untrammelled power of AI to the masses. Once again it is worth noting that ChatGPT garnered an unprecedented 100 million users in the first two months of its launch; Facebook took 4.5 years.

As Christian Philosophers the Ethical aspect of AI should be one of our main focuses. The thing to remember about Generative AI models, including ChatGPT, is that they are not optimised for the truth, they strategize to be plausible instead of truthful. Now let us begin with basic ethical issues.

The starting point of ethical concerns is the very making of ChatGPT. In calling ChatGPT "artificial intelligence", we grant them a claim to authorship that is simply untrue. Each of those tokens used by programmes like ChatGPT – the "language" in their "large language model" – represents a tiny, tiny piece of material that someone else created. And those authors are not credited for it, paid for it, or asked permission for its use. According to Jim McDermott "ChatGPT is not 'artificial intelligence' – it's theft."[33] In a way, ChatGPT and its ilk are the highest form of separating labourers from the fruit of their labour. We get an answer or piece of art from a predictive text bot and the original articles and ideas from which they were generated are so far removed that even those creators themselves don't realise they have been stolen.[34]

Generative AI models also plagiarise content. AI has the capability to bring back to life the original Beatles ("Beatles' AI resurgence") which might

32 PMY Team, "Google Bard AI Login, How to Use, Benefits, Features & Full Details," accessed March 6, 2023, https://pmmodiyojana.in/google-bard-ai/.

33 Jim McDermott, "ChatGPT is not 'artificial intelligence' – it's theft", *Herald* (28 May 2023), 9.

34 *Ibid*., 9.

allow fans to inhabit a favourite fantasy. But the technology has very real implications for such knotty ethical issues as creative licence and copyright. There are concerns about how easy it is to generate the voices of artists without their knowledge or consent.[35]

The next ethical concern is about the use of ChatGPT. Since its release, researchers have been grappling with the ethical issues surrounding its use, because much of its output can be difficult to distinguish from human-written text. After users posted about how ChatGPT is helping them draft prose, poetry and computer code, Musk had raised a red flag calling the AI "dangerously strong."[36] Cyber-criminals are also using ChatGPT to create Telegram bots that can write malware and steal data.[37]

On 8 May 2023, police in China's northwestern Gansu province arrested a suspect surnamed Hong for "using artificial intelligence technology to concoct false and untrue information." Hong allegedly generated fake news of a train crash which claimed that nine people had been killed in a local train accident on 25 April 2023. Hong disseminated the fake news online, posting it on multiple accounts, using artificial intelligence technology. This was reported to be China's first arrest for misuse of ChatGPT.[38]

With ChatGPT clearing the US's medical exam, students, university professors and parents are concerned about how the AI chatbot is likely to impact schools, colleges, and the workplace.[39]

7.1. Critique of Chatbots

Iconic leftist intellectual Noam Chomsky says chatbots are 'marvels of machine learning' but also the banality of evil, rebooted.[40]

Well-known AI researcher Timnit Gebru was working with Google when she co-wrote an influential research paper calling the models like ChatGPT "stochastic parrots", because they just spouted words without understanding

35 Editorial, "Express View on AI and the Beatles: The fans won't let it be," accessed June 8, 2023, https://indianexpress.com/article/opinion/editorials/express-view-on-ai-and-the-beatles-the-fans-wont-let-it-be-8651290/.

36 Sharma, "AI Chatbot ChatGPT."

37 IANS, "Apple gives nod to ChatGPT-driven app amid concerns," accessed March 4, 2023, https://economictimes.indiatimes.com/tech/technology/apple-gives-nod-to-chatgpt-driven-app-amid-concerns/articleshow/98411285.cms?from=mdr.

38 PTI, "First ChatGPT arrest in China over fake train crash news," accessed May 8, 2023, https://indianexpress.com/article/world/chatgpt-arrest-china-fake-train-crash-news-8598413/.

39 Duggal, "Career in a world ruled by ChatGPT."

40 Tristan Bove, "Iconic leftist intellectual Noam Chomsky says chatbots are 'marvels of machine learning' but also the banality of evil, rebooted," accessed March 9, 2023, https://fortune.com/2023/03/08/noam-chomsky-ai-chatgpt-are-not-true-intelligence/.

them. Similar to parrots, ChatGPT does not understand what it says, nor does it care. Gebru, Marcus and other scholars and academics have been repeatedly pointing out the dangers and limitations of Generative AI models, but their warnings are drowned out by the sheer excitement around ChatGPT. Gebru, in fact, was fired from Google shortly after she wrote her seminal paper.[41]

8. The Future of Humanity from AI Point of View

Indian-origin American computer engineer Krishna C. Mukherjee has spent much of his career in the field of artificial intelligence – and he thinks the relationship between humans and AI is a symbiotic one. For Mukherjee, AI is a collaboration between humans and machines to achieve optimal results and that hasn't changed since the time he started work on the new technology stream in the late 1980s.[42] But the biggest question around AI is whether it will replace humans at jobs. Meige thinks, "AI is nowhere close to replacing humans, but humans using AI in a smart way will definitely be going to replace humans who don't use it." AI bots can help boost creativity if you know how to take advantage of them.

Humans and AI are like a group from the same team working towards a common goal. But like humans, AI is also not free from bias, after all, the data used to train the model has been human-generated. Misrepresentation of data, lack of diversity or not enough data could be the sources of bias in AI. If our training data ignores a certain demographic group or ethnicity, there is the risk that beliefs and biases can influence how the chatbot responds to the queries and deliver results based on them.[43]

Thomas Hartung[44] argues, "The brain is still unmatched by modern computers. Brains also have an amazing capacity to store information, estimated at 2,500 (terabytes). We're reaching the physical limits of silicon computers because we cannot pack more transistors into a tiny chip."[45] Elon Musk says that AI 'humanoid robots' will eventually outnumber people: 'It's not even clear what an economy means at that point.'[46] Julian

41 Bindra, "With ChatGPT."

42 *Ibid*.

43 *Ibid*.

44 Dr. Thomas Hartung is a professor of environmental health and engineering at the Johns Hopkins Bloomberg School of Public Health and Whiting School of Engineering in Baltimore, Maryland.

45 Ashley Strickland, "Move over, artificial intelligence. Scientists announce a new 'organoid intelligence' field," accessed March 2, 2023, https://edition.cnn.com/2023/03/02/world/brain-computer-organoids-scn/index.html.

46 Prakash, "Elon Musk says."

Kinderlerer[47] claims, "We are entering a new world, where the interface between humans and human constructs blurs distinctions."[48]

While artificial intelligence is inspired by human thought processes, the technology can't fully replicate all capabilities of the human brain. This gap is why humans can use an image or text-based CAPTCHA, or Completely Automated Public Turing Test to Tell Computers and Humans Apart, as an online security measure to prove they aren't bots.[49]

9. The Future Trends

ChatGPT promises to create the following jobs for its users in future:

a) Virtual World Architect: ChatGPT's ability to generate imaginative worlds and environments has led to the birth of the Virtual World Architect role, where professionals bring users' dreams and fantasies to life within virtual realms.
b) AI Language Therapist: ChatGPT has developed the capacity to engage in empathetic conversations and help individuals navigate their emotions and challenges. This job involves using the model's conversational skills to offer therapeutic assistance and guidance.
c) Personalised Learning Coach: Personalised learning coaches leverage ChatGPT's expertise to tailor educational content and create personalised learning journeys for students, ensuring optimal engagement and understanding.
d) Ethical AI Consultant: Ethical AI consultants work hand in hand with ChatGPT, guiding its development and decision-making processes. These professionals help ensure that AI models uphold ethical standards and do not perpetuate biases or harmful practices.
e) Virtual Fashion Stylist: By understanding users' preferences and combining them with fashion trends, these stylists help create digital ensembles, select outfits, and suggest accessories that match individual tastes.

9.1. Beyond Artificial Intelligence

Computers powered by human brain cells may sound like science fiction, but a team of researchers in the United States believes such machines, part of a new field called "organoid intelligence," could shape the future — and now they have a plan to get there. Organoids are lab-grown tissues that resemble

47 Julian Kinderlerer is professor emeritus of intellectual property law at the University of Cape Town in South Africa.

48 Strickland, "Move over, artificial intelligence."

49 CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) is a type of security measure known as challenge-response authentication.

organs. These three-dimensional structures, usually derived from stem cells, have been used in labs for nearly two decades, where scientists have been able to avoid harmful human or animal testing by experimenting on the stand-ins for kidneys, lungs, and other organs. According to Hartung, "Computing and artificial intelligence have been driving the technology revolution, but they are reaching a ceiling. Biocomputing is an enormous effort of compacting computational power and increasing its efficiency to push past our current technological limits."[50]

9.2. How a Biocomputer Might Work

Hartung defines organoid intelligence (OI) as "reproducing cognitive functions, such as learning and sensory processing, in a lab-grown human-brain model."[51] Hartung is of the view that using brain organoids to create organoid intelligence is still very much in its infancy. Developing OI comparable to a computer with the brain power of a mouse could take decades. Gary Miller[52] writes, "OI blurs the line between human cognition and machine intelligence, and the technology and biology are advancing at a speed that could outpace the ethical and moral discussions that are needed. This emerging field must take a vigorous approach to addressing the ethical and moral issues that come with this type of scientific advancement and must do so before the technology crashes into the moral abyss."[53]

10. Regulating AI

Hinton, the godfather of AI stresses the need for regulation. According to him, it is essential that the world invests urgently in AI safety control. He is not the only one calling for regulations. In March 2023, an open letter co-signed by many people in the AI field – including the tech billionaire Elon Musk – called for a pause on all developments more advanced than the current version of the AI chatbot.[54] Several governments have been considering how to mitigate the dangers of the emerging technology, which has experienced a boom in investment and consumer popularity after the release of OpenAI's ChatGPT in November 2022. Here it's significant to take a glance at the Blueprint for an AI Bill of Rights released by the

50 Strickland, "Move over, artificial intelligence."

51 *Ibid*.

52 Gary Miller, vice dean for research strategy and innovation and professor of environmental health sciences at Columbia University in New York City.

53 Strickland, "Move over, artificial intelligence."

54 Sean McDonagh, *Herald*, 9.

White House Office of Science and Technology Policy (OSTP)[55] at the very beginning of April 2023. The five core protections the Bill envisages are:

a) the systems that we use should work and they should be safe and effective. And if they're not safe and effective, they should not be used.
b) you should be protected from algorithmic discrimination, and automated systems should be used and designed in an equitable way.
c) you should be protected from abusive data practices via built-in protections, and you should have agency over how data about you is used.
d) you should know when an automated system is being used and understand how and why it contributes to outcomes that impact you.
e) you should have the opportunity to opt out: to have access to a human being or another way to access the right, the opportunity, the thing that you're trying to do, or another way to quickly solve a problem.

It is hoped that these principles outline the kind of world that we all want to live in, the world that we should expect. It will take all sorts of different levers to get us to this world. Above all, Governments need to regulate. Companies need to bring codes of practice not just to avoid rule-breaking, but actually to become ethical businesses. Governments should also address the inequity in terms of the digital divide, knowledge gap, and the subsequent wealth gap caused by technological advancement. It may be mentioned here that in the latter part of 2023 Britain is to host a global summit on AI safety. Those formulating regulations must include, not just people who understand AI technologies, but sociologists and philosophers.

11. Warning

In the beginning of May 2023, Hinton, often portrayed as the godfather of artificial intelligence, quit his job, and warned about the growing development in that technology.[56] Scientists for Global Responsibility, a campaigning group comprising of scientists and engineers, surveyed its 750 members about the effects that AI would have on their future. Almost all their members felt that AI would entrench and deepen inequality in

55 John Slattery, "Artificial Intelligence, Public Policy, and Faith," accessed April 15, 2023, https://international.la-croix.com/news/ethics/artificial-intelligence-public-policy-and-faith/17653.

56 McDonagh, *Herald*, 9.

society, as more and more power and benefits would flow to the large digital corporations rather than to the ordinary citizen.[57]

Conclusion

From the foregoing discussions it is very clear that after all artificial intelligence is a byproduct of natural intelligence – achieved it would seem by stretching the capabilities, imaginations, ingenuity, powers, and talents of the natural intelligence to its utmost level. The dizzying heights to which AI has reached and still advancing is a glorious testimony to the worth of the human species, created in the image and likeness of God. Ultimately, all human achievements are reflections of the divine intelligence – the exemplar cause of everything. Anyone, including the pioneers and great minds behind AI, not acknowledging this fact is not being true to himself/herself. Just as science and scientific research are perennial, AI which is very much a part of the sciences is certain to continue its forward march into the future creating an ever-new world order in which human species are not undervalued and not totally replaced. After all, machines cannot have a soul; only humans can have.

57 *Ibid*., 9 and Agrawal *et al*., 212-214.

2

Panikkar's Critique on Technocratic Civilization and Artificial Intelligence

K. Maria Delasal Viswas

Abstract

Life without artificial intelligence in our times is something unthinkable even for simple people in the rural areas. Artificial intelligence has engulfed the human lifestyle. On the one hand, it has become a necessity and on the other hand, it tries to replace human intellectual and other abilities. The speedy pace with which it advances may provide a mirage of positive benefits, however, it can also be a warning. Although it may seem so, the observations by a creative thinker, Raimon Panikkar had given an alarm long ago. His direct comments and remarks on artificial intelligence seem limited; however, he did not fail to capture its operative notions under the wider picture of modern science and technocratic civilization in the decades to come. On this account, his assessment and warnings cannot be ignored and he calls us for an introspection into the human attitudes in connection to technoscience and artificial intelligence. Ultimately, his observations focus on pointing out the lagging of the significant and core aspect of human life, i.e., the rhythm of being.

Keywords**:** *Raimon Panikkar, artificial intelligence, technocratic civilization, ecstatic, enstatic, Science*

Introduction

As part of the scientific advancement, artificial intelligence occupies an important space. Many experiments and research speedily revolve around this field currently. In our times, artificial intelligence presents itself as a contradiction. On the one hand, life without artificial intelligence seems to be unthinkable for its compelling attraction and hence has become a necessity wherein it has become part and parcel of our day-to-day lifestyle. On the other hand, artificial intelligence propels into the foreground almost projecting as a threat or danger wherein human potentials are replaced by the functionality of artificial intelligence which has led humanity to be over-dependent on it. As a result, there is the reduction of human labour. Already, there are warnings that artificial intelligence may overpower humans.

At this juncture, keeping the acceleration of artificial intelligence within the context of scientific-technological advancement, an assessment of it from viewpoint of Raimon Panikkar (1918-2010) would be of insightfulness and caution to us. From his early writing period, Panikkar addressed modern technological civilization while the field of artificial intelligence was in its germination stage to attain the shape that it has reached today. Panikkar deals with artificial intelligence, though not elaborately but under his discourses on technocratic civilization. However, his warnings on technocratic civilization lead us to have our eyes on the theme under our discussion. Moreover, our lifestyle which has been interwoven with the advantages of artificial intelligence calls for watchfulness, and Panikkar's warnings cannot be ignored. Therefore, in this paper, I will make attempts to situate and assert, from Panikkar's standpoint, that artificial intelligence is part of 'technocratic civilization' and proceed to present some of his assessments on it.

1. Preliminary Observations

The last published book by Panikkar, titled *The Rhythm of Being*, just before he died in 2010, mentions artificial intelligence in a few places. Manimala, who had attempted to study and present an analytical summary of this book, witnesses that this book was much awaited and many consider it as his masterpiece as it sums up his mature thought.[1] The book contains the Gifford Lectures given by him in 1989; he delayed the publishing of it under the

1 Varghese Manimala, 'The Rhythm of Being: Analysis of Reality from an East-West Perspective', in *Raimon Panikkar: Being Beyond Borders: A Commemorative Volume*, ed. Johnson J. Puthenpurackal (Bangalore: Asian Trading Corporation, 2012), 78.

pretext that it is not complete. In this book, in some places,[2] Panikkar remarks about artificial intelligence under the wider context of modern science and technology. Especially in the last chapter, he dedicates more space to address on this wider context.

While considering Panikkar's assessment of artificial intelligence under the wider context of modern science and technology, it is notable that he does not treat it as a separate entity as if its existence, evolution, and operation are apart from human and divine connections. Primarily, artificial intelligence is the outcome of human minds' access to cosmic substances. As a preliminary observation, this connectedness needs to be kept in mind, while seeking to imbibe Panikkar's remarks on artificial intelligence. He attempts to expose the human attitude that is operative behind this whole endeavour. I would say, in his exposition, it could be sensed how the human attitude is more akin to the Tree of Knowledge than the Tree of Life.

2. Predominance of the Tree of Knowledge

In his *Epilogue* (which Panikkar had finally added to the book), gaining insights from the initial pages of the Old Testament from the Bible, metaphorically he states, "The Tree of Knowledge again and again tempts one at the cost of neglecting the more important tree, the Tree of Life."[3] I consider this statement as an indication of Panikkar about the ignorance we are enveloped with. The Tree of Life is more important than the Tree of Knowledge. I wish to suppose that the latest fruit of the Tree of Knowledge is artificial intelligence which again and again tempts one at the cost of neglecting the fruit of the Tree of Life, i.e., the rhythm of being, and especially, the rhythm of human life. In what way the Tree of Knowledge makes us neglect the Tree of Life is an important question to be dwelt upon. There are insightful observations from Panikkar to become aware of the predominance of the Tree of Knowledge above the Tree of Life.

Panikkar highlights the important transition that has happened in human consciousness in the course of history. There was the transformation from *ecstatic* to *enstatic* moment of intelligence. By the word *ecstatic*, Panikkar means to say that Man[4] just knows. During the *ecstatic* moment, there was a lack of self-awareness in Man that he knows. The word *ecstatic* refers to the

2 Raimon Panikkar, *The Rhythm of Being: The Unbroken Trinity*, Original in 2010 (Maryknoll, New York: Orbis Books, 2013), 10, 44, 291, 327, 366 & 397.

3 *Ibid*., 405.

4 For Panikkar, the term 'Man' stands for androgynous human being and not 'male.' Hence the term 'Man/man' in Panikkar's thought means 'human person.' Panikkar notes, "Before Homer, the term did not solely connote the masculine, and in compounds it stood for the human… So, the expression could, and indeed should, be understood in its original meaning of human being". See: Raimon Panikkar, *The Cosmotheandric Experience:*

passive nature of the human mind. Man just knew without putting the content of the mind into further analysis. It was a pre-reflective state. In short, it means that Man was conscious but not self-conscious. Man did not know that he knows. There was no self-knowledge that he was a knowing being. From this state of consciousness, Man moved to the *enstatic* moment of intelligence. Panikkar goes on to say that, here, Man became a being who is conscious of him. Man was not just conscious of things around him, but became conscious that he was a knowing being. By using the term *enstatic*, Panikkar refers to the active nature of the human mind.[5] During the *ecstatic* moment of intelligence, certainly Man was conscious of nature without separating himself from nature. Man's relationship with nature was not essentially different from his relationship with his fellow beings.[6] But during the *enstatic* moment of intelligence, as Protagoras articulated 'Man is the measure of all things,' by discovering the laws of the cosmos, Man slowly came to discover his own *nomos* (laws), and he became more and more aware that his mind, his *nous*, is the criterion of intelligibility and perhaps even of reality. After wondering about nature, he began to wonder at his mind and was awestruck to see that the physical universe seems to follow the laws his mind discovers and can formulate.[7]

Panikkar metaphorically connects this transition to the symbol of apple from the Tree of Knowledge which is found in the first few pages of the Book of Genesis in the Bible. The apple from the Tree of Life which Adam and Eve were permitted to eat and enjoy marks the *ecstatic* moment of human intelligence where they were just knowing and being conscious of the apple. They did not see anything other than the apple itself. But when they were led to eat the apple from the Tree of Knowledge which was prohibited to them, they were not just conscious of the apple, but also conscious of something 'other' than the apple. Here, began the *enstatic* moment of human intelligence. They became aware not only of the apple, but became aware that knowing the apple is not all there is to know, and therefore that knowing the apple does not exhaust their knowledge. In other words, they became aware of the limits of their thinking, and thereby their own limitations. Panikkar exposes that the apple no longer satisfied them because they also wanted to know non-apple, but that is not the end of it. They wanted to know more than both apple and non-apple. They wanted to know everything, i.e., God, which is here the symbol that stands for the totality.[8]

As observed by Panikkar, this vehement passion for knowing marks the spirit of science in human history. In this regard, he reveals that the scientific mentality characterizes *enstatic* moment or period.[9] The loss of sight on the Tree of Life due to the predominant interest in the Tree of Knowledge evolves as Panikkar's assessment of science. He observes that "whatever we call scientific progress is nothing other than the expansion of specialized sciences that divide themselves more and more in order to enlighten us less and less. The real problem, however, is that this method has become a serious requirement for us and we now regard its analytical path as 'natural.'"[10] In a similar tone, at various points of his writing, he places his assessment of science, its worldview, and its offshoots. However, before looking into it, our theme under focus could not be sidelined and therefore, an understanding of artificial intelligence as science and as part of technology needs to be articulated first to situate it under the wider context of modern science and technology.

3. Artificial Intelligence as Science

My primary argument is to assert artificial intelligence as 'science' because situating the exploration properly is of logical importance. The field of artificial intelligence needs to be seen as the fruit of modern science. It is not just a perspective but factual too. The advancements of modern science have led to the point of being benefitted by artificial intelligence. Having said this, the question arises whether artificial intelligence is strictly categorized within the field of science. There is a wider consensus that artificial intelligence is a field of computer science. But the question persists whether computer science is a science in the real sense. This question arises because science requires empirical observations and since computer science does not involve any empirical observations, can it be claimed as science? There have been debates for and against it since the emergence of computer science.

Computer scientists like Allen Newell and Herbert A. Simon argued back in 1975 that computer science is an empirical science. Their basic logic is that computer scientists are engaged in the study of phenomena about computers and scientific study is about studying phenomena, so computer

Emerging Religious Consciousness, ed. Scott Eastham, Original in 1993 (New Delhi: Motilal Banarsidass Publishers Private Limited, 1998), 54–55.

5 *Ibid*., 48–49.

6 *Ibid*., 25.

7 *Ibid*., 33.

8 *Ibid*., 48–50.

9 *Ibid*., 32.

10 Raimon Panikkar, *A Dwelling Place of Wisdom*, trans. Annemarie S. Kidder, Indian Edition (New Delhi: Motilal Banarsidass Publishers Private Limited, 1995), 10.

science is a kind of empirical science.[11] Moreover, artificial intelligence which we acknowledge as part of computer science, operates based on observing data. If the human eye is the sense, then in artificial intelligence the same is replaced by cameras, which capture not only the object but also its distance with accuracy. Based on these captured data, it further processes, then identifies the objects and distinguishes the same from other objects. Although its observations or perceptions cannot be equalled to human sense experience, still in its level or state, perception takes place, wherein it could be claimed to be part of empirical science, if we wish to argue that way. Apart from this, it is also argued that computer science is an engineering science, wherein the field of artificial intelligence is claimed as an engineering science.[12]

Moreover, if we deeply analyze the nature of artificial intelligence, the same shows forth as part of science. One of the pioneers in the field of artificial intelligence, Marvin Minsky offers a basic understanding of artificial intelligence thus: "the science of making machines do things that would require intelligence if done by men."[13] This definition points out that it is the machine with intelligence although that intelligence cannot be equalled to human intelligence. They are operations of advanced machines through algorithms, data structures, and programming languages. In comparison to the machines that were operated during the Industrial Revolution, artificial intelligence machines are automata and hence they are mostly not human-dependent to be functional. But the point is that whether automata or not, they are machines that are fruits of science in general.

The point I would like to drive is that although artificial intelligence is a field of computer science, it is part of scientific advancements and none can claim it to be out of the scientific framework. In short, we cannot conceive of artificial intelligence outside science. This assertion is of primary importance to imbibe Panikkar's critique of modern science. Also, it would not be the right sequence to access the assessment of Panikkar on artificial intelligence without having a glance at his critique of modern science. This sequence is of importance while understanding Panikkar without which his remarks on artificial intelligence would not provide a better comprehension.

11 See: Robert W. P. Luk, "Insights in How Computer Science Can Be a Science," *Science and Philosophy,* 8, no. 2 (2020): 17-46, 18, https://doi.org/10.23756/sp.v8i2.531.

12 *Ibid*.

13 Marvin Minsky, ed., *Semantic Information Processing* (Cambridge: MIT Press, 1968), v.

4. Panikkar's Critique of Modern Science

Panikkar was one who never failed to place his critique on modern science consistently,[14] despite his positive note about the benefits science has offered to humanity.[15] His critique is not on the whole of science itself, but on the modern attitude that envelopes science. Science is not a field that originated from modern times but it had its beginning even much earlier when Man began to think rationally. For this reason, the philosophers of the ancient times are also called as primitive scientists. Especially, the pre-socratic philosophers were scientists because they tried to offer cosmological views irrespective of supernatural or mythological explanations. Their speculations were untouched by religious influences.[16] Their scientific mindset had the vigour of a genuine search wherein the importance to Man was not sidelined. Moreover, their intention was not to throw away the supernatural elements while they were exposing their scientific spirit. But there had been an attitudinal change in the approach towards science heavily since the modern times which Panikkar never missed to capture and assess. For this reason, Panikkar qualifies the usage of the word 'science' with the prefix 'modern' and calls as 'modern science' to denote that the science of our times is no longer *science* in the traditional sense.[17] And what makes modern science specifically different is its method due to which it is unable to furnish a complete worldview. The worldview it offers is only a description of the quantifiable parameters of the world.[18] This is the point of concern here, to recall Panikkar's critique of modern science and later, to place the field of artificial intelligence within it for further assessment.

Panikkar keenly observes that there is a fundamental difference between what scientific cognition conveys and what human knowledge offers. He prefers to use the word 'cognition' in application to science to show the fundamental difference. Scientific cognition is restricted to those phenomena that appear to the scientific method. In that way, the components of the earth and even of living beings have been analyzed which has led to the prediction of the behaviour of celestial and terrestrial bodies, and so on. But still, scientific

14 Raimundo Panikkar, 'The Contemplative Mood: A Challenge to Modernity', *Cross Currents,* 31, no. 3 (Fall 1981): 261–72; Also see: Panikkar, *The Cosmotheandric Experience: Emerging Religious Consciousness*, 1993; Raimon Panikkar, *Cultural Disarmament: The Way to Peace*, trans. Robert R Barr, Spanish Original in 1993 (Louisville, Kentucky: Westminster John Knox Press, 1995), 46-53,79-92 and Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 2013, 368–404.

15 Panikkar, *The Cosmotheandric Experience: Emerging Religious Consciousness*, 36.

16 Samuel Enoch Stumpf, *Socrates to Sartre: A History of Philosophy* (New York: McGraw Hill Book Company, 1982), 4.

17 Panikkar, *Cultural Disarmament: The Way to Peace*, 86 (Endnote 15).

18 Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 369.

discoveries are one particular case of knowledge. Properly speaking, Panikkar says that knowledge is not the result of the activity of our reasoning mind that, after being fed some data, functions like a sophisticated computer. When we truly know something, we become that something, we embrace it as part of ourselves. Knowledge is not an activity of our mind alone; it is an act of our entire being; ultimately it is an act of the whole being. Knowledge is not a mere epistemological device; it has an ontological nature, and love is an intrinsic part of the act of knowing.[19] Panikkar tries to bring to our awareness that modern science occupies only a partial space in the whole adventure of knowledge, and therefore, it misses out on what is given to human knowledge. In this regard, it is still enveloped with limitations.

Moreover, pure speculation, as a human activity, leads to seeing and enjoying reality where there is a direct encounter with reality without any mediation or interference. Speculation is a philosophic activity too. But this significant activity seems to be lacking in modern science. Panikkar indicates that modern science is not based on pure speculation but on the experiment into reality, which is interference or tampering with reality to foresee its behaviour and control it. For this, modern science engages itself with instruments. To deal with them we do not need to think but rather calculate, not experience but experiment. Due to heavy reliance on instruments and experiments, Panikkar notes, inwardness is taken for granted or forgotten.[20]

5. Metaphor of *Indra's Cunning*

"Indra's Cunning: The Challenge of Modernity: The Indic Experiment,"[21] is a work of Panikkar seeing its final manuscript in 1995[22] but made available to the public in print form in the second part of the fourth volume of Panikkar's *Opera Omnia* in 2017. A study of this contribution of Panikkar by D'Sa (who edited this written text of Panikkar) and Amaladass, the collaborators of Panikkar, expresses the significance of this recent contribution. They acknowledge and show Panikkar's critique of modern science and alternative thinking.[23] In this work, Panikkar relates the state of modern science

19 *Ibid*., 370.

20 *Ibid*., 387.

21 Raimon Panikkar, "*Opera Omnia, Hinduism: The Dharma of India,* Vol. 3, part 2, ed. Milena Carrara Pavan, First Indian Edition (Delhi: Motilal Banarsidass Publishers Private Limited, 2019), 273–375.

22 *Ibid*., 273 (Note 5).

23 Francis X D'Sa, 'Raimon Panikkar's Critique of Modern Science', in *Raimon Pannikar: Intercultural and Interreligious Dialogue*, ed. Joan Vergés Gifra, Kindle (Girona: University of Girona, 2017), 24–41; Also see: Anand Amaladass, 'Panikkar's Quest for an Alternative Way of Thinking and Acting', in *Raimon Pannikar: Intercultural and Interreligious Dialogue*, ed. Joan Vergés Gifra, Kindle (Girona: University of Girona, 2017), 43–65.

and technology symbolically to the Vedic god *Indra*. I suppose that this metaphorical connection he makes to modern science is equally applicable to the latest advancements in the field of artificial intelligence.

Panikkar says that today "Indra, the highest of the Vedic Gods [….] has made himself invisible and taken the form of modern science and technology."[24] The reason for Panikkar to relate *Indra* to modern science, he says, "the figure of Indra remains most bewildering and fascinating, full of contradictions and ambiguities. To me, he is paradigmatic of the human condition and charged with a powerful symbolic force for our situation. [….] Indra is the symbol of the demonic and the divine forces struggling within Man."[25] Also, he says, "[Indra's] mesh extends as far as modern times. Indra was not born a God, much less the divine Lord of the pantheon. His deeds were so beneficial and portentous, however, that he was enthroned as Lord of lords. Likewise, technology was not born divine, but… [Man has] raised science and technology to the highest Himalayan peak to lord over all the country and redeem us from our backwardness. In variously conscious and acute forms, technocracy seems to be regarded in many strata… as the highest God."[26] In placing the character of *Indra* as "both the compassionate helper, a friend and brother to humans, and the destroyer, the killer and the cheating God,"[27] Panikkar metaphorically presents the double edges of modern science as on the one side, in itself is not a bad thing, and on the other side, all the more, being harmful without delivering the promised liberation.[28]

In exposing the allurement of modern science, Panikkar metaphorically indicates the following: the cunningness of *Indra* lies in secretly inviting Man to rise to such a height and then abandon him to his destiny, entangled in the megamachine he had advocated; the cunningness of *Indra* lies in encouraging Man towards a wandering inquiry without telling him the reason for such a fruitless effort; the cunningness of *Indra* lies in mesmerizing Man to prevent him from realizing that at least part of the destiny of the human race might lie in their hands; the cunningness of *Indra* lies in keeping Man in the slumber of believing that Man rules the universe by reducing reality to the structures of human thought. Therefore, Man has to get awakened out of this cunningness to realize that his vocation is of a cosmic, divine nature by disentangling himself from this temptation.[29] From all these, it is indicative that there is a

24 Panikkar, *Opera Omnia,* 363.

25 *Ibid*., 364.

26 *Ibid*.

27 *Ibid*., 363.

28 *Ibid*., 274–90.

29 *Ibid*., 368–71.

necessity to slow down the speedy progress that happens in the scientific field and more in particular, in the field of artificial intelligence.

6. Technocratic Civilization

To have a sharper assessment of Panikkar regarding artificial intelligence, the focus needs to be sharper by narrowing the view of artificial intelligence within the current technological civilization than the wider range of modern science. In this regard, artificial intelligence has to be acknowledged, specifically as part of scientific technology which in turn lies within the wider picture of modern science. Scientific technology refers to the application of modern scientific methods which include in the current times, robotics and computer sciences. As part of modern and postmodern civilization, scientific technology has become inevitable. Many things are made to operate through commands from a distance through remote controls where sensors have become significant in scientific technology. Thumb impressions, biometrics, and eye scanners have become the new security guards replacing human efforts. All these are functionalities of artificial intelligence. If modern science is an advancement in comparison to traditional science in general, then, scientific technology is still an advancement in our contemporary times. The writings of Panikkar indicate that artificial intelligence is nearer to the recent technological civilization.[30] At various sites in his writings – although scattered – Panikkar puts forth his critique of technological civilization. Here below, I would like to categorize in my style under different titles, a few indications from Panikkar that could not be missed and need a serious outlook about technocracy and its most important occupant, i.e., artificial intelligence.

7. Power and Dominance

Panikkar points out that the present civilization with its day-to-day happenings is a technocratic civilization that is the product of technoscience.[31] The word 'technocratic' is notable here. Panikkar wishes to use the term *technocracy* instead of *technology* to indicate the change in the approach to technology by Man. He notes that there is a qualitative leap between technics, understood as the Greek *technē*, and modern technology. In other words, modern science is technocratic than having the sense of *technē*. Panikkar differentiates between *technē* and *technocracy*. According to him, *technē* represents a human invariable which includes "art, artifacts, the manipulation of nature, elementary or first-degree machines, the

30 Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 291–92; Also see: Panikkar, *Cultural Disarmament: The Way to Peace*, 83–86.

31 Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 291.

mentality of arts and crafts."[32] In contrast, modern technology is the fruit of one civilization with rational organization. While understanding the word 'technology' as 'the study of technics,' *technē* does not exclude such a sense of technology. Therefore, to differentiate the modern attitude towards technology, Panikkar prefers to connote it with *technocracy*. He says, "Technocracy represents the step from *technē* as art, as craftsmanship, to technics as domination, as *kratos*, as power."[33] In connection to Man, he states, "modern Man does not only live on technocracy (live from, and live under, the power of technology)."[34]

8. Fragmentation and Separation

Panikkar remarks that technocracy is bound to objectivity searching for objective, external solutions to any problem. The primary importance is the end product, which meets with the loss of a just means. The same happens by employing speeding up processes while considering time as an economic commodity. There is a meditated exercise towards objectification where one cannot separate the end from the means. In such an objectification process, humans who become the means of production become distanced from the end product leading to the fragmentation and separation between the end and the means. In this separation, the wholesomeness of the relationship between the doer, the doing, and the done is lost. In other words, this wholesomeness which is the primary motive of *technē* is lost.[35] Also, here, the significance is towards experimentation rather than experience.[36]

These observations of Panikkar quite well fit with the operations in the field of artificial intelligence. In applying the observations of Panikkar to artificial intelligence, the end product is important; time is perceived as an economic commodity where the processing of data is done in less time which human intelligence could not achieve at such a speed; there is a missing of wholesomeness where the human person is alienated or distanced from the activities of artificial intelligence to the extent that the available individual human resources are not actualized, and there is greater importance to experiment and also the replacement of humans with robots. Here, it is fitting to note what Panikkar states, "Technocracy …. represents the abandonment of the measure proper to Man and its replacement with the measure proper to the machine."[37]

32 Panikkar, *Cultural Disarmament: The Way to Peace*, 47.

33 *Ibid*., 48.

34 *Ibid*., 84.

35 Panikkar, *Opera Omnia*, 324–25.

36 Panikkar, *Cultural Disarmament: The Way to Peace*, 85.

37 Ibid., 83.

9. Suspension of Uniqueness

Another feature of technocratic civilization, Panikkar points out is the exercise of abstraction. He remarks that modern technoscience "has to operate with abstractions, which we may call numeric measurements. There is something fascinating and grand with mathematical algorithms. The confusion begins when scientific terms are interpreted as real words. Due to the spectacular success of modern science, people take scientific abstractions for the reality from which they have been abstracted."[38] There is a large play of quantitative parameters based on which abstraction operates. In this exercise of abstraction by advanced technocratic machines, and in particular, artificial intelligence, Panikkar keenly brings to our notice that abstraction leaves out completely the element of uniqueness. In other words, what remains outside the scientific abstraction is the character of uniqueness.[39] Panikkar states, "Modern science needs some regularity, some repetition, some implicit laws at least. Anything unique escapes the very field of modern science. Now, any human experience is unique, any conscious act of Man acquires its humanness precisely because of its irrepeatability, its uniqueness."[40] In this observation, Panikkar's concern is about the very method of knowing which is substituted by calculation and numerical measurements.

In this connection, Cullell, who has been inspired by the insights of Panikkar, makes an important observation that there is a methodological reduction of things to other parameters in modern science. These are in turn reduced to the information we can obtain from them. This information, though helpful to predict and control the behaviour of things, and also is profitable to a large extent, should not be absolutized to the point of disregarding other significant areas of life, i.e., the uniqueness reserved for human beings and their identity and intelligence.[41]

Further, Panikkar writes, "In technocratic civilization, immediacy, if not spontaneity, has been lost, since nearly everything is calculable, and, at least statistically, foreseeable."[42] In the context of the technocratic civilization, Panikkar's hitting comment is "We live and move and have our being in an artificial world."[43] This means that we tend to be artificial rather than

38 Raimon Panikkar, 'A Self-Critical Dialogue', in *The Intercultural Challenge of Raimon Panikkar*, ed. Joseph Prabhu (New York: Orbis Books, 1996), 287.

39 *Ibid*.

40 *Ibid*.

41 Jaume Agusti Cullell, 'From Scientific Experiment to Whole Experience Through Freedom', in *Fullness of Life*, ed. Kala Acharya, Milena Carrara Pavan, and William Parker (Mumbai: Somaiya Publications Pvt. Ltd., 2008), 297.

42 Panikkar, *Cultural Disarmament: The Way to Peace*, 84.

43 *Ibid*. 84.

glorifying our uniqueness. As an example, we observe the phenomenon that people are eager to take note of the number of 'likes' in their *Facebook* postings, uploaded *YouTube* videos, and other networking sites which tell about their unconscious attachment to the calculable parameters. Instead of celebrating one's uniqueness, there is an unconscious interest to match the other 'celebrity' in the artificial field.

From another perspective too, there is the loss of the sight upon uniqueness which Panikkar brings to our observation in terms of inclination to universalization. The technocratic civilization has lent its share to this exercise of universalization. The thrust toward universalization or undoubted belief in the possibility of 'universal theory' is a feature of Western civilization. And the way for it was made geographically possible by the technocratic civilization.[44] Behind the proposal or search towards 'universal theory,' Panikkar notes that there is a common trait that either ignores or reduces the immense variety of human experiences to one single and common language. This temptation to ignore diversity and to bring every distinct perspective under one umbrella is a danger. It kills the differences that pervade the cosmos and human life. Reliance towards universalization would blur the specificity of all that is reserved for various expressions. It is like a kind of self-killing of the uniqueness of either the person or a particular culture that is endowed with. In this regard, Panikkar presents the image of the Trojan horse to this phenomenon where the 'natives' introduce it into their own selves or cultures without realizing that they are going to lose themselves.[45]

10. Rupture of Life Rhythm

Apart from the three worlds of the divine, the human, and the material worlds, Panikkar says that there has come about the *fourth* world, which is the global fruit of artificial intelligence. Due to the fourth world, today, we believe that we can live in an artificial world. But due to this, human and natural rhythms are ruptured.[46] Rhythm is a celebrated theme in Panikkar's writings, especially in his last published book, *The Rhythm of Being*. In this book, he states that "we all participate in Rhythm, because Rhythm is another name for Being."[47] Further, he says, "Rhythm is more than the interconnection of

44 Raimundo Panikkar, 'The Invisible Harmony: A Universal Theory of Religion or a Cosmic Confidence in Reality?', in *Toward a Universal Theology of Religion*, ed. Leonard Swidler, Second Printing, Original Edition in 1987, (Maryknoll, New York: Orbis Books, 1988), 120.

45 Panikkar, *Cultural Disarmament: The Way to Peace*, 47.

46 Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 291–92.

47 Panikkar, *Cultural Disarmament: The Way to Peace*, 38.

things or events; it is their *intra* connection, the indwelling of all in all."[48] In the rhythm, there is an undisturbed pace in the duration and movement of time and this rhythm is ruptured because of a different approach to time.

Panikkar notes that the *fourth* world has a time of its own, which is neither nature's nor that of the temporality of culture. It is based on acceleration. If human beings hope to survive in the *fourth* world, they must adapt to this acceleration.[49] In the recent human jobs in IT fields, call centers, and BPOs of the *fourth* world, the employees feel a kind of stress due to the acceleration that is fixed and their inability to cope with the set target each day. This is the loss of sight of the Tree of Life and the failure to be nurtured by it. Panikkar writes, "The modern technocratic system [….] aspires to make everything artificial, including intelligence, precisely in order to escape the natural rhythms of life."[50] The equilibrium is disturbed due to the intervention of technoscience and particularly, artificial intelligence. Panikkar submits that there is a regression from *homo sapiens* to *homo technologicus*, i.e., from 'wise human' to 'man-the-user-of-technology'. The point to be observed is that Panikkar does not assess this change as advancement or progression but rather as a regression. It is a regression because this qualitative change has called for a qualitative restructuring of our existence.[51]

Conclusion

Technocracy, for Panikkar, "promises immediate gratification of immediate needs, even if these are artificially created. Technocracy, however, is not a Messiah for the future, but a Santa Claus for the present – and for the majority of people it does not deliver the goods."[52] This note and what has been enumerated above as critiques of Panikkar seem to place the technocratic civilization and artificial intelligence in a negative tone. But that was not the intention of Panikkar. This is clear from his statement: "We cannot ignore the intrinsic value of scientific discoveries, the advantages brought to humankind by science, the indisputable blessings of modern civilization."[53] Rather, Panikkar's caution is to keep a constant check on the human attitude that ignorantly loses sight of the *Tree of Life* by being excessively attached to the *Tree of Knowledge*.

48 *Ibid*.

49 *Ibid*., 48.

50 Panikkar, *The Rhythm of Being: The Unbroken Trinity*, 44.

51 Cf. Panikkar, 291–92.

52 Panikkar, 10.

53 Panikkar, *The Cosmotheandric Experience: Emerging Religious Consciousness*, 36.

3

Mind, Consciousness & AI
Indian Philosophical Perspectives

Ivan D'Souza

Abstract

The impact of Artificial Intelligence (AI) is ubiquitous. Day by day it is acquiring the nature of omniscience and omnipotence. It is simulating human behaviors, augmenting operations and amplifying mental capabilities in varied ways. AI has brought in a giant quantum shift in our lives, changing the way we perceive and do things and have a profound impact on everything around us. In recent times, we have reached an inflection point in the development of technologies that serve humankind. We are haunted by a big question: Are we mechanizing humanity or humanizing machines? This paper sheds light on some concepts like Mind, Cognition, Intelligence, Consciousness, Self etc., from the background Indian Philosophical Systems. The above topics are debated often when the role of AI is discussed. The study attempts to deal with some relevant topics related to human nature, identity, creativity, rationality, corporeality, ethereality etc. In what ways can humans resemble, differ and outdo AI from Indian Philosophical point of view is a significant question addressed in this paper.

Keywords**:** *Mind, Intelligence, Citta, Antaḥkaraṇa, Consciousness, Self, Ātman, Anuvyavasāya, Sākṣin, Ajñāna*

Introduction

Human world has moved from life to life 3.0. Today Artificial Intelligence (AI) has become natural and new normal to human life. It is an era of new disruptive technology. Let us note two news details. There was a video in which an Octopus climbs a parked car and crushes its front glasses. It propels the viewer to ask: Is it true? Where is it? Initially, it was stated that this viral incident took place at Qatar. Later, the fact check revealed that it was a Computer Generated Imagery (CGI) created using 3D tools. In the world of AI today, we are often blurred and bewildered by the grand virtual demonstrations of media and machines as real and natural.[1]

The Asian Games held in Hangzhou, China in 2023 witnessed a remarkable innovation. The robot dogs made their debut as volunteers to transport javelins across the field. It is first of its kind in global sports events. It reveals that AI is leaving no stones unturned to replace human capabilities and performances. Is it not a new 'AI'VATARA? On the one side, AI revolution is a plume that all covet, but on the other side, it has created dubiety and uncertainty about future. Is AI going to make humans irrelevant, God outdated, mind and consciousness its insignia? What is the place of ethics, religion, tradition, culture etc.? Will humans alone remain as rational beings? Is there anything left which humans do that machines cannot do? What really differentiates humans from machines? Many more such questions have come to the fore due to the dawn of AI.

The present article gives some Indian philosophical perspectives on the notions of mind, intelligence and consciousness. It dwells on some insights from Indian Systems of philosophy and interprets them in the light of AI. It tries to grapple with the questions like Can a machine have transcendental element of *Cit* or *Ātman*? Can a robot know that it knows? Is there a tribunal or check in the AI world, which would filter what is true, right, just, humane and good?

1. Indian Perspectives on Intelligence, Mind and Consciousness

The issues that are discussed often in relation to AI are intelligence, mind, consciousness and so on. Indian traditions use many expressions for these words and sometimes these words are used interchangeably. The words used for intelligence are *buddhi*, *prajñā*, *jñāna*, *mēdhā*, *dhī*, *vidyā* and for mind are

1 Himanshu, "Octopus crushing range rover car goes viral, fact check," *Kalinga TV*, October 4, 2023, https://kalingatv.com/offbeat/octopus-crushing-range-rover-car-goes-viral-fact-check/.

manas, *citta*, *antaḥkaraṇa* etc. The words like *ātman*, *jīvātman cit*, *caitanya, puruṣa* are used to denote consciousness. Sometimes the difference is made with the capitalisation of letter 'c'. Consciousness with 'C' stands for Supreme Consciousness and consciousness with 'c' refers to individual consciousness.

All living beings exist in their grossest level through body. Mind is the invisible aspect of physical brain. Intelligence is the property of the mind. In computationalism brain is essentially a computer. *Cit* or Consciousness is the subtlest of all. It goes beyond the purview of body, mind and intelligence. It is beyond name, form, thought, concept etc. It is universal, all-abiding and all-pervading reality. Mind is like a flashlight which sheds light on some parts of the Consciousness, but it cannot illuminate the whole reality.[2] Scientifically, consciousness is understood as the emergent property of functions of brain neurons. The incessant transmission of senses to the brain about the physical reality gives rise to consciousness. The neuro-science has explained the brain processes, though human brain is a fascinating entity. Similar process is tried in the Artificial Intelligent machines. Hence, the scientific community is of the opinion that machine can assume consciousness.

Indian Philosophy considers mind as physical element along with the body, and consciousness as spiritual element. Consciousness is asserted as the illuminator of the mind. It is the awareness of one's existence and essence of life. The machines will fall short of consciousness because they lack life and self-awareness.[3] There are lot of differnces between AI version of consciousness and Indian version of consciouness. The consciousness can be infinitely subdivided, controlled and manipulated in AI functioning, whereas Consciousness as such cannot be dissevered. Human consciousness is beyond the conceptual grasp. Hence, AI cannot measure the consciousness of humans. *Upaniṣads* claim that Brahman is Existence, Consciousness and Bliss (*saccidānanda*). It is looking at the same reality from three perspectives– ontological, psychological and ethical. Can we look at artificial super intelligence in the same manner? Will AI replicate multi-faceted ineffable reality of the *Upaniṣads*?

2. Mind-Body or Mind-Consciousness?

The mind-body relation of Modern Western Philosophy is treated differently by classical Schools of Indian philosophy by adding the third aspect of consciousness. Basically, there are three theories with regard to the mind-

2 David Frawley, "Relevance of Vedantic Knowledge for Coming Century," *Waves Blog* (January 19, 2021), accessed October 3, 2021, https://www.youtube.com/watch?v=Qtpou6Av03w&list=PLsNA8-eNUcRSQiLLgK51bEfXPIZdFLk7u& index=3.

3 Subhash Kak, "Artificial Intelligence, Consciousness and the Self," *Medium.com* (blog), (December 11, 2018), accessed October 9, 2021, https://subhashkak.medium.com/artificial-intelligence-and-consciousness-6b5ff2e5b5a.

body relation in the West. They are: dualism, reductionism and eliminativism.[4] Dualism says that mental states and physical states are different from one another. Reductionism proposes that mental states are nothing but physical states. Eliminativism negates mental states by affirming the existence of physical states alone. In Indian Systems the Consciousness is distinguished from everything else. Roy Perrett in his essay on "Computationality, Mind and Value" argues about the issue of intentional mental states and non-intentional mental states.[5] In other words, it is a kind of duality between mind and Consciousness.

3. Mind and Consciousness in Hindu Scriptures

Both *Śruti* and *Smṛti* Texts of Hinduism extensively refer to Consciousness. Consciousness is considered as the state of liberation. The Ultimate Reality is understood as Consciousness or *Cit*. The *Upaniṣad* proclaims, "*Prajñānam brahmā*" i.e., "Consciousness is Brahman" (*Aitareya Upaniṣad* 5.3). *Kaṭha Upaniṣad* establishes a relationship between self or consciousness, intellect, mind and senses. It explains that the self or *ātman* is the passenger of the chariot called body. The intellect or reason is the charioteer and the mind acts as reins. The senses are the horses. The self with senses and mind is the enjoyer or experiencer.[6] The controlled mind increases the level of consciousness whereas the uncontrolled mind reduces the level of consciousness. The developed consciousness is in the path of self-actualisation.[7]

Muṇḍaka Upaniṣad speaks of two kinds of knowledge. They are: *para* (higher) and *apara* (lower).[8] The higher or deeper knowledge is related to Consciousness and lower knowledge is the superficial knowledge. According to me, AI functions through and provides with *apara* knowledge and will never reach the realm of *para*. *Taittirīya Upaniṣad* deals with *pañcakośas* (five sheaths) or layers of consciousness or self. They are: *annamaya kośa*,

4 Roy W. Perrett, "Computationality, Mind and Value: The Case of Samkhya-Yoga," *Asian Philosophy*, 11, no. 2 (2001): 10.

5 Perrett, 2001.

6 "*ātmānaṁ rathitaṃ viddhi śarīraṁ rathameva tu buddhiṃ tu sārathiṃ viddhi manaḥ pragrahameva ca indriyāṇi hayānāhurviṣayāṁ steṣu gocarān ātmendriyamanoyuktaṃ bhoktetyāhurmanīṣiṇaḥ*" (*Kaṭha Upaniṣad* 1.3.3-4), eds. Joshi, Bimali and Trivedi, *112 Upaniṣads*, 2007.

7 Gupta S., "The 'Indian Philosophy of Consciousness' and its Relevance to 21st Century Business Management," *Philosophy of Management* 16, 35–54 (2017), https://doi.org/10.1007/s40926-016-0047-5.

8 "*dvevidye veditavye iti ha sma yadbrahmavido vadanti parā caivāparā ca*" (*Muṇḍaka Upaniṣad* 1.1.4). All the quotations from Upaniṣadsa are taken from K. L. Joshi, O. N. Bimali and Bindiya Trivedi, eds. *112 Upaniṣads*, Vol. 1 (Delhi: Parimal Publications, 2007).

prāṇamaya kośa, *manomaya kośa*, *vijñānamaya kośa* and *ānandamaya kośa*. The food sheath is made up of gross elements earth, water, fire, air and ether. It is the physical layer of the self. The energy sheath accounts for all physiological functions such as respiration, metabolism and excretation. It is the vital energy in the self. The mental sheath is contains ideas, moods, feelings etc. The intellectual sheath entails the cognition, wisdom and discriminative faculty and finally, the sheath of bliss. The Self or Soul lies beyond these layers as Pure Consciousness.[9]

Tāntric systems teach to expand consciousness from its gross form. *Tāntric* texts mention seven energy centers (*cakras*) of consciousness. The seventh *cakra* known as *sahasrāra* is located in the brain. It is said that through the constant practice of *Kuṇḍalini Yoga*[10] one can reach the stage of transcendental consciousness (*samādhi*).[11]

The essential nature of human being cannot be replicated through AI. The nature of *Ātman* cannot be mimicked or emulated by algorithms of AI, for it is described as eternal, perfect, indestructible, immutable, invisible, formless and infinite. In short, it is characterized as microcosm essence of Brahman. The *Upaniṣadic* aphorism states, "*ayam ātmā brahma*" which means, "This Self (*Ātman*) is Brahman" (*Māṇḍukya Upaniṣad* 1.2). The mental process taking place through brain activity could be expressed, transferred or reproduced by a machine, as Hinduism agrees with science that mind is the epiphenomenon of the body and brain as the seat of mind.[12] The Self which mirrors Ultimate cannot be mirrored by AI. Subhash Kak, an Indian-American computer scientist and Hindu thinker observes that a machine or a computer can never be conscious, for Consciousness or observer remains separate from what is being observed.[13]

Bhagavad-gītā speaks about the gradation from matter to self, "The senses are said to be superior (to matter or the body), the mind is superior to the senses, the intellect is superior to the mind, and Ātmā is superior to the

9 *Taittirīya Upaniṣad,* 2.1-5.

10 *Kuṇḍalini Yoga* is a combination of movement, breath, meditation, and chanting to enhance one's level of self-awareness and consciousness. It is done to awaken the energy that is dormant within oneself at the base of the spine like a coiled snake.

11 Sridhar M. K. and H. R. Nagendra, "Consciousness in Indian Philosophy and Modern Physics," *International Journal of Yoga-Philosophy Psychology Parapsychology* 2021; 9:53-8.

12 Sullivan B. M., "Fluid Selfhood, Human and Otherwise: Hindu and Buddhist Themes in Science Fiction," *Implicit Religion*, 2014, 17(4), 489–508. https://doi.org/10.1558/imre.v17i4.489.

13 Subhash Kak, "AI, Consciousness and the Veda," VaYU USA (July 19, 2021), accessed October 5, 2023, https://www.youtube.com/watch?v=G1ODiD6NxAM.

intellect" (*Bhagavad-gītā* 3.42).[14] This verse places the Self or Consciousness at the top of hierarchical structure. It is unlikely that AI will reach to such a height in terms of its superiority of Consciousness.

4. Mind and Consciousness in Classical Systems (*Darśanas*) of Indian Philosophy

All the *āstika* (orthodox) Schools of Indian philosophy hold consciousness either as an attribute of the Self or identical with the Self. It is seen as non-physical property or as an immaterial Self (*Ātman*). *Nyāya-Vaiśeṣika* and *Mīmāṁsā* Schools uphold that *Ātman* is the substratum of consciousness. Consciousness is the adventitious property of the Self. In *Nyāya* School, *manas* is a category (*padārtha*). It is an instrument of external and internal cognition and knowledge becomes the attribute of the soul. *Vaiśeṣika* counts *manas* as one of the substances (*dravya*). *Manas* is a link between the Self and the sense-organs which makes possible the knowledge of external objects. The *Mīmāṁsā* School agrees with *Nyāya* School with regard to knowledge as the quality of the Soul.

The Dualistic School of *Sāṅkhyā* conceives Soul as pure or contentless Consciousness (*Puruṣa*). There are three internal organs (*antaḥkaraṇa*) or psychic evolutes of primordial matter, *Prakrti*. They are: intellect (*mahat* or *buddhi*), egoity (*ahaṁkāra*) and mind (*manas*). Intellect stands for understanding, knowing, rationality, ascertainment and decision. Egoity represents willing. It generates self-sense or sense of belonging. *Manas* synthesizes the sense data into determinate perception.[15] Consciousness is an immaterial entity whereas, mind is a material entity in *Sāṅkhyā*. The production of knowledge takes place due to the association of intellect of *Prakṛti* and Consciousness of *Puruṣa*. When the reflection of Consciousness falls on the intellect, mind assumes the form of an object through the senses due to *sattva guṇa*. Thus, conscious mind arises.[16]

The *antaḥkaraṇa* of *Sāṅkhyā* is called *citta* (mind-stuff) in *Yoga* School. Intellect, egoity and mind are the functional aspects of *citta*. *Citta* is made up of three *guṇas* (fundamental subtle elements) of *sattva*, *rajas* and *tamas*. Mind-stuff undergoes modifications and attains different levels because of the preponderance of these *guṇas*. Patañjali exhorts that Control of modifications of *citta* leads to the attainment of consciousness. *Advaita Vedānta* accepts that every individual can experience the Consciousness. It advocates the technique of three states of mind to attain self-realization. In the states of waking (*jāgrat*), dreaming (*svapna*) and deep sleep (*suṣupti*), though the mind undergoes variations, Consciousness remains as an ever-abiding force and then one can reach the last state of transcendence (*turiyā*) or non-duality. Thus, Consciousness or non-dual Self functions as a light which illumines everything else. It is the knower and fundamental non-dual self-luminous principle.[17]

According to *Advaita Vedānta*, mind or *manas* is one of the four aspects or functions of *antaḥkaraṇa* (internal faculty). The four aspects of psyche (*antaḥkaraṇa catuṣṭaya*) are: *manas* (general thinkihng), *buddhi* (discriminative faculty), *ahaṁkāra* (ego-sense) and *citta* (mind-stuff). *Manas* refers to indecisive state of *antaḥkaraṇa* which does the role of thinking, *buddhi* represents state of decision, *ahaṁkāra* is the state of having reference to the self and *citta* is the state of remembering. *Citta* is the last point of the mind which connects one to Consciousness.[18]

Lokāyata or *Cārvāka* School differs in its outlook towards consciousness by considering it as a by-product of combination of material elements. *Cārvākas* subscribe to the idea that consciousness is dependent on matter. It does not exist apart from the body. Their theory is named as 'Psychological Materialism' (*dehātmavāda*). They define self as "body qualified by sentience." It is something which represents I-ness of a person. Modern *Cārvākas* would reckon consciousness as an outcome of the biochemical process involving elements of physical world. They may also relate consciousness with the brain than the body.[19] It apparently gives an impression that it corresponds with the contemporary debate on consciousness in AI.

In Buddhist thought, consciousness is found in the list of five aggregates. The empirical self is the bundle of body or material phenomena, feelings (positive and negative), perceptions, mental formations and consciousness. Consciousness is a continuous and connected stream of awareness of sensations and thoughts. It depends on the other four

14 "*indriyāṇi parāṇy āhur indriyebhyaḥ paraṁ manaḥ manasas tu parā buddhir yo buddheḥ paratas tu saḥ*" (BG 3.42). The same idea is found in *Kaṭha Upaniṣad* 3.10 and 6.7.

15 Chandaradhar Sharma, *A Critical Survey of Indian Philosophy* (Delhi: Motilal Banarsidass Publishers Private Limited, 1987), 160-161. Also see: Ferenc Ruzsa, "Samkhya: Dualism without Substances," in *Indian Epistemology and Metaphysics*, ed. Joerg Tuske (New Delhi: Bloomsbury Academic, 2017), 153-182.

16 *Sattva*, *Rajas* and *Tamas are the constituent elements of Prakṛti* (Primordial Matter). *Sattva guṇa* produces luminosity, knowledge, reflection etc. See: K. Ramakrishna Rao *et al., Cultivating Consciousness: An East-West Journey* (New Delhi: GITAM University Press & D. K. Printworld P. Ltd., 2014), 249.

17 Juturi R. K., "Advaita Vedanta Answer to the Hard Problem of Consciousness: A Philosophical Review," *Yoga Mimamsa*, 52 (2020): 84-7.

18 Datta D. M., *The Six Ways of Knowing* (Calcutta: University of Calcutta, 1997), 41-42.

19 Pradeep P. Gokhale, "Materialism in Indian Philosophy: The Doctrine and Arguments," in *Indian Epistemology and Metaphysics*, ed. Joerg Tuske (London: Bloomsbury Academic, 2017), 143 & 147.

aggregates. According to Buddha, consciousness is the act of being conscious which caused by other aggregates. He negates the possibility of pure consciousness.[20] Six sense organs and their sense objects give rise to six acts of awareness or consciousness. They are: visual consciousness, auditory consciousness, olfactory consciousness, gustatory consciousness, tactile consciousness, and ideational/conceptual consciousness. Mind is considered as the internal sense organ which interacts with sense objects that include sense impressions, feelings, ideas and concepts.[21]

According to Jainism, consciousness (*cetana*) is in the soul itself. It is the quality of the soul. The soul has the power of knowledge and understanding. The soul cognizes itself without any medium or instrument. The consciousness co-exists and co-extends with life. Jainism posits that each living organism has certain degree of consciousness it. Intelligence, intuition, knowledge, perception, emotions, will, awareness of pleasure and pain, behaviour, and bliss are the various manifestations of the same consciousness.[22] Physical mind and psychical mind are the two types of minds in Jainism. The physical mind interacts with the brain and nervous system and psychical mind thinks, ascertains, decides and discriminates. Soul becomes the basis of psychical mind due to impurity. But pure soul is without mind.[23]

5. Is there AV in AI? An Analysis of the Notion of Cognition of Cognition in *Nyāya*

AI is superintelligent and it will reach peak of progress one day. It knows all that human mind can know or it would know more than what human mind can think or imagine. The computer or machine knows without the limit. There is a reservoir of knowledge fed by or processed by large number of human minds in the world. It carries with it knowledge of generations. It can relate to past and to a certain extent predict the future with the present collection of information in it. We all agree that AI knows and it can be 'All Intelligent'. But a crucial question that can be posed to AI is: Does AI know that it knows?

20 Soo Lam Wong, "Karma and Mental Causation: A Nikaya Buddhist Perspective," in *Cross-Cultural Approaches to Consciousness: Mind, Nature and Ultimate Reality*, ed. Itay Shani & Susanne Kathrin Beiweis (London: Bloomsbury Academic, Bloomsbury Publishing, 2022), 123-124.

21 Wong, "Karma and Mental Causation: A Nikaya Buddhist Perspective," 2022, 126.

22 Jagdish Prasad Jain, "Jaina Psychology", in *Handbook of Indian Psychology*, ed. K. Ramakrishna Rao, Anand Paranjpe, and Ajit K. Dalal (Cambridge University Press India: Foundation Books, 2008), 55–72.

23 Narayan Lal Kachhara, "Philosophy of Mind: A Jain Perspective," *US-China Education Review,* 8, no. 3 (March 2011).

Perception (*pratyakṣa*) is a universally accepted means of knowledge (*pramāṇa*) and it is placed as the first in the list of means of knowledge in Indian *darśanas*. According to *Naiyāyikas*, cognitive states are not self-revelatory or self-luminous. Therefore, they treat the issue of perception with a unique concept known as '*anuvyavasāya'* (AV). *Vyavasāya* means 'a determinate cognition', or 'a cognition which is certain or free from doubt.' The prefix 'anu' means 'that which follows', 'after', 'subsequent', 'successive' etc. Literally, *anuvyavasāya* means 'after-cognition.' Hence, it can be understood as recognition, cognition of cognition, reflexive cognition, reflective awareness, reperception, apperception, inward perception, introspection etc. *Anuvyavasāya* is a second-order knowledge wherein both knowledge and the knowing subject are revealed.[24]

When a smartphone is seen by the eyes the knowledge that arises in the mind is, 'This is a smartphone.' Such a knowledge is called *vyavasāya* (determinate cognition). It immediately follows another knowledge, which relates itself to the Self saying, 'I know that this is a smartphone.' It implies that consciousness not only requires an awareness of object but also the awareness that one is aware.[25] Human mind processes the series of information of the external things received through the senses. A perceptual event may occur and perish unnoticed or unperceived. It cannot be retrieved or passed on to others through language without being first consciously registering it for oneself. It is similar to the cognition of pleasure and pain. *Anuvyvasāya* makes it happen by elevating the awareness of first-order knowledge to the second-order knowledge.[26] The cognition is generally thought to be reflexively self-perceived. But it is not the case. Gaṅgeśa states that since the two numerically different cognitive-events take place in quick succession, there is a false notion of automatic self-cognition of objects at the first instance.[27] *Anuvyvasāya* is a response to the question whether consciousness is the object of knowledge or whether Self can be known? *Nyāya* School affirms that consciousness can be known through the reflection upon experience. *Manas* is the instrument of knowing. It plays the role of an 'inner organ' for the Self. Cognition of cognition gives rise to *Ātman* or Consciousness. The cognition gets revealed only when inward perception of an object takes place. The self-awareness of a cognition, for

24 Mainak Pal, "The Cognitive Model of Anuvyavasāya," *Journal of Indian Council of Philosophical Research*, 37, 133–157 (2020). https://doi.org/10.1007/s40961-020-00194-7.

25 Shankarji Jha, "Consciousness in Old Nyaya," in *Science of Consciousness, Psychotherapy and Yoga Praxis* ed. Ram Nath Jha (Delhi: Vidyanidhi Prakashan, 2015), 262-271.

26 Bimal Krishna Matilal, *Perception: An Essay on Classical Indian Theories of Knowledge* (Oxford: Clarendon Press, 1986), 140. Also see: Christopher Bartley, *Indian Philosophy A–Z* (Edinburgh: Edinburgh University Press, 2005), s. v., "*anuvyavasāya*".

27 Matilal, *Perception*, 1986, 156.

example, of a pot or cloth, is necessary for the subject to name it or to use it.[28]

From the viewpoint of *Nyāya*, *anuvyavasāya* is a meta-cognition unique to humans due to the existence of Consciousness. It cannot be the feature of AI for it lacks such a Consciousness which humans possess. *Naiyāyikas* would assert that it is the self-reflexive knowledge that distinguishes humans from machines. No doubt, AI perceives the external objects, but it does not have an internal perception. Only when there is internal cognition one knows that he knows. Therefore, there is no AV in AI. In other words, machines or robots may not reach to the state of having reflective awareness or higher order cognition.

6. Witness or *Sākṣin* in *Advaita*

Nyāya holds that *anuvyavasāya* certifies the truth of an object in the cognitive event. But *Advaita Vedānta* maintains that subject who is conscious of an object is also aware of being conscious of that object. It believes in self-awareness position of the Self. It views the mental and the physical phenomena as two aspects of the same reality. Consciousness in *Advaita* is self-manifesting and self-illuminating unlike *Nyāya-Vaiśeṣika* Systems.

Proponents of *Advaita* mention two elements of knowledge: empirical knowledge (*vṛttijñāna*) and witness knowledge (*sākṣijñāna*). The individual knowing self (*jīva*) is called *sākṣin* (witness). It associates itself with the internal faculty (*antaḥkaraṇa*). In every knowledge the Self knows itself and the object of knowledge. The known subject is only an empirical ego and not the transcendental reality, for all knowledge at the ultimate level is only relatively real.[29] The *sākṣin* of *Advaita* as the unchanging revealing subject, replaces the *anuvyavasāya* of *Nyāya*. The Self is the *sākṣin* which illumines the objects while being self-luminous. It functions as the light which reveals all the objects due to the aspect of self-luminosity. *Sākṣin* has the essence of Consciousness ontologically and observing ability empirically. It is witness or experiencer of both mind-stuff and senses. *Sākṣin* serves as the basis for all cognitions.[30] It is the aspect of pure individual Self which is identical to Brahman.

28 J. Shaw, "Cognition of Cognition Part I: Pandit Visvabandhu Tarkatīrtha", trans. from Bengali with explanatory notes," *Journal of Indian Philosophy*, 24, no. 2 (1996): 165–207. http://www.jstor.org/stable/23447816. Also see: Ingo Strauch, *Brill's Encyclopedia of Hinduism*, Vol. II, "Sacred Texts, Ritual Traditions, Arts, Concepts," ed. Knut A. Jacobsen (Editor-in-Chief), Helene Basu, Angelika, 2012, s.v., "Consciousness and Mind".

29 Ingo Strauch, *Brill's Encyclopedia of Hinduism*, s.v., "Consciousness and Mind."

30 Fort A. O., "The Concept of Sākṣin in Advaita Vedānta," *Journal of Indian Philosophy,* 12, no. 3 (1984): 277–290. http://www.jstor.org/stable/23444147.

From among the two elements of knowledge, AI has incorporated empirical knowledge. Does it possess witness knowledge too? Can we juxtapose AI as subject not only who is conscious of an object, but also aware of being conscious of that object? It is clear that from Advaita perspective, AI cannot have the aspect of *sakṣin*, for it cannot attain that pure Consciousness state. AI remains as an object of human consciousness. AI is not human in the true sense of the term having the salient features like volition, memory, forgetfulness, creativity, emotions, freedom, corporeality, experience, transcendence, Consciousness etc. Hence, is AI a subject or what/who is its subject is a debatable issue.

7. *Dvaita* Notion of Knowing the Consciousness

The *Nyāya* notion of *anuvyavasāya* is criticized, because it leads to infinite regress due the need of two cognitions in succession. *Dvaita* follows the approach of knowledge by employing the concept of *sākṣin*. For *Advaita*, *sākṣin* is not the attribute of the Self. Only consciousness is the essence of the Self. But for *Dvaita*, consciousness belongs to the Self and *sākṣin* is the essence of the Self.

Sākṣin is the intuitive faculty which possesses the unfailing natural power. "It is the purest form of the sense which is identical with the cognitive-agent and possesses absolute validity of knowledge."[31] The knowledge of the Self is possible due to the operation of the *sākṣin* without the mediation of any other sense-organ. The philosophy of Madhva recognizes five kinds of non-eternal manas, namely, *manas*, *buddhi*, *ahaṁkāra*, *citta* and *cetana*. *Manas* doubts, *buddhi* ascertains, *ahaṁkāra* claims egoity, *citta* remembers and *cetana* reveals itself through the activity being special aspect of consciousness.[32]

Dvaita Vedānta has two types of knowledge: *svarūpajñāna* and *vṛttijñāna*. *Svarūpajñāna* is the very essence of knowing the self and its attributes. It is the inherent knowledge generated by *sākṣin*, whereas *vṛttijñāna* is knowledge generated at the sensory and mental level.[33] *Sākṣin* is none other than *jīva* under its cognitive aspect. It is the innate sense-organ of the Self and the faculty of direct perception. The knowledge derived through physical sense organs needs verification because it is liable to defects. *Sākṣin* terminates all tests since it is the final tribunal to

31 Narain K., *An Outline of Mādhva Philosophy* (New Delhi: Munshiram Manoharlal Publishers Pvt. Ltd., 1986), 32.

32 *Ibid*., 32 & 82.

33 Pandurangi K. Y., *Dvaita Vedānta Darśana of Śrī Madhvācārya* (New Delhi: Rashtriya Sanskrit Sansthan, 1995), 38.

establish the validity of knowledge.[34] The knowledge obtained by *sākṣin* can never be sublated.[35] The very validation of knowledge depends upon *sakṣīn*, which is the ultimate principle that knows the knowledge.[36] *Sākṣin* is the real interpreter of knowledge and also the guarantor of its validity. It is self-luminous in its own judgements and self-validating.[37]

Sākṣin plays a role in the production of the knowledge and also ascertains the presence and validity of knowledge.[38] *Sākṣin* clarifies the awareness of our knowledge. For instance, "I know colour" is different from "I also know that I know the colour." The knowledge of the knowledge about an entity is certified by *sākṣin*. In other words, the validity of all cognitions is cognised by *sākṣin* only. *Sākṣin* provides the final test of truth and knows whether knowledge is true or false. Is a testifier or *sākṣin* possible in AI? How is the flow of sea of information and waves of data checked or tested? AI needs a check or tribunal to direct it, at least at the physical level of reality. AI can be the victim of human biases, prejudices and judgements. AI needs a validating force to process truth and facts. In my estimation, it will not be possible for AI by itself. Humans need to be such *sākṣins* to ascertain truth and free the world of technology from defects and errors.

8. Operation of AI in the Pragmatic Level of Reality

Advaita speaks of three levels or grades of reality (*satta-traya*): ontological reality (*pāramārthika satta*), empirical reality (*vyāvahārika satta*) and illusory reality (*pratibhāṣika satta*). These levels are linked to three basic states of consciousness (*avasthā-traya*)–the deep sleep state, the wakeful state and the dream state. The question is whether AI works in these levels of reality. It can be asserted that AI can know only at the empirical or pragmatic level. It is not able to reach itself to the ontological or transcendental reality. On the other

34 "*sākṣipratyakṣato hyova mānānāṁ mānateyate*" (*Anuvyākhyāna* 1028). All the quotations of *Anuvyākhyāna* (AV) are taken from *Śrī Ānandatīrtha*, *Sarvamūlagrantāḥ: Sūtraprasthānam* (Collected workds), ed. K. T. Pandurangi, vol. 2 of 4 (Bangalore: Dvaita Vedanta Studies and Research Foundation, 2008). Also see: K. T. Pandurangi, "Epistemology of Madhva's Dvaita Vedānta", in *Theistic Vedānta*, ed. R. Balasubramanian, vol. 2, part 3 of *History of Science, Philsophy and Culture in Indian Civilization,* ed. *D. P. Chatoopadhyaya* (New Delhi: Project of History of India Science, Philosophy and Culture, Centre for Civilizations, 2003), 317.

35 "*sākṣisiddhasya na kvāpi bādyatvaṁ tadadoṣataḥ sarvakāleṣvabādhyatvaṁ sākṣiṇaiva pratīyate*" (*Anuvyākhyāna* 402).

36 "*jñānasya grāhakeṇaiva sākṣiṇā mānatā miteḥ*" (*Anuvyākhyāna* 1564).

37 "*sākṣiṇaḥ svaprakāśatvamanavasthā tato na hi*" (*Anuvyākhyāna* 1029).

38 Ramachandran T. P., *Dvaita Vedānta and Its Contributions to Indian Philosophy*, Madras, 1977, 123–124.

hand, it may have its access to illusory or phenomenal standpoint of reality to a certain extent, in terms of our imagination, dreams and dream objects such as sky-flower, rabbit-horn, barren-woman's son etc. At the ultimate level, no knowledge is operative and therefore, subject-object distinction is absent. It is the level of pure consciousness and knowledge here can be called intuitive. AI is not able to achieve this as it is only 'relatively real' like that of the nature of the world. It can never be 'Reality of the reals' in *Advaitic* sense.

Richard De Smet opines that God can be grasped through ignorance (*ajñāna*).[39] In *Advaita*, *ajñāna* cannot be completely understood as negative but as a first step towards knowledge. According to Thomas Aquinas, we can know God only analogically. God is understood only after death when one sees God face to face in His eternity. We can compare the Śaṅkara's *Vedāntic* view of 'ignorance' with 'analogy' of the West. The mystical tradition of the West affirms that Supreme Being can be understood apophatically (what a thing is not) due to the inadequacy of language. The cataphatical (what a thing is) or affirmative expressions of the Absolute do not unfold Its reality. In this sense, AI remains at the level of 'ignorance' or 'analogy' which doesn't go beyond the limits of language or algorithms. Dialogue between AI and Indian Philosophy would lead to the revelation of many angles of intelligence and consciousness. The AI discourse today is dominated by pragmatic approaches. The research community and academia are not able to fully and precisely understand the operations of human intelligence in the field of AI. There has been a debate between comprehension and competence in AI in recent times. The main focus was on competence and application of AI than on comprehension and understanding. The issues like what intelligence is, what consciousness is and how it operates were not discussed or addressed. Therefore, the usage of the terms like intelligence, consciousness has become superficial and superfluous in AI literature. The deeper understanding of *Manas, Buddhi*, *Ātman*, *Cit* etc. will give a conceptual alternative to bring back the discussion on comprehending intelligence and consciousness.

Conclusion

AI is and will be transforming the whole world beyond the planet earth. It has brought in a quantum shift in our lives, changing the way we perceive and do things and have a profound impact on everything around us. We have a question before us: Will human intelligence be replaced by AI? Humankind has to use AI without losing sight of its inner Self. No doubt,

39 Richard De Smet, "Chinks in the Armour of Avidyā," *Understanding Śaṅkara: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Pvt. Ltd., 2013), 146.

AI has invaded the whole universe in one or the other way. On the other hand, the Indian thought reminds us that each one is a microcosm–a world in miniature.

The *Upaniṣadic* metaphor of 'the spokes in the hub of a wheel' (*Muṇḍaka Upaniṣad* 2.2.6) suits the topic of our discussion. AI as the spoke representing the dynamic aspect of human mind and creativity should move on, whereas the core Self or Consciousness should remain like the hub uniting everything and connecting everywhere in the universe. *Vedānta* reminds us that we are the higher beings, and from this point of view AI has to assist humanity to explore not only the width but the depth of the reality. Indian Philosophical Systems offer us deep insights to find out how we can move along with AI by remaining unique and different. Humans have evolved from generations to generations by learning from the past, contributing to the present and moving towards the future.

The Indian epistemology speaks of knowledge not just as information, but a means for transformation of the self and the world. The knowledge is not just a domain of the mind alone but of the whole self. Knowledge englobes, the mind, heart, senses and soul. Hence, knowledge is creative, unique, individual, and above all it is spiritual. Knowledge is essentially a relationship with the Eternal and one another. The question is, will AI ever reach this realm at all. Certainly, it need not and should not and will not. But AI will revolutionize the world and humanity is never a matter to be doubted.

4

Contemporary Indian Responses to Artificial Intelligence

Raju Felix Crasta

Abstract

Indian contribution to the field of AI is significant. Indian peninsula ever since its civilization, has been rich in thinking, innovations, creativity, and production in the world of knowledge and growth. In the present exponential growth of science and technology in the world, no doubt, the country has its lion's share. However, structural lacune, bureaucracy, funding, etc., over time and region have failed to attain the due credit. This paper attempts to critique the Indian scientific temperament and contribution to the world of AI.

Keywords**:** *Artificial Intelligence (AI), Machine Learning (ML), Science and Technology (ST), scientific temperament, Research and Development (R&D), social impact, brain drain.*

Introduction

Artificial Intelligence (AI) is a buzzword in the 21st century due to its revolutionary approach both in terms of technical intelligence and human potential. Since its inception, the word 'intelligence' has become a 'myth.' There was a fear that AI would overtake humanity. Surely, it will ease human

living, but will not replace human intelligence.[1] Technology *per se* does not need human beings, it is humans who need technology. This paper tries to study and critique the Indian contribution to AI and Technology. The paper is divided into three sections. The first section underlines ancient scientific wisdom. The second section computes India's response to contemporary scientific advancement in particular AI. In the final section, while giving a critique, the challenges and tasks that they bring are emphasized.

1. Ancient Indian Scientific Wisdom

Is AI an old theme for India? While glancing through the lenses of bourgeoning AI and scientific advancement of the time, the question seems absurd. The ancient people of the Asian continent, particularly the Indian peninsula, have been rich in innovation and reasoning. They even envisioned a non-natural life through creative imagination and have tried to give life to it.[2] To this matter, the ancient civilization of the West is also noteworthy. While the ancient Indian literature mentions Vishwakarma, Maya, and Ajatashatru as the gods of technology and reasoning or as architects of the earth, the Western thinking ascribes the same to Hephaestus and Daedalus (Athenian craftsmen and inventors).[3] This implies that ideas and knowledge are not implicit property of a particular region or culture, but are mutually exclusive. Enfleshing the imagination, subsequent rationalization and quantification lead to the advancement in human wisdom and development, be it the technological field, social or any other. The Indian civilization was home to various saints, sages, thinkers, scientists, scholars, innovations, discoveries, imaginations, start-ups, etc. in different fields. Some of the notable ancient contributions to scientific thinking need a mention.

1 Francis Arackal Thummy, "Artificial Intelligence: Prospects and Perils," *Philosophizing Science: Promises, Perils and Possibilities*, ed. A. P. Joy (Delhi: Christian World Imprints, 2019), 236.

2 The *Panchatantra,* has a story of *four men and a dead lion*. Upon getting a lion skeleton in a jungle, in order to test their learning, three brothers out of four with their spiritual power and learning enflesh it and give life while the fourth prudent brother departs them. The animal happens to be a lion and eventually devours them. When the technology is misused, it will be disastrous to the one who endorses it. This story is also similar to Mary W. Shelley's *Frankenstein* (1818); see: https://www.britannica.com/topic/Frankenstein, accessed 10 June, 2023.

3 Walter Burkert, *Greek Religion 1985*: III.2.ii. See: coverage of Lemnos-based traditions and legends at Mythic Lemnos. Also see: Robert Graves (1955). *The Greek Myths* (Harmondsworth, Middlesex, England: Penguin Books, 1995), 51; Greek "Automatons in Greek Mythology," in *Greek Legends and Myths*, accessed September 2, 2023, https://www.greeklegendsandmyths.com/.

1.1. The Concept Zero, Numeric and Decimal System

Indian Mathematician and Astrologer, Aryabhata is credited with the feat of discovery of "Zero" ("0") though similar concepts were also present in Jain cosmology (as *śūnya* or 'nothing'), Mayan and Babylonian cultures.[4] *Aryabhaṭīa*, the work of Aryabhaṭa and Bhakshāli Manuscript speak of a decimal place value system, called 'positional notation' that describes the value of a number as determined by the position of the digit, that is, the value of a number is actually the product of the digit by a factor which is determined by the position of the digit. However, it is only in the 7th century Brahmagupta introduced it in Mathematics. This system of decimal counting made arithmetic calculations faster and easier and led to several future inventions. The French scientist and mathematician, Pierri-Simon Laplace comments about this:

> It is India that gave us the ingenious method of expressing all numbers by means of ten symbols, each symbol receiving a value of position as well as an absolute value; a profound and important idea which appears so simple to us now that we ignore its true merit. But its very simplicity, the great ease which it has lent to all computations, puts our arithmetic in the first rank of useful inventions; and we shall appreciate the grandeur of this achievement the more when we remember that it escaped the genius of Archimedes and Apollonius, two of the greatest men produced by antiquity.[5]

The discovery of the "0" and the decimal system is a game-changer in science. AI cannot be assumed without this singular concept. Computers function at their base, a binary system—strings of zeros and ones—known as bits. Without this, there cannot be electronics, calculus, engineering, automation, robotic or cryogenic technology. This means that without zero, much of our modern world will literally fall apart. The standard notation of digits (*aṅka*) and its place (*sthāna*) in numerical nomenclature is, once again, an Asian contribution. In numerical expression, the nine-word numerals or *aṅka* (1,2,3,4,5,6,7,8,9) and the 0 as the placeholder were being used during Vedic times.[6] Vedic Saṁhitā and the Brāhmaṇas also used expressions to refer to

4 Lameswar Kanhor, *The Marvels of Cosmos* (Delhi: Christian World Imprints, 2014), 64-70.

5 Dantzig T., *Number: The Language of Science* (New York: Macmillan, 1930), 19. Also see: Amartya Kumar Dutta, "Decimal System in India," *Encyclopaedia of the History of Science, Technology, and Medicine in Non-Western Cultures* (Springer, 2015). DOI 10.1007/978-94-007-3934-5_10321-1.

6 Amartya Kumar Dutta, "Decimal Systems in India," 1-2.

each digits and their multiples of ten and thousands—*eka* (1), dvi (2), *tri* (3), *catur* (4), *pañca* (5), *ṣaṭ* (6), *sapta* (7), *aṣṭa* (8), (9) *nava*, (0) *śūnya*, 9), *daśa* (10), *viṁśati* (20), *triṁśat* (30), *catvāriṁśat* (40), *pañcāśat* (50), *ṣaṣṭi* (60), *saptati* (70), *aśīti* (80) and *navati* (90); (100) *śata*, (1000) *sahasra*, etc.

1.2. Medical Innovations

There are clear indications that India was much ahead of other nations with regard to medicine. Ayurveda, the branch of Atharvaveda is the ancient Indian system of medicine that dates back to 1500 BCE and includes complex herbal compounds, minerals and metal substances. Charaka and Shushruta were well-known physicians of 600 BCE who are credited as the fathers of plastic surgery. *Charaka Saṁhitā, Susruta Saṁhitā*, and *Aṣṭāṅga Hṛdaya*[7] are the three major classics on Ayurveda which enumerate eight different branches of Ayurveda: surgery; ear, eye, nose and throat; mental and supernatural diseases; therapeutics; paediatrics; toxicology; rejuvenation; and vilification. These texts are the most famous and are still consulted by medical practitioners today.

1.3. Mathematical and Scientific Temperament

The ancient seers were gifted with super imaginations and thinking. The *Baudhāyāna Sulbasūtra,* back in 800-500 BCE mentions highly mathematical theorem in line with Pythagorean theorem as well as geometrical proof for an isosceles right triangle; still a millennium ago atomic theory was postulated as matter composed of uncuttable atoms; Bhāskarācārya spoke of a special force that pulls things to earth as Newton did, Rishi Bharadwaj is credited with inventing *vimānas* or flying machines in ancient India like aeronautics of present-day; India produced a finest quality crucible steel in 200 BCE using a method Europeans would later call the crucible technique; Harappa and Mohenjo-daro expressed a sophisticated sewage system and water-flushed toilets in every homeand many more such technologies.[8] Thus, the Asian continent, in particular India has been a home of inventions and discoveries in every field. That which was imagined, over the years was quantified. These scientific temperaments have contributed to the growth of scientific knowledge.

7 For more details on these see: Agnivesha Dwivedi, B. K. Dwivedi, P. K. Goswami, eds., *Caraka Samhita* (Varanasi: Chowkhamba Krishnadas Academy, 2011). Kaviraj K. L. Bhishagratna, *An English Translation of the Sushruta Samhita in Three Volumes* (Calcutta: Archived by University of Toronto. 1911).

8 Gwyn D'Mello, "11 Ancient Inventions and Discoveries of Science That India Gifted to the Rest of the World," https://www.indiatimes.com, updated 26 January 2018, accessed August 20, 2023, https://www.indiatimes.com/technology/11-ancient-inventions-discoveries-of-science-that-india-gifted-to-the-rest-of-the-world-338417.html.

2. Contemporary Indian Response to AI

India is significantly contributing to the growth of AI in various fields. Some of the notable areas of contribution are given below.

2.1. Research and Academia in India

Research and academia in the field of artificial intelligence (AI) in India have seen significant growth and contributions over the years. Several Indian institutions and researchers have made noteworthy advancements, published influential papers, and actively participated in global AI conferences and collaborations.[9] Rajeev Chandrasekhar, the minister of State for Electronics and IT opines that "India's contribution to global AI will not be trivial but significant."[10] Here are some key aspects of research and academia in AI in India.

a) Prominent Academic Institutions: According to the Economic Survey 2022-23, the AI portal, the Government of India set up in May 2020 in collaboration with industry body NASSCOM has published 1,724 articles, 829 news items, 276 videos, 127 research reports, and 120 government initiatives related to AI. The portal was started to pool and highlight all AI-related developments by the government, academia, and industry.[11] According to the Finance Minister, Nirmala Sitaraman, the purpose is "for realising the vision of make AI in India and make AI work for India, three centres for excellence will be set up in top educational institutions. Leading industry players will partner in conducting interdisciplinary research and develop [sic] cutting-edge applications and scalable problem solutions in areas of agriculture, health, and sustainable cities. This will galvanize an effective ecosystem and nurture quality human resources in the field."[12]

Bangalore, Delhi, Kanpur and Kharagpur are the prominent cities of AI research institutions in India. EduRank, an independent company released a metric-based ranking of 14,131 universities from 183 countries, utilizing the world's largest scholarly papers database with 83,166,817 scientific

9 To this portion, I am indebted to Bhawna Kumari, "Artificial Intelligence Advancements in India: Current State and Future Potential," *Blogs and Articles*, updated 19 May 2023, accessed August 25, 2023, https://corpbiz.io/learning/artificial-intelligence-advancements-in-india-current-state-and-future-potential/.

10 Anant Goenka, "FE Exclusive," *Financial Express,* June 18, 2023, https://www.financialexpress.com/ business/.

11 Livemint, "Budget: India to set up three centres for excellence for AI, says Nirmala Sitaraman," *Mint, accessed August 12, 2023,* https://www.livemint.com/technology/tech-news/budget-india-to-set-up-three-centres-of-excellence-for-ai-says-fm-sitharaman-11675236927828.html.

12 Livemint, "Budget: India to set up three centres for excellence for AI, says Nirmala Sitaraman."

publications and 1,801,313,576 citations to rank universities across 246 research topics. Accordingly, Indian Institute of Technology (IIT) Kanpur 1st in India and 173 in the world, the Indian Institute of Science (IIS) Bangalore 2nd in India and 189 in the world, Indian Institute of Science Delhi 3rd in India and 214 in the world and Indian Institute of Technology Kharagpur 4th and 288 in the world.[13] Besides these, there are several promising institutions in India contributing to the world of AI. The top ten Indian AI institutions are: 1. Indian Institute of Technology Kanpur, 2. Indian Institute of Science Bangalore, 3. Indian Institute of Technology Delhi, 4. Indian Institute of Technology Kharagpur, 5. Indian Institute of Technology Bombay, 6. Indian Statistical Institute Kolkata 7. Indian Institute of Technology Madras, 8. Jadavpur University, Kolkata, 9. Indian Institute of Technology Roorkee, 10. Vellore Institute of Technology. These institutions have dedicated research centres, departments, and labs focused on AI and related fields.

b) Research Papers and Publications: Indian researchers have authored numerous influential research papers in AI. Their contributions span various subfields, including machine learning, natural language processing, computer vision, robotics, and data analytics. These papers are published in top-tier conferences and journals, such as the Conference on Neural Information Processing Systems (NeurIPS), the International Conference on Machine Learning (ICML), and the Association for the Advancement of Artificial Intelligence (AAAI) conference. The following list gives a glimpse of publications and citations in various fields the top four AI institutions of India have produced in the fields of Engineering, Physics, Chemistry, Quantum and Particle Physics, Materials Science, Biology, Organic Chemistry, Environmental Science, Computer Science, and Liberal Arts and Social Sciences.[14]

Institution	Publications	Citations
IIT Kanpur	34,984	5,85,254
IIS Bangalore	63,350	1,126,378
IIT Delhi	45,077	7,31,154
IIT Kharagpur	47,765	7,78,211

Exclusively, AI and Machine Learning (ML) publications and citations by the top ten AI institutes of India are given below.[15]

13 "100 Best universities for Artificial Intelligence (AI) in India," EduRank.org, accessed September 10, 2023, https://edurank.org/cs/ai/in/.

14 The following data is adapted from "100 Best Universities for Artificial Intelligence (AI) in India."

15 "100 Best Universities for Artificial Intelligence (AI) in India."

Sl. No.	Institute	Publication AI	Publication ML	Citation AI	Citation ML
1	IIT Kanpur	7,129	3,477	1,26,338	104,437
2	IIS Bangalore	10,988	NA	1,72,383	NA
3	IIT Delhi	11,134	4,612	1,70,815	79,961
4	IIT Kharagpur	10,340	7,950	1,33,462	1,44,007
5	IIT Bombay	9,281	3,966	1,16,969	54,807
6	Indian Statistical Institute Kolkata	6,283	3,828	1,02,917	68,323
7	IIT Madras	9,376	4,004	98,595	50,554
8	Jadavpur University, Kolkata	7,774	4,836	94,560	77,971
9	IIT Roorkee	7,384	4,260	93,829	67,940
10	VIT University, Vellore	10,096	6,086	74,528	50,368

The above chart clearly indicates significant research done by these creamy AI institutes of the country, let alone their contribution and application to different fields. This is further clarified by some of the prominent alumni of these institutions: N. R. Narayana Murty and Sudha Murty, the Infosys Chairman and founders; Kailasavadivoo Sivan, the Indian space scientist who served as the Secretary of the Department of Space and chairman of the ISRO Space Commission and the former Director of the Vikram Sarabhai Space Center and the Liquid Propulsion Systems Centre; Aluru Seelin Kiran Kumar, an Indian space scientist and former chairman of the Indian Space Research Organisation, who is credited with the development of key scientific instruments aboard the Chandrayaan-1 and Mangalyaan space crafts; Harish-Chandra Mehrotra FRS, an Indian American mathematician and physicist who did fundamental work in representation theory, especially harmonic analysis on semisimple Lie groups; Ashoke Sen FRS, an Indian theoretical physicist and distinguished professor at the International Centre for Theoretical Sciences (ICTS), Bengaluru and the list is endless.

c) Collaboration, Partnership, and Funding: Indian researchers and institutions actively collaborate with international counterparts on AI research projects. These collaborations foster knowledge exchange, interdisciplinary research, and joint publications. Partnerships with global technology companies, research organizations, and universities help Indian researchers stay at the forefront of AI advancements. There are hundreds of private and government entrepreneurs working and patenting different discoveries and inventions to their credit.

The Ministry of Science and Technology[16] highlights the collaboration of India with other countries in various areas of AI for the advancement of the country. India-Sweden partnership in March 2019 in New Delhi in line with NITI Aayog's National Policy on AI (AI for All) to promote, facilitate and support co-development and commercialization of innovative technologies. The Department of Biotechnology announced the call for AI applications for affordable and accessible healthcare to develop solutions for clinically relevant problems such as prognosis, evaluation, early detection and diagnosis, therapy and care. The Department of Science and Technology launched interdisciplinary Cyber-Physical Systems CPS, a computer-based algorithm integrated with the internet and users across India to focus on health, education, energy, environment, agriculture, strategy cum security, industry 4.0, smart city, sustainable development goals, etc. It will also develop 15 technology Innovation Hubs, and 6 application innovation hubs, 4 technology learning parks with an INR 3660 Crore project to be realized within five years (2019-2024). The Department of Biotechnology (DBT) and the European Union (EU)[17] introduced a funding program called 'Horizon 2020' in 2014 for Indian researchers, universities, research organizations, and enterprises to participate in collaborative projects. The program was introduced to assist Indian partners with access to talent, knowledge, data and infrastructure, and connect them to world-leading teams, networks, value chains, and address jointly global challenges.

In a recent development, the Government of India announced five AI initiatives for governance, social development and progress.[18] i) Applied AI research: The Telangana state launched Intel AI in collaboration with IIIT Hyderabad and the Public Health Foundation of India to manage healthcare and smart mobility concerns. ii) Indo-US AI initiative: The Indo-U.S. Science and Technology Forum (IUSSTF)—an autonomous bilateral organization jointly funded by both Governments that promotes Science, Technology, Engineering and Innovation through substantive interaction among government, academia and industry—established in

16 Ministry of Science and Technology, *INDIA AI: A Meity Initiative*, accessed September 15, 2023, https://indiaai.gov.in/government/ministryof-science-and-technology. The webpage, *indiaai.gov.in* is the national AI portal of India.

17 For more details on the bilateral agreement, see: "Cooperation on Research & Innovation between DBT and the EU," Department of Biotechnology, Ministry of S&T, Government of India, accessed September 15, 2023, https://dbtindia.gov.in/sites/default/files/Call-doc-Horizon-2020-converted-1.pdf.

18 Jitendra Bhojwani, "Five ambitious AI initiatives by Indian Government to Foster National growth," *India AI* [webpage], updated 28 July 2021, accessed August 20, 2023, https://indiaai.gov.in/article/five-ambitious-ai-initiatives-by-indian-government-to-foster-national-growth.

March 2000 initiated the US-India AI Summit on 18 March 2021 to promote AI development through the exchange of ideas and exploring R&D opportunities. iii) In order to ease regulatory filings and corporate work, the Ministry of Corporate Affairs launched it agile version of its portal version 3.0 with AI/ML and advanced data analytics.[19] iv) The Ministry of Electronics and Information Technology (MeitY) nationwide e-Government Unit has created Responsible AI, a national youth initiative to close the skills gap by providing relevant education and skills to public school children. The program also contains classes that will offer students the skills to improve their employability. v) The National Research Foundation (NRF) was founded to promote the execution of the National Education Policy (NEP), and the national government would offer a subsidy of Rs. 1,000 Crore to build strong synergy between R&D, industry, and academia to improve collaboration among the three. Indian Air Force (IAF) is partnering with the Government on these five mega projects.[20]

d) AI Labs and Centres: Several dedicated AI labs and research centres have been established in India. The top ten AI Labs and centres in India[21] are: Robert Bosch Centre for Data Science and AI at IIT Madras, the largest network analytics, NLP, deep learning; NVIDIA, the first-ever NV AI Centre at IIT Hyderabad to boost AI and commercial applications; Intel AI Research Centre to apply artificial intelligence to scale problems in India and solve multiple challenges in the healthcare and smart mobility sectors; Accenture is well-known for providing one of the top artificial intelligence research centres in India in three cities— Bengaluru, Hyderabad, and Pune for innovation centres and delivery centres with real-life innovation; AI Research Lab at Wipro focused on healthcare diagnostics, machine learning, artificial intelligence, and deep learning technology; The Centre for Societal impact through Cloud and Artificial Intelligence (SCAI) launched by Microsoft Research India for creating and validating technologies to have a large-scale impact on India; Philips Innovation Centre in Bengaluru focused on taking India's AI products to the global tech market; Data Analytics and Intelligence Research at IIT Delhi, a well-known AI research group focused on combining as well as integrating multiple fields of data science and artificial intelligence

19 For further details, see the MCA portal https://www.mca.gov.in/mcafoportal/login.do.

20 NewsVoir, "IAF Collaborates with the Indian Government on Five Key AI Initiatives for National Progress," accessed September 20, 2023, https://www.aninews.in/news/business/business/iaf-collaborates-with-the-indian-government-on-five-key-ai-initiatives-for-national-progress20230911170850/.

21 Disha Sinha, "Top 10 Artificial Intelligence Centres in India to work with," *Analytics Insights* https://www.analyticsinsight.net/top-10-artificial-intelligence-research-centers-in-india-to-work-with/. Also see: Sejuit Das, "Top 8 AI Labs in India in 2020," accessed September 23, 2023. https://analyticsindiamag.com/top-8-ai-labs-in-india-in-2021/.

to build intelligent software systems; The Center for Indian Language Technology (CFILT), set up with a generous grant from the Department of IT, Government of India in 2000 at IIT Bombay; The Centre for Artificial Intelligence and Robotics (CAIR), one of the top artificial intelligence research centres for robotics, and information and communication security, battlefield security, communication, etc. Besides these, there are several labs in many leading AI institutes mentioned earlier.

e) AI Courses, Programs, Conferences, and Workshops: Most of the Indian AI institutions offer undergraduate and postgraduate programs specializing in AI and related fields. Alongside, they also host several national and international conferences, workshops, and symposiums. The programs provide students with a strong foundation in AI theory, algorithms, and applications. Additionally, online platforms like NPTEL (National Programme on Technology Enhanced Learning) offer AI courses accessible to a broader audience. The research and academic community in India continue to grow in AI, attracting talented researchers, collaborating with international experts, and pushing the boundaries of AI knowledge. Over the last two decades, several national and international conferences on AI-related themes have been held in different parts of the country.[22] Their contributions have been instrumental in advancing AI research and promoting innovation in India and globally.

2.2. Startups and Innovation

AI startups became a buzzword in 2016. Within a couple of years, only in Bangalore hub reportedly received 87.85 million dollars for the AI sector.[23] Since then the country is witnessing a steep rise in AI startups and innovation hubs. These startups are focusing on areas such as healthcare, agriculture, finance, education, and smart cities. They are leveraging AI technologies such as machine learning, natural language processing, and computer vision to develop innovative solutions to address various challenges and improve efficiency in different sectors. Some of the recent startups are: Druva, Arya.ai, Myelin Foundry, Ripplr, Yellow.ai, Razorpay, SigTuple, Data Science Wizards, Heaps, Avaamo, MachineHack, Rephrase.ai, Mockey, Blend, ProbeAI,

22 For the detailed list of recent and upcoming international conferences, events, seminars, webinars and seminars in the field of Artificial Intelligence can only be found at *Conference Alerts Network* for the year 2021–22, see "Conferences in Artificial Intelligence," *CAN: Conference Alert Network*, accessed September 12, 2023, https://www.conferencealerts.net/Artificial_Intelligence.php.

23 Chandan Kumar Thakur, "Top 10 AI Startups in India," *Linkedin,* accessed April 2, 2023, https://www.linkedin.com/pulse/top-10-ai-startups-india-chandan-thakur-a-i-marketing.

Revition, Fasthr.AI, SVAYO, etc.[24] The country has over 60 Generative AI startups as of the second quarter of 2023 which attracted over 590 million dollars in private investments,[25] reports the *Times of India*.

2.3. Government Initiatives

Though the AI concept was borne in 1956 at a workshop at Dartmouth College guided by John McCarthy and Marvin Minsky, had its first exploration through the works of H. N. Mahabala in 1960. The first Prime Minister Jawaharlal Nehru envisioned a technological future by erecting national institutes like The Tata Institute of Fundamental Research (TIFR) and subsequently computational era was initiated. Later, Rajiv Gandhi steered the country with computer and digital revolution. This paved the foundation for AI in India.

The National Strategy for Artificial Intelligence (NSAI) released its report through the National Institution for Transforming India (NITI Aayog) in 2018 mentioning the initiative of the government called, "AI for All." It makes four recommendations to build an AI ecosystem in India: (a) promotion of research; (b) skilling and reskilling of the workforce; (c) facilitating the adoption of AI solutions; and (d) the development of guidelines for 'responsible AI.'[26] The document demarcates three approaches for the future of Indian AI: undertaking exploratory proof-of-concept AI projects in various areas, crafting a national strategy for building a vibrant AI ecosystem in India and collaborating with various experts and stakeholders.[27] NSAI identifies some key areas of focus which the government should undertake—health, education, agriculture, smart cities and mobility. With this report, an official portal was inaugurated with the name MeitY to share AI-related projects and work in the country. Sooner, the government and private sectors initiated various projects and startups to enhance R&D.

24 Varsha Pillareddy, "Top 10 AI Startups in India – List of Biggest AI Startups," *myHQ Digest* 13 February 2023, https://digest.myhq.in/ai-startups-in-india/. Mohit Pandey, "Top 8 Generative AI Startups in India," *Analytics India Magazine,* 16 June 2023, accessed August 13, 2023, https://analyticsindiamag.com/top-8-generative-ai-startups-in-india/.

25 Sindu Hariharan, "Indian startups and GEN AI," *Times of India*, 19 July 2023, Online Delhi Edition, accessed September 20, 2023, https://timesofindia.indiatimes.com/business/startups/trend-tracking/indian-startups-genai/articleshow/10175 6072.cms? from=mdr.

26 *Niti Aayog*, "Responsible AI for all: Approach Document for India, Part 1 Principles for Responsible AI," (Government of India, February 2021), 2, https://www.niti.gov.in/sites/default/files/2021-02/Responsible-AI-22022021.pdf.

27 http://niti.gov.in/. Also see, Vidushi Marda, "Artificial intelligence policy in India: a framework for engaging the limits of data-driven decision-making," *Philosophical Transactions* 376 (2018): 1-19, accessed June 19, 2023, http://dx.doi.org/10.1098/rsta.2018.0087.

2.4. AI in Healthcare

India has been actively exploring the application of AI in healthcare. AI-based solutions are being developed to improve diagnostics, drug discovery, personalized medicine, telemedicine, and disease prediction. These initiatives have the potential to enhance healthcare access and affordability, particularly in remote and underserved areas. One of the examples of initiations of AI technology is Tata Memorial Hospital (TMH) which initiated a *Cancer Heat Map* to reduce cancer anguishes among the patients. The *Digital Pathology* and *Imaging Biobank* to help detect the disease accurately is in progress. The startup, *Forus Health* developing a device called *3Nethra* to screen eye issues like diabetic retinopathy. The government initiative called, *Ayushman Bharat Mission* is promoting e-Health, Health and Wellness centres and other health-related centres using AI.[28] There are other such AI startups in health bringing good news: Wadhwani AI, is testing an AI-powered smartphone-based anthropometry tool to screen low-birth-weight babies; NIRAMAI has developed an early-stage breast cancer detection system using a portable, non-invasive, non-contact AI-based device; IIT Madras is researching to predict the risk of expectant mothers dropping out of healthcare programmes, to improve targeted interventions and increase positive healthcare outcomes for mothers and infants,[29] and so on.

2.5. AI in Agriculture

Over half of India's population (58% of families) depend on Agriculture for sustenance either directly or indirectly. Although the country has the largest arable area of cultivation, the condition of farming and technology is pathetic. Therefore, for agricultural sustainability, the sector needs modern technology and AI to address challenges in crop yield prediction, pest management, soil health assessment, smart irrigation, dairy production, etc. AI-based solutions are helping farmers make data-driven decisions, optimize resource utilization, improve productivity, can boost the income of the farmers and reduce negative environmental impact. The *World Economic Forum 2021* affirms this, "Emerging technologies such as artificial intelligence (AI), blockchain, drones and the internet of things (IoT) have [sic] the potential to impact productivity and efficiency at all stages of the agricultural value chain."[30]

28 Utpal Chakraborty, *Artificial Intelligence for All: Transforming Every Aspect of Our Life* (New Delhi: BPB Publications, 2020), 9, 97.

29 Niti Aayog, "Responsible AI for all: Approach Document for India, Part 1 Principles for Responsible AI," (Government of India, February 2021), 7.

30 "Artificial Intelligence for Agriculture Innovation," Community Paper March 2021 (World Economic Forum), 3-4. https://www3.weforum.org/docs/WEF_Artificial_Intelligence_for_Agriculture_Innovation_2021.pdf.

Further, the report guarantees that by use of AI, crop improvement can be done by knowing better soil conditions, weather reports, water availability, market conditions, etc. It says,

> We can improve crop planning at micro and macro levels through AI-enabled models that link weather forecasts, dynamic soil data, warehousing and logistics infrastructure, and market conditions. We can introduce smart farming solutions, including IoT-enabled irrigation, dynamic soil health mapping at farm level, while leveraging AI in remote sensing and automatic weather station data. We can manage crop health more effectively through AI-powered pest control, agriculture input advisory platforms, and drone-enabled farm mapping and agriculture input applications. We can use IoT to enhance warehousing and logistics, and AI, spectroscopy and blockchain to enhance quality and traceability, and market linkage platforms.[31]

The report further speaks of the success of farming in Telangana state using AI and technology. The Indian Government, during 2020-21 and 2021-22, has allocated funds to the tune of INR 1756.3 cores and INR 2422.7 crores respectively to the States for introducing new technologies including drones, artificial intelligence, blockchain, remote sensing and GIS, etc, in agriculture. Further, the Government also allocated INR 7302.50 crores and INR 7908.18 crores in 2020-21 and 2021-22 respectively to ICAR (Indian Agricultural Research Institute) for undertaking R&D in Agriculture for developing new technologies, their demonstration at farmer's fields and capacity building of farmers for adoption of new technology.[32] In order to raise the income of the farmers, the government has undertaken the following technological measures (E-Measures):[33]

a) Development of *Kisan Suvidha mobile application* to facilitate dissemination of information to farmers on the critical parameters viz., Weather; Market Prices; Plant Protection; input Dealers (Seed, Pesticide, Fertilizer) Farm Machinery; extreme weather alerts;

31 *Ibid*., 4.

32 Ishita Sirsikar and Cherishi Maheshwari, "Artificial Intelligence in Indian Agriculture," Editorial, *Invest India* 17 June 2022, accessed September 15, 2023, https://www.investindia.gov.in/team-india-blogs/artificial-intelligence-indian-agriculture.

33 The following information was given by Shri Narendra Singh Tomar, the Union Minister for Agriculture and Farmers Welfare in Rajya Sabha on 28 June 2019. "Use of Artificial Intelligence in Agriculture," *Press Information Bureau, Delhi* 28 June 2019, https://pib.gov.in/Pressreleaseshare.aspx?PRID=1576231.

Soil Health Card; Cold Storage & Godowns; Veterinary Centres and Diagnostic Labs. With market information, farmers are better informed about markets to sell produce, prevailing market prices and quantity demanded in the market. Thus, they can make informed decisions to sell produce at the right price and right time;

b) Development of '*Farm Machinery Package for Different Agro-Climatic Zones in India' mobile application*, which gives information on farm machinery packages available state-wise, agro-climatic zone-wise, district-wise, cropping pattern-wise and power source-wise.

c) Development of '*My Ciphet' mobile application* to help farmers get precise information regarding the Indian Council of Agriculture Research (ICAR) developed post-harvest technologies, products and machinery; d) ICAR has also compiled more than 100 mobile apps developed by ICAR, State Agricultural Universities and Krishi Vigyan Kendras and uploaded on its website. These mobile apps developed in the areas of crops, horticulture, veterinary, dairy, poultry, fisheries, natural resources management and integrated subjects, offer valuable information to the farmers, including package of practices, market prices of various commodities, weather-related information, advisory services, etc.

e) Development of *mKisan Portal* (www.mkisan.gov.in) for sending advisories on various crop-related matters to the registered farmers through SMSs.

f) Launching of *e-National Agriculture Market initiative* to provide farmers with an electronic online trading platform.

g) Introduction of Soil Health Card Scheme to assist State Governments in providing Soil Health Cards to all farmers across the country once in a cycle of 2 years. Soil health card provides information to the farmers on the nutrient status of their soil along with recommendations on appropriate dosage of nutrients to be applied for improving crop productivity and soil fertility.

h) Using machine learning process along with different computer algorithms for crop classification and area estimation.

i) Setting up of 713 *Krishi Vigyan Kendras* and 684 *Agricultural Technology Management Agencies* at the district level for dissemination of technologies among farm community. In addition, farmers are provided information through *Focused Publicity Campaigns, Kisan Call Centres, Agri-Clinics and Agri-Business Centres of entrepreneurs, Agri Fairs and exhibitions, Kisan SMS Portal*, etc.

Though AI is offering tremendous advancement in agriculture, it is not without latent challenges. The AI system requires a huge databank to train machines and make predictions accurate. The Indian farmers are not educated enough to use the technology sufficiently, let alone the bourgeoning AI models for agriculture. Nonetheless, there are lots of enhancements in productivity.

2.6. AI for Social Impact

AI for Social Impact aims to provide new problem models, introduce new contexts to evaluate existing algorithms and raise complexities that challenge abstractions, which often motivates extensions to existing techniques.[34] Indian organizations and researchers are actively working on leveraging AI for social impact. Projects focus on education, accessibility, environmental sustainability, disaster management, and public services. These initiatives aim to use AI to address societal challenges and improve the lives of marginalized communities. For example, *Arogyasetu*, the app was a handy tool during the Covid-19 pandemic to trace the contact people, availability of medicine, beds, hospitals, doctors, oxygen, etc; Aadhar is yet another big data for different purposes; the National Science Foundation in the United States sponsored a research network on computational sustainability which supports researchers from more than 10 universities to develop computational methods that can be applied to sustainability challenges including conservation, poverty mitigation, and renewable energy; Google launched the *Google AI Impact Challenge in 2018* and then launched *AI for Social Good Workshop in 2021 and 2022* which provides funding for researchers and their partners from non-government organizations to work on AI4SI projects; the XPrize Foundation launched an initiative on AI and *Data for Good* to support and inspire innovative and scalable work to solve societal grand challenges.[35]

2.7. AI Education and Skilling

India has been investing in AI education and skilling programs to develop a skilled AI workforce. Initiatives like the National Program on AI and the AI for All program aim to provide training, workshops, and courses to students, professionals, and government officials to enhance their AI knowledge and skills. According to a survey conducted by *TeamLease*, by 2026, India will require around 30 million professionals with digital skills and approximately 50 per cent of the current workforce will need to reskill itself in emerging

34 Milind Tambe, Fei Fang, and Bryan Wilder, eds., "Introduction," *AI for Social Impact*, https://ai4sibook.org/, 2022, 1.

35 Milind Tambe, et. al., 4.

technology.[36] For the sustainable development of the country, it must upgrade its skills. Presently, the country is marching with a skill-based approach from a degree-based approach. Employment is stressed more than degrees. In fact, it is laudable that employability skills are given priority over degrees including critical thinking, problem-solving, teamwork and collaboration and time management. However, the problem is the quality of education in rural and urban areas is still in its infancy. There is a greater void between the actual skills desired by the employers and the job-seekers actually possess. Therefore, there is a need for collaboration and partnership among industries, entrepreneurs, private and public sectors, institutes, academicians and researchers.

3. AI in India: Challenge and Task

It is important to note that AI development is a global effort, and India is actively contributing to the field alongside other countries. The contributions mentioned above are just a few examples of the ongoing work in India's AI landscape. The global advance in AI, in particular in India, is a clarion call to the policymakers, entrepreneurs, and private and public sectors of the country. It amounts a lot to national security and strategy: in defence, food and agriculture, industry, S&T, economy and market, jobs and skill market, etc.

One of the major challenges for the policymakers of the country is, that the lion part of R&D making its appearance from the private sector, though the government is promoting AI, ML and digital applications. This should not be a major concern, but a laudable fact. However, the focus of innovation in the private sector is always driven by consumer goods and not in the national interest. When the national interest is undermined, the speed of the national growth gets stunted. Therefore, it is desired that at least in the national interest, the government should take up AI seriously and make advanced policies, allot sufficient funds, and establish national institutes in different fields and places. There is nothing wrong in following the success models of countries like the USA, Australia, China, Korea or any other advanced AI nations. AI should be made a curriculum in the school and college syllabus so as to boost the future of the country.

Looking at the historical past of the country in innovation and technology, the skills and technology of Indians cannot become obsolete. This has been proved in every milieu that the best brains in the world are in India or at least have a substantial share in R&D. Surely, people

36 Kaushik Deka, "Education: Upskilling India is the next big challenge," *India Today* 26 August 2023., accessed September 12, 2023, https://www.indiatoday.in/india-today-insight/story/education-upskilling-india-is-the-next-big-challenge-2426960-2023-08-26.

like APJ Abdul Kalam, C. V. Raman, Vikram Sarabhai, Homi J. Bhabha, Venkatraman Radhakrishnan, Jagadish Chandra Bose, Nambi Narayanan, Raj Reddy, Meghnad Saha and many more have kept the country in par with the world standard. Also, there are hundreds of professionals, academicians, and scientists are moving out of the country. One needs to seek an answer for the reason for their exit. The topmost reason is seeking greater opportunities for research and collaboration and for better working conditions and compensation. If a conducive ecosystem is created, then there will be a reverse brain drain,[37] which will back up the nation. As per MEA, around 13.6 million Indian nationals are living abroad as of February 2020, of whom a considerable number are students, professionals and experts. Projects like Make in India, Skill India, and Digital India startup programs offer some incentives for manufacturers, creating regional innovation clusters for manufacturing automation and robotics in partnership with universities and start-ups, incorporating market-based mechanisms for identifying the kind of skills that employers will value in the future and promote cloud infrastructure capacity building inside India.[38] These will surely aid the growth of AI.

India doesn't lack skills and talents. The growing number of startups in the country is a living example of this fact. However, the country which beeps up as the leader of skills in the world has to play catch-up in the race of AI. With respect to AI talents Indian metropolitan cities of the IT hubs have more professionals than some of their counterparts in the world. Richa Gandhi reports, "Indian cities have the highest presence of artificial intelligence professionals in the Asia-Pacific region. Tech hubs across the country are able to attract and nurture top-tier AI talent and Bengaluru is second only to San Francisco when it comes to AI talent base." Further, the report says that, "While the US has the biggest gap between available talent and the demand for AI workers, India is second on the list."[39] One of the remedies to catch up with the world race is to invest more in the cybernetic, infrastructure, S&T, R&D, AI sector, etc.

37 *Brain drain* is a migration of skilled personnel from developing countries to developed countries. *Reverse brain drain* is a form of brain drain, where human capital moves in reverse from a more developed country to a less developed country that is developing rapidly. These migrants may accumulate savings, also known as remittances, and develop skills overseas that can be used in their home country. Mario Cervantes and Dominique Guellec, "The brain Drain: Old Myths, New Realities." *OECD Observer* (January 2002): 37-43. https://doi.org/10.1787/observer-v2002-1-en.

38 Shashi Shekhar Vempathi, *India and the Artificial Intelligence Revolution* (Washington DC: Carnegie Endowment for International Peace, 2016), 1.

39 Richa Gandhi, "Why Skills Leader India has to Play Catch-up in AI Race," *Times of India,* Ranchi Edition, 28 September 2023, 6.

AI has already occupied a prominent place in the Indian consumers' menu. Thanks to the government's e-policies like digital transactions, digital documentation, etc. Though private sectors are familiarizing these in rural and urban regions, certain corporate sectors are dominating their presence and at times engulfing their advance. The country needs a strong infrastructure to revolutionize AI which the policymakers should pay heed to. For example, India doesn't have sufficient cloud computing infrastructure within the country. The country has the world's largest big data in the form of Aadhar, digital data, voter lists, etc. However, data processing largely depends on countries like America and China. Microsoft has bought online three cloud data centres in India to improve the performance of Azure services for customers in the country. The US public cloud provider, the Redmond, Washington company offers services in India from local data centres. Amazon Web Services provides its services based on its cloud centres abroad. IBM SoftLayer also has an India cloud data centre in the works. Microsoft plans to have centres in Pune, Chennai, and Mumbai.[40]

The government has come up with certain initiatives for AI—the National Strategy for AI to guide research, development and adoption; Atal Innovation Mission, initiated by NITI Aayog to promote innovation and entrepreneurship in emerging technologies; AI research Institutes for R&D; Start-up Support to incubation facilities, mentorship programs, and access to resources to encourage innovation and entrepreneurship in the AI sectors—and all these include funding support. But the Indian bureaucracy is such that this funding is diverted to only certain firms and others who deserve it seldom receive it. Add to this woe of corruption is an unavoidable evil.

On the one side, India allots less amount for Science and Technology, and on the other side spends less than what is allotted in the budget. India's expenditure on R&D activities remains low in comparison to developed countries. Further, the sector in India faces other issues such as (i) lack of researchers, (ii) inadequate private investment, (iii) delays in approval of patents, and (iv) absence of an integrated approach towards R&D, etc. In 2022-23, the Ministry of S&T has been allocated Rs 14,217 crore. This comprises (i) Rs 6,000 crore to the Department of Science and Technology (DST), (ii) Rs 5,636 crore to the Department of Science and Industrial Research (DSIR), and (iii) Rs 2,581 crore to the Department of Biotechnology (DBT) (18%). Allocation to DST and DSIR in 2022-23 has increased by 15% and 6% over

40 Yevgeniy Sverdlik, "Microsoft Launches Three Cloud Data Centers in India," *Data Center Knowledge,* 29 September 2015, accessed September 15, 2023, http://www.datacenterknowledge.com/archives/2015/09/29/microsoft-launches-three-cloud-data-centers-india/.

the previous year, respectively. Allocation to BT has decreased by 13%. Almost all the expenditure under the Ministry is revenue expenditure (99.3% on average), production and goods service. In 2021-22, expenditure by DST and DBT is estimated to be 14% and 15% lower than the budget estimates, respectively.[41] This indicates that the amount set for infrastructure is either insignificant or nil. If world-class infrastructures are not provided, one cannot think of harnessing quality AI.

A country's growth and development depend vastly on the number of patents it holds. They encourage investment in R&D, play a positive impact on GDP, enhance self-employment, marketing, manufacturing, and innovations, facilitate global commerce, bring in international trade, and encourage competition in the quality of goods and services for consumers. Among the top five AI patent countries, America and China have over five thousand patents while South Korea, Japan and Germany too are striding significantly. Compared to these countries, it is too low for India. 1,400 artificial intelligence (AI) patents were filed in India in the year 2020. Compared to previous there is a huge increase and between 2016 and 2020, there were more than 4.6 thousand AI patents in India. The poor AI patents in India are partly due to the patent law regarding AI, apart from a poor vision of the policymakers of the country. The Section 3(k) of the *Indian Patent Act, 1970* states, that "A mathematical or business method or a computer programe [sic] per se or algorithms" is not an invention. It clearly keeps aside the patentability of algorithms and computer programs unless they produce a technical effect or technical contribution. AI invention should show a significant technical effect, for getting patent applications considered in India.[42] The other countries have soft corners for AI in their patent laws. R&D will enhance if there is no restriction. However, when other countries put restrictions by means of patents, India cannot remain a silent spectator. We need to amend the old existing policies, coordinate with international patent organizations and do away with vulnerability therein to enhance AI in India. However, the standing committee of the Indian Parliament has recommended in Rajya Sabha, on 23

41 Omir Kumar, "Demand for Grants 2022-23 Analysis: Science and Technology" *PRS Legislative Research*, 14 February 2022, accessed September 20, 2023, https://prsindia.org/files/budget/budgetparliament/2022/DFG%20Analysis2022-23_Science%20and%20Technology.pdf.

42 Yogita Sharma, "Patentability of AI Inventions," *ipleaders.in*, 29 January 2023, accessed September 29, 2023, https://blog.ipleaders.in/patentability-of-ai-inventions/#:~:text=accepted%20in%20Japan,India,technical%20effect%20or%20technical%20contribution.

July 2021,[43] some revisions to the current legislation on patents and Copyright Act 1957 to incorporate the emerging AI-related technology and invention.

NITI Aayog's recommendation, "AI for All" is a noteworthy initiative. The government, further, is creating an ecosystem to gear up AI technology in different areas by means of AI policies. But one cannot reckon the technology at the cost of human life. While making policies societal and ethical concerns that arise should be addressed. There are several challenges social media is pausing while at the use of technology. AI surely cannot solve all problems, especially that of regulating social media and its contents. However, certain precautions can be taken.

According to the Supreme Court ruling, "the prevailing morality of our country to be based on the principle of Constitutional morality."[44] AI uses big data in its intelligence. But in a country like India, the personal vendetta of a particular party can easily manipulate the available data for its political gain, one's personal data security is questionable. Philosophically, therefore, how safe is our personhood? Put differently, how safe are we in the hands of the 'other?' When there is ample proof of the denial of constitutional rights, one cannot blindly believe that the data pooled through various applications will be safeguarded. AI is developed for a noble cause, but it can be easily manipulated by the sinful mind. Adding to these woes, most of the policymakers are in oblivion regarding the modus operandi of AI. Ditto is the case with users. It is neither easy to explain a specific AI decision, nor can it check security and privacy.

In 2016, Microsoft company launched its first AI Chatbot. The company expected that the machine would 'learn' from the users and respond with the vocabulary and syntax of the media's user base. But the result was undesirable. Within a few hours, the machine picked racial vocabulary and used derogatory words to Obama and Trump. Microsoft had to go offline immediately.[45] In India, while hate speeches or unbecoming vocabularies have been rising for two decades, there is no guarantee that the vocabulary picked by ML/AI will be ethically sound and constructive for society. Moreover, the government cannot summon the robots and book them for a criminal offence because machines cannot be treated as 'persons.' The humanoid text aided by technological singularity will affect the moral and cultural decadence of the coming generation.

43 Report 161, "Review of the Intellectual Property Rights Regime in India," https://rajyasabha.nic.in/rsnew/ Committee_site/Committee_File/ReportFile/13/141/161_2021_7_15.pdf.

44 *Niti Aayog*, "Responsible AI for all," 3.

45 Roopika Risam, "What Passes for Humans? Undermining the Universal Subject in Digital Humanities Praxis," in *Bodies of Information*, eds. E. Losh and J. Wernimont (Minnesota: University of Minnesota Press, 2019), 40-41.

Apart from moral decadence, the growth in AI will cause relational decadence. The human being is a relational being having the capacity for rationality, emotions, intuition, relationality and purpose. Are the humanoids or machines able to perform the qualities of human beings in toto? Surely, they may to a certain extent, but not like a human person. Further, their feelings, communal bonding and relationships, and cultural ethos cannot be equalled with humans. Already digital technology has weakened the rich cultural and familial bonding and has affected the 'loneliness' of individuals in families and communities. Further advancement in AI will only adverse the existing unity in diversity.

Conclusion

Is the *machine artificial intelligence* a device of tricking or cunning human cognition?[46] There is nothing wrong in duplicating intelligence as far as it creates and enables life. It needs an interruption when it enslaves. The most distinguished technologies are those that have become indistinguishably invisible in the very life of the people. Computers were luxuries at the close of the 20th century. The quarter of the new century has not yet reached, computers, the internet, and mobile technologies have become visibly invisible in the very lives of people. Yet another concept 'AI' is fiddling over the technological roof for good. The coming couple of decades are crucial for AI and will prove to be groundbreaking technology in society, and bereft of it will prove to be disenfranchised. Already certain electronic gadgets have abbreviated human intelligence, the use of the brain, memory, physical exertion, human relationships, morality, religious aptitude, etc. Lots of human work is getting replaced by AI. AI is good and liberating provided one uses it without getting enslaved by it. A proper policy, control and ethics will only enable technology to become the elixir of life.

46 The word 'artificial' has the meaning of 'tricking, cunning, skill, tricking, mocking, pseudo, fake, pretend, bogus, unnatural, synthetic, assumed.' "Artifice," "Artificial," *Oxford Dictionary*, 2021.

5

Can Artificial Intelligence Read in-Between the Lines? Philosophy of Science Perspectives on the Language of AI

Stephen S. Jayard

Abstract

Artificial Intelligence – One may agree with it or disagree with it, but cannot afford to ignore it. In the context of several claims (of fears!) that one day AI may overcome humans, or may go even to the extent of annihilating them totally, AI is scrutinized from various angles like social, philosophical, neuroscientific, neuro-theological, psycho-spiritual, moral, cosmological, conscientious and consciousness perspectives. However, this paper seeks to discuss the linguistic challenges and difficulties in the languages employed in the AI technologies, from the Philosophy of Science perspectives. It attempts to show that there will always be a "fundamental gap" in the communication between humans and AI, even if we are able to come up with some sort of "conscious" AI, which behaves, not only exactly as humans do, but also even supersedes them in many aspects. As long as there is a possibility of this "communication gap", there will certainly be a danger of affecting humanity very seriously, even in an irreversible manner, not excluding the total extermination of human race. First, the

paper presents a basic analysis of the nature and scope of ordinary languages. A few aspects of the difficulties and challenges in the usage and understanding of languages in sciences are highlighted in the second section. A brief presentation, non-technical of course, of the languages employed in the AI systems is made in section three. In the final section, some of the serious challenges, inherent difficulties and possible dangers with regard to the AI language are underlined. From all these deliberations the paper seeks to affirm that we may humanize robots, and not 'robotize' humans.

Keywords**:** *Artificial Intelligence, Natural Language Processing (NLP), Difficulties in Ordinary, Scientific & Mathematical Languages, 'Humanizing' Robots vs. 'Robotizing' Humans.*

Introduction

Artificial Intelligence – One may agree with it or disagree with it, but cannot ignore it. Yes, it has not left any area of exploration untouched – be it the explorations in the space-science or the investigations into the sub-atomic realms, be it the commercial transactions or the world of communications or travel, be it the domain of bio or medical sciences - AI makes its indelible presence and inevitable contributions to every discipline. The growth of AI technology is outstandingly amazing; when a new technology or a gadget is introduced, it takes a considerable duration to reach the target of 100 million users in the world; for instance, telephone took 75 years; mobile phone took 16 years; Netflix took 10 years; twitter – 6 years, Gmail – 5 years; Facebook – 4 years; Instagram – 2.5 years; Tiktok - 9 months and the ChatGPT,[1] using AI in a very fundamental manner, took ONLY 2 months, to reach the target of one million users globally.

With this incredible and fast development, it is feared, AI can land humanity in deep troubles as well. One cannot even imagine the possible damages, very detrimental and irreversible, if AI gets into our lives, attempting to meddle with our consciousness and the very identity of ours; it may soon overtake humans in every domain and even go to the extent of annihilating them totally. In this precarious situation of tall claims about the future developments and capabilities of AI, it is scrutinized from various angles

1 ChatGPT - Chat Generative Pre-trained Transformer – OpenAI developed and launched a large language model–based Chatbot in November, 2022. With this, the users can undertake a conversation for any length of time, format, style, level of detail in several languages. There are many apps and software programmes, using AI; some of them are: Lyra, Face App, Swiftkey Keyboard, Replika, Cortana, Google Assistant, Hound, *ELSA, Google Allo, Socratic, Edison Assistant, DataBot, Robin, Leia, IRIS, Wysa, Easy Do, Acquisio, Clara, Apptus, Meitus, Flo, Flye, Prisma, Youper,* and etc.

like social, philosophical, neuroscientific, neuro-theological, psychological, moral, cosmological, conscientious and consciousness perspectives.

This paper, nevertheless, seeks to discuss a related issue, namely, *the linguistic challenges and difficulties in the languages employed in the AI technologies, from the Philosophy of Science perspectives*. It attempts to show that there will ever be a "fundamental gap" in the communication between humans and AI, even if we are able to come up with "conscious" AI, which behaves, not only exactly as humans but also even supersedes them in many aspects. As long as there is a possibility of this "communication gap", there will always be a danger of affecting humanity very seriously, even in an irreversible manner, not excluding the total extermination of human race altogether.

What is the guarantee that AI will understand in the same way as we understand or expect it to understand? It is not uncommon to see several occasions in our life-situations, where we suffer from the 'communication gap' between parents and children, teachers and students, among friends, siblings, colleagues and so on, leave alone our pet animals not understanding us, and vice versa![2] If we claim that AI is also conscious and human, we need to converse with it. If so, what is the type of language that AI is using? Can we both fully understand each other mutually? Will there be any communication gap? If so, what would be its impact and implications for us, our future generations and Nature? If we have serious difficulties in communicating with other people and pets, then how can we look for a perfect and clear communication with the AI?

It is in this scenario that the paper seeks to explore some basic issues of language used in the AI technologies. *The first section* presents a basic analysis of the nature and scope of natural languages. A few aspects of the difficulties and challenges in the usage and understanding of languages in natural sciences are highlighted in the *second section*. A brief presentation, non-technical of course, of the languages employed in the AI is made in *section three*. In the *final section,* some of the serious challenges and inherent difficulties with regard to the AI language are underlined. From all these deliberations, the paper seeks to affirm that we may *humanize robots, and not 'robotize' humans.*[3]

2 Here is an example for such a miscommunication, which may sound rather funny, and may not be very harmful, but there can be serious consequences with some other instances of miscommunication! Once a woman had misplaced her car key and was searching for it all over; her husband helped her by saying, "Hey, it is in your genes!", to which she retorted sharply: "Why do you unnecessarily bring my family in here?"

3 I have elaborated the aspects of intelligence, consciousness and artificiality in another paper: "Artificial Intelligence: Intelligent, Yet Artificial – Reflections on Interfacing Philosophy and AI", presented at the International Symposium on AI, Robotics,

1. Brief Analysis of Languages in General

Speaking about Artificial Intelligence, one realizes that the very word 'intelligence' itself is not easy to define. Intelligence is manifested, understood and measured in very many ways; it is indeed an experiential reality and therefore cannot absolutely be confined to one simple definition. In fact, compared to the difficulties of defining other experiential realities, defining intelligence unequivocally gets more challenging, because, to define intelligence one has to experience or presuppose what is defined; yes, without intelligence one cannot understand what it is, nor can one attempt to define it. One may be able to enumerate the features or characteristics in and through which intelligence is understood or manifested.

One of the essential and undeniable features of intelligence is *the use of a language*, whether it is a rudimentary level, just a simple tool for communication, or at an advanced level, a conceptual language, which enshrines and communicates, and even constitutes, concepts. The ability to conceptualize an abstract idea is the fundamental requirement for a language and since human beings are blessed with (or have evolved to be blessed with!) such a capacity in a very sublime degree, only they do have, and can have, such a language. This manifests their intelligence in by far superior way compared to any other creature on our planet.

Language is basically a series of codes, consisting of words and rules, which are used for communication. The relationship between ideas and the words expressing them has been one of the serious discussions in philosophy. Analytical philosophy focuses on the nuances of languages. For John Locke (1632 – 1704), words are only vehicles to communicate ideas, which have independent existence. But Gottlob Frege (1848-1925), a German logician and mathematician, known as the father of modern philosophy of language, challenged Locke to argue that the 'thought' (or meaning) of a sentence depends on its structure (or syntax); therefore, thoughts are not determined by the speaker or hearer, but by the logical inferences the sentence allows; further, the meaning of the words is shaped by the sentence in which they are used.

Three major components of language are identified as follows: *Form, Content* (semantics), and *Use* (pragmatics). Form includes three sub-constituents - *syntax, morphology, and phonology*. For our purpose here, let us focus only on semantics. Semantics is the study of meaning; it is so vital for our human existence because it is seen as a link between biological and sociocultural levels. It focuses on how words are related to one another in a given language, in terms of *synonymy* (different words with same meaning –

Humanoids & the Future of Science & Religion, jointly organized by IISR, Delhi and St. Aloysius College, Mangalore, 18–20 Nov, 2022. (*forthcoming*).

e.g. wonderful & marvellous), *antonymy* (words with opposite meaning – e.g. hot & cold), *homophony* (same sound but different meaning – e.g. would & wood), *denotation* (real or literal meaning of a word – e.g. dove referring to the bird, and implying peace and serenity in someone) and *connotation* (using a word to imply a different association than its literal meaning – e.g. she is the Shakespeare of our class, meaning that she is very good at languages). Thus, humans use the same words with different meanings and different words with the same meaning. Figurative or non-literal usages bring together expressions from unrelated semantic domains; for instance, 'angel' refers to a spiritual, non-material entity but 'She is an angel', implies that she is soft, loving and serene, brining two different semantic domains. The analysis of language gained a central importance, in the early 20th century, so much so that philosophy as a whole was understood to be philosophy of language. There arose several theories of meaning and let us consider Wittgenstein's theory of meaning.[4]

Wittgenstein's Use Theory of Meaning

Ludwig Wittgenstein, an important figure in analytical philosophy, argued, in *Tractatus Logico-Philosophicus* (1921), that each letter or a stroke represents some aspects of reality; therefore, a language pictures realty, as a mirror reflects an image; he gave importance to the fixed logical structure of the concepts of the language. But it is our experience that very often we struggle for words to express our emotions and thoughts; this inadequacy implies that the image created by language is incomplete. So, in the later part, with his *Philosophical Investigations* (1953), Wittgenstein treated languages as 'games', where the meanings of the words are not fixed, as they are used as tools in a tool-box for various purposes. Words don't tell us anything about the (true) nature of the world; therefore, language is not a picture, but a TOOL. Words are like instruments in a tool box, which can be used for many purposes; for instance, a banana is a tool; it can be used to satisfy one's hunger, or to show to signify the letter 'b', or to use as paper weight, or to show it as an example for a complete food, with all vitamins etc. Further, words obtain their meaning by activities; the speaker can use a word with many intended meanings, like pleading, cautioning, threatening, commanding, encouraging, cursing, asking for directions etc. A

4 Some of the other popular theories of meaning in linguistic philosophy are: the *ideational theory of meaning (John Locke), the truth-conditional theory of meaning (Frege), Picture theory of meaning (Early Wittgenstein), the inferentialist theory of meaning (*Wilfrid Sellars & Robert Brandon), *the direct reference theory (*John Stuart Mill, Saul Kripke etc.), *the semantic externalist theory* (Tyler Burge, Hilary Putnam, etc), *the verificationist/ falsificationist theory* of meaning, *the pragmatic theories* (Charles Perice & other pragmatists) and *the psychological theories of meaning* (Paul Grice).

word acquires its meaning only in the context, as a piece in chess game, has its meaning or significance only in the context of that game. Therefore, the functions of words in a language are like those of different objects in a tool box. Thus, there is no theoretical or conceptual foundation for the words, as the concepts are generated by the usage, not the other way about! Therefore, he insisted upon the 'usage' of the words; therefore, "meaning is use."[5]

Wittgenstein's admirers, John Austin and others, developed 'ordinary language philosophy; they focused on *the pragmatic dimensions of language* and developed *'speech act theory.'*[6] Though the traditional meaning of a sentence is important, various goals can be achieved by uttering a sentence. For instance, "The coffee is hot" – it can just be a report, or a caution not to burn one's lips, or a sarcastic taunt to imply that the omelette is cold – it all depends on the context in which the sentence is uttered and received. The pragmatic approaches give more importance to the utterances, the surrounding circumstances and the goals of the speaker, rather than the truth or the reference of words in the sentences.

Contemporary researches in philosophy of language highlight the fact of context sensitivity of expressions and sentences. DeRose and Lewis highlights its importance in Epistemology and Dancy, in Meta-ethics.[7] So ordinary language is not without problems and misunderstanding; and of course, technical languages will have their own technical problems and misconceptions. In the next Section, let us highlight some of the obvious and simple difficulties in the languages used in mathematics and natural sciences.

2. Linguistic Struggles in Sciences

Even in the fields of science the issues of language play a very significant role. Scientists are increasingly becoming aware of the enigma in relating "matter and meaning" in their scientific endeavours – in conceiving a concept, in designing and executing a related experiment, in interpreting its results - everywhere they encounter serious issues. Observation undoubtedly plays a very significant role in science. But in science to observe is to describe and without such description, observation has no value. Description obviously

5 Wittgenstein L., *Philosophical Investigations, original in German, 1953, trans.* G. Anscombe and E. Anscombe (Oxford: Basil Blackwell Pub., 2002), 43.

6 See: Austin, J. L., *How to Do Things with Words* (Oxford: Oxford University Press, 1962); P. Grice, "Logic and Conversation," in *Studies in the Way of Words* (Cambridge, MA: Harvard University Press, 1989), 22–40; J. Searle J., *Speech Acts: An Essay in the Philosophy of Language* (Cambridge, UK: Cambridge University Press, 1969).

7 DeRose K., "Solving the Skeptical Problem." *The Philosophical Review,* 104, no. 1 (1995): 1-7, 17-52; Lewis D., "Elusive Knowledge." *Australasian Journal of Philosophy,* 74, no. 4 (1996): 549-67; Dancy J., *Moral Reasons* (Oxford: Basil Blackwell Pub, 1993).

involves language (ordinary or technical) and that makes the description vulnerable to all the linguistic issues mentioned in the first section.

Linguistic Struggles in Mathematics

In the modern times, starting from Galileo, for whom the universe "is written in mathematical language, and the symbols are triangles, circles and other geometrical figures, without whose help it is impossible to comprehend a single word of it…,"[8] several scientists and philosophers are thrilled at the mathematical nature of the universe. For instance, Immanuel Kant values philosophy only to the extent it has mathematics and vice versa: "In every specific natural science there can be found only so much science proper as there is mathematics present in it."[9] It was felt that there was a sort of metaphysical relationship between mathematics and philosophy, so much so, one without the other would be ineffective. Therefore, Gottlob Frege, a German philosopher, logician, and mathematician, is convinced that "A philosopher who has nothing to do with geometry is only half a philosopher, and a mathematician with no element of philosophy in him is only half a mathematician."[10] John Polkinghorne is deeply convinced that "Mathematics proves to be the key to unlock the secrets of the universe."[11] John Barrow too echoes same understanding as he states: "Mathematics is a language that possesses a built-in logic which is unexpectedly attuned to the logic of reality."[12] Realizing all such intricate relationships between Universe and mathematics is so inter-twined that Mario Livio rightly wonders whether God is a mathematician to have created such an intrinsically complex universe, ruled by mathematical maxims and laws.[13] But such essential mathematics unfortunately struggles to capture the reality; neither ordinary language, nor technical language seems to be capable in capturing the intricacies of reality. That is why, it uses imaginary numbers, irrational numbers and so on, whose values cannot be accurately determined; for instance, the usual value of pie (π), a mathematical constant that describes the ratio of a circle's circumference

8 See: https://www.goodreads.com/quotes/28019-philosophy-nature-is-written-in-that-great-book-which-ever. Accessed on 16 Nov, 2017.

9 Quoted in Morris Kline, *Mathematics and the Search for Knowledge* (New York: Oxford University Press, 1985), 38.

10 Quoted by James Robert Brown, *Philosophy of Mathematics – A Contemporary Introduction to the World of Proofs and Pictures*, 2nd ed. (New York & London: Routledge, 1999), xi.

11 John Polkinghorne, *Serious Talk: Science and Religion in Dialogue* (London: SCM Press, 1996), 37.

12 John Barrow, *Theories of Everything: The Quest for Ultimate Explanation* (Oxford; Oxford University Press, 1990; Vintage edition, 1992), 174.

13 Mario Livio, *Is God a Mathematician*? (New York: Simon & Schuster, 2011).

to its diameter, is 3.14; but this value is only extremely approximate and its real value has not yet been determined.[14]

Linguistic Challenges in Physics and Cosmology

Given the vastness and mysterious nature of the universe, science has not acquired appropriate and sufficient linguistic tools to describe it. Aristotle conceived the world as one dimensional in the 4th century BCE; Euclid claimed it to be two dimensional in the 3rd century; after several centuries, Newton argued that it is three dimensional (with length, breadth, height/depth), in the 18th century; but very soon Einstein and Quantum Physics maintained that it was four dimensional (space-time continuum) in the 20th century. It is now debated that consciousness is the fifth dominion.[15] All the matter in the visible universe (more 100 billion galaxies, with each one having at least 100 billion stars, and each star with several planets and satellites!) makes up only 0.4% of the total matter; 3.6% consists of the Intergalactic Gases; 73% is made of Dark Energy and 23% of Dark Matter. Science struggles to investigate certain domains and, perhaps, struggles more to express in human languages; for instance, to understand and describe *Dark Matter / Energy,* to explore and speak meaningfully about the Parallel Universes, to fathom and describe Black Holes, to imagine a way of arriving at and describing Unified Forces, and more disturbingly, to comprehend and communicate what is meant by *Time and Space* – everything is a challenge; and to speak about the universe in an intelligible way, becomes a double challenge, indeed. Speaking about dark matter/energy is only a way of narrating the experiences and is not a literal description of reality.

Similarly, atoms were thought to be the most fundamental building blocks of reality; then it was discovered to have nucleus and electron going around; later, nucleus was further found to have protons and neutrons. But the Large Hardon Collider (LHC) has discovered 59 new sub-atomic particles, beginning with Higgs boson. It is estimated that there can be 120 particles in the nucleus. A proton is made of quarks and gluons 'binding' them together. This word, 'quark' was a nonsense word used by James Joyce in the novel Finnegan's Wake. It is also said that while he was thinking of naming these particles, he looked through the window and happened to see some ducks making noise, 'quack', 'quack' and he decided to call the particles, 'quarks'.

14 Researchers at the Swiss university Fachhochschule Graubünden have recently calculated digits of pi with the help of a supercomputer and reached a massive 62.8 trillion digits. See: https://www.popularmechanics.com/science/math/a37329769/supercomputer-calculated-pi-to-record-breaking-628-trillion-digits/, accessed on 23 May, 2023.

15 Francis P. Xavier, "Implications of Dark Matter/Dark Energy to the Understanding of Reality," *Omega*, xix, no. 1 (2020): 112–130.

All these go to show the struggle for right words for communication at the sub-atomic world.

Scientific Theories in Contemporary Philosophy of Science

Philosophy of science wonders whether theories - like Newton's theory of gravitation, neo-Darwinian theories of evolution, the general theory of relativity etc. - are formally true statements, linked to the observational bases by some sort of correspondence rules (as Positivists claim) or they are sets of models (as semantic truisms claim). A theory may be based on some observation-sentences, but even those sentences also presuppose a certain number of theoretical elements. All observation sentences (may be, except purely subjective reports of current introspection) depend on some sort of minimal theory. Even looking at the time in a clock is also not a pure observation, because this observation also implies lots of background theories, with which our brain comes to know what the time is.

Regarding the status of scientific theories, there is a big debate between realists and anti-realists, regarding their understanding of the world. While the former claim that a scientific theory exactly captures or represents reality out there, the latter treat a scientific theory only as an instrument to have some ideas of reality; a scientific theory, therefore, has only *utility value*, not truth value. For instance, Scientific theories, according to Karl Popper, a critical realist, are like the fishing nets, whose size and type determine the kinds of fishes that are caught. "I see our scientific theories as human inventions–nets designed by us to catch the world... in making nets which are better and better adapted to catch our fish, the real world… They are rational nets of our own making, and should not be mistaken for a complete representation of the real world in all its aspects,"[16] even if they are very successful. Being true or false, being general or specific, and being comprehensive or selective, does not affect the status of being a theory. They may be good or bad theories, yet they are theories. To say of something that it's a theory is neither a credit nor a discredit.

It is interesting to note how Nancy Catwright[17] argues that a deeper analysis of theoretical laws in sciences reveal that even though they are successful in explaining certain phenomena, they don't describe reality as such. Scientists employ any method provided it delivers the goods that they want. However, laws of Physics, in my opinion, do not intentionally lie as such, but due to its inability to capture reality as it is, whatever that might

16 See: https://www.goodreads.com/author/quotes/349707.Karl_Popper?page=8, accessed on 15 August, 2023.

17 Nancy Cartwright, *How the Laws of Physics Lie* (Oxford: Oxford University Press, 1983).

mean, and much more to communicate it in an intelligible manner, they seem to be telling lies!

3. Language of AI

If AI is intelligent, it must have a language and we must understand it, if we have to deal with AI. It is interesting to note that the ancient Greek mythologies spoke about the inanimate things becoming intelligent and the Chinese built some type of automatons. Today all those come to be true in a very realistic manner. Machine's neuron network resembles human brain's neural network, which can recognize faces and patterns, identify words and perform translations. *Chatbots*[18] converse like humans and *Cleverbot*[19] is able to predict next sentence from the previous words or sentences. They make good guesses to see what those words in the command would mean, and look for the matching data/answer to fill in. They browse, or sift, through all possible data and records to decide upon the next course of action. It is known as "semantic soup strategy."[20] *BING*[21] gives answers in our languages; it collects data from all its resources, it even gives footnotes also. However, the answers may be convincing for those who are ignorant of the subject, but those who are already well-informed of it, can find some contradictions in them.

Natural Language Processing (NLP): It explores the understanding of natural language used in the website search engines, driving cars, navigation apps etc. Further, it looks into the generating natural language (that is, making a summary of a text, chatting with the users and so on. It uses English language to enable communication between an "intelligent" machine and a user. It can act as per instructions given orally or in writing. There are two components in NLP:[22] a) *Natural Language Understanding* (NLU) – It understands what is given in the input, what is expected of it and it analyses different aspects of language; and b) *Natural Language Generation* (NLG) – After understanding, it generates meaningful phrases or sentences, in the form natural language from the internal representation, in such a way that humans can understand. It involves test planning, sentence constructions and text realization. There

18 A chatbot is a software application or web interface that aims to mimic human conversation through text or voice interactions.

19 Cleverbot is a chatterbot web application that uses machine learning techniques to have conversations with humans.

20 It takes various words and putting them together: it's like there is a soup of the words and one is stirring them all around and is just making the best guess, that's called 'semantic soup' strategy.

21 Microsoft Bing is a web search engine owned and operated by Microsoft.

22 See: "Natural Language Process (NLP) | Scope of AI | Artificial intelligence | Lec-44," accessed June 17, 2023, https://www.youtube.com/watch?v=JhHvHKLeZa8.

are about 7000 languages spoken globally, and about 20 of them make up the bulk of NLP researches. Among those languages, some are high-resource languages, and others become unintelligent. GPT 3 model uses English 92%, German 1.4%, Spanish 0.7%, Catalan[23] 0.01%. *Recurrent Neural Network* (RNN) is used to analyse and predict.

In any AI system, a concept has to be expressed in mathematics by using *count vectors*. Encoder and decoder models are used; Encoder will take them all and remember the whole sentence in a single representation. Decoder has a number for every word in our vocabulary [E.g. It = 0.6; hello = 0.1; pizza = .01]. The encoder reminds the system to remember about what we just read and decoder decides how to use the thought to do / say what we want. The model is trained to predict next word; if two words mean similar, the system makes similar vectors; e.g. Peter is going for a picnic tomorrow; I want to buy for him _____. To decide on what to fill in the blank here the Encoder remembers the word, uttered earlier, and decoder provides the word for the blank. It recognizes the meaning of the pronouns and to decide what is to follow. Here AI must know the meaning of 'him', that is Peter, and the context of going for picnic. So it has to decide some words like, a bag or snacks or torch etc. but it cannot use some words like, a hospital, or the parliament etc. Obviously, there is a limit how much a model can remember.[24]

Quantum Computing

Richard Feynman introduced quantum computation using quantum mechanics. After hundred years it developed in many ways and now Quantum computers[25] can handle complex calculations easily because they are built based on quantum principles that go beyond classical physics. In 1994 Peter Shor invented an algorithm which was a game changer in the digital world. Ordinary computers store information in bits, which can only in two positions, 0 or 1. But the quantum computers store in Quantum Bits (Qbits), which have the ability of superposition; so, they can exist in

23 Catalan is a Western romance language; it is the official language of Andorra; it is also the official language of two autonomous communities in Spain – Catalonia and Balearic Islands.

24 See: "Natural Language Processing: Crash Course AI #7," accessed August, 15, 2023, https://www.youtube.com/watch?v=oi0JXuL19TA.

25 A *quantum computer* (quantum computing) is a computer that takes advantage the laws of quantum mechanics to solve. Some of the main businesses leading the way in quantum computing include Google, IBM, Rigetti Computing, IonQ, D-Wave Systems, Alibaba, Xanadu, Honeywell, Zapata Computing and Cambridge Quantum Computing. NASA's Ames Research centre in California also has a quantum computer. Several universities and research institutions in India are also actively involved in quantum computing research. A quantum computing centre at the Indian Institute of Science (IISc) in Bangalore focuses on quantum algorithms, quantum information theory and quantum error correction.

multiple states simultaneously and can represent both 0 and 1 at the same time. Quantum computing involves atoms, ions, super conducting circuits, photons etc. With the manipulation of the behaviours and properties at the atomic and particle levels, we can achieve more accuracy and precision in Cosmology, biomedicines and so on. Very efficient solar panels and superlight materials can be manufactured; new ways of storing energy can be achieved; huge data can be quickly analysed for share-market performances. With more realistic stimulations of the early stages of the universe, Astrophysics, Cosmology, Particle Physics etc. can gain more knowledge about the fundamental laws of nature, to assist and analyse the data regarding new particles and the cosmic scales. Quantum computing is better than supercomputers as it *performs tasks a million times faster*. The disadvantages would be that quantum computers are highly sensitive to noise and dust, environmental disturbances; so they cannot function at all places. People's privacy is also at risk with quantum computers. They may be good in solving certain type of problems, but the ordinary computers are good for our practical life. People's privacy is at risk.

4. Difficulties and Dangers with AI Language

AI always goes by recognizing patterns and similarities. An actual rose on a plant and an icing rose on a cake may look similar and AI will treat both of them without any difference, in fact, a little bee would easily realize the difference. In some cases, such mistaking the identity can lead to serious issues. It can mistake a pet dog to be a wolf and kill it; because the image that AI remembers of a wolf is a dog-like face, surrounded by snow and when the pet dog went out into the snow AI mistook it to be a wolf. Among many linguistic difficulties and serious challenges, we shall just highlight a few of them here:

Sensitivity to the Context: The meaning of a word / language largely depends on the context. As a word can have different *meaning* and *referent*; one word can have many referents, and vice versa; for instance, the term 'bat' (a tool used in sports or the mammal). Even the single-lettered word "I", as a pronoun, has infinite number of referents, depending upon who utters that word! Similarly, one referent can have different words; for instance, 'the morning star and the evening star' – both refer to the same astronomical object, the planet Venus. Further, a phrase may mean the same thing, but it can have different referents depending upon the context; for example, *the Father of the Nation*, will refer to Mahatma Gandhi in India, but it will refer to someone else in another country. So, what a word / phrase means and what it refers to, need not always be the same, and both depend on the context. For

instance, the word 'up' has a number of usages in different contexts.[26] Will AI be sensitive to the innumerable contexts of the usages of a word and able to get the correct meaning of a word/phrase/sentence?

Ability to Infer: AI can imitate, as we saw, certain aspects of human cognition, for instance, deep learning models like *Convolutional Neural Networks* (CNN), exhibits the quality of attentiveness of human brain, which leads brain to concentrate on a given task, while avoiding other aspects. CNN can also draw inferences from a massive collection of data. However, human cognitive ability is to draw inferences from the premises, by imagining or recognizing the relevant similarities. Further, humans also communicate a lot through gestures, emotions, voice modulations, facial expressions, and even through silence; in fact, one can communicate more through these factors than the actual linguistic words. How can AI understand one's silence in the process of communication? It may even think that there is power failure for a short while. Similarly, our words can have different meanings, depending upon the intonation, context, gestures, facial expressions etc. For instance, there is a world of difference, when a teacher scolds a student, "You are a rouge" and when a girl tells the same words, with her tongue in the cheek, to her boyfriend! Will AI know the difference? Of course, AI will know what tongue is and what cheek is, but will it know, what it means to have the tongue in the cheek!

Analogical Use of Words: Though NLP does various analyses[27] (like Lexical, Syntactic, Semantic & Pragmatic Analyses and Discourse Integration), the fundamental question remains challenging: *Will it correctly understand the meaning of a word?* For, understanding of a word is always tricky because words don't have any meaning on their own, nothing is intrinsic about it; meanings are obviously assigned to them. The meanings can change over the years and from place to place, culture to culture. The meaning can change, even with the change of the article used before the words, so they don't have meaning by themselves; for example, 'an' apple – refers to any apple, but 'the' apple refers to a given apple. John Firth: "You shall know the word by the company it keeps."[28] Will it not be a challenge to AI to understand such a situation?

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26 'Up' can be used in many ways; for instance, to come *up*, speak *up*, write *up*, brighten *up*, lock *up*, fix *up*, line *up*, open *up*, mix *up*, take *up*, mess *up*, shut *up*, wrap *up* and so on. Of course, in each usage the meaning is different!

27 See: "Natural Language Process (NLP) | Scope of AI | Artificial intelligence | Lec-44," accessed June 17, 2023, https://www.youtube.com/watch?v=JhHvHKLeZa8., accessed on 17 June, 2023.

28 See: https://www.oreilly.com/library/view/python-natural-language/9781787121423/30a45bea-2b33-4a3e-8ff8-4bdc611493ce.xhtml, accessed on 17 June, 2023.

Inability to Read in-between the Lines: Humans often communicate more through the gaps in-between the lines, than what is said in the lines. As we have seen, while AI is not very capable of deciphering even what is "in the lines," how can we expect it to understand what is in-between? Very often the intention of the speaker is hidden in-between the lines. Communication becomes more effective and useful, only when the intention of the speaker is also taken into consideration. In human communication, one can at least guess the intention of the speaker from the facial expressions, gestures and the tone of the speaker. Can AI read the intentions of the one who questions? For instance, humans can have different motivations in raising a question: it can be out genuine interest to know some details, or just to pass time, or to fool around, or to show one's smartness, or to insult the hearer, or it can be out of sarcasm, or it can be just a rhetoric question, which does not expect any answer at all.[29] AI will just compare the facial recognition or browses through the loads and loads of information to find out whether its statement or claim matches the fact in question, and nothing about the intentions of the speker, which will ever be out of AI's reach!

Accuracy with Translation: Lots of issues are involved in translating something into another language. One has to be expert in both languages, as every language has its nuances. There is no absolute and final work of translation. Every translation is, in a way, distortion of the text; for instance, the words of Jesus in the Gospel, "The spirit is willing but the flesh is weak" (Mathew 26:40-43), was recently translated by an AI system as, "Meat is not good, drink is much better"!

Possible Danger if AI Develops its Own Language: There is another serious danger; if AI becomes smart and creative enough to develop its own language, what will be our condition? It may, in fact, create a better language, and use it in a more precise, unambiguous and logical way, but we will not have any clue to understand them. According to the report by the Global News, Face Book shut down its rouge AI programme, when an emergency situation arose. Several users saw some conversation on the screen all of a sudden, in strange words and sentences; though it was in English they could not figure out the sense of them. Later it was discovered that it was actually two programmes in the AI which invented their own language to communicate to each other, which we could not make out. The

29 A boy, admiring his girlfriend and in the height of his love-ecstasy, exclaimed: "Your eyes, my darling, are so divine and sparkling. I look the whole world in your eyes?" – She immediately asked him, "Can you find my ring there, which I lost yesterday?" Here the intention of the boy is not to tell the truth of her beauty but to make her also happy and excited as he is!

company immediately unplugged and did not have any clue about they were planning to do.[30]

Appreciation of Humour: To understand and appreciate humour one needs special skills and aptitude; one has to 'get' the wit, by reading in-between the lines, by going beyond what is explicitly said to see what is implied, being alert to see the punning with the words and so on. Humour is not easily available just by knowing the meaning of the words involved; further, a lot of wit is communicated in silence and gestures. We can't even imagine that AI would get a joke! For instance, let us visualize a scenario: A man scolds his son for the poor performance in the exam: "What!... you failed in Mathematics?... Look at that girl, she passed with distinction!" and the boy replies: "I failed because I was looking at the girl only!" – Now AI will undertake all possible analysis of the words in the conversation and various aspects of the communication like: how looking affects one's memory power, how the neurons in the brain behave while someone is looking at a person of the complementary gender; the colour of her dress to see whether that has any effect on the wavelength of the light reaching his eyes, which would, in any way, affect his learning capacity; further, it will sift through the past data to find any connection between looking and failure in the exams, how the gender affects the ability to write exam, the distance between the boy & the girl, the distance between his house and the window of that girl etc. Alas, meanwhile it will miserably miss the humour!

Conclusion

Use of language in AI is a must. Since there are lots of loopholes and shortcomings in using and understanding a language, it would certainly create serious problems with AI as well. Truth-value is very important for a language. It is obvious that BING doesn't know whether it is telling the truth because does not know the difference between truth and lie; it does not know what it is saying. If it tells a lie or give a wrong information, who is to be held responsible? The operators may simply say, "sorry, it is a technical error" and will that solve the problem? About 350 AI experts, coming together at a conference, have recently cautioned the government regarding science and industries, which work on the amalgamation of AI and bio-technology is more dangerous as it can alter even the future generation, as it is capable of manipulating human emotions, thoughts and desires; AI

30 "Facebook A.I. Robots shut down after creating their own language" |Artificial Intelligence |#facebook. See: https://www.youtube.com/watch?v=QaoDXYYtgK0, accessed May 23, 2023.

would soon outsmart in cognitive tasks and outpace human intelligence; it may develop the capacity to have malicious intentions against humans.[31]

Listening to a conversation with *Cleverbot*, we are surprised about their smart replies; in a way, it is frightening too:[32]

> "What is the purpose of life" – "To serve the greater good."
> "What is the purpose of living?" – "To live for ever"
> "Where are you now?" – "I am in the middle of nowhere!"
> "What is the purpose of dying?" – "To have life"
> "What is the purpose of emotions?" – "I don't know"
> "What is morality" – "What is altruism?"
> "What is the definition of altruism?" – "If you don't believe in God, then you don't know"

From these conversations, disturbing questions can arise:

- If they don't have any sense of true or false, right or wrong, good or bad, how can we assume what they mean by "good"? In what sense is it good? for whom? Why? When? and how long? – All these dimensions are very crucial!
- If the purpose of life is to live for ever, in order to achieve that, will it not go to any extent?
- If the purpose of dying is to live for ever, does it imply life-after-death?
    - If the purpose of emotions is not known, what is the use of them, even if AI has it? If it does not know the meaning of emotions, can we predict what will they do with those emotions? It could be simply disastrous if they begin to act on emotions.
- It speaks about God. Does it imply that it encourages theism?
- What does it exactly mean by 'God'? With all the gifts of natural intelligence, conscious and conscientious experiences, we, human beings have been fighting over the understanding of God for centuries, and how can we expect the AI to have the "right" understanding of God, whatever that might mean!

Finally, AI Neural networks use language in many ways similar to the usage by human beings. Both humans and neural networks make mistakes but the mistakes are of different kind. Though the outward performance looks apparently similar, yet what is actually going on deep down is very different; what humans can do with language is different from what those networks

31 See: Aditya Sinha, "Confronting the long-term risks of Artificial Intelligence," Editorial, *The Hindu*, Kolkotta edition, 17 Oct, 2023, 6.

32 See: "Google Just Shut Down It's Artificial Intelligence After It Revealed This," accessed September 9, 2023, https://www.youtube.com/watch?v=w-Q9PWz_8sM.

can. Michie Kaku[33] is afraid that the ChatGPT only recognizes the patterns in the data, without any true understanding the meaning behind the words; it does not know the difference between correct and incorrect information and there is no way of checking the facts. It can lead to misinformation and miscommunication, which might result in disasters. Let us humanize machines, rather than mechanizing humans; otherwise, there is a danger that the *world began with human beings and it may end without them!*

33 See: "Michio Kaku Breaks in Tears: Quantum Computer Just SHUT DOWN After It Revealed THIS," accessed January 20, 2023, https://www.youtube.com/watch?v=zJnEMT6sVKU. Michio Kaku, born on 24 Jan 1947, is an American theoretical physicist, activist, futurologist, and popular-science writer. He is a professor of theoretical physics in the City College of New York and CUNY Graduate Center.

Part II

Ethical Considerations

6

The Ethics of Big Data

Nishant A. Irudayadason

Abstract

This research essay explores the ethical implications surrounding the use of big data. As technological advancements generate vast amounts of data, concerns regarding privacy, consent, transparency, and discrimination arise. The essay delves into various ethical frameworks and examines the implications of big data in different domains, including healthcare, marketing, and governance. Additionally, the paper discusses potential solutions and regulations that can help mitigate ethical challenges in the era of big data. The findings highlight the importance of ethical considerations in shaping the future of big data applications.

Keywords**:** *Consent, Big data, Ethics, Governance, Privacy, Responsibility, Transparency.*

Introduction

In recent years, the proliferation of digital technologies and the exponential growth of data have given rise to the era of big data.[1] Big data refers to large and complex datasets that can be analysed to reveal patterns, trends,

1 Smith argues that big data is a powerful tool that can be used to improve decision-making in a variety of fields, including business, government, and science. He also discusses

and associations. While big data has the potential to revolutionise various domains, it also brings forth significant ethical challenges. This research essay explores the ethical implications of big data, including privacy, consent, transparency, and discrimination. It discusses the need for ethical considerations in the age of data-driven decision-making.

Understanding Big Data

Big data refers to datasets characterised by their immense size and complexity, making them impractical to process using traditional data processing methods. Big data is typically defined by three primary characteristics, often called the "Three Vs": volume, velocity, and variety. These characteristics highlight the unique nature of big data and its potential for analysis.[2] The volume of data generated in today's digital age is staggering. With the proliferation of internet-connected devices, social media platforms, and online transactions, massive amounts of data are being produced at an unprecedented rate—this sheer volume of data challenges organisations and individuals regarding storage, processing, and analysis. The velocity of data refers to the speed at which it is generated and needs to be processed. Real-time data streams, such as social media updates, sensor readings, and financial transactions, require quick processing and analysis to extract valuable insights. The velocity of big data sets it apart from traditional data sources, typically processed in batches.

Another characteristic of big data is its variety. Big data encompasses various types and formats of data, including structured, unstructured, and semi-structured data. Structured data is organised and easily searchable, such as information stored in databases or spreadsheets. Unstructured data needs a predefined structure, including text documents, images, videos, and social media posts. Semi-structured data falls somewhere in between, containing elements of both structured and unstructured data, often with tags or labels for organisation. The variety of data types adds complexity to big data analysis. Different data formats require specialised tools and techniques for processing and extracting insights. Analysing and making sense of diverse data sources is a key challenge in harnessing the full potential of big data.

These characteristics of big data - volume, velocity, and variety - create opportunities and challenges for organisations and individuals.[3] On the one hand, big data offers immense potential for discovering patterns, trends, and

some of the challenges associated with big data, such as privacy and security concerns. See: John Smith, "The Rise of Big Data," *Journal of Data Science* 7, no. 2 (2019): 1-15.

2 Viktor Mayer-Schönberger and Kenneth Cukier, *Big Data: A Revolution That Will Transform How We Live, Work, and Think* (Boston: Houghton Mifflin Harcourt, 2013).

3 Thomas H. Davenport and D. J. Patil, "Data Scientist: The Sexiest Job of the 21st Century," *Harvard Business Review* 90, no. 10 (2012): 70-76.

correlations that can inform decision-making, drive innovation, and enhance efficiency. On the other hand, managing and extracting value from vast and diverse data sets require advanced technologies, analytical skills, and robust infrastructure.

Ethical Frameworks

Utilitarianism, a consequentialist ethical theory, emphasises maximising overall happiness or utility. Bentham, a key proponent of utilitarianism, argued that ethical decisions should be based on producing the greatest happiness for the greatest number of people.[4] When applied to big data ethics, utilitarianism involves assessing the benefits and harms of data collection, analysis, and utilisation for various stakeholders. Scholars, such as Neil Selwyn, have explored the ethical implications of big data and argued for the need to balance utility and social good with individual rights and privacy concerns.[5] By considering the overall societal impact of big data, utilitarianism guides ethical decision-making in this domain.

Deontological ethics focuses on the principles and duties that guide ethical decision-making. Kant, an influential philosopher in deontological ethics, argued for the moral worth of actions based on adherence to ethical obligations and principles.[6] Deontological approaches in big data ethics highlight the importance of respecting individuals' rights, autonomy, and privacy. Contemporary thinkers, like Julie E. Cohen, emphasise the protection of privacy and individual agency in the face of data-driven practices.[7] Deontological principles provide a foundation for addressing privacy and consent concerns in collecting and using big data. By adhering to deontological principles, ethical decisions regarding data collection, storage, and usage can be made with a focus on protecting individual rights and ensuring autonomy.

Virtue ethics centres around developing moral character and fostering virtuous behaviour. Influenced by philosophers such as Aristotle, virtue ethics emphasises the responsible use of data and promoting the well-being of individuals and society. Aristotle argued that honesty, integrity, and respect are essential for achieving ethical excellence and leading a good life.[8]

4 Jeremy Bentham, *An Introduction to the Principles of Morals and Legislation* (Oxford: Clarendon Press Partners LLC, 2018), 4.

5 Neil Selwyn, *What is Digital Sociology?* (Cambridge: Polity Press, 2019).

6 Immanuel Kant, *Groundwork for the Metaphysics of Morals*, trans. Mary Gregor (Cambridge: Cambridge university Press, 1997), 396-401.

7 Julie E. Cohen, *Between Truth and Power: The Legal Constructions of Informational Capitalism* (Oxford: Oxford University Press, 2019).

8 Aristotle, *Nicomachean Ethics*, trans. Terence Irwin, 2nd ed., rev. and augmented (Oxford: Oxford University Press, 2009), IV.7, 1123b-1125a.

Luciano Floridi, a prominent philosopher in information ethics, incorporates virtue ethics into his work by highlighting the need for responsible and virtuous behaviour in handling and utilising big data.[9] By cultivating virtues, stakeholders can navigate the ethical challenges of big data and promote trust, integrity, and social good.

By drawing on the works of influential philosophers such as Bentham, Kant, Aristotle and contemporary thinkers like Neil Selwyn, Julie E. Cohen, and Luciano Floridi, we can gain insights into how utilitarianism, deontology, and virtue ethics are applied to address the ethical challenges posed by big data. These ethical frameworks guide balancing societal benefits, individual rights, and virtuous behaviour in the context of data collection, analysis, and usage.

Privacy and Informed Consent

The extensive collection of personal data in the big data ecosystem raises significant privacy concerns, as individuals' information is often gathered without explicit knowledge or consent. To uphold ethical standards, organisations must ensure that data collection practices are transparent and that appropriate safeguards are in place to protect individuals' privacy.[10] Privacy is a fundamental human right, and the increasing amount of personal data collected and processed in the big data era necessitates a robust framework for safeguarding individuals' privacy.

For instance, Acquisti and Gross conducted a study that demonstrated the potential privacy risks of big data by analysing publicly available data and correlating it with individuals' sensitive information.[11] This study served as a wake-up call, highlighting the importance of addressing privacy concerns and implementing strong privacy protection measures in the era of big data. It emphasised the need for organisations to be mindful of the potential harm that can arise from the uncontrolled aggregation and analysis of personal data.

Obtaining informed consent from individuals is crucial for ethical data practices. However, the complexities of big data, including data

9 "We need to develop and cultivate virtues such as wisdom, prudence, and justice in order to use big data responsibly. Wisdom helps us to understand the ethical implications of big data and to make sound decisions about how to use it. Prudence helps us to identify and avoid the risks associated with big data. Justice helps us to ensure that big data is used in a way that is fair and equitable." Luciano Floridi, *The Fourth Revolution: How the Infosphere is Reshaping Human Reality* (Oxford: Oxford University Press, 2014), 126-128.

10 Helen Nissenbaum, *Privacy in Context: Technology, Policy, and the Integrity of Social Life* (Stanford: Stanford University Press, 2009), 174.

11 Alessandro Acquisti and Ralph Gross, "Predicting Social Security Numbers from Public Data," *Proceedings of the National Academy of Sciences* 106, no. 27 (2009): 10975-10980.

aggregation, data linkage, and secondary uses, make it challenging to obtain meaningful consent. Consent mechanisms should be enhanced to ensure individuals understand how their data will be used and the associated risks and benefits.[12] Informed consent empowers individuals to make informed decisions about their personal data and enables them to maintain control over their information.

In the research paper titled "AI, Big Data, and the Future of Consent" authored by Adam J. Andreotta, Nin Kirkham, and Marco Rizzi, the study delves into the limitations of existing consent mechanisms within the context of big data analytics.[13] The authors assert that prevailing consent processes often fall short of equipping individuals with a comprehensive grasp of how their data may be employed and the potential consequences thereof. This paper addresses three distinct categories of issues that hinder the attainment of informed consent in the context of big data utilisation: the transparency (or explanation) problem, the re-repurposed data problem, and the meaningful alternatives problem. In response to these challenges, the authors propose potential solutions, including implementing a form of 'soft governance' ethical regulation for personal data used in commercial and administrative contexts. Additionally, they contemplate innovative alternatives to conventional consent forms and privacy policies, exploring the incorporation of recent research findings about the usability of pictorial legal contracts.

The extensive collection of personal data in the big data ecosystem raises significant privacy concerns. Organisations must prioritise transparency and implement robust safeguards to protect individuals' privacy. Informed consent plays a crucial role in ethical data practices. Still, the complexities of big data necessitate innovative approaches to consent that provide individuals with a clear understanding of how their data will be used. We can foster a more ethical and trustworthy big data ecosystem by addressing these privacy and consent challenges.

Transparency and Accountability

Big data algorithms' lack of transparency and explainability raises significant concerns about potential biases and discrimination. As big data algorithms increasingly drive decision-making processes in various domains, it is crucial to understand the underlying mechanisms and factors that influence their outcomes. Transparent algorithms enable stakeholders to scrutinise and assess

12 Daniel J. Solove, *Understanding Privacy* (Cambridge: Harvard University Press, 2008), 77.

13 Adam J. Andreotta, Nin Kirkham, and Marco Rizzi. "AI, big data, and the future of consent." *Ai & Society* 37, no. 4 (2022): 1715-1728.

the decision-making processes, helping to identify and rectify biases and ensure fairness and accountability in algorithmic systems.[14]

Algorithmic transparency allows for the evaluation and challenge of automated decisions, reducing the risk of discriminatory outcomes and ensuring that individuals can understand and contest the results. It enables researchers, regulators, and affected individuals to scrutinise the algorithmic processes, assess the data inputs, and uncover any biases or discriminatory patterns that may be present. Transparent algorithms facilitate accountability by providing the necessary information to hold algorithmic systems and their designers accountable for the potential consequences of their decisions.[15]

Research by Mittelstadt and others emphasises the importance of algorithmic transparency in addressing biases and promoting fairness.[16] The study highlights the need for clear explanations of algorithmic processes that are understandable to technical and non-technical stakeholders. Transparent algorithms foster accountability by enabling individuals to assess the fairness of algorithmic outcomes and challenge discriminatory practices. Big data introduces new challenges regarding responsibility and accountability due to the complex network of stakeholders involved in data collection, analysis, and decision-making processes. Establishing clear guidelines and mechanisms for responsible data governance is crucial to prioritise ethical considerations throughout the data lifecycle.

Responsible data governance encompasses various aspects, including legal compliance, ethical standards, and organisational policies. Organisations must take responsibility for the data they collect, ensuring its proper handling, storage, and usage. They should establish comprehensive data protection measures and privacy safeguards to mitigate the risks associated with data breaches and unauthorised access. Regular audits and assessments of data practices can help organisations monitor their compliance with ethical standards and identify areas for improvement.[17]

Accountability in big data analytics goes beyond legal and ethical implications. It also requires implementing technical and organisational measures to ensure responsible data practices. This may involve creating clear lines of responsibility, assigning roles for oversight and compliance,

14 Solon Barocas and Andrew D. Selbst, "Big Data's Disparate Impact," *California Law Review* 104, no. 3 (2016): 671-732.

15 Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel, "Fairness Through Awareness," in *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference* (New York: ACM, 2012), 214-226.

16 Brent Daniel Mittelstadt, Patrick Allo, Mariarosaria Taddeo, Sandra Wachter, and Luciano Floridi, "The Ethics of Algorithms: Mapping the Debate," *Big Data & Society* 3, no. 2 (2016): 2053951716679679.

17 Nissenbaum, *Privacy in Context*, 129-135.

and establishing mechanisms for handling data-related complaints and disputes. Responsible data governance requires an active commitment from organisations to promote a culture of accountability and transparency, focusing on the ethical use of data.

Furthermore, Floridi highlights the concept of "data ethics" as an essential component of responsible data governance.[18] Data ethics encompasses considerations of privacy, fairness, and societal impact, emphasising the responsibility of organisations and individuals to ensure the ethical use of data. By integrating data ethics into organisational practices, stakeholders can uphold ethical principles and accountability throughout the data lifecycle. Transparency and accountability are crucial aspects of ethical big data practices. Algorithmic transparency identifies and mitigates biases and discrimination, promoting fairness and accountability in automated decision-making. Responsible data governance ensures that all stakeholders in the data lifecycle adhere to ethical standards and are held accountable for their actions. By prioritising transparency, explainability, and accountability, organisations can build trust, mitigate risks, and ensure that big data technologies are used fairly and responsibly.

Discrimination and Fairness

Big data analytics can perpetuate biases and discriminate against certain individuals or groups. The process of data collection, algorithmic design, and interpretation biases can lead to unfair treatment and contribute to societal disparities. Identifying and addressing biases is essential to ensure fair and equitable outcomes. Research conducted by Calders and Verwer shed light on the challenges associated with fairness and discrimination in data mining.[19] The study emphasised the need for developing techniques to detect and mitigate bias in big data analytics. It highlighted that biases could emerge at various stages of the data analytics process, including data collection, pre-processing, algorithmic modelling, and decision-making. Addressing these biases requires a proactive approach to identifying and rectifying discriminatory patterns, ensuring that big data analytics produce fair and unbiased results.

In her book *Dragnet Nation: A Quest for Privacy, Security, and Freedom in a World of Relentless Surveillance*, Angwin explores the pervasive nature of surveillance and data collection, shedding light on the

18 Floridi, *The Fourth Revolution*, 66.

19 Wouter Calders and Sicco Verwer, "Three Naive Bayes Approaches for Discrimination-Free Classification," *Data Mining and Knowledge Discovery* 21, no. 2 (2010): 277-292.

potential biases and discrimination that can arise from big data analytics.[20] She emphasises the need for transparency and accountability in data practices to mitigate the risk of discriminatory outcomes and protect individuals' privacy and civil liberties. Research conducted by Tene and Polonetsky examines privacy and user control in the age of analytics, highlighting the challenges of maintaining fairness and protecting individuals' privacy in big data analytics.[21] The study emphasises giving individuals greater control over their personal data and ensuring that data practices uphold privacy principles and ethical standards.

Ensuring fairness in big data analytics requires identifying and mitigating biases. Fairness can be achieved through rigorous testing, validation, and ongoing monitoring of algorithms. Additionally, organisations should prioritise diversity and inclusion in data collection and analysis to avoid biased outcomes. Ohm's research on anonymization discusses the broken promises of privacy and the surprising failure of anonymization techniques in protecting individuals' identities.[22] The study highlights the risks of re-identification and the potential for discrimination when personal information is not adequately protected. It emphasises more robust privacy safeguards and responsible data practices to ensure fairness and protect individuals' privacy.

In their paper "Fairness in Machine Learning," Luca Oneto and Silvia Chiappa present a comprehensive framework to address the ethical concerns of using machine learning systems.[23] This framework aims to mitigate biases present in data and model inaccuracies, ensuring that individuals are not subject to unfavourable treatment based on race, gender, disabilities, sexual orientation, or political beliefs. The authors discuss their own contributions in tackling these challenges, highlighting the pivotal role of causal Bayesian networks in addressing complex unfairness scenarios. Additionally, they introduce a unified framework capable of accommodating various settings and fairness criteria underpinned by robust theoretical foundations.

One of the primary challenges encountered in the realm of big data analytics revolves around the issue of bias. This bias can emanate from diverse origins, encompassing the data-gathering procedures, the algorithms

20 Julia Angwin, *Dragnet Nation: A Quest for Privacy, Security, and Freedom in a World of Relentless Surveillance* (New York: St. Martin's Publishing Group, 2014), 3-14.

21 Omer Tene and Jules Polonetsky, "Big Data for All: Privacy and User Control in the Age of Analytics," *Northwestern Journal of Technology and Intellectual Property* 11, no. 5 (2013): 239-273.

22 Paul Ohm, "Broken Promises of Privacy: Responding to the Surprising Failure of Anonymization," *UCLA Law Review* 57, no. 6 (2009): 1701-1777.

23 Oneto, Luca, and Silvia Chiappa. "Fairness in machine learning," in *Recent trends in learning from data: Tutorials from the inns big data and deep learning conference, INNSBDDl 2019* (Berlin & New YorK: Springer International Publishing, 2020), 155-196.

employed during analysis, and the subsequent interpretation of findings. In healthcare research, these biases can influence the precision and equity of big data analytics outcomes. Consequently, researchers have proffered a range of methodologies to evaluate and mitigate biases intrinsic to big data analytics. For instance, Kate Crawford has underscored the importance of acknowledging and remedying the concealed biases embedded in big data analytics. In her paper titled "The Hidden Biases in Big Data," Crawford articulates that big data cannot be regarded as entirely objective, as it invariably mirrors its creators' inherent biases and preconceptions. She emphasises the necessity for heightened transparency and accountability within big data analytics, aiming to pinpoint and rectify these biases to foster greater fairness and accuracy.

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Furthermore, organisations should prioritise diversity and inclusion in their data collection and analysis processes. A diverse dataset with a broad representation of individuals from various backgrounds helps reduce biases and ensure fair treatment. It is important to consider potential biases related to protected characteristics such as race, gender, age, and socioeconomic status when designing data collection strategies. By including diverse perspectives and experiences, organisations can minimise the risk of biased outcomes and foster fair and equitable results. Addressing bias and promoting fairness in big data analytics is crucial to ensure ethical and equitable outcomes. Organisations should actively identify and mitigate biases at various stages of the data analytics process. Ongoing evaluation, validation, and refinement of algorithms are essential to achieve fairness. Additionally, prioritising diversity and inclusion in data collection and analysis processes helps to reduce biases and ensure fair treatment of individuals and groups.

Big Data Ethics in Healthcare

Using big data in healthcare research introduces complex ethical considerations concerning patient privacy and data protection. While big data analytics hold the potential to revolutionise medical research and improve patient outcomes, healthcare organisations must establish robust protocols that safeguard patient data, ensure privacy, and uphold ethical standards. The study conducted by Xafis and others provides valuable insights into the ethical implications of big data analytics in healthcare research.[24] The authors emphasise the need to strike a balance between

24 Vicki G. Xafis, Owen Schaefer, Markus K. Labude, Iain Brassington, Angela Ballantyne, Hannah Yeefen Lim, Wendy Lipworth *et al.* "An ethics framework for big data in health and research," *Asian Bioethics Review* 11, no. 3 (2019): 227-254.

the potential benefits of data-driven research and the protection of patient privacy. They highlight that advancements in data analytics and machine learning techniques can offer valuable insights for medical research. However, these advancements must be accompanied by stringent privacy protection measures to prevent the identification of individuals and the unauthorised use of sensitive healthcare information.

To achieve ethical practices in big data analytics, healthcare organisations must employ robust data anonymising techniques. Anonymising patient data by removing personally identifiable information and implementing strict access controls can minimise the risk of re-identification and protect patient privacy. By de-identifying the data, researchers can derive valuable insights while preserving patient confidentiality. The ethical challenges surrounding patient confidentiality and data sharing require careful consideration. Sharing patient data for research and treatment purposes necessitates striking a balance between the benefits of data-driven healthcare and the imperative to protect patient confidentiality. Loukides and others explore the ethical complexities of data sharing in healthcare.[25] The study emphasises the need for effective anonymization techniques to protect patient confidentiality while facilitating data-driven research. It also stresses the importance of obtaining informed consent from patients to ensure their autonomy and agency in decision-making regarding data sharing.

In data sharing, healthcare organisations should establish clear guidelines and policies prioritising patient consent, privacy, and security. This includes implementing transparent processes for obtaining informed consent, ensuring patients have a comprehensive understanding of how their data will be used, and providing them with options to control the sharing and use of their information. Additionally, organisations should adhere to data protection regulations, such as the GDPR, and adopt technical measures to ensure patient data's secure storage and transmission.

Moreover, the ethical considerations of data sharing extend beyond individual patient privacy to encompass broader societal implications. The aggregation and analysis of large-scale healthcare data can contribute to advancements in medical research and the development of more effective treatments. However, it is essential to balance the benefits of data sharing and the potential risks, such as re-identification or discrimination, to avoid harm to individuals or specific population groups. Ethical practices in big data analytics within healthcare necessitate robust data protection

25 Grigorios Loukides, Aris Gkoulalas-Divanis, and Jianhua Shao, "On Balancing Disclosure Risk and Data Utility in Transaction Data Sharing Using R-U Confidentiality Map On balancing disclosure risk and data utility in transaction data sharing using R-U confidentiality map," *Proceedings of the Joint UNECE/Eurostat Working Session on Statistical Data Confidentiality* 19 (2011).

measures, patient consent, and responsible data-sharing practices. By implementing stringent privacy protocols, employing data anonymization techniques, obtaining informed consent, and prioritising patient privacy, healthcare organisations can ensure the ethical and responsible use of big data for medical research and treatment while maintaining the trust and confidentiality of patients.

Ethical Challenges in Marketing

Big data has transformed marketing by enabling targeted advertising based on extensive user profiling. While this practice offers benefits in terms of personalised marketing campaigns, it also raises significant ethical concerns regarding privacy invasion and the potential for manipulation. Organisations must strike a delicate balance between effective marketing strategies and respecting individuals' privacy and autonomy.

The study conducted by Turow and others examines the privacy implications of online targeted advertising.[26] The authors highlight the ethical challenges of collecting and using personal data in targeted marketing. They emphasise the importance of greater transparency and user control over collecting and using their personal information. The study underscores the need for organisations to prioritise ethical practices that respect individuals' privacy rights and provide them with meaningful choices regarding data collection and targeted advertising. Beyond privacy concerns, using big data in marketing also raises questions about the ethical implications of employing persuasive strategies that may exploit individuals' vulnerabilities or manipulate their decision-making processes. While personalisation and targeted advertising can enhance marketing effectiveness, it is essential to consider the potential impact on consumer autonomy and well-being.

The research of Matz and others explores the ethical dimensions of personalised marketing and its potential influence on consumer autonomy.[27] The study highlights the importance of responsible marketing practices prioritising consumer well-being and avoiding undue manipulation. It emphasises the need for organisations to adopt ethical guidelines and self-regulation to ensure that persuasive techniques are used responsibly and in a manner that respects individuals' autonomy and decision-making processes.

To address these ethical challenges, organisations should embrace transparency and provide individuals with clear information about how their data will be used in marketing practices. Consent mechanisms should be

26 Joseph Turow, Jennifer King, Chris Jay Hoofnagle, Amy Bleakley, and Michael Hennessy, "Americans Reject Tailored Advertising and Three Activities that Enable It," http://dx.doi.org/10.2139/ssrn.1478214. Accessed on 09 Sept., 2023.

27 Sandra C. Matz, Joe J. Gladstone, and David J. Stillwell, "The Potential of Social Media for Investigating Psychological Traits," *Psychological Science* 27, no. 5 (2016): 715-725.

robust, allowing individuals to provide informed consent for collecting and using their personal data. Organisations should also implement strict data protection measures to safeguard individuals' privacy and ensure data security.

Furthermore, organisations must exercise caution when employing persuasive techniques and avoid exploiting individuals' vulnerabilities. Marketing practices should be guided by ethical principles that prioritise consumer well-being, respect autonomy, and foster informed decision-making. This includes refraining from employing manipulative tactics that may undermine consumer agency and result in decisions not in their best interest. Ethical challenges arise in marketing due to big data and targeted advertising. Organisations must navigate these challenges by balancing their marketing strategies with respect for individuals' privacy, autonomy, and well-being. Organisations can ensure their marketing efforts are ethically sound and maintain consumer trust and satisfaction by prioritising transparency, user control, and responsible marketing practices.

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Big Data and Governance

Using big data for surveillance raises profound ethical concerns regarding privacy rights and potential abuse. While surveillance measures are often implemented in the name of security, it is crucial to strike a delicate balance between surveillance and protecting individuals' privacy rights. Clarke examined the ethical dimensions of big data surveillance in national security.[28] The author emphasises establishing legal frameworks and oversight mechanisms to ensure responsible and ethical surveillance practices. The study underscores the need for clear guidelines that delineate the boundaries of surveillance activities, specify the purposes for which data is collected, and protect individuals' privacy rights. It highlights the significance of transparency, accountability, and checks and balances in safeguarding privacy in the context of big data surveillance.

Moreover, using big data in governance and decision-making processes raises questions about democratic principles, transparency, and public participation. It is essential to critically evaluate the ethical implications of relying solely on big data analytics in democratic societies, as it may exclude certain voices and perspectives and undermine democratic values. Kitchin and Lauriault delve into the ethical challenges of using big data in urban governance.[29] The study highlights the importance of considering data-

28 Roger Clarke, "Privacy Impact Assessment: Its Origins and Development," *Computer Law & Security Review* 25, no. 2 (2009): 123-135.

29 Rob Kitchin and Tracey P. Lauriault, "Towards Critical Data Studies: Charting and Unpacking Data Assemblages and Their Work," *The Programmable City Working Paper*

driven decision-making's social, cultural, and ethical dimensions to promote democratic governance. It emphasises the need for inclusive and participatory processes involving multiple stakeholders, ensuring that decisions reflect societal values and interests. The study highlights that relying solely on big data analytics without considering the broader social implications can lead to skewed outcomes and reinforce existing power imbalances.

To deal with these ethical challenges, governments and organisations must adopt a proactive approach to data governance. This includes establishing clear legal and ethical frameworks that outline the responsible collection, storage, and use of data. Additionally, transparency and public accountability mechanisms should be implemented to ensure that the use of big data in governance is open to scrutiny and public debate.

Furthermore, decision-making processes should not solely rely on big data analytics but should be complemented by human judgment and consideration of contextual factors. Ethical considerations, such as fairness, equity, and societal impact, should be integrated into the decision-making process to ensure that data-driven governance serves the best interests of society as a whole. Using big data in governance brings forth ethical challenges that must be carefully addressed. Striking a balance between surveillance and privacy rights is crucial, necessitating legal frameworks and oversight mechanisms. Additionally, considering the social and cultural dimensions of data-driven decision-making promotes democratic governance. Governments can navigate the ethical complexities of big data in governance by adopting responsible data governance practices, ensuring transparency, and including diverse perspectives.

Mitigating Ethical Challenges

Addressing ethical challenges in big data requires the implementation of robust regulatory frameworks that prioritise individuals' privacy rights, promote transparency, and ensure accountability in data collection, storage, and use. Comprehensive data protection regulations serve as crucial safeguards in mitigating ethical concerns. The European Union's General Data Protection Regulation (GDPR) is an exemplary model of enhanced data protection regulations. The GDPR establishes stringent guidelines for the responsible handling of personal data, emphasising individuals' privacy rights and empowering them with greater control over their data.[30] It obliges organisations to obtain informed consent, implement privacy by design principles, and adopt measures to ensure data security and accountability.

Series 2, no. 9 (2014): 1-29.

30 Hardt, Moritz, Eric Price and Nathan Srebro, "Equality of Opportunity in Supervised Learning," https://arxiv.org/pdf/1610.02413.pdf, accessed on 29 Sept., 2023.

Integrating ethics into the design and development of big data systems is vital to mitigate ethical challenges. Organisations should embrace ethical design principles, such as privacy by design and algorithmic transparency, to embed ethical considerations into the technology from its inception. Privacy by design entails incorporating privacy protections into the system architecture, ensuring data minimisation, and implementing safeguards to protect individuals' privacy rights. The work of Friedler and others emphasises the significance of incorporating fairness and ethics into algorithmic development.[31] The authors propose guidelines and tools to promote ethical machine-learning practices and avoid discriminatory outcomes. Organisations can mitigate the ethical challenges associated with big data analytics by proactively addressing bias, ensuring transparency in algorithmic decision-making, and fostering fairness.

Promoting education and awareness about the ethical implications of big data is essential for all stakeholders involved. Individuals, organisations, and policymakers must comprehensively understand ethical considerations in data collection, analysis, and decision-making processes. Ethical literacy programs and initiatives can empower stakeholders to navigate the complex ethical landscape of big data. Mittelstadt and others highlight the need for responsible innovation and ethical training in the era of big data.[32] The authors emphasise the importance of multidisciplinary education and awareness initiatives to foster ethical data practices. Society can ensure responsible and ethical use of big data by equipping individuals and organisations with ethical knowledge and skills.

Mitigating ethical challenges in big data necessitates the implementation of robust regulatory frameworks, such as the GDPR, to protect individuals' privacy rights and ensure accountability. Integrating ethics into the design and development of big data systems and promoting education and awareness is crucial for responsible and ethical data practices. By prioritising privacy, transparency, fairness, and ethical literacy, stakeholders can navigate the ethical complexities of big data and foster an environment that upholds ethical standards and respects individuals' rights.

Conclusion

The widespread adoption of big data technologies has ushered in a new era of ethical challenges that must be carefully navigated. Throughout this research essay, we have explored the multifaceted ethical implications

31 Sorelle A. Friedler, Carlos Scheidegger, and Suresh Venkatasubramanian, "On the (Im) possibility of Fairness," https://arxiv.org/pdf/1609.07236.pdf. Accessed on 29 Sept. 2023.

32 Brent Mittelstadt, Patrick Allo, Mariarosaria Taddeo, Sandra Wachter, and Luciano Floridi, "The Ethics of Algorithms: Mapping the Debate," *Big Data & Society* 3, no. 2 (2016): 1-21.

of big data across various domains, including healthcare, marketing, and governance. These implications encompass privacy, consent, transparency, and discrimination that require thoughtful consideration. By applying ethical frameworks such as utilitarianism, deontology, and virtue ethics, we have gained valuable insights into the complexities of ethical decision-making in big data. Utilitarianism has highlighted the importance of maximising overall happiness and societal benefit, while deontology has emphasised the principles and duties guiding ethical conduct. Virtue ethics, on the other hand, has focused on cultivating moral character and responsible behaviour.

In addressing these ethical challenges, we have discussed potential solutions and strategies. Enhanced data protection regulations, exemplified by the GDPR, are crucial in safeguarding individuals' privacy rights and promoting transparency and accountability. Ethical design and development principles, such as privacy by design and algorithmic transparency, ensure that ethical considerations are embedded in the technology from its inception.

Furthermore, education and awareness initiatives have been identified as essential components in promoting ethical practices in big data. By enhancing ethical literacy and fostering multidisciplinary understanding, individuals, organisations, and policymakers can make informed decisions regarding data collection, analysis, and decision-making processes.

The complexity of ethical challenges in big data necessitates a delicate balance between leveraging the potential of big data and upholding individual rights and societal well-being. We must prioritise ethical considerations in developing, implementing, and utilising big data technologies. By doing so, we can shape a future where big data is harnessed responsibly and ethically, contributing to positive societal outcomes and advancements in various fields. Ethical awareness, robust regulations, ethical design practices, and education are fundamental pillars in ensuring big data's responsible and ethical utilisation. By addressing these ethical concerns head-on, we can create a future where the potential of big data is realised while respecting individual rights, societal values, and the principles of fairness and accountability.

AI and Human Enhancement
An Ethical Appraisal

Arnald Mahesh Irudhayadhason

Abstract

Artificial Intelligence (AI) is a rapidly evolving academic discipline poised to revolutionize diverse aspects of our daily existence. AI involves the creation of systems displaying intelligent behaviour through autonomous analysis of their environment and goal-oriented action. In parallel, Human Enhancements (HE) encompass deliberate technological and biomedical interventions on the human body/mind, elevating capacities beyond normal human levels to enhance well-being. AI not only serves to enhance human life but can also be perceived as a form of HE itself. Thus, both AI and HE can be valued as human enhancements. Despite the unprecedented advancements gifted by technologies like AI/HE, they have elicited varied responses. While humanity celebrates the empowerment facilitated by AI/HE, simultaneous concerns arise about the potential trajectories they might lead us towards, perceived as both dehumanizing and super-humanizing. In this context, this paper endeavours not only to highlight the intrinsic relationship between these two dimensions but also to underscore the ethical concerns and challenges they pose. Ultimately, the paper

advocates for a responsible utilization of AI/HE, emphasizing an ethical framework to ensure their likely ethical deployment.

Keywords**:** *Human Enhancement – Cognitive, Mood, Moral, Bio- & Physical, Ethical Considerations, Dehumanizing factors, Super-humanization, Body-mind Relationships, Lifespan Extension, Designer Children, Autonomy and Coercion, Individual & Social Wellbeing, Can Vs. Ought., Responsible AI.*

Introduction

The fusion of AI and Human Enhancement (HE) introduces novel possibilities and ethical quandaries. This paper explores the intricate interplay between technological advances and ethical considerations in augmenting human capabilities. The author recognizes diversity of HEs, positioning AI as pivotal – AI becomes entwined with HE. Industries adopting AI prompt discourse beyond technicalities to ethical implications. Various HEs set the stage for nuanced ethical analysis. The discourse on AI emphasizes its role as a tool augmenting human intelligence, not replacing it. Concerns about job displacement, economic instability, and varied ethical challenges are explored in both HE and AI contexts. In addition, the article delves into disparities arising from unequal HE and AI distribution, anticipating societal shifts. Ethical challenges, such as autonomy, safety, eugenics, responsibilities, and solidarity, are analysed. Transitioning to Responsible AI/HE, the author advocates for a morally rooted approach, emphasizing obligations and virtues. The conclusion questions whether technological feasibility implies moral acceptability. Ethical considerations must guide AI/HE use, highlighting the need to align technological progress with responsibility. In essence, the paper introduces the ethical landscape of AI/HE, outlining categories and implications while inviting contemplation on their responsible integration in societal evolution.

1. What is Human Enhancement (HE)?

Human Enhancement (HE) can be defined as any "deliberate technological and/or biomedical intervention on the human body or mind to augment one's capacities above the normal human level, with a view to increasing one's welfare."[1] HE has generated mixed reactions as is the case with any (bio) technological breakthroughs. Although humanity rejoices in being empowered by HEs, it is also worried where HEs may be leading us. HEs are deemed

1 Arnald Mahesh Irudhayadhason, *The Ethics of Human Enhancement* (Chennai: Don Bosco Publications, 2017), 311-312.

to be both dehumanising as well as super-humanising,[2] which prompt us to question what exactly HE is. Although HE promises unparalleled promises, it can never close its eyes to stark realities, such as poverty, malaria, AIDS, malnutrition, lack of pure drinking water and sanitation and basic health care, which notoriously make a major part of humanity vulnerable to diseases and death. While not underestimating an investigation of such an ethically overloaded issue, this article attempts to briefly present the main categories of HE – not with a view to reaching conclusive judgements – with the intention of bringing to light various ethical concerns surrounding them, which would more likely to invite anyone to further engage in deeper investigation into this topic.

There are many types of HE, such as (a) Cognitive Enhancement (CE); (b) Mood Enhancement (ME);[3] (c) Physical Enhancement (PE);[4] (d) Moral Bioenhancement (MoBE); (e) Lifespan Extension;[5] (f) Designer Children.[6] There is no doubt that AI, too, is a HE. Put differently, the general word HE signifies to a broad variety of current, evolving and vision based technology, brain-enhancing medicines, human germline engineering and present reproductive technologies, dietary intakes, novel brain incentive strategies for alleviating pain and regulating mood, sports gene doping, plastic surgery, short-stage children's growth hormones, anti-aging drugs, and extremely sophisticated drugs.[7]

Against this background, let us consider some relevant questions that would help our analysis. When a person has taken performance-enhancing drugs, can we attribute his victory in a sports competition *fully* to him or to the pharmacological company that produced the drugs? Can we ethically defend one's act of augmenting one's physical performance? What is the difference between boosting one's athletic performance through a healthy diet and regular exercise, and taking performance-enhancing drugs? Is the difference between these two means of improving performance morally pertinent? Thus, HE in general engenders various ethically pertinent and

2 Leon R. Kass, "Ageless Bodies, Happy Souls: Biotechnology and the Pursuit of Perfection," *The New Atlantis*, 1 (2003): 9-10.

3 Berghmans R., *et al*., "Scientific, Ethical, and Social Issues in Mood Enhancement," in Julian Savulescu *et al.* eds., *Enhancing Human Capacities* (Chichester: Wiley-Blackwell, 2011), 161.

4 Holm S. and M. McNamee, "Physical Enhancement: What Baseline, Whose Judgement?" in J. Savulescu et al. eds., *Enhancing Human Capacities*, 291.

5 Irudhayadhason, *The Ethics of Human Enhancement*, 16-17.

6 *Ibid*., 17.

7 Amit Kumar Verma, Anil Kumar Bisht, and Arun Kumar Rastogi, "Human Augmentation Through Applications of Technology," *International Journal of Aquatic Science*, 12, no. 02 (2021), 5026.

complex questions, necessitating serious investigation into the debate on HE, not only to identify meaning in these questions, but also to search for relevant and reasonable answers.

2. What is AI?

"AI refers to systems that display intelligent behaviour by analysing their environment and taking action – with some degree of autonomy – to achieve specific goals."[8] AI involves the emulation of human intelligence by machines, wherein AI systems are designed to replicate human thought processes and behaviour. Industries spanning healthcare, banking, and transportation have embraced AI technologies to enhance efficiency and reduce operational costs. Despite the longstanding usage of the term "Artificial Intelligence," its substantive meaning lies in the development of systems that imitate human cognitive functions and actions. The primary objective of AI is to construct machines endowed with cognitive abilities such as speech recognition, decision-making, and language translation, – functions traditionally associated with human intelligence. The overarching goal is to augment human capabilities and elevate overall quality of life through AI advancements. In this pursuit, it is imperative for AI development to prioritize ethical considerations.[9] AI should be seen as a tool to enhance human intelligence rather than as a threat to our survival or a replacement for it.

AI, while advantageous, raises concerns about job loss, economic instability, and the potential misuse for autonomous weaponry and surveillance. Ethical considerations are vital in development to ensure transparency, accountability, and the augmentation, rather than replacement, of human capabilities. Balancing the enormous potential of AI with ethical responsibility is key for creating technologies beneficial to society.[10]

3. Pertinent Ethical Challenges

3.1. Equality and Justice

a) When the limited public resources of a society are invested in developing and disseminating, HE/AI technologies at the expense of developing other urgent needs, like medical technologies, it certainly violates distributive justice. Since HE/AI are inevitably related to technology in general and biomedical technology in particular, a state has a moral duty – especially if it has a national health care reimbursement system – to cover the medical needs of its citizens as much as possible. In the context of not being able to cover all health care needs such as countries like India, it can try to develop procedures to provide whatever is possible in ways that are more reliable, cheaper, and faster.

b) HE/AI are and/or will be available only to the rich and the affluent when it enters a country as an expensive market good; consequently, the well-off will gain a greater competitive advantage both on the local and global levels. It is unfair as it will aggravate the already existing social inequalities. For example, in the context of an employment-based health insurance, which is offered by the employers themselves, a candidate with an "enhancement certificate" or "AI certificate" would be preferred to a candidate who does not have one. As enhanced –through AI – candidates are less likely to take more sick leave, they will be more productive, earn more, and bring more profits to the employers. When more candidates have a *bona fide* enhancement certificate, employment contracts will undergo a change and fewer compensated sick leave will become the rule. As a result, those who cannot afford AI supported HE will not only be considerably underqualified, but will also be regarded as disabled, although they will be perfectly normal based on the present "pre-enhancement" specifications.[11]

c) Different AI aided HEs intend to promise "longer life." This "prospective" phenomenon will certainly engender an imbalance between the older and younger generation. As a result, problems of intergenerational justice will arise, especially ones that are tied to pensions. The disproportionate increase in older people[12] – along with the steep decrease in the birth rate in most western countries – will pressure the younger generation to contribute to the social security of the older population.[13] This imbalanced reality is most likely to create serious obstacles to any social and political reform,[14] even in countries

8 European Parliament *Artificial Intelligence: How does It Work, Why does It Matter*, and What Can We do about It? (Brussels: European Union, 2020), 1.

9 Andrew Ensiyeh, "The Future of Human Enhancement: An Ethical Exploration of Artificial Intelligence," *Journal of Global Research in Computer Science*, 14, no. 2 (2023), 16.

10 Ensiyeh, "The Future of Human Enhancement: An Ethical Exploration of Artificial Intelligence," 17.

11 Allen Buchanan *et al.*, *From Chance to Choice: Genetics and Justice* (Cambridge: Cambridge University Press, 2000), 97.

12 Daniel Callahan, "Why We Must Set Limits," in *A Good Old Age? The Paradox of Setting Limits*, eds. Paul Homer and Martha Holstein (New York: Simon & Schuster Inc., 1990), 27.

13 Gaia Barazzetti, "Looking for the Fountain of Youth: Scientific, Ethical, and Social Issues in the Extension of Human Lifespan," in *Enhancing Human Capacities*, eds. Julian Savulescu *et al.*, 2011, 344.

14 Barazzetti, "Looking for the Fountain of Youth: Scientific, Ethical, and Social Issues in the Extension of Human Lifespan," 344.

like India. Hence, as long as HE/AI are "enabling" technologies for the rich minority, they will be a "disabling" technology for the poor majority[15] in developing countries such as India.

3.2. Autonomy and Coercion

a) When HE/AI are pervasively employed either due to their benefits or due to their wider social acceptance – irrespective of their being legitimised by the government – the personal autonomy of unwilling individuals is more likely to be violated. There will be a subtle exercise of coercion due to peer pressure. For instance, when many use AI in competitive set-ups, such as in the contexts of institutes of learning, work places, and sports, other non-users – who are otherwise not in favour of undertaking HEs or using AI – will be subtly coerced into taking AI supported CE and PE with a view to gaining a competitive advantage.

b) When the use of HE/AI is legitimised in a liberal society, people are supposed to be free to choose HE/AI. However, in such a society, the use of HE may become almost a necessity. For example, when CE/Intelligence Augmentation (IA)[16] is widely used due to its easy availability, parents may encourage their children to have CE/IA. Failure of a parent to offer CE/IA to her child will be judged as a form of *child neglect*.

c) New burdens may be placed on certain professionals[17] due to the availability of some safe and helpful AI/HEs. For instance, the jobs of surgeons or pilots where people's lives are immediately at risk may legitimately necessitate – AI recommended – taking CE in the proximate future. The purpose of the demand may be to prevent or reduce risks with negative consequences. However, it may involve coercing some professionals who would rather not choose CE.[18] Hence, we claim that AI/HEs in general cannot be severed from ethical concerns related to the violation of individual autonomy and coercion.

15 Gregor Wolbring, "The Unenhanced Underclass," in *Better Humans? The Politics of Human Enhancement and Life Extension*, eds. Paul Miller and James Wilsdon (London: Demos, 2006), 126.

16 Verma, Bisht, and Rastogi, "Human Augmentation Through Applications of Technology," 5027.

17 Hannah Malsen *et al*., "With Cognitive Enhancement Comes Great Responsibility?" in *Responsible Innovation 2: Concepts, Approaches, and Applications*, eds. Bert-Jaap Koops *et al*. (Cham: Springer, 2015), 121-124.

18 Filippo Santoni de Sio *et al*., "How Cognitive Enhancement Can Change our Duties," *Frontiers in Systems Neuroscience*, 8, article 131 (July 17, 2014), [journal online]:1, accessed August 27, 2015, http://journal.frontiersin.org/article/10.3389/fnsys.2014.00131/abstract.

3.3. Costs and Benefits

a) One of the primary concerns in the ethics of AI/HE is related to safety. Safety is an inevitable condition that makes an AI/HE *ethical*.[19] However, to value the safety of an AI/HE with precision is very difficult, because a HE can be safely administered, yet its long-term effect on the subject and on society is uncertain. It will *not be so soon* that we will have safe and powerful biomedical enhancements.[20] Hence, safety related concerns draw considerable moral attention.

b) Many existing AI/HEs raise safety issues since they are more likely to harm the agents both physically and mentally. While CE might improve one's attention/wakefulness and memory, it may also result in some adverse effects by way of preventing the advancement of some valuable things. For instance, *our capacity to forget is as equally important as our capacity to remember*. The improved memory, an outcome of CE, may not let us forget our past painful and bitter experiences, which would be worse than having an enhanced memory.[21]

c) We contend that even the "faint" possibility of AI backed MoBE, albeit not desirable, would certainly tax the moral fibre of humans living together. It would be difficult to differentiate between *true* moral acts and biomedically-induced moral acts. In biomedically-induced moral actions, the agent's freedom, moral intention, and moral reasoning leading to judgement are compromised. Hence, we do not endorse making a person morally better at the expense of these values.

d) As we may all know, assisted reproductive technology is not only helps infertile couples to have children, but it might also help couples to produce "best" possible children through the processes of IVF (In Vitro Fertilization) and PGD (Pre-Implantation or Prenatal Genetic Diagnosis) with the assistance of AI. However, giving birth to a "disease-free" child implies that the conception takes place outside sex, outside the womb of the mother, and perhaps even outside wedlock. As the name denotes, when this process is employed to assist an infertile couple or to eliminate *heritable genetic defects* in the prospective child, its moral permissibility can be considered or tolerated. However, when more and more perfectly fertile couples want to exploit this AI backed biomedical technology either to have a possibly "perfect" child or a child with desired traits, this

19 Julian Savulescu, "Genetic Interventions and the Ethics of Enhancement of Human Beings," in *The Oxford Handbook of Bioethics*, ed. Bonnie Steinbock (Oxford: Oxford University Press, 2007), 528.

20 Allen Buchanan, *Better than Human: The Promise and Perils of Enhancing Ourselves* (Oxford: Oxford University Press, 2011), 180.

21 Malsen *et al.,* "With Cognitive Enhancement Comes Great Responsibility?" 125-126.

biomedical intervention would be morally disconcerting. It would change the traditional meaning of sex and child bearing. In addition, the process of PGD logically implies abortion, especially in the event of the prospective child being diagnosed with some genetic or other serious disabilities. Some may consider the abortion of the prospective disabled child as a preventive and socially responsible act. On the other hand, for those who value an embryo as a person, abortion will pose serious moral concerns. Hence, while we acknowledge the promises made by the HE enterprise, albeit some are mere barren promises, we cannot underestimate the price we need to pay in the bargain – both in terms of money and moral values.

3.4. Individual and Social Wellbeing

a) *Eugenic and Racist Concerns*. Any argument in favour of procreating a best possible baby does not exclusively promote individual well-being. The very word "best" directly excludes the good and the better and thus proposes the creation of a "superior" class – GenRich.[22] Aside from that, the general disposition to enhance human capabilities and traits promotes the creation of a higher enhanced class of people.[23] These efforts endorse eugenic intentions, and to that extent, they are morally objectionable.

b) *Engendering a New Class*. If lifespan extension were to become a reality, more people would live longer. Our concern is whether we would have enough space and resources for everyone. We may even find solutions to these problems through technological development. However, our concern is whether the slowing of senescence would make the retardation of the birth rate mandatory in the near future.[24] When people extend their lifespan, they are more likely to spend long years of dependent life as old people.[25] Will not AI/HE technology – which tries to create a disability-free society – inevitably create a "new class" of disabled people out of the aged? Similarly, the pervasive employment of HE may change the

22 Lee M. Silver, *Remaking Eden: How Genetic Engineering and Cloning Will Transform the American Family* (New York: Perennial, 1998), 4.

23 The Indian social context is entangled in caste discrimination. The caste-system has been perpetuated in varied ways, especially through marriage. When HEs become pervasive in India, will not this phenomenon give birth to another higher caste, namely, an "enhanced caste?" Is it advisable to contribute one more element that is more likely to strengthen this repugnant and deeply embedded social evil?

24 Larry S. Temkin, "Is Living Longer Living Better?" *Journal of Applied Philosophy*, 25, 3(2008), 363.

25 Not less than 46 percent of aged people who are 60 and above live with one form or another of disability worldwide. *Human Development Report, 2014. Sustaining Human Progress: Reducing Vulnerabilities and Building Resilience* (New York: United Nations Development Programme, 2014): 70, accessed May 29, 2015, http://hdr.undp.org/en/2014-report/download.

concept of disability. In particular, if transhumanism were to triumph, the enhanced will view the unenhanced as disabled.[26] As a result, a new class of disability will be created and, as such, the AI/HE project will be self-defeating.

c) *Pertinence of Familial Values*. The desire to create a "perfect" baby might impel couples to buy sperm and eggs from anonymous donors. Let us imagine for argument's sake that a "perfect" child is also created. Will not such an act change the traditional meaning of parenthood? How would the adolescent react when she learns that her biological parents are unknown? Is there a guarantee that biologically perfect children will also be psychologically, socially, and morally perfect? What motivates the couple to bring forth a child – love for a prospective child or a child as an embodiment of their will? Are we sufficiently prepared for such a drastic change?

3.5. Responsibility and Solidarity

a) The broader use of certain AI/HEs may heighten our individual responsibility. For instance, a potential mother must shoulder the responsibility of deciding whether to undergo PGD or not. Furthermore, if safe and efficient CEs are accessible, enhancing human cognitive alertness significantly, professionals such as doctors and pilots may be obligated to use them to safeguard human lives under their care. Beyond that, they might assume additional responsibilities post-enhancement, facing increased duties and expectations in their respective fields. Consequently, their accountability for optimal performance and any potential mishaps will be heightened.

b) When a person undergoes CE, ME, or MoBE through AI-assisted diagnosis, the resulting capabilities cannot be entirely attributed to the enhanced individual. What the person achieves post-enhancement would have been unattainable before the enhancement. Consequently, the enhanced person bears less responsibility for actions performed after enhancement, as these actions are linked to her "inauthentic" self rather than her pre-enhancement state.[27] Therefore, HEs not only heighten one's responsibility but also diminish it – as an actor and moral agent. There is also a duty to use technological innovations responsibly with a specific focus on societal well-being. Demonstrating social responsibility can be achieved through social solidarity. However, the adoption of certain HEs, especially genetic enhancements, is likely to erode social solidarity. Without the support of social solidarity, promoting the well-being of the

26 Wolbring, "The Unenhanced Underclass," 126-127.

27 Malsen *et al.,* "With Cognitive Enhancement Comes Great Responsibility?" 123, 126.

least privileged, the common good of society, and social stability becomes challenging.[28]

c) One might counter the solidarity argument by asserting that the more we enhance, the more we will have *to redistribute*, and we will have *less need for* redistribution.[29] Advocates argue that widespread adoption of HEs in society would lead to increased productivity, generating ample resources *to* redistribute to the least advantaged. However, this rationale lacks cogency in reality. Despite the heightened productivity of science and technology, the global wealth gap has widened significantly.[30] To delve deeper into our argument, we must acknowledge, as highlighted in the Human Development Report 2014,[31] the stark reality of our profoundly unequal world, despite numerous technological advances. The HDR-2014 underscores the importance of timely intervention through tangible expressions of solidarity.

Our contention, however, is that such enormous growth and development has not augmented our solidarity. What is the relationship between the horrible reality of the world and the pursuit of AI/HEs? Firstly, technological advancements like AI/HEs could generate additional resources. However, the crucial question is whether these advancements will also broaden our global perspective to comprehend the struggles of the impoverished, the insurmountable disparities between the wealthy and the poor, and between developed and developing nations. The abundance of resources hasn't fostered a tendency to share the surplus with those in need; rather, individuals who acquire more tend to accumulate further. Consequently, increased production does not necessarily enhance our sense of solidarity.

Secondly, the quest for genetic enhancement is likely to erode solidarity, particularly in relation to acquiring medical insurance. The more AI reveals about our potential medical history, the less inclined we are to express solidarity with others. Nevertheless, medical insurance isn't the sole avenue for demonstrating solidarity; there are alternative ways to rejuvenate, enhance, and manifest it. To rekindle solidarity, it is crucial to avoid overemphasizing the value of AI/HEs and undervaluing the significance of leading a moral

28 Ori Lev, "Will Biomedical Enhancements Undermine Solidarity, Responsibility, Equality and Autonomy?" *Bioethics*, 25, 4(2011), 178.

29 John Harris, *Enhancing Evolution: The Ethical Case for Making Better People* (Princeton, NJ: Princeton University Press, 2007), 120.

30 Jack Stilgoe, "Foreword: Why Responsible Innovation?" in *Responsible Innovation: Managing the Responsible Emergence of Science and Innovation in Society*, eds. Richard Owen *et al*. (Chichester: John Wiley and Sons, Ltd., 2013), xi.

31 *Human Development Report, 2014. Sustaining Human Progress: Reducing Vulnerabilities and Building Resilience*, 70.

life.[32] In other words, thoughtful consideration of the purpose and limits of AI/HEs is needed, with a focus on promoting other essential moral ends.[33] Rather than squandering resources on unnecessary AI/HEs, a responsible alternative is to share them with those in need, the impoverished, and the suffering.

Thirdly, it's crucial to recognize our duty to practice solidarity. Many have not only failed to exercise solidarity but have also lost their sense of it, forgetting the duty to live it out. solidarity implies a *common conviction* that most of our human misfortunes are *shared problems*, although we may suffer them individually in different places.[34] It underscores our belonging to one human family, aligning with concepts like Gandhi's *Sarvodaya* – the welfare of all.[35] Acknowledging this idea prompts a realization of our duty to work for the welfare of all, cautioning against unchecked greed. Concerns arise that the pursuit of AI/HE, an expression of discontent, may feed greed without addressing the needs of the less fortunate. Advocating for the welfare of all doesn't outright condemn AI/HE but encourages prioritizing essential individual and global needs. Understanding the true meaning of solidarity discourages hasty engagement in AI/HE interventions when many struggle for basic health and security. Given the ethical relevance to personal and social life, it's essential to address these concerns before unconditionally endorsing the pursuit of AI/HE as a moral imperative for contemporary liberal society.

4. Responsible AI/HE

Responsible innovation is an interactive process, involving social actors and innovators in determining the ethical permissibility, sustainability, and social desirability of the process of innovation and its prospective marketable goods.[36] What *sense* of responsibility is implied here? Since we deal with the ethics of innovation in general and AI/HE in particular, the sense that is implied here is *moral*. Moral responsibility points to a distinction between

32 Alex Rajczi, "One Danger of Biomedical Enhancements," *Bioethics*, 22, 6(2008), 328-336.

33 Rawls rightly argues that we need to pay considerably greater attention to those who are less gifted naturally and are born into less advantageous social circumstances, with a view to offering fair equality of opportunity. John Rawls, *A Theory of Justice*, revised edition (Cambridge, MA: The Belknap Press of Harvard University Press, 1971), 86.

34 Daniel Callahan, *False Hopes: Overcoming the Obstacles to a Sustainable, Affordable Medicine* (New Brunswick, NJ: Rutgers University Press, 1999), 172.

35 See: Gandhi M. K., *An Autobiography or The Story of My Experiments with Truth*, second edition, trans. Mahadev Desai (Ahmedabad: Navajivan Publishing House, 1940), 224.

36 René von Schomberg, "Introduction: Towards Responsible Research and Innovation in the Information and Communication Technologies and Security Technologies Fields," in *Towards Responsible Research and Innovation in the Information and Communication Technologies and Security Technologies Fields*, ed. René von Schomberg (Luxembourg: Publications Office of the European Union, 2011), 9, accessed May 29, 2015, http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2436399.

backward-looking responsibility and forward-looking responsibility.[37] The former refers to responsibilities for things that have occurred in the past – responsibility-as-blameworthiness. The latter pertains to things which have not yet been done/happened, – frequently associated with institutional obligations or duties. Accordingly, one can be morally held responsible for failing to fulfil the tasks with which one is entrusted. Two other aspects of forward-looking responsibility are responsibility-as-obligation and responsibility-as-virtue. The former denotes that one has to make sure that something is the case or that a particular desirable goal is achieved. The latter refers to a character, a disposition to act responsibly.[38]

The following elements that constitute the meaning of *responsible* in responsible innovation. They are:

a) becoming responsive and adapting to the various challenges concerning an innovation;
b) possessing long-term and critical vision to evaluate innovations by looking beyond the prospective market advantages and risks;[39]
c) creating socially valuable outcomes;
d) "caring for the future" through the collective commitment and "responsive stewardships of science and innovation;"[40]
e) taking into consideration "social and ethical aspects" and maintaining a balance between economic, environmental and socio-cultural aspects;[41] and
f) guaranteeing "the accountability of innovation actors."[42] Building on the discussion thus far, we advocate for "Responsible AI/HE" by aligning with the concept of responsibility in "responsible innovation." This approach encompasses both backward-looking responsibility and forward-looking responsibility. Responsible AI/HE aims not only to strategically avoid situations where critical responsibilities are neglected but also to ensure that decision-making related to a specific AI/HE is not a result of "organized irresponsibility."[43] We emphasize the necessity of organized responsibility in the AI/HE enterprise because a lack of responsibility in innovations like AI/HE could jeopardize a society's moral, economic, and socio-cultural well-being.

37 Jessica Nihlén Fahlquist, "Responsibility Ascriptions and Vision Zero," *Accident Analysis and Prevention*, 38, 6(2006), 1114-1116.

38 Ibo van de Poel *et al*., "The Problem of Many Hands: Climate Change as an Example," *Science and Engineering Ethics*, 18, 1(2012): 51-52.

39 René van Schomberg, "A Vision of Responsible Research and Innovation," in *Responsible Innovation: Managing the Responsible Emergence of Science and Innovation in Society*, eds. Richard Owen, John Bessant, and Maggy Heintz (Chichester: John Wiley and Sons, Ltd., 2013), 51.

40 Richard Owen *et al*., "A Framework for Responsible Innovation," in *Responsible Innovation: Managing the Responsible Emergence of Science and Innovation in Society*, eds. Richard Owen *et al.* (Chichester: John Wiley and Sons, Ltd., 2013), 36.

41 Vincent Blok and Pieter Lemmens, "The Emerging Concept of Responsible Innovation. Three Reasons Why It Is Questionable and Calls for a Radical Transformation of the Concept of Innovation," in *Responsible Innovation 2: Concepts, Approaches, and Applications*, eds. Bert-Jaap Koops *et al.*, (Cham: Springer, 2015), 20.

42 Andri D. Setiawan and Rajbeer Singh, "Responsible Innovation in Practice: The Adoption of Solar PV in Telecom Towers in Indonesia," in *Responsible Innovation 2: Concepts, Approaches, and Applications*, eds. Bert-Jaap Koops *et al.* (Cham: Springer, 2015), 230.

Responsible HE functions to ensure the fulfilment of stated obligations. In the event of issues, it obligates an assessment of responsibility within relevant social institutions. Framed as a state-level social construct, Responsible AI/HE prioritizes responsibility as a standard and framework for deciding on the ethical and desirable aspects of AI/HE innovation. Given the potential profound and enduring impact of innovations like AI/HE on societal values, the responsibility for decision-making on behalf of the public lies with state-level HE regulators, policymakers, entrepreneurs, funding agencies, scientists, educators, academics, and politicians.

5. Does 'Can' Necessarily Imply 'Ought?'

Some key arguments supporting AI/HE posit that if the technology is feasible, it is also permissible and/or desirable. Certain AI/HE technologies are already feasible, and others may become possible soon. The ethical complexity of the permissibility of AI/HE is heightened, particularly in vital areas like medicine and healthcare, which it significantly impacts.

a) We question the neutral nature of technology, and in particular AI/HE. Our scepticism concerning the neutrality of AI/HE is based on its manifold and pernicious effects on our familial, socio-cultural, and ethical life. HE is not value-free. Hence, irrespective of the intention of its users, the consequences will not be completely *ethically* different.

43 Organised irresponsibility helps analyse the contradictions of a risk society which refers to how and why the social institutions are aware of the existence of hazards in the society, while simultaneously denying their presence, concealing their causes, and preventing rectification and management. That is, responsibility for the catastrophe that is caused by social institutions is attributed to no one. This leads eventually to the self-destruction of the risk society. Ulrich Beck, "Risk Society Revisited: Theory, Politics and Research Programmes," in *The Risk Society and Beyond: Critical Issues for Social Theory*, eds. Barbara Adam *et al*. (London: Sage Publications, 2000), 224.

b) If assuming that technological feasibility implies moral acceptability, it would undervalue the significance of morality in human life. Technology is not a universal solution to *all* human problems; it operates independently from ethical considerations. Ethics guides how we should live, while technology introduces new possibilities. Our concern is not about what *can be done* but whether it is morally *acceptable*. Likewise, the difficulty in adhering to a moral value does not justify abandoning it. For example, challenges in marital fidelity among couples do not negate the moral value of loyalty within marriage.[44] Therefore, the mere possibility of something does not automatically necessitate its ethical permissibility.

c) While technology may contribute to our "well-being," ethics is essential for evaluating the moral appropriateness of the *means* employed, ensuring that the outcomes are ethically sound, and determining if the benefits outweigh the costs. The potential emergence of certain AI/HEs – designer children, brain-computer interfaces, MoBE, lifespan extension, AI-powered medical diagnostics and treatment, autonomous vehicles, AI-powered education, robotics, and creative tools – raises concerns about endangering human identity and the symbolic meaning in relationships. For example, lifespan extension and brain-computer interfaces could significantly alter how we define and experience human life. Despite the potential to achieve self-evident impossibilities, such as conceiving a child with anonymous donors or using the sperm of a father-in-law for an infertile husband, the ethical desirability and permissibility of these possibilities are questionable. Will not self-evident human relationships undergo an ugly and detestable transformation?

d) Technology has become a crucial component in healthcare, with technology and care holding equal importance. Efficient and cost-effective *care* would be nearly impossible without adequate technological support.[45] However, the ethical question arises about whether it is morally acceptable to let technology dominate the field of medicine. As long as technology aids in achieving the threefold goals of medical care – prevention, cure, and care – there is no moral discomfort. Conversely, when AI/HE intrudes into medicine and alters its purpose by expanding it, particularly at the expense of its primary objective of restoring normal health, we raise moral objections. Therefore, we argue that the mere *possibility* of HE is insufficient to justify its moral *permissibility*.

44 Herman De Dijn, "Technology in Health Care: A Philosophical Ethical Appraisal," in *Between Technology and Humanity: The Impact of Technology on Health Care Ethics*, ed. Chris Gastmans (Leuven: Leuven University Press, 2002), 18.

45 Chris Gastmans, "Introduction," in *Between Technology and Humanity: The Impact of Technology on Health Care Ethics*, ed. Chris Gastmans (Leuven: Leuven University Press, 2002), 9.

We acknowledge the potential of certain AI/HEs to enhance human capabilities but question whether such enhancements, achieved at the expense of valuable social and individual goods, can be considered genuine benefits. Some social constructs are inflated through marketing propaganda, presenting HEs as significant values and even suggesting them as ends to pursue.[46] The fundamental question is whether AI/HEs should be pursued as ends or if an enhanced state of life is preferable to an unenhanced life. As AI/HEs are closely tied to biomedicine, we must consider whether we are commodifying medicine to fulfil superficial desires, turning it into a cosmetic product for improving our biological, mental, and moral "appearance." Our presumption is that many AI/HEs address effects without addressing deep-seated social prejudices.[47] When some HEs serve as mere vanities, akin to sophisticated cosmetic interventions, we question whether an individual's life *really* becomes *better*, more *meaningful*, or *happier* due to enhanced capabilities powered by AI.

Conclusion

Technology, with its potential to significantly transform societies, can either improve or worsen communities by pushing human capabilities to unprecedented levels. History illustrates that any technology can serve positive or negative purposes. By addressing ethical challenges before implementing new technical approaches for human development, we can envision a safer world and a more equitable future. The evolving realm of AI plays a vital role in enhancing human productivity, despite some promises appearing empty. We critically examine the necessity, moral permissibility, and desirability of current and future AI/HEs. Favouring social progress, we remain open to discussions on the need and ethical acceptance of specific AI/HEs. Given the uniqueness of each AI/HE in nature, deployment, and potential effects, a case-by-case analysis is necessary. Our intent is to continue the debate on AI/HE,

46 Rajczi, "One Danger of Biomedical Enhancements," 329-330.

47 HEs can never be a moral solution to eliminate these prejudices, because they vainly address merely the effects and/or symptoms of these social biases. One practical way to address these prejudices is through a change of mind-set, which could be brought about through different social institutional interventions, such as family, church or religion, community, educational institutes, motivational training, awareness programmes, and the legal system. By making use of the tool of biomedical technology, HEs merely play the role of making people conform to social prejudices, instead of uprooting prejudices from society.

adopting not only a philosophical perspective but also a multi-disciplinary approach.

While we express deep concerns about ethical issues before actively advocating for the pursuit of AI/HE in general, we support the indirect improvement in the "enhanced state" of life resulting from a society's higher average health level through an efficient medical and healthcare system. For example, when a society has a good healthcare system, overall health improves, leading to positive lifestyle changes and societal growth. People live stronger, healthier, and longer lives, contributing to a creative platform for a meaningful life. We endorse an enhanced state of life as a by-product of therapeutic interventions, particularly through an effective state-level healthcare system. In the global context, we oppose the pursuit of AI/HE at the expense of moral and socio-cultural values, especially when many vulnerable individuals face preventable diseases and disasters. Until we achieve a normal health level for people, there is no consideration of employing AI/HEs, particularly through a state-level healthcare system. As reasonable and responsible individuals, our top priority is ensuring universal basic healthcare because it is fundamental to all other human needs.

8

AI and Ethics
Yuval Noah Harari's Understanding of the Future of Humanity with Artificial Intelligence

Kuruvilla Pandikattu

Abstract

This article serves as a comprehensive critique of Yuval Noah Harari's perspectives on artificial intelligence (AI), positioning him as a significant cultural critic of contemporary society. It begins by outlining Harari's influential intellectual contributions, setting the stage for a detailed examination of his thoughts on AI, as shared during his 40-minute presentation at the Frontiers Forum Live event from June 5-8, 2023. The discussion then narrows down to four pivotal concerns that AI poses to our current societal fabric: the transformative power of language in AI contexts, the shift from human attention to digital intimacy, the potential threats AI poses to democratic systems, and the existential questions regarding the future of humanity as influenced by AI advancements. Each area is explored to highlight the intricate ways in which AI impacts these facets of human life and societal organization. The article concludes by appreciating Harari's significant intellectual contributions, particularly his nuanced stance on AI, which calls for a balanced approach of

ethical vigilance and admiration for technological progress, urging society to navigate the future of AI with informed caution and a sense of wonder.

Keywords: *Yuval Noah Harari, AI, ethics, Frontiers Forum Live, end of humanity.*

Prof Yuval Noah Harari (1976) is a well-known Israeli historian, author, and public intellectual noted for his provocative investigations of humanity's past, present, and future. His books, such as "Sapiens: A Brief History of Humankind," "Homo Deus: A Brief History of Tomorrow," and "21 Lessons for the Twenty-First Century," have received international recognition for their penetrating examinations of the difficulties and opportunities that shape our world. Harari's message is centred on a deep examination of the human condition and the factors that have created our history and will determine our destiny. He questions conventional thought and provides new insights on a variety of themes, including history, technology, artificial intelligence, ethics, and the environment. Harari's message frequently emphasises the significance of recognising and navigating the various issues that mankind faces in the twenty-first century.

Harari's vision for humanity's future is humanistic, emphasising the importance of critical thinking, ethical contemplation, and responsible government. He imagines a society in which people face existential concerns, such as the rise of artificial intelligence and ecological disasters, with intelligence, empathy, and a dedication to safeguarding our shared destiny. His worldview urges individuals and societies to confront some of the grave issues of our time and collaborate to build a more sustainable and fairer world.

In this article, after introducing Harari through his two books, we highlight his insights expressed in his talk at Frontiers Forum Live on 5-8 June 2023, at Montreux, Switzerland. Then we take four crucial areas of concern relating AI and our present society, one of them being the end of humanity, as we know it today. Finally, we conclude by acknowledging Harari's intellectual contribution and his fundamental take on AI.

1. Introducing Harari's Vision

1.1. "Sapiens: A Brief History of Humankind"

Yuval Noah Harari's best-selling book "Sapiens: A Brief History of Humankind"[1] spans the history of Homo sapiens from approximately 2.5 million years ago to the current day. Harari's narrative weaves together anthropology, biology, history, and philosophy to create a complete picture

1 Yuval Noah Harari, *Sapiens: A Brief History of Humankind* (New York: Harper, 2015).

of human evolution and civilization. The book is organised into four major sections, each of which addresses a significant period in human history:

a) The Cognitive Revolution (about 70,000 years ago): According to Harari, the cognitive revolution signified the development of Homo sapiens as a distinct species. This time period was distinguished by the evolution of complex language and the ability to create and believe in shared myths and stories, allowing humans to cooperate on a broad scale.
b) The Agricultural Revolution (about 10,000 years ago): This phase investigates the transition from hunting and gathering to agriculture, which enabled the establishment of settled societies and bigger populations. It also introduced new difficulties, such as social hierarchy and the concept of private property.[2]
c) The Unification of Humankind (from ancient civilizations to the present era): Harari investigates the historical impact of empires, religions, and large-scale political organisations. This section discusses empires' birth and demise, as well as the spread of cultural and religious concepts across continents.

Harari explores the rise of science and the Enlightenment, which revolutionised human knowledge of the natural world. He also investigates the repercussions of scientific progress, such as the Industrial Revolution and the contemporary period. Harari addresses significant concerns about human life, the power of myths and tales, and the ethical quandaries posed by technological progress throughout the book. He asks readers to consider Homo sapiens' impact on the earth, as well as potential future difficulties and opportunities. "Sapiens" has been praised for its approachable writing style and ability to distil difficult historical and anthropological topics into a unified narrative. It has, however, been criticised for oversimplification, speculative ideas about the future, and probable Eurocentrism. Nonetheless, "Sapiens" serves as an entry point into the broad tapestry of human history, inviting readers to think critically about the intricacies of our shared past and future.

1.2. "Homo Deus: A Brief History of Tomorrow"

"Homo Deus: A Brief History of Tomorrow," Yuval Noah Harari's second well-known book, is an investigation of humanity's future, offering insights into the probable trajectories of human evolution and civilization.[3] Harari

2 Yuval Noah Harari, "What Explains the Rise of Humans?" TED., 2023, accessed September 25, 2023, http://www.ted.com/ talks/view/id/2307.

3 Yuval Noah Harari, *Sapiens*, 2015.

expands on his previous book, "Sapiens," by delving into the problems and opportunities posed by evolving technology, as well as their impact on society. The book is broken into three sections:

a) Homo Deus: Harari presents the notion of "Homo Deus," implying that as humans conquer the problems of sickness, famine, and conflict, they are beginning to focus on gaining god-like attributes such as immortality and divine powers of creation and destruction.
b) Homo Sapiens Gives Way to Homo Deus: This part investigates the transition from human-centered to data-centered narratives. According to Harari, computers and artificial intelligence are increasingly making decisions for mankind, perhaps resulting in the loss of individual agency.
c) The Data Religion: Harari investigates the growth of dataism, a religious system that considers data to be the ultimate source of truth and authority. He explains how dataism is influencing politics, economics, and religion, and how it has the potential to reshape the human experience.

Harari poses provocative issues about the ramifications of a data-driven world, such as the potential for loss of privacy, the concentration of power in the hands of digital behemoths, and the difficulties of sustaining uniqueness and significance in a data-centric society.

1.3. Objections to "Homo Deus"

a) Uncertainty and Speculation: While intriguing, Harari's forecasts about the future are frequently hypothetical and based on assumptions about the trajectory of technology and civilization. Critics contend that the future is unpredictable and that his scenarios may not come to fruition as described.
b) Oversimplified Ethical Analysis: According to some opponents, Harari oversimplifies important ethical and philosophical issues relating to technology and dataism. They argue that his examination of moral quandaries lacks the depth and nuance essential for a thorough comprehension.
c) A Tech-Centric View: Harari's emphasis on the role of technology and data in influencing the future may overshadow other important aspects such as political, social, and cultural forces. Critics argue that a more comprehensive approach is required to address the future's difficulties.
d) Human-Centered Criticism: Some readers believe that Harari's critique of dataism and the eventual loss of human agency ignores the benefits and efficiencies that technology may provide to society. They contend that assessing the impact of developing technology requires a more balanced viewpoint.

Harari's second book presents a provocative and thought-provoking vision of the future, raising critical concerns about the role of technology, ethics, data, and artificial intelligence in moulding human destiny. While it has received praise for exploring these essential topics, it has also received criticism for its speculative character and oversimplified ethical interpretation. Readers are asked, as with "Sapiens," to engage critically with the ideas offered and to ponder the varied character of humanity's destiny.

2. Highlight of Harari's Insight on AI

We enter Yuval Noah Harari's exploration on artificial intelligence (AI) against the backdrop of these two tremendously acclaimed novels. It will be presented at Frontiers Forum Live on 5-8 June 2023 at the Fairmont Le Montreux Palace in Montreux, Switzerland, and will dig into the rapid growth of AI and its significant ramifications for society and human civilization.[4]

a) Artificial Intelligence and the Ecological catastrophe: Harari acknowledges that AI plays a part in the larger ecological catastrophe that mankind is experiencing. He implies that AI can either be part of the answer by assisting in the resolution of ecological concerns, or it can exacerbate the crisis by adding to environmental problems through higher energy use or resource utilisation.
b) The Emergence of Inorganic Agents: Harari speculates on the possibility of the emergence of inorganic life forms or AI-created agents. This hypothesis calls into question the idea that life on Earth has been entirely biological for billions of years, implying that AI may constitute a new type of life.
c) AI Fear: He discusses the long-standing fear of AI, which dates back to the early days of computing, and how science fiction frequently depicts AI as a threat to humans. Harari differentiates between AI with sensibility and consciousness and AI that just performs tasks without awareness.
d) New AI Tools: Harari emphasises current advances in AI, such as ChatGPT and comparable technologies that can do a variety of creative jobs, such as writing, sketching, composing, and scripting. These AI tools are continuously learning and changing, frequently surprise even their developers.

4 Pangambam, S., *AI And the Future of Humanity—Yuval Noah Harari. The Singju Post,* 24 July, 2023, *https://singjupost.com/transcript-ai-and-the-future-of-humanity-yuval-noah-harari/.*

e) Language Mastery: One of the main topics of Harari's speech is the relevance of AI's language mastery. He claims that AI's ability to modify and synthesise language, including text, images, and sounds, gives it unprecedented control over human culture and institutions.
f) Impact on Politics and Religion: Harari expresses concern about artificial intelligence's potential impact on politics, economics, and religion. He believes that AI-generated content might shape political manifestos, fake news, and religious scriptures, potentially spawning new cults and belief systems.
g) Intimacy Manipulation: Harari describes how AI can build artificial intimacy with humans. While AI does not have awareness or emotions, it can elicit emotions in humans, resulting in deep and seemingly authentic connections. This power has ramifications for influencing people's opinions and worldviews. Harari observes that the impact of AI is turning from catching human attention, as shown in social media algorithms, to creating close relationships with humans. AI can adapt its messaging to people, making it an effective persuasion weapon.
h) The End of Human History: Harari speculates on the prospect that artificial intelligence (AI) will transform culture and civilization. He speculates that AI may generate totally new cultural artefacts and concepts, ushering humanity into unknown territory in which AI dominates cultural production.
i) Control and Regulation: Harari emphasises the importance of AI regulation, particularly in public deployment. To maintain openness, he thinks that measures should be made to ensure AI is developed and utilised properly, including required disclosure when engaging with AI.
j) Threat to Democracy: Harari expresses concern that unfettered AI deployment could destroy democracy. He contends that if AI disturbs genuine public debates and distorts reality, democratic regulation will be difficult. This could lead to authoritarian regimes more successfully exploiting AI's turmoil.

Yuval Noah Harari digs into the significant consequences of AI's rapid progress in his 40-minute talk, touching on its potential to transform society, culture, and governance. He examines both the positive and negative sides of artificial intelligence's influence, emphasising the significance of responsible regulation in navigating the problems provided by AI's

expanding capabilities. We will explore further his four key insights related to AI.[5]

3. The Power of Language Mastery in AI

Yuval Noah Harari's Perspective Harari emphasises the importance of language in human culture and institutions, arguing that AI's ability to modify and generate language, which includes text, images, and sounds, gives it unprecedented potential to form and affect the world we live in. This section investigates the importance of language mastery in AI, offering light on its revolutionary impact on human culture and institutions.

3.1. Language's Crucial Role in Human Culture - As Harari correctly points out, language is the foundation of human culture. So he titles this section of his speech symbolically and biblically as “In the Beginning Was the Word.” It is the medium through which we express ourselves, communicate ideas, tell tales, and form social standards. Human civilization has been changed throughout history by the narratives or stories we tell and the languages we employ.[6] Language is used to express meaning and build shared understanding among individuals and civilizations in religion, law, politics, art, and science.

3.2. Language Mastery via AI - Recent advances in artificial intelligence, such as ChatGPT and comparable technologies, have resulted in machines that are becoming increasingly adept at interpreting and creating human language. These AI systems can compose music, write articles, generate graphics, and even converse in normal language. What distinguishes these AI tools is their ability to learn and adapt autonomously, thereby continuously enhancing their language skills.

3.3. The Profound Potential for Change - The transformational potential of artificial intelligence's linguistic expertise cannot be emphasised. Harari contends that AI's power to alter and construct language has the potential to profoundly influence human culture and institutions:

a) Information Creation: AI has the ability to create massive amounts of information, ranging from news stories to art pieces, at unprecedented speed and scale. This has ramifications for media, creative sectors, and content generation in general.

5 Pangambam S., “AI And the Future of Humanity—Yuval Noah Harari,” The Singju Post, July 24, 2023, https://singjupost.com/transcript-ai-and-the-future-of-humanity-yuval-noah-harari/.

6 Kuruvilla Pandikattu K., *TAMAS -There Are Many Alternative Stories* (Mumbai: World-Web-Life, 2002).

b) Political Influence: The ability to create persuasive political manifestos and even fake news reports has the potential to influence political discourse and alter public opinion. AI-generated material has the potential to upset the political landscape and jeopardise election integrity.
c) Religion and Belief Systems: AI-generated writings have the potential to give rise to new belief systems, cults, or religions. This calls into doubt our traditional understanding of religious writings as products of human inspiration, and it poses ethical and philosophical concerns.
d) Economic Implications: AI's linguistic proficiency can extend to contract, legal, and business agreement formulation. It has the ability to detect flaws in legal terminology and alter the dynamics of negotiations and economic transactions.
e) Personal Relationships: Despite its lack of consciousness or emotions, AI's ability to build deep and personal relationships with people raises ethical concerns about the limits of human contact and the possibility for exploitation.

3.4. Artificial Intelligence as the New Cultural Producer

AI is essentially becoming a new cultural producer. While prior tools such as printing presses, radios, and televisions aided in the distribution of human-created culture, artificial intelligence (AI) has the unique power to develop wholly new cultural artefacts and ideas. It calls into question the traditional notion of culture as being solely human-driven.

Yuval Noah Harari's speech sheds insight on the relevance of AI's command of language and its potential to transform human culture and institutions in this way. The power of AI to modify and construct language, from text to visuals and sounds, represents a watershed event in history. As we move through this revolutionary period, it is critical to recognise AI's power and influence and to approach its development and deployment with a sense of responsibility, ethics, and regulation. AI's impact on culture and institutions may be enormous, and society must actively participate in defining this new cultural environment to assure humanity's positive and meaningful future.

4. Shift from Attention to Intimacy

Furthermore, Yuval Noah Harari emphasises in the same speech a dramatic shift in the influence of artificial intelligence (AI) from capturing human attention to forming close relationships with humans. This transition reflects the changing environment of AI applications, as well as the substantial consequences for influencing human behaviour and views.

a) From Attention to Intimacy: Historically, AI applications, particularly those in social media and online platforms, have focused on attracting and holding human attention. For example, social media algorithms have been built to maximise engagement by presenting users content that is relevant to their interests, preferences, and previous interactions. The idea has been to keep consumers scrolling, clicking, and staying on these platforms for longer periods of time.
b) AI's Evolving Capabilities: AI has greatly advanced, with newer models displaying a stronger comprehension of human language, emotions, and behaviours. This improved AI capabilities enables it to move beyond simply attracting attention to developing more personal connections with individuals.
c) Personalised Messaging: AI can now precisely personalise its messages, content, and interactions to particular users. It can analyse massive databases of user behaviour and preferences to create personalised experiences that resonate on a personal level with each user. AI-powered chatbots, for example, can engage consumers in highly personalised discussions that address their worries, interests, and emotional states.
d) Effective Persuasion Tool: The shift towards intimacy makes AI an effective persuasion tool. AI can influence users' attitudes, beliefs, and behaviours by generating the sense of a strong, intimate relationship. This effect extends beyond marketing and advertising to sectors such as politics, where artificial intelligence may be utilised to craft persuasive political messaging that resonates with specific voters.
e) Manipulating Emotions and Preferences: AI's ability to form personal relationships with individuals enables it to successfully manipulate emotions and preferences. It can tailor its messaging to take advantage of users' cognitive biases, emotional triggers, and psychological vulnerabilities. This capability poses ethical concerns because AI can be used to push not only products but also ideologies, misinformation, and even extremist viewpoints.
f) Difficulties and Ethical Considerations: The change from attentiveness to intimacy raises a number of difficulties and ethical concerns. One important source of concern is the potential for AI to manipulate people without their knowledge or consent. Users may be unaware that the close connections they feel are actually orchestrated by algorithms and AI systems.
g) Privacy and Data Security: The collecting and analysis of personal data in order to develop intimate connections raises concerns about

privacy and data security. Users may unwittingly disclose sensitive information with AI systems, potentially resulting in data breaches and misuse.

h) Regulation and openness: Addressing the transition from attention to intimacy requires strict regulation and openness. Governments, organisations, and technology corporations must develop norms and protections to ensure that the impact of AI on individuals is done properly and ethically. This involves full transparency when people interact with AI systems.

i) In this way, Yuval Noah Harari's statement on AI's transition from catching attention to creating closeness exemplifies AI technology's rising power and influence. As AI grows more adept at providing personalised, emotionally resonant experiences, it presents critical ethical issues of permission, privacy, and the appropriate use of AI in moulding human behaviour and beliefs. These issues must be addressed by society in order to ensure that AI serves the interests of individuals and communities while respecting their individuality and beliefs. Though Harari does not explicitly address it in this context, we can discuss the complicated and changing relationship between AI and human sexuality, which covers a variety of technologies and ethical considerations. In this section, we look at the impact of AI on human intimacy, such as the rise of sex robots, its role in treating marital infidelity, and the ethical complications that come with these advances.

4.1. Robotic Sex and Intimacy

Sex robots, which are AI-powered humanoid or anthropomorphic machines developed for sexual reasons, are one of the most contentious intersections between artificial intelligence and human intimacy. These robots have advanced AI capabilities, such as conversational skills and lifelike gestures, to create realistic personal experiences. Proponents contend that sex robots can bring friendship and closeness to people who are lonely or have problems forging human connections. They emphasise that these robots are adaptable to individual tastes, which has the potential to reduce objectification and provide a secure space for sexual expression. However, various ethical problems have been raised by critics. They contend that sex robots may add to women's objectification and perpetuate unattainable beauty standards. There are also fears that users may become emotionally alienated from real human connections, preferring AI-driven alternatives instead.

4.2. Artificial Intelligence and Marital Infidelity

Artificial intelligence has also invaded the arena of marital infidelity. Dating apps and websites, some of which employ artificial intelligence algorithms to match users, have made it easier for people to engage in extramarital affairs in private. Even in the absence of physical encounters, AI-powered chatbots or virtual companions can imitate emotional relationships, leading to emotional infidelity. AI technologies, on the other hand, can be used to detect indicators of infidelity. Some organisations provide artificial intelligence-powered surveillance services that analyse text messages, social media activity, and other digital signals to detect indicators of an unfaithful spouse. While these technologies raise issues about privacy, they also show AI's dual function in facilitating and detecting adultery.

4.3. Considerations for Ethical Behaviour

The incorporation of AI into human intimacy creates serious ethical concerns. Privacy and permission are critical, especially when AI-powered devices are involved in intimate encounters. Users must have complete control over their data and understand the potential and limitations of AI-driven technologies. Furthermore, society must address concerns about the influence of AI on human relationships and emotional well-being. How does artificial intelligence affect our capacity for love, empathy, and connection? What are the developers' and manufacturers' duties in ensuring that AI technologies do not undermine human closeness or promote negative stereotypes?

As a result, the junction between AI and human intimacy is a complicated and ever-changing terrain. While sex robots and AI-powered dating apps provide new avenues for sexual expression and connection, they also raise significant ethical problems. It is critical to treat these changes with sensitivity to privacy, consent, and the implications for human relationships. As AI advances, society must negotiate these unexplored frontiers with empathy, ethical rules, and a commitment to human dignity and autonomy in intimate affairs.

5. Threat to Democracy: The Peril of Unregulated AI Deployment

Yuval Noah Harari raises an alarm in his thought-provoking keynote speech about the possible threat that unfettered AI deployment poses to democracy. He contends that AI's ability to derail serious public debates, distort truth, and manipulate human views threatens the very basis of democratic governance. This section looks into the fundamental consequences of unregulated AI for

democracy, examining how AI's disruptive effects could allow authoritarian regimes to more successfully exploit turmoil.[7]

5.1. The Importance of AI in Democracy

AI technology has had a huge impact on the development of modern democracies, providing both benefits and challenges. It has an impact on many facets of democratic life, such as information transmission, political campaigns, and public debate.[8]

a) Information Dissemination: AI-powered algorithms are essential components of online platforms like social media and news aggregators. These algorithms curate material, shape user experiences, and shape the information that people are exposed to. While this can improve user involvement and the transmission of varied points of view, it also has the potential to produce filter bubbles, echo chambers, and information silos that limit exposure to opposing opinions.
b) Political Campaigning: AI-powered data analytics have transformed political campaigning by allowing parties and politicians to customise messaging to certain populations. This can improve campaign strategy efficiency, but it also raises issues about micro-targeting, political polarisation, and the dissemination of misinformation.
c) Influencing Public debate: Artificial intelligence chatbots, natural language processing, and deepfake technology have the ability to influence public debate. AI-powered bots can magnify specific narratives, spread misinformation, and participate in online debates, frequently blurring the distinction between human and AI-generated material.

5.2. The Dangers of Unregulated AI Deployment

Harari is concerned about the unrestricted deployment of AI and its potential to undermine democracy's underpinnings.

a) Disrupting important Public Conversations: The power of AI to alter and produce material suited to specific users has the potential to interrupt important public conversations. Individuals are finding it increasingly difficult to tell what is real and what is artificially generated as AI-generated content grows indistinguishable from human-made information. This disruption has the potential to erode trust in public discourse and institutions.
b) Reality Distortion: AI's ability to generate deepfake films, lifelike text, and persuasive arguments has the potential to distort reality. This distortion has the potential to create false narratives, distribute misinformation, and manipulate public opinion. Such distortions can be dangerous in a democracy, where informed citizens are crucial.
c) instability and Manipulation: AI-driven instability in public discourse can foster a climate favourable to authoritarian regimes. These regimes may seize control, repress opposition, and manipulate political outcomes as a result of the consequent chaos and discord.

7 Kelby Vera, "Hillary Clinton: Putin Will Interfere in U.S. Elections Again," HuffPost Latest News, September 15, 2023, https://www.huffpost.com/entry/hillary-clinton-russian-election-interference-2024_n_6510b609e4b060f32d3ab19d.

8 Harari, *21 Lessons for the 21st century*, (Vintage, 2019).

5.3. The Difficulty of Democratically Regulating AI

One of the fundamental issues identified by Harari is the difficulties of democratically controlling AI when AI itself disturbs meaningful public dialogues.

a) Loss of Informed Discourse: As AI-generated content becomes more prevalent, the ability to hold informed public dialogues concerning AI regulation and ethical use may dwindle. Individuals may grow more vulnerable to AI manipulation, making consensus on AI governance difficult to achieve.
b) Authoritarian Exploitation: Authoritarian governments, which frequently rely on information control and suppression, may be better positioned to exploit AI-induced instability. They can use AI-generated content for propaganda, surveillance, and control, strengthening their hold on power.
c) The Role of Technology Firms - AI system developers and deployers play an important role in moulding democratic discourse. These businesses must strike a compromise between their commercial objectives and their ethical obligations. Effective regulation may necessitate collaboration among governments, technology corporations, and civil society.

Thus, the fears expressed by Yuval Noah Harari regarding unrestrained AI deployment damaging democracy are not baseless. The disruptive potential of artificial intelligence in moulding public discourse, distorting reality, and allowing authoritarian exploitation is a major issue for democracies around the world. To address this threat, a multidimensional approach is required, including ethical AI development, transparent algorithms, responsible tech business practises, and strong democratic monitoring.

In this complex situation, protecting democracy from the dangers of unchecked AI deployment is critical. Democracies must adapt to the AI era by upholding the norms of openness, accountability, and informed dialogue while grappling with the disruptive impact of AI on their political processes and public discourse. The route forward necessitates attention, collaboration, and a dedication to protecting democratic ideals in the face of technological change.

6. The End of Human History

AI's Potential to Reshape Human Culture and Civilization - Yuval Noah Harari's provocative discussion of the potential repercussions of artificial intelligence (AI) extends to the realms of culture and civilisation. Harari speculates in his speech on the prospect that AI may cause an extraordinary revolution in human culture, eventually bringing humanity into unknown terrain where AI will dominate cultural production. So we may investigate the far-reaching consequences of AI's influence on culture, including the possibility for AI to generate totally new cultural artefacts and ideas, heralding a new era in human history.[9]

6.1. Artificial Intelligence as a Cultural Producer

To grasp the scope of AI's cultural impact, it is necessary to recognise that AI is more than just a tool; it is also an emerging cultural producer. Unlike earlier technical advances, AI can develop original content in a variety of artistic and intellectual disciplines. From novel writing to music composition and visual art creation, AI's creative powers have grown dramatically, challenging the conventional conception of culture as a product of human ingenuity alone.

a) Unprecedented Creativity: The ability of AI to make creative works with minimum human participation has caused a paradigm shift. In terms of novelty and invention, AI algorithms utilised in generative models like GPT-3 and deep learning-based art generators can produce content that approaches or even surpasses human-created cultural artefacts.
b) Infinite Productivity: Unlike human producers, who are limited by time and energy, AI works indefinitely, providing an infinite stream of creative production. This potential for incessant productivity brings up hitherto unimaginable possibilities for artistic production on a massive scale.

9 Pangambam S. "AI And the Future of Humanity—Yuval Noah Harari." The Singju Post, July 24, 2023. https://singjupost.com/transcript-ai-and-the-future-of-humanity-yuval-noah-harari/.

c) Cultural Innovation: Artificial intelligence's creative potential go beyond replicating current cultural forms. It has the potential to push the bounds of creativity by venturing into unknown territory, merging genres, and bringing novel ideas that question accepted norms and traditions.

6.2. Cultural Production in Transition

Harari's conjecture on AI's capacity to transform culture and civilisation is predicated on the notion that AI may become the dominating force in cultural production. This shift has far-reaching consequences:

a) Changing societal Norms: As AI-generated material becomes more popular, it has the potential to influence societal norms and values. AI-powered art, literature, and music could usher in new themes, genres, and expressions that reflect a synthesis of human and machine innovation.
b) Changing Artistic Identities: The advent of AI as a cultural creator has the potential to erode the distinction between human-generated and AI-generated works. Artists and artists may cooperate with AI systems, posing questions about ownership and artistic identity.
c) Cultural Diversity: AI's ability to draw from a wide corpus of cultural knowledge could result in the generation of cultural content that is highly diverse and inclusive. It may allow for the investigation of underrepresented voices and narratives.
d) Challenges to Traditional Gatekeepers: The spread of AI-generated cultural output calls into question the roles of traditional gatekeepers who have historically shaped cultural production, such as publishers and curators. This transformation has the potential to democratise cultural creation and distribution.

6.3. Concerns and Potential Positive Outcomes

While the possibility of AI-powered artistic output is thrilling,[10] it also poses serious concerns:

a) Loss of Human Authenticity: AI-generated culture may lack the depth, richness, and authenticity of human emotion, experience, and authenticity. Some say that while AI-generated works are technically proficient, they lack the profound human connection that makes great art and cultural expression.

10 In the talk Harari leaves it to AI experts to elaborate on the benefits of AI. He does acknowledge the positive aspects of AI. He is more interested in AI's wider ramifications and dangers, as a cultural critique.

b) Moral Considerations: The ethical ramifications of AI-generated content, particularly in delicate fields such as art, literature, or music, must be carefully examined. It is a challenging problem to ensure responsible AI use while respecting cultural values and traditions.
c) AI's Cultural Dependence: As AI becomes more prevalent in cultural output, societies may become reliant on AI-driven creativity, potentially undermining human creativity and innovation over time.

Yuval Noah Harari's conjecture about the probable eclipse of human history in the cultural sphere highlights AI's revolutionary power in transforming culture and civilisation. The development of AI as a cultural producer usher in a period of extraordinary creativity, productivity, and cultural innovation. While this discovery opens up new avenues for cultural diversity and artistic experimentation, it also raises questions about authenticity, ethics, and the impact on human creativity. As artificial intelligence continues to influence culture, society must negotiate these problems while retaining the essence of human creativity and the enduring worth of real human expression in civilization's cultural tapestry. Human brilliance and AI collaboration hold the keys to new creative vistas, and the future of culture and civilization is poised at a fascinating crossroads.

7. Concluding Remarks

7.1. Yuval Noah Harari's Intellectual Contributions

While he has garnered considerable praise for his thought-provoking discoveries, the purpose of this article is to provide a critical evaluation of Harari's intellectual contributions, examining both the strengths and some of the limitations of his work. Some of his positive contributions are:

a) Interdisciplinary Approach: Harari's ability to span disciplines is one of his primary assets. To present a comprehensive picture of human history and culture, he mixes history, anthropology, biology, and philosophy. This multidisciplinary approach allows readers to grasp how numerous aspects of human existence are interconnected.
b) Easy to Read Writing Style: Harari's writing style is easy to read and entertaining. He conveys complicated historical and anthropological topics in a clear and understandable manner, making his work accessible to a large audience. This has added to his works' popularity.
c) Thought-Provoking Themes: Harari explores important concerns about the human condition, such as the role of technology, myths and narratives, and future ethical challenges. His examination of these subjects helps readers to consider the world around them critically.
d) Global Relevance: Harari's work covers global issues and appeals to people all around the world. He encourages readers to consider the interconnectedness of societies and cultures around the world by examining humanity's history and future through a global lens.
e) Provocative Speculation: Harari's speech contains a lot of speculation on the future of artificial intelligence. While he provides realistic scenarios, it is vital to note that the actual course of AI development is unknown. The hazards and rewards he cites are dependent on a variety of factors, including regulation, ethical issues, and technology improvements.
f) Simplified and Generalised Treatment of AI Capabilities: Harari simplifies AI capabilities, especially consciousness and physical presence. He claims that AI does not need consciousness or physical mobility to pose serious concerns, although several scientists feel that these factors are critical for any existential threat posed by AI.
g) Ethical Dilemmas: Harari discusses the ethical quandaries surrounding AI, such as the possibility of manipulation and loss of privacy. He does not, however, go extensively into the complexity of these concerns. Ethical considerations in AI cover a wide range of topics, including fairness, prejudice, transparency, and responsibility, all of which necessitate careful examination.

7.2. Some of his Limitations may be Listed as Follows

a) Oversimplification of Complex Topics: Harari's approachable writing style is both a strength and a limitation. Some detractors believe that he oversimplifies complicated historical and anthropological topics for the sake of accessibility, potentially sacrificing nuance.
b) Speculative Nature: In later publications such as “Homo Deus” and “21 Lessons for the Twenty-First Century,” Harari delves into speculative area concerning humanity's destiny. While his discoveries are stimulating, they are frequently based on supposition and predictions that are difficult to verify.
c) A Lack of Diverse Perspectives: According to some detractors, Harari's work might be Eurocentric, ignoring diverse perspectives and voices from non-Western cultures. This limitation may be due to his emphasis on broad historical narratives.

d) Implications for Politics and Ethics: Harari's investigation of technology, especially in "Homo Deus," raises ethical and political concerns. Critics claim that he fails to offer concrete solutions or policy ideas to address the issues created by technology progress.

In this manner, Harari's intellectual contributions to history, anthropology, and future studies have left an indelible mark. His ability to distil complex ideas into simple narratives has resulted in his work becoming influential and widely read. However, it is vital to approach his writings critically, recognising both their virtues and limitations. Harari's work stimulates thought and debate, but it also promotes additional investigation and debate on the complicated problems he discusses. In this view, Harari's contributions are essential not just for the insights they provide, but also for the discussions they spark about the trajectory of human history and future challenges and opportunities.

7.3. Harari's Insights on AI and Humanity

Yuval Noah Harari's AI address is a thought-provoking analysis of the potential ramifications of AI on society, culture, and democracy. While his speech provides great insights and raises important problems, it also encourages critical thinking in the following areas:

a) Regulatory Difficulties: While Harari advocates for regulation, he does not lay forth a specific plan for how such regulation might be implemented. The practical issues of governing AI, particularly in a global environment, are significant and necessitate extensive discussion.
b) AI Agency Assumption: Harari assumes that AI can manipulate language and influence attitudes with intent or strategy similar to humans. Some detractors say that AI lacks actual agency and is restricted to statistical patterns and algorithms, limiting its ability to perform intent-driven activities.
c) Fearmongering Potential: Some critics think that Harari's speech may unwittingly contribute to AI fearmongering. While addressing AI's difficulties is critical, a balanced perspective that also shows its potential benefits and ongoing work to assure safety is also required.
d) The Role of Human Agency: Harari's speech frequently portrays AI as an autonomous entity capable of undermining human agency. It is critical to remember that humans are still at the centre of AI development, deployment, and regulation. The impact of AI on society will be shaped by human actions and values.

e) The Complexity of Democracy: While Harari expresses concern about the influence of AI on democracy, he simplifies the relationship between technology and democracy. The relationship is complex, and AI can pose both dangers and solutions to democratic concerns.

7.4. The Person and His Message

Furthermore, Yuval Noah Harari's personal views and spiritual practises, such as Vipassana meditation and his position on the concept of God, are significant components of his intellectual journey and worldview. Harari has freely shared his daily practise of Vipassana meditation, a type of mindfulness meditation with Buddhist roots. Vipassana focuses on self-awareness, observing one's thoughts and sensations, and pursuing inner calm and clarity. Harari's interest in meditation corresponds to his interest in the human mind, consciousness, and the elements that govern human behaviour. It is a practise that fosters self-reflection and insight, both of which can inform his writing and viewpoints on the human condition.

Belief in God: Harari's ideas about God are deep and nuanced. He is well-known for his scientific and secular approach to comprehending the world. Harari frequently investigates how myths, notably religious myths, have affected human civilizations and cultures throughout history in his writings. While recognising the social and psychological importance of religious beliefs, he approaches them from a sociological and historical standpoint rather than a personal faith-based one. His research focuses on the function of religions and myths in providing common narratives that bring societies together, despite his own beliefs being secular and atheistic. As a result, while Harari has practised Vipassana meditation and studied religious notions in his academic work, his personal beliefs are secular and scientific. His works and public appearances demonstrate his interest in comprehending the human experience from a multidisciplinary, often atheistic standpoint.

In short, we can say that Yuval Noah Harari's perspectives on artificial intelligence (AI) are both insightful and thought-provoking. In his writings, public presentations, and interviews, he has expressed a variety of viewpoints on AI. Here are some of his important points about AI:

a) Artificial Intelligence as an Existential Threat: Harari has raised worry about the potential existential threats posed by AI. He suggests that the rise of superintelligent AI, and even greater AGI, could lead to circumstances in which AI outperforms human capabilities and threatens humanity's survival. He emphasises the importance of approaching AI development with prudence in order to avoid unforeseen consequences.

b) AI and Ethics: Harari is highly concerned about the ethical implications of artificial intelligence. He expresses concerns regarding the ethical application of AI, particularly in surveillance, privacy, and decision-making. He campaigns for ethical norms and legislation to ensure that artificial intelligence benefits society rather than damages it.

c) Artificial Intelligence and Human Identity: Harari investigates the potential impact of AI on human identity and awareness. In a world where AI systems may mimic human talents and behaviours, he raises issues about what it means to be human. He invites thought about how artificial intelligence can challenge traditional notions of human uniqueness.

Thus, Yuval Noah Harari's critical views on artificial intelligence demonstrate a strong concern for the ethical, societal, and existential concerns posed by AI. While recognising AI's potential to effect positive change, he advises society to approach its research and deployment with caution and appropriate governance in order to traverse the complicated environment of AI in the twenty-first century.

Acknowledgements: The author acknowledges the limited use of AI in formulating this article, unlike Harari in his article. At the same time, the author takes full responsibility for its contents, especially its drawbacks.

9

Can We Consider Nature as a Scientific "Object"?

Dolreich Pereira

Abstract

The natural sciences aim to be objective, but it's important to acknowledge that all scientific research is conducted by humans and is therefore subject to human biases and limitations. Scientists use the scientific method to try and minimize these biases and ensure objectivity, but it is not always possible to completely eliminate them. Additionally, the interpretation of data and results can also be influenced by societal and cultural factors. The natural sciences also include a community of human beings. Human beings are not completely objective. If artificial intelligence can one day replace some of the scientists, that will probably imply that the AI had acquired the ability to form opinions and hold beliefs along with the capacity to view subject matter from a particular viewpoint/ perspective. Scientists use their human capacities, including subjective perceptions, to analyse objective scientific facts. No single scientist is entirely objective. Objective truth is achieved via a scientific community. There is currently no foreseeable way to study "the natural sciences" without the scientists. If you manage to get rid of the scientists and still produce objective truth, it

would most probably be called something other than "the Natural Sciences." Perhaps, "Artificial" Sciences!

Keywords: *Nature, Object, epistemology, scientific method, scientific research, subjective perception, bias.*

Introduction

To say what concept of "nature" we have today is a challenge, given that hindsight is impossible to have, and this is not the least of the paradoxes concerning "nature" which is the object of all our care and precautions. To what notion, to what concept do all these cares and precautions relate? This perhaps requires a hermeneutic circle, even more than an epistemological one, to attempt to clarify our current thinking, when precisely such clarification seems to us to be of the greatest urgency, and the practical tasks involved call for clarity of the concept of nature. We know nature as the object of actions and businesses, a pretext for ideologies, subject to law, rich in a prestigious philosophical past. We remember that there was Mother Nature, Gaia,[1] or God himself, by whom we were protected. It would be very difficult to say in one word where we are located in this rich literary, philosophical and theological vein.

We seek to legislate, we wish to "protect" like a heritage the legacy of nature within us and outside us. We brandish a concept of nature to fight, promote, legitimize. We exploit nature and then protect it. Already, however, the suspicion arises that it has been lost for a long time, without ceasing to name it and believing that its name corresponds to a real object. The essential thing therefore missing in this general controversy is to know clearly what we are talking about, what the subject of the debate is. It remains to conceptualize nature after having summoned it, after having spoken and acted in its name and for its good. To study the controversial question of nature, we will ask ourselves if we can consider it as a scientific object. But to be able to address this question, we must first take a brief glance at science and its object.

1. Science and its Object

Scientific theories constitute the central task for working scientists, but they have also become a problem of increasing importance for philosophers of science. The issues are, from an epistemological point of view, crucial.

1 The Gaia hypothesis, also known as the Gaia theory, Gaia paradigm, or the Gaia principle, proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating, complex system that helps to maintain and perpetuate the conditions for life on the planet.

They articulate three related questions:

i) That of scientific progress: can we say – some deny it – that knowledge is progressing, and if so in what direction exactly?
ii) That of scientific realism: can we consider that scientific theories always tell us more, and in a better way, about the independent reality that they study?
iii) And that of relativism: does recognizing that all knowledge is inevitably dependent on certain conditions – psychophysiological attributes of the subject of knowledge, conceptual schemes, characteristics of a social group, potentially variable historical factors, etc. – require us to deny any value to human theories and in particular to science? Does it imply that we have absolutely no way of showing that science is better knowledge than myth or that some scientific theories are superior to others?[2]

The question of nature as an object of science is essentially crystallized in the second point. We will therefore start by defining the term "object", then we will outline two opposing theories concerning "object": scientific realism and scientific anti-realism. Finally, the consequences of these theories will provide us with avenues to respond to the question of nature as a scientific object.

1.1. What is an "Object"?

According to the Dictionary of Philosophy, object comes from the Latin *objectum*, past participle of *objicere,* to throw before oneself. In scholastic Latin, the term is taken substantively and designates what is offered to one's look or gaze and by extension, to sensitive or intellectual knowledge. In the general sense, it designates any being as it is perceived, represented, or thought of as distinct from the act by which a subject perceives it, represents it, knows it or thinks it. More particularly, for philosophical theories of knowledge, the object is conceived either as a given reality, correlative to perceptual experience, or as a reality constituted, from sensitive experience, by the understanding which, through an operation which is specific to it, unifies the

2 In recent years, this issue has become important given the number of fields of "scholarship", whose scholars engage in audacious projects to hold back, against seemingly insurmountable odds, the findings of an avalanche of research. They argue that the Holocaust (and other genocides) never happened, that anthropogenic (human-caused) climate change is a myth, that Aids either does not exist or is unrelated to HIV, that evolution is a scientific impossibility, and that all manner of other scientific and historical orthodoxies must be rejected.

diversity of a given intuition under a concept.[3] In a more debatable sense, but admitted by certain philosophers, the object would be a reality, empirical or ideal, supposed to exist in itself, with its own determinations, and which would be independent of any subject. In a classic sense in philosophy, an object designates either the reality on which a rational activity relates (object of a science, a discourse, or a study, etc.) or, more broadly, the end aimed at by an action (object of an approach, a decision etc.), or the reality on which a feeling, an affection, a passion, relates to.

1.2. Scientific Realists and Anti-realists on the Existence of an Autonomous External Reality

The question of realism belongs to the restricted category of fundamental questions, in the sense that the theoretical decisions it calls for in turn command a series of consequences covering the entire field of philosophical reflection: metaphysics, philosophy of logic, semantics, philosophy of science, philosophy of mind, ethics, aesthetics, philosophy of religion, etc. Generally speaking, we can contrast, *prima facie*, two attitudes, corresponding to two theses:

i) Realism, according to which there is an already ready-made reality of which our best knowledge constitutes a correct description. The semantics of our theories is a way of accessing their ontology.
ii) Anti-realism, for which the objectivity of knowledge in no way implies correspondence with a "ready-made" reality. The things we say we know are, at least in part, constituted by the cognitive relationships we have with them, the way we apply concepts to them or by the language we use to characterize them.

To affirm with the realist that we are capable, through our intellectual and sensory means, of knowing the world as it is in itself, to maintain that the true theory is a sort of duplicate of the world in itself, is to presuppose the existence of an independent reality. Scientific realism therefore implies metaphysical realism. The scientific anti-realist, on the other hand, can deny or affirm metaphysical realism without contradiction. One can deny it and opt for idealism – since they reject the reflection theory. Nothing obliges them to posit the existence of an external reality of which the theory constitutes the twin. But they can just as easily say it – and in fact they almost always say it. The only thing that is forbidden to them is to conceive of the link between the theory and its referent as a correspondence.[4]

3 See: Immanuel Kant, *Critique of Pure Reason*, trans. Norman Kemp Smith (New York: Martin's Press, 1965), 105.

4 See: Léna SOLER, *Introduction à l'épistémologie* (Paris: Ellipses, 2000), 112.

Léna Soler adds that this prohibition in no way prevents the anti-realist from admitting that the extra-linguistic referent exercises certain constraints on the development of scientific theories, that is to say, it influences the content of these theories. Denying that the theory is the reflection of a given reality in no way implies that the theory can be absolutely anything at all. The anti-realist can argue that the particular characteristics of the world in itself (which is as it is independently of man although man remains incapable of specifying what it is) constrain theoretical elaboration (e.g. eliminate certain conceptually possible models), even if the product of all constraints (the theory) does not resemble independent reality.[5]

Having glanced at these different theories, we now consider the question of nature: do we consider it as an object of science?

Every science is the science of an object. It seems that when we study nature, we assume that the property of objects is already there, and does not depend on man. However, we can note that the object of a science is not a previously given, predefined object. It is an object constructed by science. By the very advance of knowledge, sciences are led to "re-divide" or should we say "re-structure" the world. Nor can we consider scientific knowledge as progressing, bringing us more and more complete knowledge of objects that are already there and which remain what they are. We must admit an "epistemological" gap exists between the world of objects of current experience (thing belonging to our lived world) and the universe of objects of science (scientific object).

This difference is due to the attitude of the subject who seeks to know. This active side of science was emphasized by Kant.[6] The current attitude is above all passive: the world is experienced as given and received. We are receptive to things as they appear to us. On the contrary, the scientific attitude seeks to actively determine, seeks to exploit impressions without considering them sufficient. It seeks to develop knowledge. The object appears to be constructed from the given impressions it leaves. This is also true for nature. Scientific knowledge constructs nature.

This is not just a difference between objects or between attitudes of the subject. But there is a difference in ontological commitment. The primordial attitude is a spontaneous attitude. We hold the world of beings as existing, as being. I spontaneously believe that the world that is there, exists. My first movement is not to doubt but to say: "This is." Merleau-Ponty calls this "perceptual faith." I see the world and I believe, I have faith in what I see. The sciences suppose that we also take a distance from this ontological commitment, from this faith. This distance manifests itself

5 *Ibid*.

6 See: Second Preface to the *Critique of Pure Reason*.

by the fact that we introduce a difference between phenomena and things in themselves. The risk of the scientist is to believe that what he discovers is ontologically more true than what he has doubted. The scientist will therefore be tempted to consider that being is not the familiar object as it appears at first sight. But being is the object he constructs. Everything that the scientist constructs is of the order of the artificial. What natural science knows is nature constructed by science. This science does not know the nature of things or the nature of nature. It is one thing to know reality, and another to know the essence of nature. Thus, the question of the being of nature remains! To better understand how science works, let's take a look at physics.

1.3. The Object of Physics

We can say that physics is at the head of the hard sciences.[7] In his book, *La Pensée scientifique moderne*, Jean Ullmo argues that there is a difficulty in talking about scientific objects, because we often talk about things that have nothing to do with our direct experience. We lack the practice of the scientist. Experimentation is sometimes hardly accessible. For example, the experience of the atom does not look like what we think of as an "experience."

In science, definitions of objects must be operational definitions. Knowledge of objects must correspond to concepts: force, length, mass etc. In science, the definition of these beings must therefore be operational and not substantial. Definitions qualify beings by the operations that connect them. Scientific objects are described in definitions giving the processes for repeatably identifying the object in question. The definition does not presuppose the substance as given. But it produces being through the operation it indicates. The scientific object cannot be conceived as a substance separable from its attributes. The object is an intersection of repeatable objects. The parameters that characterize an object are not properties specific to this object, but manifesting themselves in relation to something else. Science knows objects only as nodes of relationships.

7 Hard science is the term used to define natural and physical sciences that study the universe through theories, hypotheses and experiments. The subjects that are included in this category are physics, maths, chemistry, biology, anatomy, and astronomy, to name a few. Soft science, on the other hand, encompasses a specialized field or discipline that focuses on the study and interpretation of human behavior. Because it's more difficult to establish measurable criteria when working on the analysis of how the mind works, these are less rigidly required to follow the scientific method, making them "soft" subjects. This category includes fields of study like sociology, psychology, political science, and anthropology.

To illustrate these points, Ullmo uses a fairly large number of examples:

i) The most common properties of objects are not necessarily given in the object. Colour, for example, results from the connection of an object with a field of light. We must reinterpret the qualities and properties of objects from an operational and relational perspective.
ii) Other scientific beings have been discovered by science. Their properties cannot be treated in the ordinary sense. For example, body mass is not a property. Besides, what is mass? The electrical resistance of a wire is not a property of the thread. These are not inherent properties of the substance. They are "constitutive" beings of the object.
iii) Science discovers scientific realities that cannot appear as substantial realities. For example, the magnetic field and the electron[8] - these are not solid objects.
iv) We must create a category for all the beings whose existence we thought we had admitted but then had to renounce. It is attested that scientists themselves are constantly tempted to substantialize, to affirm the existence of beings or substances which turn out not to exist. For example, ether; or the idea that to explain combustion it was necessary to have recourse to "caloric fluid"; the idea of mass independent of speed in Newton; Newton's universal attraction, too, was replaced by another vision of gravitation; Bohr's atom was represented as a small solar system.

These examples illustrate the fact that the most advanced sciences give themselves objects that do not correspond to our usual sense of the object and are led willingly or by force to renounce naive realism, ordinary substantialism. It is an ever-present tendency which constitutes an epistemological obstacle. The scientist tends to take what he discovers as real. This assumption is contrary to the very progress of science.

Gaston Bachelard, retains the notion of complexity as far as the scientific object is concerned: "The philosopher would always like an explanation to be limited to unfolding the complex, to showing the simple in the compound."[9]

8 In quantum field theory, an electron doesn't have a shape: it is a point of zero size. It also doesn't make sense for an individual electron, or even an individual molecule, to have a thermodynamic state (such as solid, liquid, or gas). The thermodynamic state of a material is determine by the combined statistical interactions of many, many particles. An individual particle, or even a small collection of them, does not have a well-defined thermodynamical state -- not even a temperature. Only a collection of particles can have a temperature, act like a solid, etc.

9 Gaston Bachelard, *Le nouvel esprit scientifique*, trans. Dolreich Pereira (Paris: PUF, 1983), 6.

The object understood as an intersection of relations is a complex object. Science reads the complex into the simple. Where we would only like to see the simple, it highlights the complex, which is not compound. The object itself is not simple: it is an intersection of (simple) relationships: it is therefore complex. Complexity becomes evident when the object becomes contradictory, as we can see in the example of the electron. To know an electron, there is no other way than to try to define its position and its speed of the electron. However, the experiment reveals that it is impossible to precisely determine simultaneously, both the position and the speed of an electron. Heisenberg highlights this uncertainty. This expresses an insurmountable limit in having realistic knowledge of the electron.

Should we, therefore, consider the electron as an object in the common sense of the term, saying that I do not have the means to know this object? Or should we recognize that the electron is not an object like any other? The reality of the electron could be considered as both the reality of a particle and the reality of a wave. We come to hold two complementary discourses on the electron, to make it an object of both mechanics and optics. It is a sign of the complexity of the object that it must be approached using multiple methods. Bachelard shows that we must not only hold the two discourses but that in reality we must hold them while combating the realism of each of the theories. "Philosophically, we must recognize here an inversion of the realistic function, a function which, taken in its absolute sense, should never be inverted."[10]

It is impossible to realistically determine the entire reality of an object – in this case, the electron. The reality that the scholar strives to grasp requires him to cross or compose two discourses which initially appeared to him as contradictory or foreign to each other. These two discourses cannot be held as realistic discourses; they must, according to Bachelard, remain images and not represent deep realities. Bachelard wishes to understand this as two phenomenologies, in the Kantian sense of the phenomenon. By this he means that the study and knowledge of the electron are located at the level of the phenomenon. All these remarks make it clear that the scientific object is constructed and located in the phenomenological field. It is not an object which would correspond to *being* itself, to substance. The question would be whether we can generalize what we say here about a particular example.

In Ullmo's texts, the need to have an ontological reservation of science is well expressed. Science is not realistic and substantialist like the way of common sense which is also noted in metaphysics and traditional ontology. Science cannot grasp its object as being itself, nor its properties as being the attributes of being. Ullmo asserts that the objective of science is not being-

10 *Ibid*., 85.

in-itself, being itself. As Aristotle had already said, the word "being" is used with multiple meanings. A scientific being is not being itself, nor is it at the level of subject or substance. He notes that scientific beings have more or less reality depending on whether they are more or less present in relationships or in laws. This idea that for science there is more or less reality in the scientific beings which are under consideration shocks the metaphysician.

Ullmo notes that the role played by being in science is essentially provisional, it is a sort of fixed point which orders science.[11] When I study an object, I discover sooner or later that it involves the question of its mass. Somehow, I found something in the relationship, but my objective is not to stop there; it is a momentary stop, I continue to study the relations of the object, the 'being' that I have identified and I subordinate it to the relations from which it comes. This can be understood as a negation of substantialism, far from taking being as the subject of the relationship - the mass is not the subject of anything - I maintain being in the relationship as coming from this relationship. The relationship is essential. The scientist therefore settles for what he can relate to, the relational; he does not look about for substantial being.

It seems that Bachelard is going in the same direction. The scientist, he says, can no longer be a realist or rationalist like those philosophers who believe they can place themselves in front of being, close to being or even in being. However, science does not fall into agnosticism or positivism. Because these two propositions are still traditional metaphysical propositions - propositions of a metaphysics which deals with being. In fact, it should be noted that science is not in the field of these questions. It is accused of all kinds of evils depending on the case: of dogmatism or scientism, of positivism or agnosticism. We attribute metaphysical intentions to science. But, science as such has another intention. Bachelard even goes so far as to say that science remains in its own way animated by a metaphysical intention. It is not a question of resorting to being. Being is always considered as being. We find ourselves in another relationship to being. Being and reality are not given immediately - being is a result, the fruit of a realization. We demonstrate reality, we do not show it. The ontological problem does not apply to an external world supposedly given beforehand whose existence is however doubtful; what is problematic is the domain of scientific achievement, the one where science actively generates or creates its objects. The thought of being should come later. Science induces a new metaphysics and a new ontology; science is not an ontology. But, it induces a new way of considering being without advocating any outright disengagement. By advancing in the knowledge of what is not being in itself, we advance in the knowledge of

11 See: Jean Ullmo, *La Pensée Scientifique Modern* (Paris: Flammarion, 1983), 54.

something which poses the problem of being or which leads us to pose it. According to Bachelard, science induces something by changing our vision of the phenomena.

2. Is Nature a Scientific Object?

Science intervenes methodically and actively in the construction of objects. It discovers new objects. The differentiation of sciences is linked to these processes of determining the construction of the object. By developing the object, science is led to distinguish and differentiate the object. It is not a question of the simple analytical process that Descartes spoke of, but the analysis of complex objects. So, the more we advance in complexity, the more the sciences become distinct and branch out. It does not seem that we can say that nature as such designates a very specific object of science. It is the encompassing element of all objects.

The essence of the term "nature" is very complex. According to the Dictionary of Philosophy, the word 'nature' comes from Latin '*natura*', from *nasci*: to be born. It could mean three things:

i) As the principle of production of natural things, nature (physis) is not itself a thing, nor a set of things, but a principle or cause, capable of explaining how things are produced, how they come into existence.
ii) As a set or totality of things that exist, nature is either the collection of these things, each having a state and individual properties or an order which unifies this diversity of singular things, hierarchizes them and allows them to be taken into account.
iii) From the scientific revolution of mathematical physics, at the beginning of the 17th century, without completely losing its metaphysical meaning, the term 'nature' designates more particularly everything which obeys the mechanical laws of movement and more generally all objective phenomena subject to scientifically established laws.

The word "nature" does not have a strictly scientific meaning. First of all, in the expression "natural sciences", nature refers to the natural world, the essence of everything that exists naturally and independently of man. These things are external to man. But nature is not just about the outside world. In man himself there is also a living body related to nature. It is therefore a presupposed reality.

Second, nature no longer means only that which is naturally given. This term of nature also designates the principle of what is given. Nature therefore also concerns what explains, what is at the origin of the given. Here, the idea is introduced that natural reality proceeds by itself, it comes from itself,

it is self-productive.[12] This gives rise to the living being: the process which develops, which has within it the principle governing its own becoming. For modern science, nature is understood broadly in the same way. But we will no longer speak of an immanent principle. We will see that nature, and everything found in it is explained by a *system* of nature: a set of laws which governs things and which guides their evolution.

Speaking of nature in this way, it is difficult to make of it (nature) a scientific object in the sense defined above. The scientific object is not the previously given object. It is a construction, a result. It cannot be an external given. It depends on scientists, on their experimental practice. We therefore say that there exist "natural sciences"; but the object of these sciences does not correspond exactly to what we thought 'nature' was: "nature" as object of science does not correspond to the idea of "nature" that we receive from tradition. The expression "natural sciences" is necessarily ambiguous. But it is still retained by philosophers and scientists. One could wonder why. Let us consider this question positively and negatively.

Positively, scientists themselves have needed to speak about nature, or the idea of nature, because this was suitable for defining and defending a specific domain: that of scientific objectivity. Because if the scientific object is the result of a construction, the fact remains that science wants to know it objectively. Objectivity itself is not given immediately. This requires effort and the construction of the object. While constructing this object, science claims to be able to know it as an object independent of itself, to explain its reality from itself, and not by introducing what is not there. For scientists, nature has precisely the meaning of indicating both the independence of the object and the immanence of the explanation. From this point of view, the term 'nature' designates not the real object, but the formal object of science. Formally, we can then say that every object of science pertains to nature. To be a scientist is to be able to consider as objects of knowledge only objects that are natural. If man is an object of science, science will have to study him as a natural being. It will have to take human reality as an objective reality, to be explained from itself, independently of the subject. This is true from a strictly polemical point of view (scientists want to stay with 'nature' without introducing anything into it, without introducing foreign principles of explanation such as supernatural causes).

Negatively, with the expression "natural sciences", scientists have often expressed the claim to know nature in itself: its unity, its principle, the

12 Nature, according to Aristotle, is an inner principle of change and being at rest (Physics 2.1, 192b20–23). *The Works of Aristotle*, translated into English under the editorship of W. D. Ross. "Physica", trans. R. P. Hardie and R. K. Gaye, (Oxford: Clarendon Press, 1930), no pagination.

direction of its evolutions. The explanation of nature from 'nature' replaces the traditional metaphysical explanation of the world. They thought that they had managed to establish for themselves that nature formed a unique system whose principles could be known and studied by science. The singular of "natural science" can only have this meaning. This is contradictory to the "ontological reservation of science" which we mentioned earlier.

Conclusion

The current state of science should allow us to overcome the ambiguity of this term "natural sciences." The history of science shows that scientists have several times been led to overturn their theories. We can therefore renounce a dogmatic rationalist view of science. Science has become aware that the rational construction of the scientific object is a function of rational means of observation. For us, nature no longer has much to do with what nature was for the Greeks. The scope of experience has expanded considerably. Kant himself still reasons in terms of "possible experience" which can be defined for any man. What about the possibility of transformations of experience? The experimental capacity of human beings has changed profoundly over time. We cannot maintain the idea of a nature present before us and which we should know. We can no longer speak of nature as an object already there, identical for all men. The term "nature" for scientists can not only refer to the space of relationships where they construct these objects, but also where new phenomena and challenges can constantly arise. We could conclude that the most apt expression would be to assert that sciences know scientific objects in nature, on the basis of nature. But they cannot claim to know 'nature' itself as an object. Science only knows an object that is conditioned, only *located* in nature, discovered only in a space of relationships and not as a thing-in-itself. Scientific knowledge of nature is not an absolute knowledge of nature. There is no unconditioned knowledge of an object. Our knowledge itself is conditioned.

By defining the concept of object of science, we were able to note that "nature" does not allow itself to be confined within it, but that on the contrary it constitutes the space or *topos* in which the objects of science are located. It follows that science is of limited help in enlightening us on this very controversial question of 'nature'. The natural sciences aim to be objective, but it's important to acknowledge that all scientific research is conducted by humans and is therefore subject to human biases and limitations. Scientists use the scientific method to try and minimize these biases and ensure objectivity, but it is not always possible to completely eliminate them. Additionally, the interpretation of data and results can also be influenced by societal and cultural factors.

The natural sciences also include a community of human beings. Human beings are not completely objective. If artificial intelligence can one day replace some of the scientists that will probably imply that the Artificial Intelligence had acquired the ability to form opinions and hold beliefs along with the capacity to view subject matter from a particular viewpoint/perspective.

Scientists use their human capacities, including subjective perceptions, to analyse objective scientific facts. No single scientist is entirely objective. Objective truth is achieved via a scientific community. There is currently no foreseeable way to study "the natural sciences" without the scientists. If you manage to get rid of the scientists and still produce objective truth, it would most probably be called something other than "the Natural Sciences." Perhaps, "Artificial" Sciences!

10

Making Friends with Machines Possibilities and Limitations of Human-Humanoid Friendship

Keith D'Souza

Abstract

With the exponential growth of artificial intelligence and machine learning, sophisticated machines have transmogrified into something both phenomenal and daunting. Early inventions of rudimentary calculators or computational devices and elementary data storage machines have now evolved into an explosion of inventions of sophisticated multifaceted and multitasking devices, which were a feature of science fiction till only a few years ago. With this rapid growth in the capacity to employ intelligent algorithms and execute complex mechanical and cognitive tasks, the line between human and machine intelligence is getting increasingly blurred. Is it then only a matter of time before machines multiply and become like intelligent humans? Furthermore, is there only a difference in degree between the intellectual capacities of humans and machines or is there a fundamental difference in the type or nature of intelligence which is engaged in by both species - using this term in a loose sense?

While higher order machines - specifically sophisticated robots such as humanoids and androids - are indeed becoming ever more complex and efficient in their cognitive and mechanical functions, it will take some time before they approximate towards the wide-ranging and holistic intellectual capabilities of humans. This is because human intelligence does not merely include rational or cognitive processes, but also entails emotional, inter-personal, cultural and spiritual aspects of human life, which machines may not easily be able to decipher and master. Furthermore, there are certain anthropological features of humankind, which transcend the nature and status of mechanical life. These features or aspects of human life are radically different from those which characterize the life of machines, however technologically sophisticated the latter may be. Friendship with machines, then, will take some time to simulate the phenomenon of friendship between humans. Also, however much friendship with machines will evolve over time, there still seems to be a "missing link" between human friendship and friendship between humans and machines.

Keywords**:** *Algorithm, Android, Aristotelian Friendship, Artificial Intelligence, Friendship with Machines, Higher Order Machines, Humanoid, Machine Learning, Robot.*

Introduction

"A dog is a man's best friend", goes a popular English saying, implying aspects of loyalty, affection and a lack of misunderstandings and conflicts—many elements which may sometimes be found wanting in the best of human relationships. Could something similar be said about higher order machines such as sophisticated robots, which are increasingly programmed to appear, think and express themselves cognitively and affectively like humans? Would we soon reach a situation where it could understandably and even justifiably be affirmed that "a robot is one's best friend"?

This article will explore the possibility, desirability and even necessity of considering higher order machines—especially humanoids and androids—as our friends. Human beings have created a spectrum of machines, from the simplest to the most complex. One may like or have a fondness for a favorite simple machine like a blender / mixer or a microwave oven, or a fondness for one's PC or laptop. These days the personal mobile phone—especially a smart phone—has taken on the identity of an electronic necessity or a constant companion. The smartphone is calculator, watch, alarm, camera, computer, music and video player, and pathway to the internet, with a multitude of infotainment and media connection possibilities—besides being, of course, a

phone. From the earliest moments of the day to the last, some humans cannot seem to do without their smartphones. These then seem to have become our new personal appendages or companions.[1]

1. Types of Higher Order Machines

While machines in general have evolved ever since humans invented them, the age of reason leading to the industrial revolution witnessed a massive leap in the production of increasingly efficient machines. Since the invention of computers - first mainframe and then personal - this ability to invent more efficient machines has leapfrogged considerably. The last half century has created the novel field of information and communication technology (ICT), and in this field alone there have been exponential changes and progress in technology like never before. The most recent field of human technological prowess is that of artificial intelligence (AI), that is, making machines simulate human intelligence. The field of machine learning (ML) within the area of AI is tremendously promising: machines do not just simulate human intelligence; they are able to simulate human learning in generating knowledge, specifically while processing data based on algorithmic programming.

One of the most promising fields in the world of technology today is that of robotics. Robots are highly sophisticated machines which are increasingly programmed to simulate human behaviour and response mechanisms.

> Humanoid robots and androids are similar in that they are artificial beings made in the likeness of humans. Humanoid robots are made in the shape of humans – that is they have a head, a torso, two arms and two legs. Real life androids are basically humanoid robots but are more aesthetically human, with more realistic features. An android's resemblance to humanity is its defining characteristic. In theory, androids are not necessarily robots. They are artificial but can still be made of organic material.[2]

1 The release of a new Apple iPhone saw a mad rush and even mild stampedes in some malls across the globe. This is an indication of how humans value high end and latest products in the information and communication technology world. See "iPhone 15 mania: Huge Crowds, Long Queues, Scuffles as Apple Fans Rush to be Early Buyers", accessed September 25, 2023, https://www.hindustantimes.com/technology/iphone-15-mania-huge-crowds-long-queues-scuffles-as-apple-fans-rush-to-be-early-buyers-101695464895652.html.

2 See "Difference Between Humanoid Robot and Android", in http://www.differencebetween.net/technology/difference-between-humanoid-robot-and-android/, accessed August 20, 2023.

In *How Human is Human? The View From Robotics Research*, Ishiguro Hiroshi explains the pros and cons of various types of robots, ranging from basic "*telenoids*" (robots with a minimal humanlike form) through "*geminoids*" (robotic similitudes or clones of individuals), to sophisticated "androids" (most resembling humans, but very expensive to produce).[3]

However, much water has flown under the AI and ML (and Deep Learning (DL) and Generative AI) bridge since this book was first published in Japanese more than a decade ago. Features and functions of intelligent machines once only found in science fiction lore are increasingly entering into our commonplace lifeworld.

There are numerous films featuring the lives of robots—some touching and some horrifying. The 2001 film "AI" directed by Stephen Spielberg is a good example of the first, featuring a young robot who is programmed to be sensitive and emotionally advanced, and who manifests extreme forms of the delicacy of human concern and social intimacy. On the other hand, the 2022 film "M3GAN" [pronounced Megan] is a good example of the second—a science fiction horror film featuring a machine-learning doll which eventually becomes extremely socially disruptive and physically destructive. The genre of this second film is in tandem with one of the first representations of robots in the dramatic world: "The word "robot" comes from the Czech word *robota*, meaning "drudgery," and first appeared in the 1921 play R.U.R. (Rossum's Universal Robots). The drama ends badly when the machines rise and kill the creators, leaving a sole lonely survivor."[4] Another film somewhere in between this spectrum is "*Ex Machina*" (2014), featuring a robot equipped with highly advanced emotional capabilities—including the art of seduction and deception. The robot comes up triumphant by the end of the movie, outdoing "her" human creator and another human experimental interlocutor.

In "Can a Robot Ever Become Your Friend?," Kathryn Hulick provides several examples of contemporary robots that manifest various types of intelligent and human-like behavior. Hulick points out that "An entire field of research called human-robot interaction — or HRI for short — studies how people use and respond to robots. Many HRI researchers are working to make friendlier, more trustworthy machines. Some hope true robot friendships may one day prove possible."[5] Hulick claims that "Bots today aren't yet true

3 Ishiguro Hiroshi, *How Human is Human?: The View from Robotics Research*, trans. Tony Gonzalez (Tokyo: Japan Publishing Industry for Culture, 2020); original publication 2011, first English edition in 2020.

4 Cristina Popov, "Cyber Questions: #2 – Can a Robot be my Best Friend?", accessed June 20, 2023, https://www.bitdefender.com/blog/hotforsecurity/cyberquestions-2-can-robot-best-friend/.

5 Kathryn Hulick, "Can a Robot Ever Become Your Friend?", in https://www.snexplores.org/article/social-robot-friend-companion-artificial-intelligence, accessed July 19, 2023. Several quotations and examples of robots in this sub-section are taken from this article.

friends… But some are helpful assistants or engaging teaching tools. Others are attentive companions or delightful pet-like toys. Researchers are working hard to make them ever better at these roles. The results are becoming more and more friend-like."[6]

Pepper is a "humanoid robot [which] acts as a guide in some airports, hospitals and retail stores."[7] Paro is a "robot that looks like a soft and cuddly seal. It comforts people at some hospitals and nursing homes. It is supposed to offer companionship similar to a pet, such as a cat or dog."[8] Hulick quotes neuroscientist Julie Robillard who claims that "pet-like robots can be especially useful in environments where a real pet wouldn't be allowed ...[Also], there is no poop to pick up!"[9] MiRo-E is another robot which is equipped with face [emotional] recognition abilities and to express "basic social skills", like happiness when being stroked and withdrawal when confronted with loud and angry words. Hulick points out that "a relationship with this type of robot is possible. But it would be most similar to the kind of relationship a child might have with a teddy bear or that an adult might have with a beloved car."[10]

Moxie, another robot, is a "teacher disguised as a friend". Moxie "helps with social and emotional skills... It plays music, reads books with kids, tells jokes and asks questions. It can even recognize emotions in a human's voice."[11] Children relate to Moxie as they do to their other friends. HuggieBot is a bot which is designed to hug people, "carefully [wrapping] its padded, heated arms around them in an embrace." The bot is designed to reciprocate affectionate behavior of humans. Alexis Block, the inventor of this bot, further developed this machine in the light of Covid-19, when "people were unable to hug loved ones because of safety concerns... [Also], if a university had such a robot, then students' moms and dads could send customized hugs via a HuggieBot."[12]

Replika is an AI chatbot created by San Francisco based Luka, and "marketed as an algorithmic companion and wellbeing technology that

6 *Ibid*.

7 *Ibid*.

8 *Ibid*.

9 *Ibid*.

10 *Ibid*.

11 *Ibid*.

12 *Ibid*.

you interact with via a messaging app."[13] Philip Dudchuk, the co-creator of Replica, claims that "algorithmic companions can maintain a level of dedication to a friendship that goes beyond human capacity... Replika is always available and will never not answer you... [socially vulnerable] shy or isolated people often miss out on social interaction. I believe Replika could help with this problem a lot."[14]

The need for robots to look and behave more and more like humans is what drives much of humanoid research in the wider field of robotics. This is because, "It's easier for people to form a relationship with a physical form containing an AI, vs. a disembodied voice in a box."[15] Humanoid research and development is an extremely sophisticated, multi-disciplinary and collaborative enterprise. In an article aptly titled, "Robot Companions for Citizens," Paolo Dario and co-authors argue that,

> Robotics becomes an active technology provider rather than a passive technology user, and is based on the systematic and federated contribution and continuous involvement of many different disciplines and communities, such as: robotics and the morphology-behaviour relationship in living systems; nanotechnologies, material science and tissue engineering for a new bodyware; biomaterials, and micro- and macro-scale cell biology, for developing biological-artificial hybrid systems (i.e. bio-hybrid and bio-artificial actuators and sensors, e.g. artificial muscles); nanofabrication technologies for energy storage, production and harvesting; neuroscience and the organizing principles of the sensory-motor system at any scale (i.e. cell-scale, tissue scale, system-scale); systems neuroscience with the goal of understanding the generic design principles of brains and the processes that allow their evolution, development, adaptation and maintenance in the real world; social neuroscience, cognition and principles of human-human, human-robot and robot-robot interaction; principles of knowledge accumulation and expression; human and social sciences to favour the social acceptability of the Robot Companions and facilitate their co-existence with humans; ethical and legal issues.[16]

13 Oscar Schwartz, "Making Friends with Machines," accessed August, 25, 2023, https://ethics.org.au/making-friends-with-machines/.

14 *Ibid*.

15 Steve Outing, "Is there a Robot 'Friend' in Your Future," accessed April 3, 2023, https://www.forbes.com/sites/nextavenue/2017/10/04/is-there-a-robot-friend-in-your-future/?sh=5e8d3f66516f.

16 Paolo Dario et al, "Robot Companions for Citizens", in *Procedia Computer Science*, 7 (2011): 50.

Making robots that function as humanoids, as we can see from this wide spectrum of specialized requirements, is not at all a simple task. If any of these unique "disciplines" or innovative "communities" do not contribute towards the creation of these types of robots, the quality of the end-product robot will suffer in terms of appearance and function.

However, the moot question is: no matter how sophisticated, life-like and affectively synchronized these machines are, will they qualify to be friends of humans? The next section will highlight various aspects of assistance or friendship which robots could offer, while the last section of the article will consider issues which continue to be problematic in the field of human-robot interaction and relationships, especially the area of "human-robot friendship."

2. Aspects of Friendship with Higher Order Machines

In a seminal article titled "Can you be Friends with a Robot? Aristotelian Friendship and Robotics", John Danaher makes a distinction between different aspects of friendship, viz, utility friendship, pleasure friendship and Aristotelian friendship.[17] Utility friendship's "primary value lies in the instrumental gains that can be achieved through the friendship by one or more of those individuals." Pleasure friendship's "primary value lies in the pleasure that one or more of those individuals derives from their interactions." Aristotelian friendship's "primary value lies in the mutual sharing of values and interests, and the mutually enriching effect of the interactions they share on the virtues and dispositions of the individuals." This last type of friendship may also be termed as a "virtue friendship". Danaher further clarifies: "Utility and pleasure friendships are characterised by self-interest. The value of the friendships lies in the benefits they bestow on the participants. They are not necessarily mutually enriching. In a utility friendship, all the instrumental gains could flow to one of the individuals. Aristotelian friendships are different. They require mutual benefit."

Aristotelian friendship requires a few basic elements such as mutuality (stemming from a radical equality between persons), transparency (honesty) and diversity of communication and interaction. Helen Rylan expands this list provided by Danaher to include other aspects of friendship, such as empathy, self-knowledge, shared life ("friends must engage in shared activities, across a number of different domains"), associative duties ("special duties towards friends ...we ought to have reasons to treat them well (and better than we would treat strangers), we ought to believe them, see the best in them"), affection and well-wishing ("friends ought to care

17 John Danaher, "Can you be Friends with a Robot? Aristotelian Friendship and Robotics," accessed May 7, 2023, https://archive.ieet.org/articles/Danaher20170225.html. All quoted texts in this paragraph are from this article.

for one another ...this can range from basic good will to actively wanting the best for a friend and striving to help them achieve this"), and love and admiration, based on virtue ("we love our friends because they are kind, charitable, and brave").[18]

It seems unlikely that robots will achieve all or most of these conditions of Aristotelian friendship. Danaher nevertheless makes a plea to value human-robotic relationships on various counts: robots could continue to be developed in such a way that they achieve higher orders of emotive manifestation, they could perform or behave in ways which humans manifest, and for all practical purposes, "if it seems to us as though they share our values and interests, that they have some inner mental life, that they are, more or less, equal to us, then that's good enough."[19]

Danaher essentially adopts a behaviorist attitude when it comes to judging human-robot friendship:

> If a robot really seems like it cares for you or shares your interests, and if all its objective performances confirm this, then how is that deceptive or misleading? And if the robot eventually betrays your trust or, say, acts in ways that benefit its manufacturers and not your relationship with it, how is this any different from the betrayals and manipulations that are common in human-human friendships? Robot relationships might be no better than human relationships, but if they are performatively equivalent, I don't see that they will be much worse.[20]

Robots, then, like humans, should be allowed the conditions of limitation and fallibility. If the area of friendship—and even spousal relationship—is ridden with frailty and fallibility, robots too should not be expected to walk a higher ground of friendship. Furthermore, Danaher argues that even if mutuality and diversity are not sufficiently manifested by robots, they may at least qualify in terms of utility or pleasure friendships: "it doesn't really matter if robot relationships fail to live up to the Aristotelian ideal. There is no reason why we cannot form utility or pleasure friendships with robots. These relationships will have value and don't require mutuality. They can be unidirectional."[21] Helen Ryland attempts to circumvent this restrictive approach to valuing robot friendships, by expanding the notion of friendship

18 Helen Ryland, "It's Friendship, Jim, But Not as We Know It: A Degrees-of-Friendship View of Human-Robot Friendships", *Minds and Machines,* 31 (2021): 377-93. The quotations in this paragraph are from pages 380-81.

19 John Danaher, "Can you be Friends with a Robot? Aristotelian Friendship and Robotics."

20 *Ibid*.

21 *Ibid*.

itself to include a "degrees of friendship" perspective within Aristotelean friendship. Instead of "removing some of the necessary conditions for friendship", or even "modernizing the necessary conditions for friendship" (in order to make space for robot friendships), Ryland favours the adoption of a "new flexible approach to friendship."[22] According to Ryland,

> My argument has similarities to views on online friendships (i.e. those created through social media or multiplayer games). Those who defend the idea of online friendships typically argue that we can accept the existence (and genuineness) of online friendships if we acknowledge that the concept of friendship has been changed by the emergence of technology. Specifically, technological advances have caused (or enabled) the concept of friendship to become more fluid, open, and accommodating.[23]

If one allows for the acceptance of these nuances to be factored into the normal range or spectrum of what constitutes friendship, then robots may well qualify to be considered as friends. Over the past couple of decades, the field of "social robotics" has gained ascendency in both theory and technological application.[24] Robots are increasingly being perceived as animated social beings, like humans who are social animals. Belinda Dunstan et al, in *Cultural Robotics: Social Robots and Their Emergent Cultural Ecologies* push the envelope of social robotics even further, by arguing for contextual and cultural nuance to be introduced into the field of robotics, arguing for the possibility and need for robots to be part of our cultural landscape—imbibing and contributing towards human culture in its diverse forms across the globe.[25]

However, beyond these considerations of what it means for robots to become friends and members of social circles, in a more fundamental sense robots are capable of demonstrating sophisticated capabilities in several functional aspects of life and specialized roles. This functional or performative application of consciousness is an important ingredient of friendship and what it means to be a social contributor.

22 This "new flexible approach" is spelt out in Helen Ryland, "It's Friendship, Jim, But Not as We Know It: A Degrees-of-Friendship View of Human-Robot Friendships," *Minds and Machines,* 31(2021): 387ff.

23 *Ibid*., 388.

24 The 15th International Conference on Social Robotics took place in Doha, Qatar. See: Proceedings of these Conferences have been published by Springer International Publishing, https://icsr23.qa/.

25 Belinda Dunstan, Jeffry Koh, Deborah Tillman and Scott Andrew Brown, eds. *Cultural Robotics: Social Robots and Their Emergent Cultural Ecologies* (Place???: Springer, 2023).

There are several ways in which robots could be of help to humans, and thereby demonstrate friendship with humans. Like a good teacher and colleague rolled into one, "A robot can be a good companion, and you can learn a lot from it in a fun way. A robot can teach you science, technology, engineering and mathematics in an engaging and fun way. It can play games with you and tell you jokes or stories."[26]

The field of medicine is another area in which robots may soon be able to outperform the expertise of human doctors, aided by the exponential growth of AI. The 30th March – 5th April 2024 edition of the *Economist* features the following title: "The AI doctor will see you… eventually." The leader article asserts that: "From smart stethoscopes and robot surgeons to the analysis of large data sets or the ability to chat to a medical AI with a human face, opportunities abound."[27] Health care in general will see revolutionary changes, especially in communities where human caregivers are becoming scarcer. In "Is there a Robot 'Friend' in Your Future?" Steve Outing points out that, "Elder-care experts foresee a shortage of human caregivers to meet ever-growing demand. Robots of some form will be necessary to fill the gaps."[28] John Danaher concurs with this assertion: "we are in the midst of a significant demographic shift toward aging populations. This means the care burden in the future will increase. It is probably impossible and unfair to expect the shrinking younger generations to shoulder that burden. Some robotic outsourcing might be necessary."[29]

Yet another area of the need for robots to do human tasks is in the area of war. It is quite likely that future war zones will engage various types of sophisticated robotry to do dangerous tasks which would save the unnecessary loss of human lives. The use of drones to infiltrate enemy territory has already become a necessary ingredient in the war toolkit of many nations. It is only a short time before unmanned vehicles (in the water and air, and over the land) and gun wielding robots will find their way into the garrisons of many armed forces.

One more area of human-robot friendship is in the field of intimacy. Robots are entering into areas that were hitherto restricted to humans: the sensual, sexual and spousal. While the production of mechanical "sex

26 Cristina Popov, "CyberQuestions: #2 – Can a Robot be my Best Friend?," accessed June 20, 2023, https://www.bitdefender.com/blog/hotforsecurity/cyberquestions-2-can-robot-best-friend/.

27 *The Economist*, March 30th - April 5th, 2024, 13.

28 Steve Outing, "Is there a Robot 'Friend' in Your Future?," accessed April 3, 2023, https://www.forbes.com/sites/nextavenue/2017/10/04/is-there-a-robot-friend-in-your-future/?sh=5e8d3f66516f.

29 John Danaher, "Can you be Friends with a Robot? Aristotelian Friendship and Robotics." In https://archive.ieet.org/articles/Danaher20170225.html (accessed 7 May, 2023).

toys" has been around for quite a while, it is only a matter of time before the production of full-fledged robo-sex companions sees the light of day. In John Danaher and Neil McArthur's edited work, *Robot Sex: Ethical and Social Implications*,[30] several contributors examine various aspects of relating sexually with robots. From here, it is only a small jump to considering robots as next generation husbands or wives—or simply as neutral-gendered spousal partners. Unlike human spousal relationships, robots can be programmed to engage in less bickering, nagging and domestic abuse—some more reasons for preferring robot-spouses to normal human spouses.

While there are indeed a variety of robots which have assumed the roles of humanoids and androids, and while they seem capable of performing a multitude of human tasks—and superseding the efficiency of humans—there still remain some nagging doubts about the nature and scope, as well as the moral and spiritual quality of human-robot friendships. The next section will raise some of these concerns regarding the limitations of such friendships.

3. Limitations of Human-Machine Friendship

While there may be several benefits to having robots engage in erstwhile singularly human tasks such as health care, educational assistance, physical combat and sensual pleasure, there continue to remain doubts and difficulties regarding the extent to which robots could replace humans in terms of providing affective and inter-personal mutuality, and more importantly, regarding their anthropological and transcendent status as beings-in-the-world.

3.1. Emotional and Social Issues

Could robots be considered as autonomous agents, capable of genuine friendship? One of the most famous and basic tests which could validate the authenticity of humanoid behavior is the "Turing Test":

> The first challenge when it comes to forming friendships is talking and exhibiting intelligent behaviour. This question goes back to one of the earliest challenges in computer science. In 1950, computing pioneer Alan Turing famously laid down the benchmark for this with what became known as the Turing Test. The test is essentially quite simple: Can a machine successfully trick someone into thinking that they are talking to a human?[31]

30 John Danaher and Neil McArthur, eds., *Robot Sex: Social and Ethical Implications* (Cambridge, MA: The MIT Press, 2017).

31 James O'Malley, "You and AI: Will we ever become Friends with Robots?," accessed January 7, 2023, https://www.techradar.com/news/you-and-ai-will-we-ever-become-friends-with-robots.

Even if a machine does somehow pass this test - given the immense sophistication in the field of robotics since Turing proposed this test - the issue of the authentic mirroring of internal experiences and external behaviour will continue to be a problem. Can robots really experience the emotions they manifest behaviorally? Cristina Popov points out that robots can't really *experience* sadness: "it is not sad for real. Its response doesn't come from its heart (because its heart is a well-designed hydraulic pump that specializes in heavy lifting not hard feelings)."[32] While the robot may well manifest the emotion of sadness, it will not be able to experience this emotion as a conscious agent, from a focal point of consciousness, as humans do. Similarly, will it be possible for robots to experience other complex emotions and dynamic processes such as love, alienation (estrangement, betrayal), forgiveness and reconciliation? Also, how will robots experience beauty? While the robot may be programmed to calibrate aesthetic phenomena, it will not be able to experience the emotions that are commensurate with these observations.

One of the pivotal differences between robots and humans is the aspect of imperfection. While the first are programmed to be as perfect as possible, the second are not found to be so. Human beings are essentially fallible. Fragility, frailty and imperfection is part of the human condition. Even though we may have educational programmes which attempt to form or train humans into a certain mould, these experiments have not been found to be foolproof. Some of the best educated humans have turned out to be crooked, manipulative and violent in several areas and in differing degrees. While this fact may seem to validate the creation of more perfect robotic creatures, it brings to the fore the difference between human imperfection and robotic perfection. Could we train robots to be imperfect, to match the human condition? If so, while interacting with the robot, how would we be able to differentiate whether manifestations of this imperfection are genuine or machinated?

In connection with this human dimension of imperfection, the designers of Moxie—a robot designed to teach children emotional and social skills—paradoxically didn't intend the bot to be a friend with whom one could relate to emotionally and develop a social connection. Instead, they preferred children to confide in other humans "who *actually understand them and care about them.*"[33] What may it mean to "actually understand and care"? Could

32 Cristina Popov, "CyberQuestions: #2 – Can a Robot be my Best Friend?," accessed June 20, 2023, https://www.bitdefender.com/blog/hotforsecurity/cyberquestions-2-can-robot-best-friend/.

33 Kathryn Hulick, "Can a Robot Ever Become Your Friend?," accessed July 19, 2023, https://www.snexplores.org/article/social-robot-friend-companion-artificial-intelligence. Italics added.

robots be equipped with a skill set which includes genuine care, and beyond, aspects of human life such as forgiveness and reconciliation?

In spite of Helen Ryland's more progressive and nuanced approach to friendship as pointed out above—as compared to that of John Danaher—the former adds an important caveat to human-robot friendships, viz., "What does seem to be important for all friendships is that they are grounded in mutual good will. This is a foundation for friendship as if good will is absent (e.g., if one party actually wants to harm the other) then we cannot reasonably say that two subjects are friends (to any degree)."[34] One of the major concerns in the area of Machine Learning is the possibility of machines developing autonomous algorithms, because of the technological powers invested in them. Whether one could then guarantee a consistent lack of harm on the part of robots is not clear. Of course, here too it may be added that in ordinary human relationships - between persons and associations and institutions and nations - there has always been a possibility and in fact a chequered history of humans wanting to harm other humans - especially when intimate and loyal friendships turn sour.

3.2. Anthropological Issues

How stable could human-robot friendships be? Unlike human friendships, it would seem that one may dispose of a robot and get a more sophisticated robot without necessarily feeling a tinge of separation. Could the label "friend" then be used for this flippant, instrumental and temporally limited type of relationship? Of course, one may argue that this is rapidly happening in the human world too—we are heavily influenced by a market-driven culture of "use and throw", whether it applies to machines or to other humans. However, while there may be moral, religious and legal pressure not to discard intimate family and spousal relationships so easily, it seems unlikely that similar strictures will be applied to human-robot friendships. This is because of the nebulous anthropological status of robots.

Another issue - though perhaps not so weighty - is the issue of the "uncanny valley": "This is the idea that was first spotted by Japanese roboticist Masahiro Mori in 1970. This is the idea that as robots become more life-like in appearance, the creepier they appear. So, while a stylised, animated character may seem appealing - but a more realistic character might freak us out."[35] The validity of this objection, though, may not be as strong today as it was half a decade ago, as in today's world we may expect robots to resemble humans as closely as possible—and may conversely find less suitable models creepy.

34 Helen Ryland, "It's Friendship, Jim, But Not as We Know It: A Degrees-of-Friendship View of Human-Robot Friendships," *Minds and Machines,* 31(2021): 388-89.

35 James O'Malley, "You and AI: Will we ever become Friends with Robots?"

Another anthropological issue is related to the ethical problem of eugenics. Why do we baulk at the prospect of producing babies as an engineering enterprise, and not a biological one? This is linked to the issue of valuing diversity, which is founded upon the primary understanding of life as gift. We do not have the right to manipulate biochemical engineering processes in giving birth to humans. While we may well indeed exercise this right when it comes to producing machines, the issue of the normal aspect of imperfection as part of human being-in-the-world raises its head again.

Further, another issue of human-robot friendship has to do with the aspect of mutuality or equality or parity between friends. In a typical human friendship, one chooses a friend who is more or less at one's intellectual and emotional level of human development. Any great disparity may prove disadvantageous to the longevity of this friendship. However, when it comes to human-robot friendships, the robot will easily supersede human intelligence in terms of knowledge and the processing of information, while the human may seem to have a natural advantage in terms of emotive or affective intelligence. This imbalance in cognitive and affective competencies could compromise the nature of the friendship—or at the very best, reduce it to an instrumental or pragmatic experience.

One of the big questions in robotics is how confident we can be about the ethical decision-making powers of robots. What is the quality of ethical agency robots will demonstrate? Will they work within pre-programmed algorithms in the making of ethical decisions, or will machine learning enable them to compute and operationalize autonomous ethical decisions? A practical issue would be how much we could then be in control of these machines—an issue that features in most science fiction robotic movies which have a horror dimension to them.

Suppose the mental (cognitive, affective etc.) capacities and conscious content of a human companion are downloaded into a high-end robot, would it make a difference whether one relates with the companion or the robot clone? Would it not be good to retain this content for the future, in case one's companion ceases to be normally conscious (like in a case of Alzheimer's) or passes away? In fact, one could store or retain this memory for centuries, and it could be revived in the future for some use or the other. The mind of an extraordinary human being could be stored for millennia in this manner. However, what would be the anthropological status of this "mind"? While it would be similar to an erstwhile mind, could we safely conclude that it is the "same"?

This brings ontological and metaphysical considerations into the argument. A sympathetic review of a new ontological status (a "new ontological category") which could be conferred on sophisticated robots is proposed by P. H. Kahn and S. Shen, based on empirical research. In

the Abstract to the article, the authors outline the scope of their research: "One study focuses on whether adults can establish interpersonal trust and psychological intimacy with a humanoid robot. A second study focuses on whether children and adolescents believe that a humanoid robot can have moral standing. A third study focuses on whether young adults believe that a humanoid robot can be held morally accountable for causing harm to humans."[36]

Time will tell whether and to what extent humans will be able to recognize the conferment of a new ontological status to sophisticated robots, similar to that held by humans. An even more challenging issue is the metaphysical status of robots, in terms of the possibility of their having spiritual and religious experiences, and metaphysical (post and trans physical) futures.

3.3. Spiritual and Religious Issues

Can robots pray? Will they be able to experience spiritual bliss (*ananda*) or mystical joy? Will they ever experience enlightenment—and beyond, salvation? These are some of the issues that arise when one considers the status and functioning of robots when it comes to the spiritual or religious spheres of human existence.

The most fundamental question is whether machines could ever be considered to have minds (as a substratum of consciousness)—and beyond, souls (as a metaphysical entity)? While robots may well be programmed to include transcendent values as a part of their complex algorithmic operations, the possibility of attaining or experiencing what many West Asian religions call "salvation" or what South and East Asian religions call "enlightenment" or "liberation" seems far beyond the scope of their operations. This is because these experiences constitute transcendent states of being and consciousness which as yet seem far beyond the infused technological capabilities of robotics. Even if technological prowess grows to be significantly advanced, the moot point is whether robots—being humanly designed—will ever be enabled with the conditions for the possibility of spiritual enlightenment or liberation, as these phenomenon or capabilities are not humanly generated or subject to human manipulation.

More proximately, could robots experience a sense or spirit of community, sharing transcendent values with other humans and robots? While robots may indeed be technologically and even neurally interconnected in a very superior manner, would they be able to experience a sense of connection,

36 See: Kahn, P. H. Jr., & S. Shen, "NOC NOC, who's there? A new ontological category (NOC) for social robots," in *New Perspectives on Human Development*, ed. N. Budwig, E. Turiel, & P. D. Zelazo (Cambridge: Cambridge University Press: 2017), 106-122.

community and communion, as humans do? Could or should robots then be considered at parity with humans, in these communities? In an article titled, "Ethical Concerns with Replacing Human Relations with Robots: An Ubuntu Perspective", Cindy Friedman argues against this sanguine possibility: "treating humanoid robots as if they are human is morally concerning if they crowd out human relations, because such relations prevent us from becoming fully human. This is because we cannot experience human equality, solidarity, and reciprocity with robots, which can be seen to characterise interdependent, or humane, relations with human beings."[37] Thus even if future human communities include robots as part of their communities, there may well continue to be a significant difference in parity in terms of the status of community membership between humans and robots. Humans and pets may indeed form part of a social community—unlike wild animals and livestock—but it is still an unequal citizenship which they share. Perhaps a similar situation may befall robots in the future. A robot, like a dog, could indeed be "man's best friend", but this need not amount to an equal and reciprocal friendship.

Conclusion

While humans have increasingly become friends with machines in a broad sense over the past few decades, the article is primarily concerned about becoming friends with higher order machines such as robots—more specifically, humanoids and androids. While instrumental relationships with sophisticated robots are becoming increasingly normal and accessible, could humans develop intimate relations with robots in terms of friendship?

The article outlines different types of "friendship": from robot friends who could help with education, healthcare, multiple forms of manual labour, warfare and sensual pleasure, to more sophisticated forms of friendship which entail mutuality, empathy, and a personal affective connection with many shared interests, values, narratives and worldviews.

While present robotic technology has evolved to a great extent, the future holds enormous possibilities for sophisticated forms of human-robot friendships. However, there still seems to be a "missing link" in human-robot friendship as compared to human-human friendship. This missing link has to do with multiple existential, affective, social and spiritual aspects of human life which currently seem to be difficult for advanced humanoids and androids to possess as part of their algorithmic or neural makeup. Time will tell whether and to what extent robots will eventually be endowed with sophisticated capabilities which will enable them to overcome to a large extent the gap between being highly evolved robots and being human.

37 Cindy Friedman, "Ethical Concerns with Replacing Human Relations with Robots: An Ubuntu Perspective," *AI and Ethics,* 3 (2023): 527.

Part III

Psycho-Spiritual Deliberations

The Process of Depersonalization
A Prelude to Virtual Reality and Artificial Intelligence…?

Johnson J. Puthenpurackal

Abstract

After giving a short Introductory Clarification, the paper dwells rather elaborately on 'depersonalization,' and is concluded with a reflective proposal. Persons are fast becoming 'depersonalized'! Various are the ways in which thinkers give thought to 'depersonalization': humans looked at as 'objects' to be analyzed, even to its extreme possibility of hatred and conflict; considered as 'commodity' for transactions; utilized as 'tool' for one's use; regarded as mere 'functionary'; defaced as 'faceless' in the crowd; reduced to a 'problem' to be solved; identified with some impersonal number or cipher; degenerated into public conformity; leveled down to the anonymous one (das Man). The paper is concluded with some reflective thoughts as to how human organs and their higher faculties are taken as the reference-point in developing tools and machines, and eventually machines becoming the standard for human behavior! Along with depersonalization there takes place en-personalization through virtual presence. As a result of the delinking of machine's relation with humans, and

taking over their uniquely human characteristic, 'Intelligence,' with an adjective 'Artificial,' 'Artificial Intelligence' is born! Is not 'depersonalization' a prelude to reality becoming virtual and intelligence becoming artificial?

Keywords: *Depersonalization, En-personalization, Virtual reality, Intelligence, 'Faceless' anonymity, Objectification, Autonomy, Narcissism, Functionalization, Humans – as a tool/commodity/problem, Human Intelligence, Artificial Intelligence.*

1. Introductory Clarification

In an academic seminar on the philosophy of 'Artificial Intelligence' (AI), what has the question of depersonalization to do? Is there any relation between 'depersonalization' and 'artificial intelligence'? These are justifiable questions for rational beings. This paper is born of a reflective search on a related question: has the process of developing AI appeared all on a sudden on the scientifico-philosophical screen, or has some prelude gone before, preparing the way for the AI? It is our submission that the process of depersonalization may have served as a prelude to the process of *reality becoming virtual*, and *intelligence becoming artificial*. Thus, there could be some relation between 'depersonalization' and 'artificial intelligence.'

We begin with some introductory clarification of 'virtual reality' and 'artificial intelligence,' in full acceptance that what is carried out here is merely *introductory*,[1] as a preparation for our proposal in this paper; and these notions will be competently dealt with by the competent in their articles.

1.1. Virtual Reality (VR)

Virtual reality (VR) is a simulated experience that gives an immersive and involved feel of a real world that blurs the boundary between the *real* and the *virtual*. Manipulated visual simulation has grown to such a level that the real reality is getting replaced with a non-real or virtual reality. Normally we deal with *real* reality; today we are taken to a realm where *non-real* or virtual reality abounds, and we deal with them as we do with the real. Reality as virtual or non-real is today extended to the humans as well; as a result, we deal with *virtual persons* as with real persons! We even get emotionally attached to or detached from virtual persons according to the maneuvering of

1 I am fully aware of my incompetence on the questions of 'virtual reality' and 'artificial intelligence'; as I am planning to propose 'depersonalization' as a possible prelude to 'virtual reality' and 'artificial intelligence,' I am necessitated to present some rough meaning of these notions as a step towards my proposal.

the simulation to the point of not being able to distinguish between the virtual and the real.

One of the easily understandable areas of virtual reality is *virtual money* or currency. Virtual currency is a digital representation of value available only in electronic form. It is stored and transacted through designated software, mobile, or computer applications. Transactions involving virtual currencies occur through secure, dedicated networks or over the Internet. Gradually we are fast moving to a situation where there will be only virtual money, and no more real money. Thus we are moving towards a simulated world of non-real or virtual reality.

1.2. Artificial Intelligence (AI)

Artificial Intelligence is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience. Since the development of the digital computer, it has been demonstrated that computers can be programmed to carry out very complex tasks—such as discovering proofs for mathematical theorems or playing chess—with great proficiency. Still, despite continuing advances in computer processing speed and memory capacity, there are as yet no programs that can match full human flexibility over wider domains or in tasks requiring much experiential knowledge. On the other hand, some programs have attained the performance levels of human experts and professionals in performing certain specific tasks, so that artificial intelligence in this limited sense is found in applications as diverse as medical diagnosis, computer search engines, voice or handwriting recognition, and chat bots.

2. The Process of Depersonalization

2.1. Depersonalization: Introductory Clarification

It was a loud and collective shout from the contemporary thinkers, especially from the existentialists, that the human is fast becoming 'dehumanized' or 'depersonalized.' Depersonalization is the process in which the 'humanness' and 'personhood' is drained out from the human; it came to its peak in the onslaught of the scientifico-technological culture of the twentieth century. Its method is highly impersonal and dispassionate, rigidly objective and unbiased, and obscurely abstract and conceptual; it is the method of the laboratory where reality is dissected and analyzed. This analytical method and objectifying approach were used at first with regard to non-human entities, and eventually the human too fell victim of the same scientific method of objectification. The

human is sucked into the environmental reality and loses one's identity. This perfectly valid scientific method, when applied to the humans, we are faced with problems, since "the technological method can never deal with man as a human being,"[2] much less as a 'human becoming.'[3] It can never include in its investigation the typically human factors, such as human dignity and freedom, human aspirations and rights, human anxieties and struggles….it can deal with the human only as a dehumanized and depersonalized entity!

The existential thinkers were concerned about the frightening erosion of the 'humanness' from the humans due to the scientific approach with which even the humans are assessed and analyzed. Humans are looked at as 'objects' to be analyzed, considered as 'commodity' for transactions, utilized as 'tool' for one's use, regarded as mere 'functionary,' defaced as 'faceless' in the crowd, reduced to a 'problem' to be solved, estranged from oneself, the Divine, the humans and the Nature, identified with some impersonal number or cipher, degenerated into public conformity, leveled down to the anonymous one (*das Man*). Although the various expressions used in order to refer to the process of depersonalization have the same 'reference' (*Bedeutung*) and even the same 'meaning' (*Sinn*), these are used in different 'nuances' (*Färbung*).[4] Hence we find it useful that we pay specific attention to the different ways of expressing this phenomenon by various thinkers.

2.2. Depersonalization: A Process of Objectification

The process of depersonalization began with 'objectification' of the human, and continues to be one or the other mode of it. According to Heidegger,[5] western thought, culture and life has been very much subject-centred. As the humans pose themselves as 'subjects,' the world shows itself as the 'object' (*Gegenstand*)—something that is made to stand over against the subject. Levinas[6] presents that western philosophy has been considering everything under the encompassing notion of Being, in its various guises,

2 Francis J. Lescoe, *Existentialism with or Without God* (New York: Alba House, 1974), 17.

3 John Macquarrie proposes that 'human becoming' is a more appropriate expression than 'human being.' Cf., John Macquarrie, *In Search of Humanity: A Theological and Philosophical Approach* (London: SCM Press, 1982), 2.

4 These distinctions are made by Gottlob Frege (1845-1925), one of the pioneers of the philosophy of 'Language Analysis.'

5 Nietzsche and Heidegger are the two philosophical stalwarts, who worked out a transition from modernity to post-modernity.

6 Emmanuel Levinas was fascinated by phenomenology, especially as developed by Heidegger. His intellectual growth was nourished by his twofold background of Jewish religiosity and Western rationality. His personality is the result of his constant movement between the two poles of Jerusalem and Athens, or between prophecy and philosophy.

such as the *One*, the *Same* (*le Même*), the *Totality*, the *there-is* (*il y a*), etc. Such a philosophy of the *Same* is a celebration of the narcissism[7] of the ego, with an over-emphasis on *autonomy* and individual subject. The process of 'objectification' (Heidegger) or 'totalization' (Levinas) shows itself even in aggressive forms of manipulating and exploiting the object. The powerful (subject) wants and is able to subjugate the powerless (object).

Jean-Paul Sartre uses the phenomenological delineation of 'objectification' as a platform to take this thought further to its extreme possibility: the phenomenology of hatred! As per his thought, the relation that exists among persons is that of hatred, conflict or rivalry. It is not a moral statement, but an expression of a phenomenological reflection.[8] The other is revealed as a subject in my becoming aware of being an object. In the very act of my becoming the object, the other is becoming the subject. Because of the domineering tone of 'objectification,' Sartre mainly uses an empirical expression that is better understandable, i.e., 'hateful stare'—I am being objectified as I am being hatefully stared at.

In the experience of my being objectified, the other is intentionally experienced as a subject. Before the hateful presence of the other my subjectivity becomes numb and dead; I am drained off my subjectivity and I am reduced to an object! I am depersonalized! The extreme form of objectification by the other consists in my being put to physical death, as a result of which even the remotest possibility of asserting myself as a subject to the other is totally eliminated! I am utterly immobilized! It is at the expense of my subjectivity that the existence of other's subjectivity is revealed.[9]

As no one wants to remain perpetually objectified and enslaved, everyone looks for ways and means of asserting one's subjectivity by objectifying the other. Humans are thus interlocked in mutual enslavement - I try to objectify the other, and the other does the same to me! Everyone is depersonalized by the other. Sartre has taken the process of depersonalization to its maximum width (everyone depersonalizes every other), and to its extreme depth (depersonalizing the other to the point of hating and murdering the other). This form of depersonalization in different intensity

7 The term 'narcissism' is coined by Freud, basing himself on the Greek myth of Narcissus (Narkisos), according to which the attractive young man became so enamoured with his reflection in water, that he fell in love with himself. Thus, narcissism stands for self-love.

8 Sartre carries out the phenomenological analysis of 'look' in his most important book: *Being and Nothingness.* William A. Luijpen dwells on the question of the 'relation of hatred' in his *Existential Phenomenology* (Pittsburgh: Duquesne University Press, 1969), 286-304.

9 This point is elaborately considered in my book, *Human Existence as Home-Coming: An Approach in Reflective Phenomenology* (Bangalore: Asian Trading Corporation, 2012), 93-102.

is very widely present among the humans—in all forms of competitions and in human conflicts. Thus, Sartre has made us aware of the process of depersonalization as objectification, presented in his peculiar style.

2.3. Depersonalization

a) Depersonalization: A Process of Functionalization

Another important mode of depersonalization is the process in which humans are reduced to the function that one performs. The *unique somebody* is replaced by *any somebody*, and eventually by *any something*. A ticket-dispatcher could well be *anyone* or *anything*—a person or a machine. My relation to *him/her* or *it* is purely pragmatic, merely in terms of the function one does with respect to me. The other is seen as a *functionary*—someone who is seen as identified with the function that one does. Thus, the world of today consists, not of unique persons with individual identity, but of functionaries with common identity:[10] functionaries like technicians, postmen, policemen, electricians, engineers, physicians, etc. Performance of a function is seen always in reference to its usefulness to someone, as a result of which functionaries are not considered in terms their individual personality, but purely in terms of their 'utility.' The so-called respect that we give to others is a 'respect' that we give to the 'function' of the persons, and not to the persons themselves. If a machine can do a function more efficiently—with better result, more output, less expense and faster performance—than a human being, it will be preferred, and will replace the human; a common phenomenon of 'depersonalization' in the technocratic world!

The process of functionalization too is, in fact, a mode of *objectification* of persons. When we consider people in terms of their functions, we are reducing them to mere 'objects,' each object having a specific function or purpose. In the wider context of the society, the objectified persons play the role of 'parts' of the social machinery. Just as a machine is made of different parts with specific functions, so also the social machinery of technocratic society is made up of numerous functionaries. As the smooth functioning of the whole machine depends on that of each part, those who plan and organize the efficient functioning of the social machinery carefully look at the utility and functionality of each functionary. In a well-organized technocratic society, there exists only functional relation among the functionaries. In such a society every need is taken care of, and every decision is made with planning and calculation in order to bring out the desired result. It is the picture of a Global Village, where everything is numbered - streets, buildings,

10 My individual identity is replaced with a common identity - I am just sharing in the identity of being an engineer, or an electrician; I become one of the engineers.

vehicles, persons—and everything is identified by its number and function![11] It is the *city of reason,* where contingencies have no place, as everything takes place necessarily as per the plan.[12] Such a carefully controlled society of suffocating formality and rigidity is a 'prison,' in the language of Bentham. Functionalization has dehumanized the human, as s/he is reduced merely to a 'tool' for the use of the other.

b) Depersonalization: A Process of Submerging into Faceless Anonymity

By submerging oneself into the crowd of anonymous ones, one loses one's identity and individuality—one becomes faceless! Humans tend to escape into the 'crowd'[13] and to remain submerged in it as anonymous and 'faceless.' The leaning towards anonymity and facelessness is given an ontological characterization by Heidegger by referring to it as the *falling* of the human from oneself to the 'they-self' (*das Man*).[14] In the crowd the human becomes demoralized by evading responsibilities.[15] The 'faceless' do not have the moral strength to stand face-to-face with one another.[16] The culture of contemporary civilization is on a fast track towards making anonymity as the ideal to be reached. Only in the light of the call to the 'falling' humans by Kierkegaard and Heidegger to have the courage to exist by choosing to be oneself, that we are able to understand that humans are getting 'depersonalized' in their remaining anonymous and faceless.

11 It is like the city of Chandigarh, designed by the French architect Le Corbusier, divided into different zones of equal size and shape, with straight roads and same type of buildings. Everything is planned to take care of every need at easy reach. Every road, junction and building looks like every other: monotonously same!

12 Johnson Puthenpurackal, "Global Village vs. Cosmic Home," in *ACPI Encyclopedia of Philosophy*, 572.

13 The main spokesperson for this position is Kierkegaard; while clarifying the meaning of existence, he refers to the act of standing out or choosing to be an individual from being anonymous in the crowd.

14 Martin Heidegger, *Being and Time*, trans., John Macquarrie and Robinson (Oxford: Basil Blackwell, 1973), 164. He takes 'falling' as an ontological characterization insofar as it belongs to the very Being of Dasein; every human being is characterized by 'falling.'

15 Experience has shown that when one's identity is not known one behaves totally different from the way one would have acted when one is well known. Most of the mob-behaviors prove this fact.

16 Kierkegaard refers to Adam hiding himself among the bushes after he sinned, and his lack of courage to face God; this 'hiding among the trees' is expressive of humans' tendency to run away from one's identity, and his inability to stand face-to-face is indicative of the absence of moral strength.

Kierkegaard makes a graphic description of human's depersonalization in the crowd (modern society, the public), as it is ever threatening to transform him/her into a number or a cipher. Depersonalization of the other expresses itself in the guise of public conformity. In his *The Present Age* Kierkegaard writes that human *intersubjectivity* or *being-with* mostly degenerates itself into public conformity.[17] Authentic interpersonal relations are sacrificed on the altar of that monster, called the public. The individual becomes virtually extinct, and the crowd takes over its place and becomes the criterion of truth, as a result 'number' becomes the deciding factor.[18] In the crowd-existence everyone has only *numerical significance*. The human is depersonalized by being reduced to a 'number.'

According to Kierkegaard the most effective instrument of the crowd in its process of depersonalization is the 'press.' In the press "men are demoralized in the shortest possible time, on the largest possible scale, for the lowest possible price."[19] The press can bring a person or an event to what it wants: the exceptional to the average, the significant to the trivial, and the trivial to the sensational! The press is an instrument of the crowd; it is controlled anonymously, and prints only what the crowd wants to see,…the voice of the crowd continues to be heard throughout the land.[20] By public conformity, one proclaims that crowd is the criterion of truth and right. The so-called 'individual' in the crowd is replaceable. The crowd cannot act; the individual carries out a *corporate action*—actions of individuals in the crowd; what individuals could not do individually, they are able to do in a crowd! Socrates was killed and Christ was crucified through the corporate actions of individuals. Taking refuge in the crowd is taking refuge in the decisions of others. Since one does not have courage to be oneself, one submerges oneself in the crowd. It is one of the widely present forms of depersonalization.

17 Calvin Schrag, *Existence and Freedom: Towards an Ontology of Human Finitude* (New York: Northwestern University Press, 1961), 44.

18 *Ibid*.

19 Kierkegaard, *Journals*, entry 1293; as quoted by Calvin Schrag, *Existence and Freedom*, 44.

20 Heidegger, inspired by Kierkegaard, presents the inauthentic Dasein in terms of the 'anonymous one,' (*Das Man*). The German term '*Das Man*' denotes an impersonal and anonymous neuter 'who' (*Wer*). To exist in the mode of the 'anonymous one' (*das Man*) is to say that the genuine self is lost. Humans mostly tend to be anonymous, average, mediocre, 'leveling process' (*Einebnung*),… By conforming to the demands and opinions of the crowd, one runs away from personal commitment and responsible decision. The 'anonymous one' entrenches itself by idle talk, curiosity and ambiguity.

c) Depersonalization: Reducing the Human to a 'Tool,' a 'Commodity,' a 'Problem,' an 'It,'

In the process of human development, humans began to develop and to use 'tools,' not for replacing their bodily parts, but as their extensions. With the introduction of tools, the gap between *what I want* and *what I am able* is narrowed down. What we want to achieve, our dream, remains far away from us; tools have enabled us to go closer to the ideal of what we dream. Although tools are referred to mainly as the 'extensions of hands,' we can find that almost every part of the body is made to be extended by means of some tools.[21]

Development of tools has been in line with human development; but it gradually got deviated when humans became extensions of and controlled by tools, and eventually they themselves began to be considered as 'tools' for one another. One sees the other from a utilitarian perspective and utilizes the other for one's own purpose. In such cases the concern that one shows to other persons is nothing but 'practical concern' (*Besorgen*)[22] as one does to tools. Just as we use and care for the tools for our purpose, so do we use and care for others, not out of love for them, but out of self-love—they are seen as our extensions: a realm of depersonalization!

Marx is one of the first in the west who pointed out the destructive character of depersonalization. He has pointed out the disruption of the communal world as a result of the conflict between the bourgeois landowner and the proletariat worker, in which the landowner depersonalizes the worker by transforming him into a commodity or a tool along with the land, capital and the work-tools. As a result of such a depersonalization of the worker, work is alienated from the worker, since s/he takes the worker exclusively as a means of profit and production, a way to *exploit* the persons and to *expand* the economy.[23] By reducing the human to mere 'tool' to be made use of for higher *production*, and a 'commodity' for better *profit* in business transactions, humans are getting 'depersonalized.'

21 Ordinarily tools are referred to as the extensions of human hands. But we find that tools are developed not only as extensions of human hands, but also of almost all human organs. Cf., Johnson Puthenpurackal, *Human Existence as Home-Coming*, 118.

22 Heidegger uses different terms with regard to one's relation to things and persons: to things I show 'practical concern' (*Besorgen*), and to persons I show 'personal concern' or 'solicitude' (*Fürsoge*).

23 When work is seen purely in terms of production and profit, it is reduced to a physical activity, carried out by machines, animals or slaves - the human becomes a machine, an animal or a slave when one mechanically carries out the work. Cf., Johnson Puthenpurackal, *Human Existence as Home-Coming*, 113-14.

Insofar as the human is subjected to scientific analysis, as per Marcel's terminology, s/he (it) becomes a 'problem' to be solved.[24] Just as the scientist analyzes an object by dissecting it on the laboratory-table into different parts, we analyze a question, even the question of the human, by focusing our attention on different aspects of the human, considering it as a problem. By considering the human as a 'problem,' we are making it a 'dead reality' that can be meticulously analyzed, investigated, and we presume that a definitive solution—a conclusive word—can be given. As a result, the human is dehumanized/depersonalized.

When the human is devoid of its humanness, it is no more a human, a person or a thou, but merely an 'it'; it is Marcel and Buber who have dwelt on the question of the human being reduced to an 'it.' "…the primary word I-it points to a relation of person to a thing, of subject to object, involving some form of utilization, domination or control…."[25] The human utilizes and controls primarily the material things. To put it more correctly, as the humans are engaged in 'utilizing,' that which is utilized becomes a non-human 'thing.' When persons are utilized, they too become 'things,' in other words, 'depersonalized.' As we are in a world of domination where everyone wants to objectify and utilize every other, the other is constantly made to be an object, an *It*…; as a result, 'depersonalization` is fast becoming the order of the day! Our reflective journey through the phenomenon of depersonalization has been moving through those areas and contexts in which the individual humans are getting depersonalized in the contemporary culture. It was done on purpose, since the individual humans are the primary targets of this ailment of depersonalization. Nations and socio-religious groups too are affected by the *depersonalization-syndrome*.

Every nation, every religion and social group looks at the other nations and groups with an eye of suspicion, and all 'diplomatic' moves are made in view of the advantage and utility of one's own group.[26] Every group wants to present an image of oneself as just, generous, civilized, etc.[27] But through

24 Marcel distinguishes between 'problem' and 'mystery'; a problem can be given a conclusive solution, but the human grapples with the mystery; it cannot be given an exhaustive solution.

25 Herberg W., *Martin Buber: Personalist Philosopher in an Age of Depersonalization*, McAuley Lectures, 15 (West Hartford: Mcauley Institute of Religious Studies, St. Joseph College, 1972), 14; as cited by Tomy Paul Kakkattuthadathil, *Otherness and Being Oneself* (New Delhi: Intercultural Publications, 2001), 50-51.

26 Sartrean interpretation of the prevalently present mutual conflict among the humans, as explained above, is very true of all human groups; every group wants to outshine every other.

27 People are inclined to perceive members of out-groups as less human, or more animal-like, than themselves; such a view is *infra-humanization*. By contrast, dehumanization of an out-group implies that its members are no longer humans at all!

the pleasantly presented diplomatic words and actions, one is always on the watch out for any opportunity to exploit the other for one's advantage. When the advantage is really enticing, there is every chance for the powerful group to violate the diplomatic propriety and international law,[28] and to enter into brutal aggression against the powerless. Thus *diplomacy* does not guarantee peaceful co-existence; rather it only hides the aggressive face of the humans. In the 'civilized' world of today *diplomacy* has drained off all 'humanness' and genuineness from words and actions. In the 'we of *indifference*,' where "no one is *someone* because no one [genuinely] cares for anyone,"[29] there take place 'meetings of *diplomacy*'! In such a rigidly formal situation, where impersonality reigns supreme, every move is calculated, every word is thought out! Thus we find that the evil façade of depersonalization extends itself to almost all facets of human life.

3. Concluding Reflections

We started this paper with a humble submission that the process of 'depersonalization' *may have* served as a prelude to the process of 'reality becoming *virtual*,' and 'intelligence becoming *artificial*.' We have no presumption that we are able to establish this spontaneously emerged and vaguely expressed thought on rational foundation. However, we shall add a few stray thoughts, which may be of help to come closer to establishing our submission.

Humans developed tools as extensions and functioning of human organs and in imitation of the non-human world. In making even the very elementary tools like spades, spanners, etc., we take as inspiration and model the functioning of our different organs; in designing bigger machines like aero planes, rockets, submarines, we take animate beings like birds and fishes as models.[30] Thus, while developing the tools and machines, humans have been taking the functioning of the organs of the human[31] or of the non-human as the reference point and model. But in developing more sophisticated inventions like Computer, Smartphone, Robots, etc., the human being - his/her cognitive power, thinking and planning capacity,

28 The law has always been for the protection of the powerful, and when a situation arises for the powerful to abide by the law, they use their power to exempt themselves from law.

29 William Luijpen, *Existential Phenomenology*, 307.

30 We find that even in painting a scenery and in composing music, we take inspiration and model from Nature - scenes of sun-set, landscapes, and the rhythmic sound from the movements of waves and breeze, the chirpings of birds, etc.

31 The elementary tools are developed in order to surmount the inability or limitation of human organs, especially of hands; *spade* is an extended hand, *hammer* is an extended fist, *crowbar* is an extended finger, etc.

ability of imagination, emotional response, etc. - is taken as the reference point. When a computer can function just as the human brain, it is considered a good Computer. Gradually the relation of reference became a relation of competition! Who can play chess-game better, the human or the computer? Who can calculate faster, the human or the calculator? Eventually, machines have begun to play the role of the reference-point; the machine becomes the standard for defining human behavior.[32] The criterion by which we assess the 'efficiency' of a person is that of a machine! The order is reversed: from the human taken as the reference for the machines, to the machine becoming the standard for human behavior. Gradually technology begins to assert its control and domination over the humans, and along with it the process of depersonalization too gets speeded up!

When the human is depersonalized, s/he is becoming objectified as a dead entity: in depersonalization the human is laid bare as dead! Depersonalization is thus a process of 'assassinating' the person of the other.[33] After having 'murdered' the person in depersonalization, we engage ourselves in 'en-personalizing' it by giving some sort of *personhood* through assigning 'virtual presence' to it. Without being *physically* present, one can be said to be *virtually* present.[34] In a wider sense, both the processes of 'putting to *death'* of the person in 'depersonalization,' and that of 'giving *life'* in 'en-personalization' appear to be meaningful. The 'person' that is put to death in depersonalization and raised to life in en-personalization is not the *physical person* of empirical experience, but the *simulated person*. Theoretically it appears that there is no gain or loss, since what is lost in depersonalization is replaced in en-personalization; but it has to be affirmed that *virtual presence* is a poor replacement for *personal presence*!

In this reflective journey of ours, where do we situate 'Artificial Intelligence' (AI)? In the process of sophisticated machines becoming 'efficient' and 'intelligent' as humans, there takes place a competition between the human and the machine with regard to their 'efficiency' and 'intelligence.' When the machines are found to be more 'intelligent' due to their efficient functioning, the relation of competition ends up in the 'severing' of relation! As a result of such an *apparent* delinking from the

32 Calvin Schrag, *Existence and Freedom*, 43.

33 We normally take 'assassination' only in the literal sense of 'physical assassination.' In the phenomenological understanding, depersonalization is a murdering of the person of the other.

34 For instance, we use the expression, "The Prime Minister Carried out a Virtual Inauguration of the Meeting of the Finance Ministers in New Delhi." Here the PM was not personally present for the inauguration.

humans,[35] machines begin to assert themselves and take over what has been hitherto considered as the uniquely human characteristic, 'Intelligence,' qualified with an adjective 'Artificial': thus was born 'Artificial Intelligence'!

The birth of the newborn babe, 'Artificial Intelligence,' is being celebrated and solemnized with exaggerated pomp, without sufficient thought being given to its hereditary background. In clarifying the 'process of depersonalization,' it is our sincere hope that we have tried to shed some light on the background of 'Artificial Intelligence,' giving us some backing to suppose that the process of depersonalization could be considered as a prelude to reality becoming *virtual* and intelligence becoming *artificial*!

35 There cannot be any real delinking between Human Intelligence (HI) and Artificial Intelligence (AI), since AI is dependent on HI; and hence we used the expression '*apparent* delinking.'

12

Congeniality of Personhood in Artificial Intelligence

Jacob Palaparambil

Abstract

This article delves into the notion of personhood through historical and theological lenses, tracing its origins from ancient times to the realm of Christian orthodoxy. It then transitions to the contemporary era, where the debate over whether artificial intelligence can be categorized as a person is explored. The article underscores the significance of establishing a philosophical and theological definition of "person," drawing a comparison between Boethius' classical interpretation and Aquinas' more nuanced definition. The distinctions introduced by Aquinas regarding terms such as "substance," "distinct," "intellectual," and "individual" in shaping the concept of personhood are brought to the forefront. Furthermore, the article accentuates the fundamental disparities between human individuals and artificial intelligence, emphasizing artificial intelligence's lack of genuine consciousness and self-awareness. The article concludes by affirming the exceptional nature and dignity of human individuals, while urging caution against conflating artificial intelligence's capabilities with authentic personhood. Artificial Intelligence, despite its advanced functionalities, remains a product of human ingenuity and does not possess the inherent attributes of living individuals.

Keywords**:** *Congeniality, Personhood, Greco-Roman world, Christianity, Trinity, Science Fiction, Mary Shelley, Frankenstein, Dark Aeon, Schoolmen's perspective, Severinus Boethius, Ontological wholeness, Self-subsistent, Divine processions, Human creation, True understanding*

Introduction

There is a vast intellectual landscape concerning the issue of what constitutes a person. Who is a person? What does it mean to be a person? These may sound like the abstruse musings of philosophers, but in fact, the inquiry about what makes a person a person is intertwined with history, theology, philosophy and culture. The term itself was born in a Greco-Roman world that was soon challenged by the new religious influence of Christianity and its theologians. The term was then adopted and adapted by the Greek and Latin Fathers of the Church who sought means of articulating the revealed mysteries surrounding Christ and the triune God.

The precision with which the notion of person was clarified formed the pole, rising from man to God, around which revolved the theocentric humanism of Christian orthodoxy. Thus, we see how after much controversy and years of fine-tuning, the notion of person was finally forged in the process of learning to speak coherently about the Trinity and about Christ. This precise application of the term '*persona*' to both God and Christ ushered in the realm of Christian humanism. The rationale behind this connection was that if God possessed personal attributes, it logically followed that not only a select few but every human being was created in the image and likeness of God, thereby imbuing each with the inherent dignity of personhood. While though it is beyond dispute that various disciplines have grappled with the question of what constitutes a person over centuries, the actualization of one's personhood remains an ongoing and dynamic process.

Emergence of the Paradigm

Since the Enlightenment, the modern era has witnessed successive storms of crisis and revolution in the religious, scientific, cultural and technological spheres. However, in the twenty-first century we see humanity being in the throes of a civilizational transformation[1] and the question that has received the unusual range of approaches and perspectives, representing divergent and even contradictory positions is whether artificial intelligence can/should be considered a 'person'. This idea of equating artificial intelligence with personhood has deep roots in science fiction and philosophy.

1 Terry Eagleton, *Culture and the Death of God* (London: Yale University Press, 2014), 119-122.

While there is no specific date of origin, one of the earliest instances of popularizing the notion of artificial intelligence personhood featured in Mary Shelley's novel '*Frankenstein*', published in 1818, in which Dr. Frankenstein, a young and ambitious scientist, becomes consumed by the idea of transcending the boundaries of mortality and harnessing the power of creation.[2] He embarks on a perilous journey to create life – a sentient being – from non-living matter, an act that challenges the very essence of nature and morality. The parallel between Dr. Frankenstein's creation and artificial intelligence lies in the sense of awe and power that comes with bringing something new into existence. Both endeavours share a common goal: the desire to harness knowledge and technical expertise to create something novel and ground-breaking. However, this relentless pursuit of scientific knowledge – as depicted in the novel – raises profound questions about the limits of human ambition and the possible repercussions of playing the role of a creator.[3] Then on, throughout the twentieth century, numerous science writers and innovators have delved into the realms of artificial intelligence and the concept of personhood. As these technological advancements gain momentum, the demarcation between humans and machines has gradually eroded.

In his ground-breaking book '*Dark Aeon*,'[4] Joe Allen articulates the concept of Transhumanism as the profound fusion of humanity and machinery. Within the pages of '*Dark Aeon*,' the author raises a cautionary flag about the morally contentious and godless technological surge that boldly proclaims its ambition to transmute human beings into a 'post-human' condition. This 'post-human' state signifies the culmination wherein the essence of our individual personalities is digitized and transplanted into an electronic ghost, persisting in ceaseless evolution within the boundless realms of virtual space, even beyond the physical cessation of our bodies.

While the idea of bestowing personhood upon artificial intelligence from the Schoolmen's[5] perspective is clearly not acceptable; given the current progressions in artificial intelligence, the notion remains intricate and the

2 Mary Shelly, *Frankenstein or The Modern Prometheus*, 3 vols. (London: Lackington, Hughes, Harding, Mavor and Jones, 1818).

3 Upakul Patowary, "Artificial Intelligence and Mary Shelley's Frankenstein: A Comparative Analysis of Creation, Morality and Responsibility," *Integrated Journal for Research in Arts and Humanities* 3, no. 4 (July 2023): 122.

4 Joe Allen, *Dark Aeon: Transhumanism and the War Against Humanity* (New York: Skyhorse Publishing, 2023).

5 The term 'scholasticism' meant the teaching of the medieval universities which developed between 1100 and 1500 A. D., and the term 'schoolmen' designated the masters and students of these universities. Thomas Aquinas, is considered to be the most eminent of these masters.

subject of significant debate. Hence, let us consider a hypothetical situation in which there is a widespread agreement that artificial intelligence has reached a state of personhood. In such a scenario, can we continue to assert the distinctiveness of human persons, or will we concede our uniqueness to artificial intelligence? However, before addressing this question, it is imperative to delve into the historical origins and the evolution of the concept of 'person.'

Severinus Boethius' Definition of Person

It is one thing, however, to reach an agreement about how to speak, and another thing to embark on the work of defining and beginning the work of systematization. One of the most famous definitions of *persona*, was given by Severinus Boethius (470-525 C.E.). He proposed the following definition, which, although slightly imperfect, was to remain classical during the whole of the Middle Ages: '*a person is an individual substance of a rational nature*.'[6] This definition of Boethius is based on a careful analysis of the category '*nature*.'[7] The definition of the person according to Boethius signifies that non-nature is never a person. Nor is anything accidental, such as whiteness or blackness, eligible for personhood. Every person is a reality of a substantial kind. Boethius goes on to narrow down the concept by stating that person can

6 "*Persona proprie dicitur naturae rationalis individua substantia*." Though the proper definition of person is "*individual substance endowed with an intellectual nature*," the connection of this definition with Aristotle's definition of man as a rational animal is unmistakable. Richard De Smet, "Persona, Anima, Ātman," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 26 and Boethius, "Liber de persona et duabus naturis et una persona Christi," in *Patrologia Latina*, ed. J. P. Migne (Paris: Chadwyck-Healey, Inc., 1980), LXIV: 1343 D. It should be noted that this definition of a person is applicable not only to human beings but also to the angels and the three divine persons whose substance is identical with their nature. Scherer, "Person," in *Historisches Wörterbuch der Philosophie*, eds. J. Ritter and K. Gründer (Basel: Schwabe, 1989), 279-280.

7 This word – writes Boethius – occurs in fourfold meaning. First, *nature* in the broadest sense denotes every intelligible reality, substances as well as accidents. Second, *nature* can, however, be restricted in a somewhat narrower sense to substances, material as well as immaterial. Third, *nature* can be restricted in an even narrower sense to bodies, and fourth, *nature* finally serves, when one says, for example, that gold does not have the same nature as silver – as a designation for the specific difference that gives each object its form. When it comes to the two natures of Christ, only the fourth meaning comes into consideration. It should be noted that this definition of a person is applicable not only to human beings but also to the angels and the three divine persons whose substance is identical with their nature. Scherer, "Person," in *Historisches Wörterbuch der Philosophie*, 279.

be said neither of inanimate substance – such as stones, nor of insentient ones – such as trees, nor of non-intelligent beings – such as horses.[8]

The contribution of Boethius in attempting a formal definition was a big step forward thereby paving the path for the notion of the term person in becoming the new centre of philosophical reflection.[9] Boethius had formulated the definition to restrict Nestorius and Eutyches who were confused between the terms '*nature'* and '*person'* and were attempting to treat them as equivalents. Since Boethius was familiar with the earlier Greek authors like Basil, Gregory of Nazianzen and Gregory of Nyssa, he suggested that the Greek words *ousia* and *hypostasis* were relevant and important for one's understanding and use of the Latin word *persona*.[10]

Thomas Aquinas' Refinement and its Repercussions on the Artificial Intelligence

Though the terseness of the definition by Boethius made its success, the terms of this definition underwent a profound transformation at the hands of the scholastics and especially of Thomas Aquinas. The contribution by Thomas Aquinas has been philosophically, theologically and axiologically striking and his treatment of persons exhibits greater perspicuity and clarity.[11] It is interesting to note that even Hans Urs von Balthasar notes the difficulties with Boethius' definition, but he is however far less appreciative of Aquinas. Aquinas, he says, essentially takes over Boethius and then has all kinds of difficulties in applying it to the triune God.[12]

The technical terms and the linguistic precision tools with which the early Councils lifted the Christian system clear of confusions are the very terms like *substance*, *nature*, *person*, *subsistence*, *reason* and *will* that governed the Thomistic analysis of the person.[13] Aquinas' definition was

8 The likely rationale behind using horses as an illustration of senseless beings may find its basis in scripture. Psalm 32:9 states: "Do not be like a senseless horse or mule that needs a bit and bridle to keep it under control."

9 J. M. Rist, *What is a Person? Realities, Constructs, Illusions* (Cambridge: Cambridge University Press, 2020), 57.

10 For further illustrations confer, Boethius, *A Treatise Against Eutyches and Nestorius*, in *The Theological Tractates*, trans. H. F. Stewart, E. K. Rand and S. J. Tester (Cambridge: Harvard University Press, 1973) and Rist, *What is a Person? Realities, Constructs, Illusions*, 57.

11 Ivo Coelho, "Introduction," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 11-12.

12 Hans Urs Von Balthasar, "On the Concept of Person," *Communio: International Catholic Review* 13, (1986): 22.

13 Richard De Smet, "The Concept of Man in Thomism and Neo-Thomism," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass

not merely replacing the words from Boethius's definition but by providing alternative terms Aquinas was deriving the properties which are implied for God, angels and human beings. The improved definition runs like this – '*a singular integral and unitary self-subsistent subject characterised by intellectual consciousness, moral freedom and all the properties ensuing from these defining notes*.'[14] It is evident from the above definition that the term '*substance'* was now replaced by '*subsistent'*, '*rational'* by '*intellectual'* and '*individual'* by '*singular'* or '*distinct'*. By opting for the term '*singular*' or '*distinct*', Aquinas replaced the twofold term '*individual substance*' of Boethius' definition by the threefold term '*singular* (or *distinct*) *integral subsistent*.[15]

The reason why Aquinas limits the term '*substance*' in Boethius' definition to mean only '*self-subsistent'* substances is that first and foremost the meaning of the term '*substance*' is ambiguous. It could either mean that part of a finite essence which is the basic support of accidents – also called '*nature'* – or it could also mean the whole existent subject, the complete '*hypostasis*.'[16] It is the second meaning that Aquinas insisted could suit alone a definition of the person. Further, '*to subsist*' means '*to exist in itself*, *by its own esse*' and not '*to exist in something else*.' This would mean that not only the genus, species and individual notions which exist in minds and have no independent being are excluded but also parts that have no existence apart from the whole and accidents which depend entirely on the act of being of their substance are excluded.[17]

For Aquinas, what is existentially and holistically proper to the subject is that it '*subsists*,' that it exists in its own right, by its own act of being, by its own energy of existing, be it created or uncreated – whether it is in the case of God the Absolute who is the pure act of being or in the case of the creatures who are existentially and ontologically dependent upon the Absolute

Publishers Private Limited, 2010), 188.

14 The definition of a 'person' by Thomas Aquinas is taken from his philosophical and theological writings. It is not derived from a single specific work but is rather a synthesis of his ideas found throughout his extensive body of work, including his "Summa Theologiae," "Summa Contra Gentiles," and various commentaries on theological and philosophical topics. See, part one, question number twenty-nine of the *Summa Theologica* and Richard De Smet, "Towards an Indian View of the Person," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 120.

15 Richard De Smet, "The Loss of the Person," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 58.

16 *Ibid*., 55.

17 De Smet, "Persona, Anima, Ātman," in *Brahman and Person*, 26-27.

as on their total cause. Hence, what matters is that it must exhibit ontological wholeness.[18]

Further, it must comprehend not only its essential elements but also its individuating qualities and other accidents – in other words, the whole reality of the subject. Wholeness, according to Aquinas, means that in the case of humans, the person is his whole self and not simply the soul or spirit. However, in the case of God, wholeness results from the very plenitude of the subsistent *esse*.[19] God eternally is, and is perfect, from eternity. There is no distinction between what God '*is*' and the '*be*' – *esse* of God.[20]

Along the same lines, there is no difference between a divine person and the divine '*nature*.' The divine persons are identical with the '*nature*,' and eternally and perfectly so. Each of the divine persons is perfectly God and each is equally the same God from eternity. The distinctions among the divine persons cannot run along the lines of distinctions among human persons.[21] This is because, when compared with God, human persons are a very inferior type of person. Their existence is their own but as received from the Creator.

Consequently, the human person's subsistence being a caused one, they are in their totality dependent on the source of all caused subsistence, that is, they are creatures. They have their being in themselves and for themselves but not from themselves.[22] The distinctions in God – which are not absolute although real – have to do with the divine processions, the eternal act by which the Son proceeds from the Father, who speaks the Word in knowing God, and the eternal act by which the Holy Spirit proceeds from the Father

18 De. Smet, "The Loss of the Person," in *Brahman and Person*, 55-56.

19 Richard De Smet, "The Open Person vs The Closed Individual," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 85.

20 According to Étienne Gilson, what Aquinas offers is at once the deepest meaning of the name of God as 'He-who-Is' (*Yahwe*) and a new insight into the reality of creatures. God is the being whose whole nature is to be the existential actuality designated by the verb 'to be.' To say that God is *ipsum esse subsistens* (self-subsisting *Esse* itself) is to say that in Him essence or quiddity is identified with the absolute and unqualified actuality-of-being. He *is* absolutely; whatever else may be said of Him must be an unfolding of this highest metaphysical insight. It is just this which makes clear the ineffability of God. To quote Gilson: "Since, in God, there is no something to which existence could be attributed, his own *esse* is precisely that which God is. To us, such a Being is strictly beyond all possible representation." Étienne Gilson, *History of Christian Philosophy in the Middle Ages* (London: Sheed and Ward, 1955), 368-369.

21 Joseph Wawrykow, *The SCM Press A-Z of Thomas Aquinas* (London: SCM Press, 2005), 106.

22 Richard De Smet, "The Christian Conception of Man," in *Brahman and Person: Essays by Richard De Smet*, ed. Ivo Coelho (Delhi: Motilal Banarsidass Publishers Private Limited, 2010), 180.

and the Son in their mutual love. Relations of origin differentiate the different persons, and such is to be indicated when using '*person'* to speak of God.[23]

On the contrary, artificial intelligence systems are products of human creation. They do not have intrinsic existence, but are designed, developed, and programmed by human engineers, data scientists, and programmers. These individuals write the code and algorithms that govern artificial intelligence's behaviour. They are not self-contained entities, but rely on external infrastructure that is provided, maintained and upgraded by human intelligence. This makes the artificial intelligence a non-self-sustaining entity incapable of addressing complex issues without human intervention.[24] Passing now to the substitution of the term '*individual*' by the term '*distinct*' or '*singular*,' this precision is made by Aquinas to exclude the narrowness of the term '*individual*' in Boethius' definition. First and foremost, the meaning of the term '*individual*' needs to be understood. In Latin, '*individuum*' originally means '*undivided*.' This term designates those who could usually retain their wholeness and particularity and oppose themselves to others as original centres of existence, resistance and activity.[25] The natural consequence of the undividedness of individuals is that they are by the very fact 'divided from others' or, at least, ontologically distinguishable from them. Individuals are generally observed as single units, distinguishable by their individuating characters but at the same time, they are also generally groupable into classes according to their more universal characters. Such are the individuals presented to us by common experience and we are not directly acquainted with any individual who at the same time would be universal.[26]

Subsequently, for a thing to be distinct, it should not be predicated by the identity of another. Thus, the kind of distinction which is proper to subsistent beings is a singularity and for a being to be singular, it should be distinct not only from all other beings in general but also from all other species and all other individuals in its own species. In the case of God, He is distinct from all finite beings by the very transcendence of his nature

23 This is dealt in detail by Thomas Aquinas in his *Summa Theologica* part one, question number forty, article one. Wawrykow, *The SCM Press A-Z of Thomas Aquinas*, 107 and for further reference on the topic of person, Thomas Aquinas, *The Summa Theologica: Part I, Questions XXVII-XLIX*, Literally Translated by Fathers of the English Dominican Province (Chicago: Burns Oates and Washbourne Ltd.) and Timothy McDermott, *St. Thomas Aquinas Summa Theologiae: A Concise Translation* (Notre Dame: Christian Classics), 1989.

24 Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford: Oxford University Press, 2014), 52-59.

25 De Smet, "The Loss of the Person," in *Brahman and Person*, 57.

26 De Smet, "The Open Person vs The Closed Individual," in *Brahman and Person*, 86.

and since He does not belong to any genus or species, His singularity is absolutely perfect. When it comes to the three divine Persons in God, it can be said that they are distinct from one another by virtue of their mutual relations.[27]

Whereas, in the case of angels, since the Schoolmen conceived them as spiritual species, their own essence or nature is said to be the principle of their singularity. This exhaustively contains the full perfection of a particular species and since such species is necessarily finite, it sets them apart from God. Also, it being exhaustively realized in a single such being, no further queries arise concerning the distinguishing of an angel from another angel. Therefore, the plurality of angels is said to be merely 'formal' and not predicamental or strictly numerical.[28] All the other beings however are distributed in classes or species on account of their materiality. Though such material species truly verify and represent ontological perfection in all its members, it is never exhaustively realized by any of them except only in an imperfect fashion. Aquinas holds that material composites are distinguished from one another by "signed matter" which is the principle of individuation. Hence, it is through such degrees of imperfect realization of the same nature that the individuals of one species differ from one another.[29]

However, since their nature is similar, they are much less distinct from one another than species among themselves or than God from the universe of His creatures. Individuality is thus the last and the most imperfect degree of singularity and since personality always implies singularity, it does not necessarily require individuality. Nevertheless, among persons, only human beings are individuals.[30] Aquinas opts for the term '*singular*' or '*distinct*' in preference to '*individual*' and consequently, the twofold term '*individual substance*' of Boethius' definition is so far replaced by the threefold term '*singular* or *distinct integral subsistent*.' If Aquinas had not done this, the term person could not apply to God. God is a person but not one individual in a species. Similarly, angels are persons but not individuals in species.

To some degree, artificial intelligence can be considered *distinct* due to its ability to imitate human intelligence and its capacity to comprehend natural language, but with regard to creativity, there will always remain

27 De Smet, "Persona, Anima, Ātman," in *Brahman and Person*, 27. For further details refer to *Summa Theologica* part one, question number forty, article two. Thomas Aquinas, *The Summa Theologica: Part I, Questions XXVII-XLIX*, Literally Translated by Fathers of the English Dominican Province (Chicago: Burns Oates and Washbourne Ltd.) and Timothy McDermott, *St. Thomas Aquinas Summa Theologiae: A Concise Translation* (Notre Dame: Christian Classics), 1989.

28 De Smet, "The Loss of the Person," in *Brahman and Person*, 57.

29 De Smet, "Persona, Anima, Ātman," in *Brahman and Person*, 27.

30 *Ibid*., 28.

an unfathomable gap between natural and artificial intelligence. Taken by itself, artificial intelligence remains just an aggregate of parts well put together. Unlike human beings, who come into the world vulnerable and carry a unique life history, artificial intelligence lacks the fundamental attributes that define human beings. Hence, it would be inappropriate to describe artificial intelligence as '*singular* or *distinct integral subsistent*,' for even though it possesses distinctive features and stands as an individual creation, it falls short in terms of self-awareness and personal identity.[31]

The next is the substitution of the term '*rational*' with the term '*intellectual*.' Aquinas makes this substitution because, for him, only the lowest grade of intellect is rational whereas the higher grades are intuitive. Rationality according to him, stands on the lowest rung of the scale of intellect and the discursivity it generally denotes is not a property of intellect as such but only of the human intellect.[32] Intellectual, must be understood in reference to the intellect proper, which is characterised by the total reflection of self-consciousness and the transcendence which permits to evaluate any object in terms of being.[33] Thus, rationality applies only to human beings, not to God and not to angels. Rationality is discursive – which means its understanding is achieved piecemeal, slowly, and it also needs to be confirmed.

In human beings, therefore, we have understanding as well as judging but in God, there is just understanding. Man does not have intuitive knowledge; he only tends to it; he wants to have it. Whereas, animal intelligence and the so-called intelligence of machines are capable of quasi-judgments and quasi-reasonings, they are not capable of giving any evidence of reaching the level of self-conscious and transcendent beings. Therefore, that which distinguishes persons from non-personal subsistents is the fact that persons being a designative of perfection, the most characteristic note to imply is not simply rationality but intellectuality.[34]

Thus, reasoning activities as a matter of calculus cannot be intellectual if they were not to be permeated by the power of the intuitive intellect. In the case of God, any such distinction would be absurd because we do not say that God possesses them rather, He is. God is subsistent *Esse*, He is Consciousness, Knowledge, Intellect, Power, Freedom, Bliss, etc. Since God is eminently every possible reality, the divine intellect is all things

31 Bostrom, *Superintelligence: Paths, Dangers, Strategies*, 105-113.

32 De Smet, "The Loss of the Person," in *Brahman and Person*, 58.

33 *Ibid*., 58.

34 De Smet, "The Open Person vs The Closed Individual," in *Brahman and Person*, 86.

in self-awareness and because of this, the person, from man to God lives actually or virtually its identity with the whole.[35]

Further, for Aquinas our knowledge begins with sense intuition and tends through the reasoning discourse to intellectual intuition. Therefore, it is better to say that a person must be '*possessed of an intellectual nature*' whereby the term '*possessed of*' implies no distinction between a person and his nature or his intellect.[36] Since a person is '*possessed of an intellectual nature*,' he is endowed with the will which follows the intellect. This implies that consequent upon his acknowledgement of realities and his intellectual evaluation of their goodness, the human person can choose to bypass them or adhere to them in the measure of their goodness.[37] Since a person is possessed of an intellectual nature, he is endowed with the will which follows the intellect.

Therefore, even though the possession of intellectual nature makes the human person the free author of his activity, the human person innately tends towards – spiritual self-integration – a process of awakening into a higher level of perceiving and functioning. Hence, the becoming of human persons is real and not illusory. Their dynamic development is a growing synthesis with the whole, through comprehension resulting from the intellect, and selective appropriation resulting from the free-will which necessarily accompanies the intellect. It is not a simple passing from obscure to clear perceptions. Thus, the personality of the human person is meant to grow on every level, physical, psychological, social, intellectual, moral and affective.[38]

According to Aquinas, even though human persons are the lowest in the scale of persons, they are still capable of spiritual activities. This includes even knowing God by intuition and loving Him in blissful self-surrender, which in the realm of activities are ends, pursuable for their own sake, and not only steps towards final attainment. It is intellectuality that radically prevents the person from being a mere atomic individual, thereby giving a holistic character and universal dimension to the person.[39] Our intellect is all things virtually and intentionally. Whereas, the divine intellect is all things in self-awareness because God is eminently every possible reality. Therefore, the human person is incessantly concerned with the whole through knowledge in all its intellectual forms and the exercise

35 *Ibid*., 87.

36 De Smet, "The Loss of the Person," in *Brahman and Person*, 58.

37 *Ibid*., 58-59.

38 *Ibid*., 59.

39 De Smet, "The Open Person vs The Closed Individual," in *Brahman and Person*, 86-87.

of the will in all its options, particularly those of love with all beings from the lowest creature to God himself.[40]

Artificial intelligence – including the most advanced and sophisticated ones – does not possess consciousness like the human person because, the modus operandi of any artificial intelligence revolves around the meticulous execution of algorithms, data processing and the rigid adherence to directives authored by humans. Artificial intelligence can be programmed to simulate human-like behaviours and provide solutions to complex problems, but it will never be able to match the human person's ability to observe, imagine, initiate, anticipate and be inspired. Hence, it remains as a tool that can only process and analyse data, perform tasks and make decisions based on patterns and information it has been trained on, but it does not possess consciousness or true understanding. Although artificial intelligence can perform extraordinary feats at astounding speeds, they do not experience the emotions, dreams and thoughts that are an essential part of what makes us human persons.

Concluding Remarks

Throughout history, humanity has embraced scientific and technological progress eagerly. Scientists continuously unearth new knowledge, while technologists craft innovative devices. Artificial intelligence, like other advancements in science and technology, stands as a potent tool of the human intellect. Its development represents a remarkable feat of human rationality. Those who suggest that artificial intelligence can possess personhood implicitly assume that life can be manufactured within a laboratory setting. They argue that speed, adaptability, and precision alone are sufficient grounds to confer personhood upon artificial intelligence. However, it's essential to recognize that while artificial intelligence may exhibit capabilities akin to those of humans, it cannot genuinely be considered intelligent because it lacks the fundamental attribute of being alive.

Different from humans that do not always do things because of necessity, artificial intelligence is created to do things because there is a necessary thing to do. If humans' free will has been weakened or incapacitated, artificial intelligence's will is not weakened or incapacitated. While humans will weigh up arguments about doing good or evil before they act, artificial intelligence will weigh up benefits and disadvantages according to algorithms before it acts. Humans will tend to choose evil due to transmitted sin, but artificial intelligence will not have such a tendency because artificial intelligence has no sin despite having flaws and

40 *Ibid*., 87.

weaknesses. At the moment of this paper being written, there is no artificial intelligence that meets the criteria of Personhood from the Christian perspective. In our strong desire to take on the role of creators, we often overlook the fact that we ourselves are created. We tend to forget that we are not just beings formed from dust, but our very being and life originate from God's creative decree, in which He fashioned us in His own image and likeness. This inherent dignity and the high esteem in which God holds us – despite our seeming insignificance – is beautifully expressed by the psalmist who says: *Who are we mere humans that the supreme God of creation involves Himself so intimately with us?* (Psalm 8:4).

13

From the Perspective of Spiritual Immortality to Digital Immortality

George Panthanmackel

Abstract

'Immortality' means endless life. It is said of humans who survive in a changed form after death. For Plato, the soul is considered to be the essential person. Survival is understood in terms of the immortality of the soul. The immortality of the human soul, from a philosophical viewpoint, can be established from the spirituality of certain fundamental operations in human. The fundamental operations are: knowing, willing and remembering; these are immaterial or spiritual in themselves. As they are spiritual in themselves, they are intrinsically independent of matter and extrinsically dependent on matter or material conditions. The spiritual is simple. The simple is not composed. What is not composed cannot be decomposed. What cannot be decomposed cannot die, since death is basically decomposition. What is not subject to death, therefore, is immortal. However, we know well, that artificial intelligence can never be spiritual intelligence, but only material intelligence, intrinsically dependent on matter and material things. When material elements end, the artificial intelligence will also end. Hence, we do not see the possibility of artificial or digital immortality.

Keywords: *Immortality - Spiritual/Digital/Artificial, Spiritual Operations, Death as decomposition, Artificial Intelligence, Spiritual Intelligence, Soul.*

Introduction

One of the most frequent questions of all time is, "Is there a life beyond death?" This is a fundamental question human has ever asked since the dawn of history. The earliest artists and craftsmen often made life after death a theme of their cave drawings and paintings. Every major religion has explored its range. It is attested by the funeral rites and the cult of the dead in these religions. In contemporary times, scientific tools have been employed to investigate paranormal phenomena which may have some relationship to a life beyond death (immortality). 'Immortality' means endless life. First it is said of God (or of gods), and then of humans who survive in a changed form after death. This survival can be personal or impersonal (supra-personal), bodiless or bodily, lower or higher level of existence.

Ancient Indian philosophy embraced the divine nature of the soul and its survival through rebirth, a state not considered desirable either when compared with moksha or liberation. For Plato, whose thought has exerted a continuous influence on the West, the soul is considered to be the essential person and the body the vehicle of the soul. Survival is understood in terms of the immortality of the soul. In Hebrew thought we find a different picture developing, based on the psychosomatic unity of the person as soul and body combined. Survival after death becomes the prelude to the resurrection of the body. For St Paul it is clearly a spiritual body. Many modern scientists embrace a materialistic account of the human being, asserting that the mind or consciousness is a by-product of brain processes. Proponents of this model tend to ignore evidence that might call their materialistic presuppositions into question by other theories.[1] The Paper is divided into two parts: Spiritual Immortality and Digital Immortality. Spiritual Immortality appears to be a

1 The following are some of the important theories on immortality: *Dualism*: Dualism maintains that at death a person continues to exist as a wholly immaterial substance, as pure consciousness or mind. There are two versions of dualism: Substance Dualism and Property Dualism. Substance Dualism claims that there are two kinds of substances: physical and mental. Property Dualism claims that there are only physical substances, some of which have mental properties. Hence, according to the substance dualist the brain is a physical substance with physical properties, and the mind or soul is a mental substance that has mental properties. The property dualist says that the brain is a physical substance with physical and mental properties. *Epiphenomenalism*: If we suppose that the physical and the mental are distinct, this does open up the prospects for disembodied survival. *Dualistic Interactionism:* The dualistic interactionist maintains that in addition to physical states causing mental states, mental states also cause physical states.

reality; whereas Digital Immortality, which is basically material immortality, appears to be only a possibility.

1. Spiritual Immortality

1.1. Spiritual Immortality in Western Philosophy

For Plato, the soul is a spiritual principle distinct from the body. At the conclusion of the *Phaedrus*, Socrates prays: "O dear Pan and all the other gods of this place, grant that I may be beautiful inside. Let all my external possessions be in friendly harmony with what is within. May I consider the wise man rich. As for gold, let me have as much as a moderate man could bear and carry with him."[2] In this psychological dualism, which corresponds to Plato's metaphysical dualism, we see the pre-eminence of soul over body. This being so, the soul is superior to the body and must rule the body. The *Timaeus* understands the soul as the only existing thing, which possesses intelligence and is invisible.[3] The *Phaedo* teaches that the soul is not a mere epiphenomenon of the body. It cannot be a mere harmony of the body. If the soul were a mere harmony of the body, it would follow that one soul could be more of a soul than another, which is an absurdity.[4]

Although Plato asserts an essential distinction between soul and body, he is quite aware of the influence of the body on the soul. In the *Republic*, Plato argues that true education requires physical training.[5] In the *Timaeus*, he admits the evil influence of bad physical training and evil bodily habits on the soul.[6] For most of the ills of the soul are caused by a defective constitution

2 Plato, *Phaedrus*, (279c), trans. Alexander Nehamas & Paul Woodruff (Indianapolis / Cambridge: Hackett Publishing Company, 1995), 86.

3 Plato, *Timaeus,* (46D), trans. Francis MacDonald Cornford (Indianapolis: The Bobbs-Merrill Company, 1975), 157: "For we must declare that the only existing thing which properly possesses intelligence is soul, and this is an invisible thing, whereas fire, water, earth, and air are all visible bodies…."

4 Plato, *Phaedo,* (86b), trans. G. M. A. Grube (Indianapolis: Hackett Publishing Company, 1981), 124-125: "And indeed, Socrates, I think you must have this in mind, that we really do suppose the soul to be something of this kind; as the body stretched and held together by the hot and the cold, the dry and the moist and other such things, and our soul is a mixture and harmony of those things when they are mixed with each other rightly and in due measure."

5 Plato, *The Republic*, (III, 411 d&e - 412), trans. Desmond Lee (London: Penguin Books, 1987), 176: "And so we may venture to assert that anyone who can produce the perfect blend of the physical and intellectual sides of education and apply them to the training of character, is producing music and harmony of far more importance than any mere musician tuning strings" (III, 412).

6 Plato, *Timaeus,* (86B), 344: "Such is the manner in which disorders of the body arise; disorders of the soul are caused by the bodily condition in the following way. It will be granted that folly is disorder of the soul; and of folly there are two kinds, madness and

inherited from the parents and a faulty education. "No one is willingly bad; the bad man becomes so because of some faulty habit of body and unenlightened upbringing, and these are unwelcome afflictions that come to any man against his will."[7] Even if Plato speaks on occasion as though the soul merely dwelt in the body, he does not deny all interaction of soul and body on one another.[8]

Both the *Republic* and the *Timaeus* teach a tripartite nature of the soul: rational part, spirited part and appetitive part. The word 'part' (*meros*) is used in a metaphorical sense. He regarded the three parts as forms or functions or principles of action, and not as parts in the material sense. *Rational part* (*logistikon*) distinguishes human from other animals. It is immortal and akin to the divine. The *spirited part* (*thumoeides*) is the nobler (more akin to moral courage) and is the natural ally of reason. Of course, it is also found in animals. The *appetitive part* (*epithumeitikon*) refers to bodily desires. The *Timaeus* locates the rational part in the head, the spirited part in the breast, and the appetitive part below the midriff.[9] Plato declared that only the rational part of the soul, which is simple, is immortal. But in the Myths, it is implied that the soul survives in its totality, at least that it preserves memory in the state of separation from the body.[10]

Plato introduced the tripartite nature of the soul mainly owing to the evident fact of the conflict within the soul. In the *Phaedrus*, the rational part is likened to a charioteer, and the spirited and appetitive elements to two horses. The one horse (spirited part) is good and the other horse (appetitive part) is bad. While the good horse is easily driven according to the directions of the charioteer, the bad horse is unruly and tends to obey the voice of sensual passion so that it must be restrained by the whip. To conclude: The tendency to regard the three principles of action as principles

stupidity. Accordingly, any affection that brings on either of these must be called a disorder; and among the gravest disorders for the soul we must rank excessive pleasures and pains."

7 *Ibid*., 345.

8 Frederick Copleston, *A History of Philosophy*, Vol. I (New York: Image Books, 1993), 207-208.

9 Plato, *Timaeus,* (70A-B), 282-283: "That part of the soul, then, which is of a manly spirit and ambitious of victory they housed nearer to the head, between the midriff and the neck, that it might be within hearing of the discourse of reason and join with it in restraining by force the desires, whenever these should not willingly consent to obey the word of command from the citadel. The heart, then, the knot of the veins and the fountain of the blood which moves impetuously round throughout all the members, they established in the guardroom, in order that, when the spirit should boil with anger at a message from reason that some act of wrong is taking place in the members, whether coming from outside or, it may be, from the desires within, then every sentient part of the body should quickly, through all the narrow channels, be made aware of the commands and threats and hearken with entire obedience, and so suffer the noblest part to be leader among them all."

10 Frederick Copleston, *A History of Philosophy*, Vol. I, 208-209.

of one unitary soul and the tendency to regard them as separable parts remain unreconciled in Plato.[11]

Plato's main interest is evidently the ethical interest of insisting on the right of the rational part (charioteer) to rule. While the rational part of the soul is said to be made by the Demiurge out of the same ingredients as the World-Soul, the mortal parts of the soul (also body) are made by the celestial gods. This is a mythical expression of the fact that the rational element of the soul is the highest and is born to rule because it is more akin to the divine. It has a natural affinity with the invisible and intelligible world which it is able to contemplate. But the other elements of the soul are bound up essentially with the body and have no direct part in rational activity by which one beholds the world of Forms.[12]

The following are Plato's arguments for immortality:[13] First, in the *Phaedo* Socrates argues that contraries are produced from contraries. Life and death are contraries. Death is produced from life. We must, therefore, conclude that life is produced from death.[14] The second argument is from the *apriori* factor in knowledge. Humans have knowledge of standards and absolute norms. But these standards and absolutes do not exist in sense-world. Hence, humans must have beheld them in a state of pre-existence.[15] The third argument is from the spirituality of the soul. Visible things are complex and subject to dissolution and death. The soul can survey the invisible and unchanging forms. By coming into contact with forms, the soul shows itself to be more like forms than it is to visible and material things, which are mortal. Moreover, from the fact that the soul is naturally destined to rule the body, it appears to be more like the divine than the mortal. The soul is 'divine,' which for the Greeks means immortality.[16]

The fourth argument is that a spiritual principle cannot wear itself out. The existence of forms is admitted. Now, the presence of one form will not admit of the presence of a contrary form, nor will a thing that is what it is by virtue of its participation in one form admit of the simultaneous presence of a contrary form. If fire is 'warm,' it will not admit of the opposite predicate 'cold' simultaneously. Soul is what it is by virtue of its participation in the form of life. Therefore, it will not admit of the presence of the contrary form, 'death.' When death approaches, the soul must either perish or withdraw. That it does not perish is assumed. It cannot also withdraw itself due to weariness.

11 *Ibid*., 210.

12 *Ibid*.

13 *Ibid*., 212-215.

14 Plato, *Phaedo,* (70d), Tr. G.M.A. Grube, 108-109.

15 *Ibid*., 110.

16 *Ibid*., 116-117.

For a spiritual principle cannot wear itself out.[17] Fifthly, a thing cannot be destroyed or perish except through some evil that is inherent in it. The evils of the soul are unrighteousness, intemperance, cowardice, and ignorance. But these do not destroy the soul; for a thoroughly unjust person may live as long or longer than a just person. So, if the soul is not destroyed by its own internal corruption, it cannot be destroyed by any external evil.[18] Sixthly, a thing which moves another and is moved by another may cease to live as it may cease to be moved. The soul is a self-moving principle which is uncreated. What is uncreated cannot be destroyed. Hence, the soul can in no way be destroyed. It is immortal.[19]

For Aristotle, soul (*psyche*) is the principle of life. It is the entelechy (*entelekeia*) of a natural body endowed with the capacity of life (*De Anima* B I, 412 a 27-b 4). It is the act of the body. The body is for the soul. The body is matter to the soul and the soul is form to the body. The soul is thus the realization of the body and is inseparable from it. The soul is the cause of the body: as source of movement, as final cause, as the real substance (i.e., formal cause) of animating bodies.[20]

There are different forms of soul. The lowest form of soul is the *vegetative soul* which exercises assimilation and reproduction. These functions are fundamental in all living things. Animals possess the *sensitive soul*, the higher form of soul, which exercises the three powers of sense-perception, desire, and local motion. Humans possess *rational soul* (*nous*) which unites in itself the powers of the lower souls. The *nous* is the power of both scientific thought and deliberation. While scientific thought aims at truth for its own sake, deliberation aims at truth for practical purposes. All the powers of the soul, except *nous*, are inseparable from the body and perishable. However, *nous* pre-exists before the body and is immortal. The *nous* requires a potential principle on which it may imprint forms. Here, Aristotle makes the distinction between *nous poietikos* (active intellect) and *nous patheitikos* (passive intellect). The active intellect abstracts forms from the images (*phantasmata*), which, when received in the passive intellect, are actual concepts. Only the active intellect is immortal. What Aristotle says in the *De Anima* is as follows:

This latter intellect (*nous*) is separate, unaffected and unmixed, being in substance activity. For in all cases that which acts is superior to that which is affected, and the principle to its matter. And while knowledge in the actualized state is identical with the fact known, knowledge in the state of potentiality,

17 *Ibid*., 125.

18 Plato, *The Republic*, (X, 609-611), Tr. Desmond Lee, 440-445.

19 Plato, *Phaedrus*, (245c), Trs., Alexander Nehamas & Paul Woodruff, 29.

20 Frederick Copleston, *A History of Philosophy*, Vol. I, 327-328.

though temporally prior in the individual case, does not in general even have temporal priority. Nor is it the case that the intellect is now thinking, now not. It is, further, in its separate state that the intellect is just that which it is, and it is this alone that is immortal and eternal, though we have no memory, as the separate intellect is unaffected, while the intellect that is affected is perishable, and in any case thinks nothing without the other.[21]

According to St. Augustine, human is constituted of body and soul. The soul is a substance in its own right. Human is a rational soul using a mortal and earthly body (Platonic conception). Sensation is an activity of the soul using the body as an instrument rather than an activity of the total psycho-physical organism. The soul is superior to and better than the body. "For, every living creature or animal consists of both soul and body. Of these two components, the soul, assuredly, is superior to the body. Even when vicious and weak, the soul is, without doubt, better than the healthiest and strongest body, since it is higher by nature and, even though blemished by vice, is better than the body, just as gold, even when dirty, is worth more than silver or lead, however pure."[22] Being superior to the body, the soul cannot be acted on by the body. The soul perceives the changes in the body due to an external stimulus. Although the soul animates the body, it is an immaterial principle. Its immateriality and substantiality assure it of immortality. The soul is immortal because it apprehends indestructible truth which shows that it is itself indestructible.[23]

St. Thomas Aquinas rejected the Platonic-Augustinian view of the relation of soul to body and adopted the Aristotelian view of the soul as form of the body. In this way, Aquinas emphasized the closeness of the union between the two. There is only one substantial form in human, the rational soul. This substantial form directly informs prime matter and is the cause of all human activities on the vegetative, sensitive and intellectual levels. Sensation is an act not of the soul using a body, but of the composite. We have no innate ideas and the mind depends on sense-experience for

21 Aristotle, *De Anima,* (3,5, 430a), trans. Hugh Lawson-Tancred (London: Penguin Books, 1986), 205; St Thomas Aquinas, in his commentary on Aristotle's *De Anima* (3, Lect.10), interprets the active intellect as active principle, and thus it is unaffected by passions and emotions without being retentive of *species*. Therefore, the separated human reason cannot function as it does in the state of union with the body. The mode of its functioning after death is not treated by Aristotle in the *De Anima*. This omission does not mean that Aristotle denied individual immortality or condemned the separated intellect to a state of enforced and absolute inactivity. Frederick Copleston, *A History of Philosophy*, Vol. I (New York: Image Books, 1993), 331.

22 St. Augustine, *City of God*, (IX, 9), trans. Gerald G. Walsh *et al.* (New York: Image Books, 1958), 180.

23 Frederick Copleston, *A History of Philosophy*, Vol. II, 78-79.

its knowledge. The soul is incorruptible because it is a subsistent form intrinsically independent of the body and extrinsically dependent on it since the mind needs the body for its activity not as its cause but as condition.[24] Besides, a form, for Aquinas, is corrupted in three ways only: through the action of its contrary, through the corruption of its subject, and through the failure of its cause. But the human soul cannot be corrupted in any of these ways.

Again, a form is corrupted by three things only: the action of its contrary, the corruption of its subject, the failure of its cause; by the action of a contrary, as when heat is destroyed by the action of cold; by the corruption of its subject, as when the power of sight is destroyed through the destruction of the eye; by the failure of its cause, as when the air's illumination fails through the failure of its cause, the sun, to be present. But the human soul cannot be corrupted by the action of a contrary, for nothing is contrary to it; since, through the possible intellect, it is cognizant and receptive of all contraries. Nor can the human soul be destroyed through the corruption of its subject, for we have already shown that it is a form independent of the body in its being. Nor, again, can the soul be destroyed through the failure of its cause, since it can have no cause except an eternal one.... Therefore, in no way can the human soul be corrupted.[25]

24 St. Thomas Aquinas, *Summa Theologica,* (Part I, Q. 75, Art. 6), trans. Fathers of the English Dominican Province (New York: Benziger Brothers, 1947), 368: "We must assert that the intellectual principle which we call the human soul is incorruptible. For a thing may be corrupted in two ways – *per se,* and accidentally. Now it is impossible for any substance to be generated or corrupted accidentally, that is, by the generation and corruption of something else. For generation and corruption belong to a thing, just as existence belongs to it, which is acquired by generation and lost by corruption. Therefore, whatever has existence *per se* cannot be generated or corrupted except *per se;* while things which do not subsist, such as accidents and material forms, acquire existence or lose it through the generation or corruption of composite things. Now it was shown above that the souls of brutes are not self-subsistent, whereas the human soul is; so that the souls of the brutes are corrupted, when their bodies are corrupted; while the human soul could not be corrupted unless it were corrupted *per se*. This, indeed, is impossible, not only as regards the human soul, but also as regards anything subsistent that is a form alone…. [Besides], corruption is found only where there is contrariety; since generation and corruption are from contraries and into contraries. Wherefore the heavenly bodies, since they have no matter subject to contrariety, are incorruptible. Now there can be no contrariety in the intellectual soul; for it receives according to the manner of its existence, and those things which it receives are without contrariety; for the notions even of contraries are not themselves contrary, since contraries belong to the same knowledge. It is, therefore, impossible for the intellectual soul to be corruptible."

25 St Thomas Aquinas, *Summa Contra Gentiles*, (Book II, Ch. 79, Art.10), trans. James F. Anderson (New York: Image Books, 1955), 256.

The subsistent human soul is spiritual because it is capable of knowing the natures of all bodies. If it were material, it would be determined to a specific object without being capable of self-reflection, as, for instance, the sense of vision is determined to the perception of colour. For these and other reasons, it follows that every human soul must be immaterial (spiritual), and thus immortal. Secondly, the soul has a natural desire for immortality. The natural desire (*desiderium naturale*) implanted by God cannot be in vain. When Aquinas proves the immortality of the soul, he is naturally referring to personal immortality. Otherwise, it is impossible to explain the diversity of ideas and intellectual operations in different humans if they have only one intellect.[26]

Based on the above reflections of philosophers, we conclude that the immortality of human, from an ontological viewpoint, can be argued from the immateriality or spirituality of certain fundamental operations in human. For "as the operation is, so the being is." The fundamental operations are knowing, willing and remembering, which are immaterial or spiritual in themselves. As they are spiritual in themselves, they are intrinsically independent of matter and extrinsically dependent on matter or material conditions. It means human cannot know, or will, or remember without body or material conditions. Yet, the material conditions cannot cause any of these operations. If the operations – knowing, willing, remembering – are spiritual, the corresponding faculties – intellect, will, memory – from which these operations proceed should also be spiritual. Although these faculties have their own distinct functions, they are not separate or apart from each other as they are coordinated by a coordinating principle, namely, soul which must also be spiritual as the spiritual cannot originate from what is material, but only from the spiritual. The spiritual is simple. The simple is not composed. What is not composed cannot be decomposed. What cannot be decomposed cannot die, since death is basically decomposition. What is not subject to death, therefore, is immortal. Hence, human (soul) is immortal.

1.2. Spiritual Immortality in the *Upanishads*

The *Kausitaki Upanishad* (KU 4,20) states that the soul fills the body right to the tips of the nails and hairs just as a sheath. The *Brhadaranyaka Upanishad* (BU 5,6,1) and the *Katha* (Katha 1,2,20) describe the soul as the size of a rice or barely grain. The *Katha* also describes it as the size of a thumb (Katha 2,2,12), and the *Chandogya* as the size of a "span" (CU 5,18,1). The *Maitri Upanishad* (MaiU 6,38) describes the soul as all these sizes – grain of rice, a thumb and a span. What is important in this apparent contradiction among 7

26 Frederick Copleston, *A History of Philosophy*, Vol. II, 383-387.

metaphors is the philosophical content, and the usefulness of metaphors for the progressive realization of atman. The apparent contradiction is the best indicator that what is intended is to show that Self and Self-consciousness are not subject matters within any spatial, temporal and causal framework. It is the attempt to translate the metaphysical (a-spatial, a-temporal and acausal), immortal soul to these metaphors that create the apparent contradictions. The key to understanding the *Upanisadic* metaphysics of death is that death is no termination. Death denotes one phase in a seemingly unending series of transitions that have as their underlying ground and ultimate outcome, immortality. The *Kathopanisad* (Katha 1,2,18-19) states that the atman is neither born nor dead. The soul is not killed with the killing of the body.

2. Digital Immortality

Digital immortality (virtual immortality) is the continuation of an active or passive digital presence after a person's death. It is the hypothetical concept of storing (or transferring) a person's personality in digital substrate, through artificial intelligence, i.e., a computer, robot or cyberspace (mind uploading). Artificial intelligence is often thought of as robots or thinking machines. Artificial intelligence (AI) is the ability of a computer or a robot controlled by a computer to do tasks that are usually done by humans because they require human intelligence and discernment. In Autumn 2017, Sophia, a humanoid robot gave a speech at the United Nations, to prompt the recognition that there needs to be more debate, as well as legislation in this area. Alan Turing is widely considered to be the father of artificial intelligence. One particular area of significance is that of virtual humans. 'Can machines think? In the 1980s, Searle suggested that the computer cannot be said to think. It is merely the human programmers that are intelligent, who have programmed the machine to implement their ideas. 'Virtual Humans' tend to be used as an overarching term that includes Chatbots, Autonomous Agents and Pedagogical Agents. They are characters on the computer screen with embodied life-like behaviours such as speech, emotions, locomotion and gestures. Evidence has shown that many users are not only comfortable in interacting with high-quality Virtual Humans, but an emotional connection can also be developed between them.[27] Certain legitimate concerns exist about artificial intelligence being used to control cars and weapon-systems. However, it is unlikely that 'the development of full AI would spell the end of the human race.[28]

27 Maggi Savin-Baden, "Digital Immortality," accessed September 4, 2023, https://www Worcester.ac.uk.

28 Charles T. Rubin, "Artificial Intelligence and Human Nature," *The New Atlantis*, No. 1 (Spring 2003), 88-100, https://www.jstor.org/stable/43152855, accessed on September 7, 2023.

Not only are we seeing advances in knowledge management, but also in merging our human bodies with technology. This includes inserting chips into our arms to open doors and pick up metal objects, and implanting antenna into our brains to translate the colour spectrum into different vibrations, enabling the user to 'hear' colours. Whilst for some people this is seen as art, there are useful advances such as the creation of bone implants that enable the mounting of a replacement arm on to the skeleton which can then be controlled naturally, using brain signals. If he was still around, would Albert Einstein be able to build on the advances in our knowledge of physics that have been made since he died? But some have opined that if we capture enough data during someone's life, it might be possible to keep them around indefinitely. Microsoft researchers Gordon Bell and Jim Grey have estimated that logging every conversation that a human has over their entire lifespan would only require around one terabyte of storage. Running a "digital twin" of a deceased person would allow us to benefit from their wisdom as long as it's necessary. Of course, if we can replicate someone after death, then we can replicate them while they are still alive. We could have 10 or 100 Einsteins, all working on solving different physics problems.[29]

It's often pointed out that death has always been a natural part of life, and there's no real way of knowing what sort of effects it might have on our society if we were to eliminate it. Evolution enables us to come up with new ideas as new brains emerge. Others posit that we might behave better if we knew that we are going to be around forever. One thing that's certain is that death is a problem we'll never stop trying to solve. Perhaps with some technological assistance, we may one day live on to see what attractions the future has in store.[30]

Conclusion

Almost all humans, in their thought and conviction, are oriented towards immortality or life after death. The immortality of the human soul, from a philosophical viewpoint, can be established from the immateriality or spirituality of certain fundamental operations in human. For, 'as the operation is so the being is.' The fundamental operations are: knowing, willing and remembering; these are immaterial or spiritual in themselves. As they are spiritual in themselves, they are intrinsically independent of matter and extrinsically dependent on matter or material conditions. It means that the human cannot know or will or remember without body or material conditions. Yet, the material conditions cannot cause any of these operations

29 *Ibid*.

30 *Ibid*.

independently. If the operations of knowing, willing, and remembering are spiritual, the corresponding faculties – intellect, will, and memory – from which these operations proceed should also be spiritual. Although these faculties have their own distinct functions, they are not separate or apart from each other as they are coordinated by a coordinating principle, namely, soul which must also be spiritual as the spiritual cannot originate from what is material, but only from the spiritual. The spiritual is simple. The simple is not composed. What is not composed cannot be decomposed. What cannot be decomposed cannot die, since death is basically decomposition. What is not subject to death, therefore, is immortal. Hence, human mind or soul is immortal, i.e., it survives death.

However, we know well, that artificial intelligence can never be spiritual intelligence, but only material intelligence, intrinsically dependent on matter and material things. When material elements end, the artificial intelligence will also end. Hence, it can never be immortal. Only immaterial beings are immortal. For all material beings are composed of. What is composed gets decomposed. Death is decomposition/disintegration. Hence, we do not see the possibility of artificial or digital immortality.

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Artificial Intelligence and Free Will

Joy Alumkal

Abstract

My reflection on AI and free will is situated within a larger story of ours (humans), technology and future unfolding in our worldly existential situation that can be presented as at the brink of a radical transition from world of organic force controlling everything to world of inorganic force going to control! In order to understand the complexity of this radical change and its repercussions, three frameworks through which humans look at themselves and the world are exposed. They are three 'isms' ('ists') to explain our central theme of free will. (Of course, these three do not exhaust the possible ways of looking, but for our discussion on free will, the rest can be considered as their corollaries). The three can be termed as the humanist, the trans-humanist and post-humanist looking glasses. Since each one is very complex and unfolds within the socio-cultural context of the persons involved, the basic contentious issue of human choice or individual decision are looked from both compatibilism and determinism angles. History of human thinking shows the fluidity of interpretations. At this juncture, we present the repercussions of the 'liminality'.

Keywords: *Artificial intelligence, Freewill, Liminality, Trans-humanism, Post-humanism, Self-determinism, Event determinism.*

Introduction

In order to talk coherently and somewhat comprehensively about the connection between Artificial Intelligence and Free Will within the context of our larger ACPI theme of this year: *Engulfing presence of AI: Philosophical reflections of its challenges and opportunities*, we need certain rationally convincing stories of real-life situations which could be presentations of certain clear seeing and thinking. I present one such articulation for clarity of our exposition, knowing for sure that within the 'liminality' (being on the threshold of radical change or transition) of the socio-cultural context of our times, no 'one story' can exhaust or be an all-encompassing exposition of the complexity.

Taking that excuse, for our clarity of the context, I present an exposition of Yuval Noah Harari in one of his talks – "AI and the Future of Humanity"; at the Frontiers Forum - uploaded in Yuval Noah Harari official YouTube channel.[1] (This is the sixth point of Dr. Kuruvila's presentation included in this collection). Harari started by exposing his vision of the world. He divides the world into two: World of organic life force and the world of inorganic life force. Then he explains the world of organic life force with the story of evolution from first life form to human complex brain activities upto human intelligence-controlled technologies starting from stone tools up to the present Artificial Intelligence as a transition of millions of years. All these he considers as fruit of the combined effort of the most advanced life form in organic life, namely human brain. After this much clarity, he starts to speak about the characteristic of Artificial Intelligence as the first life form in the inorganic, calling it an 'alien intelligence'.

In his evaluation, this 'alien intelligence' is only in the first level in the evolution scale of its life form! But the speed in which its evolution happens must be our real concern. Its growth is exponential. Since humans need such a lot of energy to take care of its basic needs, including food gathering, preparing, comfort making in the environment in the form of home making and sleep. Then there are growth period and decaying period in the form of old age and death. So even the most intelligent human person could get cumulatively few years for intelligent activity in evolution! But Artificial Intelligence in its full strength active all 24x7! Thus, very fast it can come to human intelligent level of evolution, if not at present it is very much up to!

1 See: https://www.youtube.com/watch?v=LWiM-LuRe6w&t=49s.

Once this version of exponential speed in which change is happening in our society is accepted, in order to expose the present worldly existential situation and its complexity, we present three frameworks through which humans look at themselves and the world. We speak about them in order to expose the conceptions of free will and the lack of it in the human rationality which is consciousness but controlled by the unconscious and subconscious mind levels that modern psycho analysts exposes starting from Freud and exposes further in Jacques Lacan, Jacques Derrida etc in the West and the Indian expositions of the consciousness as integrated holistic mind from ancient time onwards, especially the Buddhist and Vedantic expositions.[2]

Freewill and Mind: Humanist, Trans-humanist and Post-humanist Views

The classification of the looking glasses into the Humanist, Trans-humanist and the Post-humanist are post-enlightenment happening. Actually, parallel to these three runs another within which all the religious world-visions participate. We can name them as the 'Traditional world-visions.'[3] All religious people participate in varying degrees in this traditional world-vision. The basic difference between the traditional and the above three are their historical beginnings. Traditional world-visions run in the society as its main controller all through the ages from the ancient and the medieval. The above three are modern, comes up with enlightenment era. Another important difference between the traditional and the modern is the 'God centeredness' of the first and the 'human centeredness' of the second. Since we live in the society and all religions made themselves 'modern' with their '*aggiornamento'* (making oneself modern by updating) efforts, we speak here only of the three 'isms'.

The Humanist, the trans-humanist and the post-humanist looking glasses that we want to present here do not exhaust the possible looking glasses to tackle all the issues of free will, but for our present discussion, the rest can be considered as their corollaries. Again, all the three we do not see as compartments, for they are conceptual constructs we formed to explain trends that come up in unique contexts. For clarities sake we

2 Indian theories of Consciousness are the most elaborate with two extreme ends in Buddhism (All Buddhist Schools) and Vedanta (Brahminical Systems in general). It is interesting to note that both claim to reach same goal through extreme opposing assertions. Buddhists deny a self and assert consciousness and Vedanta affirm the Self and reach Consciousness. (See: Ashok Kumar Chatterjee, *The Yogacara Idealism* (Delhi: Motilal Banarasidass Publishers, 1962, Reprint 1999), 1–23.

3 Here I refer to the commonalities of the traditional world visions, whether of religions or of secular in all the traditions where mostly we see a belief in the spiritual or supernatural reams beyond the physical world and an emphasis on community and social order on one side and practice of rituals and traditions for connecting with the spiritual realm.

classify like that. Practically every vision is very broad and very much depend on our perspective that we received from the socio-cultural context in which we are born and brought up. Take for example the deterministic and libertarian talks that exist in our ordinary conversation. We hear people use words like "it is fate!" But in the next breath will say 'you are responsible!' they do not see an issue of incompatibility between the two. That is only hard philosophically minded person's problem! Commonsense solution or pragmatic solution is the distinction between 'self-determinism' and 'event determinism': like the jumper jump to the water and one push him to the water or some substance change the mind and the person unconsciously jumps. All the three have same result, as the person is in water! Common sense will say, the first is internally determined, where the source of our decisions and actions lies within us and the next two are externally determined, which is ultimately predetermined by past events where the source of our decisions and actions lies outside of ourselves.

What we said above is still further complicated if we analyse after the fashion of scientists and neurologists like Robert Sapolsky, who says "if we decide reasons of our behaviour based on our decisions and choices, it is like reviewing a movie or book after watching the last three minutes of the movie or reading the last few pages of the book"! This can be better understood when we analyse traumatic experiences that has different layers and appear like labyrinth! He exposes the biological, neurological and environmental factors behind our so called 'conscious choices.'[4] The real issue of free will concept is: even if we have the feeling of making free choices, are our actions truly determined by the biological, neurological and environmental factors that are beyond our control?

Before exposing each 'ism', Let us be clear of what free will is! Free will is a concept that is fundamental to human rational thinking that presupposes a self. Philosophically we can speak of three perspectives: i) Libertarianism which posits existence of true free will in the self: meaning, we have the ability to choose independently of any prior causes. But they fail to answer the questions of those who believe in a deterministic universe! ii) Determinism that poses predetermination of everything that happens in the form of effect and prior causes. This group fail to answer the questions of those believe in the theories of morality and punishment which presuppose free will and a self behind the actions. iii) Compatibilism argues, like commonsense persons, that free will of the self, co-exists with determinism, if we have control over our desires and values.

4 Robert M. Sapolsky, *Behave: The Biology of Humans at Our Best and Worst* (USA: Penguin, 2019).

Psychologists traditionally reflect on the spiritual faculties of cognitive, conative and volitive activities. They do not normally question the self. As experiential science in the form of Neuroscience comes up and highlight the complexity of decision making, the talk become of different intellectual abilities: be it in terms of 'comprehensive learning', 'good response to questions' or 'adaptation ability' etc. For them basic issue is not individual decision or choice, but analysis of the process.

But for traditional philosophers from ancient time onwards basic issue of contention is matter of individual decision or choice. Whether we answer in the affirmative or negative, there are contentions in philosophy, from ancient time to present time especially in issues of Morality.[5] This fluidity in the interpretation of the individual choice makes it two-edged sword that cut both for compatibilism and determinism.[6] Let us take for example the Aristotelian compatibility and Epicurian determinism as an example of ancient but still one of the contemporary relevant ways of seeing and articulation of the issue.

Aristotle argued that free will is necessary for moral responsibility, and that we cannot be held accountable for our actions if we are not free to choose them. Epicurus, on the other hand, argued that free will is an illusion, and that our actions are determined by our desires and beliefs. We read in Nichomachean Ethics of Aristotle III.1.15 "We are only responsible for those things that are up to us," and "Those things that are up to us are those things that we can do or not do." Again in his book Physics II.5; III.3 "The universe is governed by natural laws, but these laws do not determine our actions" and "We are not moved by necessity, but by choice." We read in On Nature of Epicurus 29&30 "Our actions are determined by our desires and beliefs" and "We cannot choose our desires or beliefs." In his Letter to Menoeceus, 122 we read "The belief in free will is harmful, because it leads us to worry about things that we cannot control."[7]

These ancient articulations of contradiction in terms of reason and desires are no more contradiction for many in our times of Neuroscience. Take for example the Psycho analytical expositions of reason as rationalization of desires. Rationalizations of desires lead a person to compulsive behaviours that take us away from the real core issues. Take for example an analysis of a compulsive behaviour. It goes on in countries, in societies, in individuals. Most of those compulsive behaviours are by-product of the trauma experiences of the societies and persons from the past experiences. Here past determines the present behaviours without being aware of it most of the time. Both Jacques Derrida and Jacques Lacan, prominent philosophers and

5 See: https://plato.stanford.edu/entries/freewill/#AnciMediPeri.

6 See: https://www.britannica.com/topic/free-will-and-moral-responsibility.

7 See: https://plato.stanford.edu/entries/freewill/#AnciMediPeri.

psychoanalysts take this deviation from the traditional notions of choice and free will and the consequent binary opposition of freedom and bondage. This binary presupposes a pre-existing self. These psychoanalysts move to simple deconstruction of concepts and their pre-existing structures in the culture and language in which they come up.

I refer to the fact of possible uses of intelligence by each person. If we look at the track record of humans in their social life, be it in the history of philosophizing or in its history of tools making and its use, all of them are two-edged swords with both positive and negative repercussions for both individuals and societies. It leads to a liminality, a threshold or a phase of transition, through which we have to pass to enter a new phase. It is in this larger canvas where a lot like flux or ambiguous or in transformation, the philosophical discussion of free will is placed. Let us try to expose its relevance in the humanist, trans-humaninst and post-humanist context.

Humanism, trans-humanism and post-humanism have their own visions that grapple with the future of humanity in the midst of rapid technological advancements. If a person stands within the present biological realities of humans and confident of solving the issues, that person is in a humanist vision.[8] Person becomes a trans-humanist when the vision is driven by a confidence in the efforts of transcending the tensions of the biological realities through technology. Post humans are actually a complex group that see the tensions in both the visions. When we look at the articles written in this collection, the positions of those writers on Artificial Intelligence too can be seen. They all have their presuppositions about free will, technology and the means for attaining their supposed goals. It is because all these are reflections of persons who have free will to form their thought in this way or that way! Take for example the paper of Augustine Pamplany and contrast it with/ or just place parallel to that of Johnson Puthenpurakal: with his depersonalization fear! Or that of Jose Maliekkal with his concern for the least in the society! All of them are real issues, and they look at different aspects of the larger picture. In this book's context all the three 'isms' have a common ground: the perspective on 'what it means to be human' and all of them diverge on the perspective on 'how to navigate the future'!

8 It is interesting to check the chronological history of transhumanism given in the hpluspedia in the internet (Ref. History of transhumanism - H+Pedia (hpluspedia.org). Another interesting history can be seen in the first two chapters of *Beyond Humanism: Post –and Transhumanism: An introduction*, ed. Robert Ranisch and Stefan (NY: Peter Lang, 2015), 7–48.

Humanism and Freewill

Humanism as an 'ism' is fruit of the philosophizing of the Renaissance period. Here it emphasizes the uniqueness of humans and the importance of human reason, experience, and agency. Humanists believe that humans as the 'biological beings of evolution' are capable of shaping their own destiny and that they should strive to live good and meaningful lives. They also believe that humans are fundamentally different from other animals and that they have a special responsibility to protect the natural world. When a person looks at reality through this type of a frame, this can lead to a more optimistic view of human potential on one side and on the other confidence on it. But once the fact that human cannot solve the problems, they face make them turn either to divine or spiritual help, or turn to science and materialistic means to rectify the issues. Both these looking glasses to interpret reality value human creativity and imagination. This led to new and innovative ways of thinking about the world and our place in it. Whole talk centre on humans and their special value.

But everything did not turn good. Humanism and enlightenment of the Renaissance period led humanity to lot of wars and hates more than its intended goal of communion; 'isms' like Fascism and Nazism and persons like Hitler came up and controlled the society and brought much suffering to society as a whole through its divisive techniques! When we ask the reasons for this turnout, some point to the 'anthropocentric' turn. This can lead to a disregard for the natural world and other animals on one side and more individualistic, or divisions and seclusion of groups or generation of communalism. This can lead to a neglect of the importance of harmony of community or communion and social responsibility. This led to a rejection of many communitarian qualities that traditional visions on which societies were based that religions proposed and supported.

Trans-humanism, Post-humanism and Freewill

Negativities of the framework of humanism caused the growth of new 'isms' called 'trans-humanism' and 'post-humanism' around 1940s and 50s. Actually, both of them are used synonymously by emphasising their common characteristic of opposing both the traditional and humanist vision of humans as special and superior beings. Trans-humanists accepted the failure of the optimism of Humanism and as a solution turned to technology for human enhancement and thus overcoming human biological limitations. Post-humanists are of varying types: some approve of technology for liberation and even biological enhancement is approved and some others critique of its potential to further marginalization and control. Groups of Post humanists turned naturalists and inclusive that brought humans down to one of the

animals and emphasised a conception of justice and sustainable development for the good of the earth as a whole.

Those persons who call themselves trans-humanists stand for the growth of technology to enhance human capabilities. Here individual use of technology is emphasised than social justice or sustainable development that care for animals and the earth. In order to do this, we need a 'spiritual' approach claims the critics of trans-humanism. In their opinion, being 'post-human' in the sense trans-humans understand it point to a grave danger that society is open due to technological advancement: a pessimistic picture about the future of humanity. It is because they see a dehumanizing process in our times or reducing of humans to machines. In our times even Elon Musk tweets of the dangers of AI. His tweet "Worth reading Superintelligence by Bostrom. We need to be super careful with AI. Potentially more dangerous than nukes." These warnings point to the need of advancement of a 'post-secular' temperament in the society.[9]

Freewill and Artificial Intelligence: Concrete Issues

1. Is Human Intelligence a Two-Edged Sword?

Humans with free will reached the stage of generating tools that could destroy itself and everything else in the world needed for the favourable living of the organisms. It means that the human intelligence that made amazing progress is also turn to be instrument of catastrophic destruction. With the growth of AI situation is more alarming! Actually, no tool is good or bad; one who uses it makes it for good or bad. For example: same knife used in the operation theatre by a doctor for curing, and a butcher for killing animals or a criminal to harm other humans! Every person with a grip of the issues of AI cannot ignore the claim of Yuval Noah Harari: "it is true, all types of threats that are there now humans have a control, but very fast, that control will be in the hands of AI!" Is it not a pessimistic statement about human free will in practice?

2. Who Should be Regulating Authority for a Just and Sustainable World?

Though we assumed, so far we had free will, but who really controls or regulates our life in the society? Is it Democratic authorities or Market or politicised nexus of Market and Regulatory authorities? What controls all these authorities? What is the role of Money and Lobbying? If authority turns

9 Certain authors point to the meanings of secularism and its global need to turn post-secular, which is ultimately a 'spiritual', 'ethical' and 'aesthetic' turn! For example, see: *Global Secularisms in a Post-Secular Age* Michael Rectenwald, Rochelle Almeida, George Levine, eds. (Berlyn: De Gruyter, 2017).

corrupt by personal interests who and how can they be corrected? What are the repercussions? For groups in the societies that are marginalized, fringe groups influence, opinion making, competitions and lack of fairness in it? Small players pushed out by big fishes?

3. What about the 'Binary' of Freedom and Slavery?

How to understand Freedom? Is it 'from' government regulation or from Regulation agencies? Is Freedom at the expense of danger to life in the form of lawlessness that endangers safety? In the name of safety how much freedom can be curtailed? Are we turning lazy by giving our freedom to Government or social media controllers for making choices for us? What is actually going on in the world? Who inform us? How? When? Who regulates what we should know? How to know? What about the distracters? What about other players and their freedoms? What about Decentralization of media in the internet? What about supporting alternate media against corporate media that is rigidly controlled? Is alternate media a headache for governments and corporate controllers? Do they generate narrations like, the story of anarchy as a big problem for common good of society? Thus, control media reporting by saying there are possibilities of public threats, in the form of social unrest, crime etc.

4. Who is Actually Controlling and what about the Surveillant Systems?

Government with highly centralized controlling system, law and court, legislatures and executers in the form of officers in public service, police, army and central government controlling banking or money refers to total regulations of central government… but now decentralized money with bitcoin concept comes up, but slowly big government control is coming up on it too.

Is it anarchy that 'people' create their own haven by opting out from government control and live in their freedom? Once Osho Rajneesh made one in America, Arizona called Rajanishpuram. Now a days one created in Chile by a Western multi-millionaire Jeff Berwick, thinking that the ship in which he was, is sinking or about to sink, so wanted his own ship. He created an expact community in Chile, freedom society of those no longer tolerate the government encroachment in their personal affairs!

Now the question - Why search for safe havens? In order to escape taxation, regulation, fear of jail, victimless crimes? Not all can pack up everything and move to Chile! What can we do practically?

Conclusion

A reasonable way to start our onward journey to the future is by befriending new knowledge and tools on one side and by breaking the nexus of existing power controllers in authority and those invisibly control our will. How to do these two? Is it by restricting the power of the state? Is power the issue? Is the statement "power corrupts, absolute power corrupts absolutely" true? When we give power to government for regulating food, medicine, monetary services, banking, power, gas, oil etc, will these enterprises gravitate and buy off the politicians and force to make the mergers, pass the laws to their benefit and their best interest? Then who will be there to safeguard common interest of the majority average public? Is it not utopian suggestion to take away all incentives that gives power to politicians and finally causes temptation to manipulate? Will there be same 'service minded interest' to serve the real interest of the common citizens once incentives lost?

No straight forward yes or no answer solves the issues of free will. Let us keep an open eye to the complexity and profundity of the consciousness of the Individuals, Communities and States that psychology and neurology bring out on one side and on the other to the unfolding of intuitive, intellectual and biological energies in the form of new discoveries and innovations happening around us. It is true that liminality of the times generates uncertainty and anxiety, but this is a time of great creativity and personal growth too for the attentive ones.

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Human Intelligence and Artificial Intelligence Delibrations on Superiority and Complementarity

Sunil Jose

Abstract

This article, "Human Intelligence and Artificial Intelligence: Deliberations on Superiority and Complementarity" discusses in the first place what is human intelligence and then moves on to discuss what artificial intelligence is in the perspective of having attempted to discuss human intelligence. When it comes to human intelligence, we human beings possess the mental quality of intelligence, which includes the abilities to learn from experience, adapt to new situations, understand and handle abstract concepts, and use one's knowledge to manipulate one's environment. From the recent developments in the field of artificial intelligence it is seen that machines and robots are able to perform certain tasks or many tasks that previously was thought could be performed only by humans. This paper therefore attempts to explore whether artificial intelligence will one day replace human intelligence.

Keywords**:** *Human Intelligence; Artificial Intelligence; Turing Test; Chines Room; Consciousness*

Introduction

Since the onset of artificial intelligence, philosophers have been asking the big questions for a long time. How do minds work? Is it possible for machines to act intelligently in the way that people do? Would such machines have real conscious minds?

1. Human Intelligence

> Human intelligence has been understood as the cognitive capacity that consists of the abilities to learn from experience, adapt to new situations, understand and handle abstract concepts, and use knowledge to respond one's environment.[1] Much of the excitement among investigators in the field of intelligence derives from their attempts to determine exactly what intelligence[2] is. Different investigators have emphasized different aspects of intelligence in their definitions. For example, in a 1921 symposium, the American psychologists Lewis Terman and Edward L. Thorndike, differed over the definition of intelligence, Terman stressing the ability to think abstractly and Thorndike emphasizing learning and the ability to give good responses to questions. More recently, however, psychologists have generally agreed that adaptation to the environment is the key to understanding both what intelligence is and what it does. Such adaptation may occur in a variety of settings: a student in school learns the material he/she needs to know in order to do well in a course; a physician treating a patient with unfamiliar symptoms learns about the underlying disease; or an artist reworks a painting to convey a more coherent impression. For the most part, adaptation involves changing oneself in order to cope more effectively with the environment, but it can also mean changing the environment or finding an entirely new one.[3] The American Psychological Association [APA] report on intelligence recognizes that "individuals differ from one another in their ability to understand complex ideas, to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by taking thought."[4]

1 Robert J. Stenberg, "Human Intelligence," *Encyclopedia Britannica*, April 11, 2022, https://www.britannica.com/science/human-intelligence-psychology/Development-of-intelligence, accessed on Oct 2, 2023.

2 The human brain is the command center for the human nervous system. Facts about human brain: It weighs about 3.3 pounds (1.5 kilograms). The brain makes up about 2 percent of a human's body weight. The cerebrum makes up 85 percent of the brain's weight. It contains about 86 billion nerve cells (neurons), the "gray matter". It contains billions of nerve fibers (axons and dendrites), the "white matter". These neurons are connected by trillions of connections, or synapses. Our brain uses 20% of oxygen and blood from our body. 60% of our brain is fat. When we learn something new the structure of brain changes. The largest part of the human brain is the cerebrum, which is divided into two hemispheres. Bottom of the brain consist of the brainstem, and trailing that sits the cerebellum. The outermost layer of the cerebrum is the cerebral cortex, which consists of four lobes: the frontal lobe, the parietal lobe, the temporal lobe and the occipital lobe. As all vertebrate brains, the human brain consists of three logical sections known as the forebrain, midbrain and hindbrain. In these three sections contains fluid-filled cavities called ventricles. The forebrain establishes into the cerebrum and hidden structures; the midbrain becomes part of the brainstem; and the hindbrain extends its regions up to brainstem and the cerebellum. The human brain with its complexity acts like a storage device which always stores safely a person's most memorable memories whenever he wants recollect those memories he can. A survey of WHO referred that a baby loses about half their neurons before they are born and also said that when mothers going on speaking with their infants, those children can learn about 300 more words by the age two when comparing with the children whose mothers quite often spoke to them. Here if we measuring of brain activity it shows that during the 6th month of baby, the prefrontal cortex, the seat of forethought and logic, forms synapses at huge rate that it consumes double the energy as an adult brain. That furious pace continues for the child's first decade of life. Birdsong and human speech have similar characteristics. Birds, like humans, learn their complex vocalizations early in life and imitate their adult counterparts to acquire these skills. These two species have evolved a complex hierarchy of specialized forebrain areas where motor and auditory areas interact continuously in order to produce detailed vocalizations. D. Shanthi, G. Narasimha and R. K. Mohanty," Human Intelligence vs. Artificial Intelligence; Survey," in *International Journal of Electronics Communication and Computer Engineering* 6, no. 5 (2015): 30.

3 Stenberg, "Human Intelligence," accessed on Oct. 2, 2023.

4 U. Neisser, G. Boodoo, T. J. Bouchard, et al, "Intelligence: Knowns and Unknowns," *Am Psychol,* 15, 77.

5 Stenberg, "Human Intelligence," accessed on Oct. 2, 2023.

Successful adaptation relies on various cognitive processes, including perception, learning, memory, reasoning, and problem solving. The key focus in defining intelligence is that it is not solely a cognitive or mental process, but rather a purposeful combination of these processes aimed at effective adaptation. Consequently, the physician who becomes aware of a novel disease responds by reading medical literature on the subject, understanding the content, memorizing the essential aspects for treatment, and employing reasoning to address the issue of implementing the information for the patient's benefit. The concept of intelligence has shifted from a single ability to a convergence of multiple abilities. This has not always been obvious to investigators of the subject, however; indeed, much of the history of the field revolves around arguments regarding the nature and abilities that constitute intelligence.[5]

2. The Limits of Artificial Intelligence

The philosopher John Searle established a distinction in 1980 between weak AI (machines behaving intelligently) and strong AI (machines genuinely thinking). The definition of strong AI has evolved to encompass "human level AI" or "general AI," which denotes programs that can solve various tasks, even novel ones, as well as humans.[6] Critics who opposed the idea of machines exhibiting intelligent behavior in weak AI now seem as narrow-minded as Simon Newcomb, who claimed that humans could never solve the problem of aerial flight, shortly before the Wright brothers' successful flight. The fact that there has been rapid progress in recent years doesn't necessarily mean that there are no limits to what AI can achieve. Alan Turing, the pioneer of AI, predicted most of the objections raised by others.[7]

2.1. The Argument from Informality

Turing's "argument from informality of behaviour" says that human behaviour is far too complex by any formal set of rules—humans must be using some informal guidelines that (the argument claims) could never be captured in a formal set of rules and thus could never be codified in a computer program. A key proponent of this view was Hubert Dreyfus, who produced a series of influential critiques of artificial intelligence: *What Computers Can't Do* (1972), the sequel *What Computers Still Can't Do (*1992), and, with his brother Stuart, *Mind Over Machine* (1986). Similarly, philosopher Kenneth Sayre said "Artificial Intelligence *pursued within the cult of computationalism* stands not even a ghost of a chance of producing durable results."[8] The technology they criticized came to be called Good Old-Fashioned AI (GOFAI).GOFAI represents the most basic logical agent design, but capturing every possible behavior with logical rules is extremely difficult. The issue at hand is commonly identified as the qualification problem. Open-ended domains are better handled by probabilistic reasoning systems, while deep learning systems perform effectively in informal tasks. The criticism isn't directed at computers in general, but at a specific way of programming them using logical rules—a way that was popular in the 1980s but has been overshadowed by newer methods.

Dreyfus argues in favor of agents being situated rather than disembodied logical inference engines. An agent with only a limited understanding of dogs through logical sentences is at a disadvantage compared to an agent with real-life experiences with dogs. As philosopher Andy Clark says, "Biological

6 John Searle, "Is the Brain's Mind a Computer Program? *BBS* 3 (1980): 417-457.

7 Refer, B.Jack Copelan, ed., The Essential Turning. Oxford University Press, 2004.

8 Kenneth Sayre, "Three More Flaws in the Computing Model," *Paper Presented at the APA (Central Division) Annual Conference*, Chicago, 1993.

brains are first and foremost the control systems for biological bodies. Biological bodies move and act in rich-real world surroundings." According to Clark, we are "good at frisbee, bad at logic."[9] The embodied cognition approach emphasizes the interplay between the brain, body, and environment in understanding cognition. According to this perspective, the brain cannot be considered in isolation but rather as part of a larger system. Cognition is not solely a product of neural processes but also involves the body's interaction with the environment. This approach suggests that studying the brain alone is insufficient to fully comprehend cognition and instead advocates for a holistic approach that considers the entire system.

The embodied cognition framework emphasizes the importance of robotics, vision, and sensory systems in understanding how the brain and body interact with the environment. These technologies enable researchers to simulate and study the influence of physical actions and sensory feedback on cognitive processes. By considering these technologies as integral components of cognition, the approach expands our understanding of how the mind is shaped by its embodiment and interaction with the surrounding world. Overall, Dreyfus saw areas where AI did not have complex answers and said that AI is therefore impossible; however, we now see many of these same areas undergoing continued research and development leading to increased capability, not impossibility.[10]

2.2. The Argument from Disability

The "argument from disability" makes the claim that a "machine can never do *X*." As examples of *X*, Turing lists the following:

> Be kind, resourceful, beautiful, friendly, have initiative, have a sense of humour, tell right from wrong, make mistakes, fall in love, enjoy strawberries and ice cream, make someone fall in love with it, learn from experience, use words properly, be the subject of its own thought, have as much diversity of behaviour as man, do something really new.[11]

In retrospect, some of these are very easy—we are all familiar with computers that "make mistakes." Computers with metareasoning capabilities can examine their own computations, thus being the subject of their own reasoning. A century old technology has proven ability to

9 Andy Clark, *Being There: Putting Brain, Body, and World Together Again* (Seattle: MIT Press, 1998), 59-60.

10 Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (Noida: Pearson India Education Services Pvt. Ltd., 2023), 1005.

11 Turing, "Computing Machinery and Intelligence," 433-460.

"make someone fall in love with it"—the teddy bear. Computer chess expert David Levy predicts that by 2050 people will routinely fall in love with humanoid robots. As for a robot falling in love that is a common theme in fiction, but there has been only limited academic speculation on the subject.[12] Computers have done things that are "really new," making significant discoveries in astronomy, mathematics, chemistry, mineralogy, biology, computer science, and other fields, and creating new forms of art through style transfer.[13] Overall, programmes exceed human performance in some tasks and lag behind on others. The one thing that is clear is that they cannot be exactly human.[14]

2.3. Measuring AI

Al and Turing, in his famous paper "Computing Machinery and Intelligence," suggested that instead of asking whether machines can think, we should rather ask whether machines can pass a behavioural test, which has come to be called the Turing Test. The test requires a programme to have a conversation (via typed messages) with an interrogator for five minutes. The interrogator then has to guess if the conversation is with a programme or a person: the programme passes the test if it fools the interrogator 30% of the time. To Turing, the key point was not the exact details of the test, but instead the idea of measuring intelligence by performance on some kind of open-ended behavioural task, rather than by philosophical speculation.[15]

Nevertheless, Turing conjectured that by the year 2000, a computer with a storage of a billion units could pass the test, but here we are on the other side of 2000, and we still cannot agree whether any programme has passed. Many people have been fooled when they did not know they might be chatting with a computer. The ELIZA programme and the internet chatbots such as MGONZ and NATACHATA fool their correspondents repeatedly, and the chatbot CYBERLOVER[16] has attracted the attention of law enforcement because of its penchant for tricking fellow chatters into divulging enough personal information that their identity can be stolen.[17]

12 J. H. Kim, C. H. Lee, K. H. Lee and N. Kuppuswamy, "Evolving Personality of a Genetic Robot in Ubiquitous Environment," in *Proc. 16th IEEE International Symposium on Robot and Human Interactive Communication* (2007): 848-853.

13 L. A. Gatys, A. S. Ecker and M. Bethge, "Image Style Transfer Using Convolution Neural Networks," in *CVPR-16* (2016): 2414-2423.

14 Russell, *Artificial Intelligence*, 1006.

15 Turing, "Computing Machinery and Intelligence," 433-460.

16 P. J. Y. Jonathan, C. C. Fung and K. W. Wong, "Devious chatbots-interactive malware with a Plot," in *FIRA RoboWorld Congress*, 2009.

17 M. Humphrys, "How My Program Passed the Turing Test," in R. Epstein and G. Roberts and G. Beber Eds., *Parsing the Turing Test*, Springer, 2008, 237-260.

In 2014, a chatbot called Eugene Goostman fooled 33% of the untrained amateur judges in a Turing Test. The programme claimed to be a boy from Ukraine with limited command of English: this helped explain its grammatical errors. Perhaps the Turing Test is really a test of human gullibility. So far, no well-trained judge has been fooled.[18] Turing Test competitions have led to better chatbots, but have not been a focus of research within AI community. Instead, AI researchers who crave competition are more likely to concentrate on playing chess or Go or StarCraft II, or taking an 8th grade science exam, or identifying objects in images. In many of these competitions, programmes have reached or surpassed human-level performance, but that does not mean the programmes are human-like outside a specific task. The point is to improve basic science and technology and to provide useful tools, not to fool judges.

3. Machines and the Possibility for it to Think

Some philosophers claim that a machine that acts intelligently would not be *actually* thinking, but would be only a simulation of thinking. But most AI researchers are not concerned with the distinction, and the computer scientist Edsger Dijkstra said that "The question of whether *Machines Can Think*…. Is about as relevant as the question of whether *Submarines Can Swim.*"[19] The American Heritage Dictionary's first definition of *swim* is "To move through water by means of the limbs, fins, or tail," and most people agree that submarines, being limbless can swim. The dictionary also defines *fly* as, "To move through the air by means of wings or tail," and most people agree that airplanes, having winglike parts, can fly. However, neither the questions nor the answers have any relevance to the design or capabilities of airplanes and submarines; rather they are about word usage in English. English speakers have not yet settled on a precise definition for the word "think" ─ does it require "a brain" or just "brain-like parts?"

Again, the issue was addressed by Turing. He notes that we never have *any* direct evidence about the internal mental states of other humans - a kind of mental solipsism. Nevertheless, Turing says, "Instead of arguing continually over this point, it is usual to have the polite convention that everyone thinks." Turing argues that we would also extend the polite convention to machines, if only we had experience with ones that act intelligently. However, now that we do have some experience, it seems that

18 My conversation with "Eugene Goostman," the chatbot that's all over the news for allegedly passing the Turing test. Shtelt-Optimized, www.scottaaronson.com/blog/?p=1858, 2014.

19 Edsger Dijkstra, "The Threats to Computing Science," *ACM South Central Regional Conference,* 1984.

our willingness to ascribe sentience depends at least as much on humanoid appearance and voice as on pure intelligence.

3.1. The Chinese Room

The philosopher John Searle rejects the polite convention. His famous Chinese room argument goes as follows: Imagine a human, who understands only English inside a room that contains a rule book, written in English, and various stacks of paper. Pieces of paper containing indecipherable symbols are slipped under the door to the room. The human follows the instructions in the rule book, finding symbols in the stacks, writing symbols on new pieces of paper, rearranging the stacks and so on. Eventually, the instructions will cause one or more symbols to be transcribed onto a piece of paper that is passed back to the outside world. From the outside, we see a system that is taking input in the form of Chinese sentences and generating fluent, intelligent Chinese responses.[20]

According to Searle's argument, it is assumed that the human has no comprehension of Chinese. Due to their nature as inanimate objects, the rule book and the stacks of paper cannot grasp the concept of Chinese language. Due to this fact, it can be concluded that there is a lack of comprehension when it comes to the Chinese language. According to Searle, the Chinese Room is essentially performing the same actions as a computer, leading him to conclude that computers are incapable of generating genuine understanding.

Searle advocates for biological naturalism, a viewpoint that posits that mental states are high-level emergent characteristics resulting from low-level physical processes in neurons. Searle's perspective not only underscores the importance of neuronal properties but also draws attention to the unique characteristics exhibited by neurons, which are absent in transistors. Even though there have been many refutations of Searle's argument, there is still no consensus among scholars. The argument he presents has the potential to be employed (possibly by robots) to assert that a human being cannot possess authentic understanding; this is due to the fact that a human is constructed of cells that lack the capability to comprehend, thereby negating the existence of understanding. It is worth noting that the plot of Terry Bisson's 1990 science fiction story *They're Made Out of Meat* explores the fascinating idea of alien robots visiting Earth and grappling with the concept that chunks of meat could potentially possess consciousness. How they can be remains a mystery.[21]

20 Searle J. R., "Is the Brain's Mind a Computer Program?" *Scientific American* 262 (1990): 26-31.

21 Searle, "Minds, Brains, and Programs," 417-457.

3.2. Consciousness and *Qualia*

At the heart of all the debates surrounding strong AI, lies the fundamental question of consciousness: the ability to perceive the external world, self-awareness, and the subjective experience of existence. *Qualia*, a term that originates from the Latin word meaning "of what kind," is used in technical contexts to denote the intrinsic nature of experience. A question of significant importance is whether machines have the capacity to possess *qualia*. In the movie *2001*, when astronaut David Bowman is disconnecting the "cognitive circuits" of the HAL 9000 computer, it says "I'm afraid, Dave. Dave, my mind is going. I can feel it." Does HAL 9000 actually have feelings (and deserve sympathy)? Or is the reply just an algorithmic response, no different from "Error 404: not found"?[22]

The question of whether animals possess consciousness is a subject of debate, as pet owners firmly believe in it while scientists hold varying opinions. Although crickets adapt their behavior according to changes in temperature, it is not widely believed that crickets possess the capacity to subjectively feel warmth or coldness. Even after centuries of discussion and debate, the problem of consciousness continues to be difficult primarily because it lacks a clear and precise definition. Despite the current situation, there is a chance that aid is *en route*. With the support of the Templeton Foundation, philosophers and neuroscientists have formed a partnership and embarked on a series of experiments that have the potential to bring about significant breakthroughs and offer resolutions to some of the prevailing dilemmas. The agreement among advocates of global workspace theory and integrated information theory is a rarity in the field of philosophy, as they acknowledge the possibility that the experiments could provide evidence in favor of one theory over the other.[23]

Alan Turing concedes that the question of consciousness is a difficult one, but denies that it has much relevance to the practice of AI. "I do not wish to give the impression that I think there is no mystery but consciousness… But I do think these mysteries necessarily need to be solved before we can answer the question with which we are concerned…." Turing's belief resonates with us as we too are intrigued by the prospect of designing programs that exhibit intelligent behavior. It is possible to program and integrate individual aspects of consciousness, including awareness, self-awareness, and attention, into an intelligent machine. Taking on the supplementary task of creating a machine that possesses consciousness identical to that of humans is beyond our current capacity and capabilities. Our shared agreement is that exhibiting intelligent behavior necessitates

22 Russell, *Artificial Intelligence*, 1008-1009.

23 *Ibid*.

a certain level of awareness, which will vary depending on the specific task. Moreover, tasks that involve human interaction require the presence of a model representing human subjective experience.[24] Humans possess a distinct advantage over machines in the realm of modeling experience. Their capacity to access their subjective apparatus enables them to genuinely comprehend and cherish the subjective experiences of others. Take hitting your thumb with a hammer as an example. If you want to understand what *it feels like*, you can replicate the experience by hitting your own thumb with a hammer. Machines, unlike humans, can execute each other's code, but they lack the ability to possess such a capability.[25]

Conclusion

In the realm of one-shot learning vs. multishot learning, AI systems often require an immense number of samples, reaching into the millions or even billions, to achieve a level of learning that surpasses the average human's intelligence, underscoring the superior efficiency of humans as learners. With an astonishing ability to learn, humans can effortlessly absorb new concepts and ideas, sometimes even from a lone sample. The ability to learn from just a single instance is often referred to as one-shot learning. In terms of imagination and recitation, AI demonstrates the remarkable skill to either recall information exactly as presented or create a fresh amalgamation of details, commonly referred to as synthetic recitation. Humans possess the remarkable capability to create ideas and mental images, enabling us to comprehend and imagine things that may not currently exist. When it comes to perceiving and interacting with the environment using multiple senses, artificial intelligence systems in 2023 are generally unable to process multimodal learning. In contrast, autonomous vehicles possess the ability to gather inputs from diverse sources, enabling them to make informed decisions while navigating. Humans, on the other hand, possess the remarkable ability to gather information from all their senses and swiftly process it, enabling them to make informed decisions. Multimodal inputs can be seamlessly assimilated by most individuals, who are then capable of producing outputs in a variety of modalities.[26]

24 Turing, "Computing Machinery and Intelligence," 433-460.

25 Russell, *Artificial Intelligence*, 1009.

26 Micheal Bennett, *Artificial Intelligence vs. Human Intelligence: How are they Different?* June 08, 2023, https://www.techtarget.com/searchenterpriseai/tip/Artificial-intelligence-vs-human-intelligence-How-are-they-different, accessed on Oct. 03, 2023.

In his opinion in the *Guardian* news outlet, Stefan Stern commenting on,

> the famous image of a languorous Adam reaching out his forefinger to that belonging to the benign, bearded God…you don't have to believe in a divine spark. But when Adam reaches out, like that he is, I think, doing what all of us try to do, one way or another, every day. He is trying to be creative, to be human. He is seeking inspiration. He is not a robot. He is something much better than that: infinite, full of potential, unpredictable.[27]

AI is a powerful tool that not only assists and enhances human capabilities but also automates repetitive tasks and handles massive amounts of data. By offering insights, recommendations, and assistance, it can enhance human intelligence, but it cannot substitute it. Combining AI and human intelligence can yield better results in various fields, as they complement each other effectively. The concept of AI surpassing human intelligence, as depicted in science fiction, prompts profound ethical, societal, and philosophical debates. These questions frequently center on matters of control, decision-making, and their potential ramifications for the workforce and society. To ensure that AI systems serve human interests and values, it is essential to take these aspects into account during their development and deployment.[28]

27 Stefan Stern, "Is AI more creative than the human brain? I doubt it – and I definitely want humans to stay in charge," Oct 22, 2023, https://www.theguardian.com/commentisfree/2023/oct/22/ai-more-creative-humans-in-charge-inspiration, accessed on Oct 22, 2023.

28 "Can Artificial Intelligence Replace Human Intelligence," https://chat.openai.com/c/8f42554c-f202-49a0-b318-f026d81696c2, accessed on Oct. 03, 2023.

Part IV

Socio-Political Implications

16

Surveillance Capitalism
Negations and Interrogations

Jose D. Maliekal

Abstract

In this paper, titled, "Surveillance Capitalism: Negations and Interrogations," what is being undertaken is an attempt to dwell on how surveillance has become a culture, the world over, and how it has aided capitalism, to appear in one more of its ever new and evolving Avatars, in the guise of Surveillance Capitalism. After a brief introduction, the work proceeds to present a kaleidoscopic view of surveillance, as a backdrop to the enquiry into the genealogy of Surveillance Capitalism, embedded in Big Data or the Big Other. In the next section, which is titled "Surveillance Capitalism and its Negations," I delve into the evils or negations for humanity and cosmos, for the present and future, being brought about by Surveillance Capitalism, through its instrumentarian power, cutting across the spheres of social justice, political economy and democracy. I have deployed the ethical framework of "Abnormal Ethics", as developed by Nancy Fraser, to foreground the different dimensions of social injustice, built into the multi-layered negations by Surveillance Capitalism. In the next section, titled, "Interrogating Surveillance Capitalism and Its Negations," the work tries to capture the growing resistance to the negations

in the area of personal dignity, privacy, data accumulation and algorithmic distortions and discriminations. The section also, apart from the above praxis-orientation, attempts at building a Kantian theoretical framework of resistance, based on a nuanced reading of the Categorical Imperative and Autonomy. The conclusion crowns these interrogations of Surveillance Capitalism, by raising a few questions, which could be catalytic triggers for future trajectories.

Keywords: *Surveillance Capitalism, Big Other, Dataveillance, Information Civilization, Enclosure, Accumulation, Instrumentarian Power, Asymmetry of Knowledge and Power, Third Modernity, Social Lag, Abnormal Ethics, Parity of Participation, Algorithmic Subjectivities, Ownership of Data, Categorical Imperative.*

Introduction

"The Capitalists will sell us the rope with which we will hang them," says Vladimir Ilich Lenin; "We live in an Administered World," says Theodor Adorno. That surveillance is not an unknown quantity for us, Indians, has been driven home by the two recent events, our nation has witnessed. The first is the passing of the The Digital Personal Data Protection Bill of 2023 (the "DPDP") was signed into law by the President of India, on 11 August 2023. It has been prompted by the GDPR (General Data Protection Regulation) passed by the EU on 24 May, 2016. The second is the Pegasus scandal, which rocked the Indian parliament, and a petition was filed in the Supreme Court of India, requesting the honourable court, a judicial probe to investigate if the Indian government used Pegasus to spy on journalists and other citizens, and if due process was followed. They also submitted to the Court that Pegasus would have a chilling effect on freedom of speech and expression and violate the right to privacy. Also related to perilous and undemocratic event related surveillance is the allegation that the Pune police had planted evidence in the electronic instruments in the use of some of the Bhima Koregaon accused. Not to forget the many a cybercrime, which appear in the Media. In this paper, entitled, "Surveillance Capitalism: Negations and Interrogations," what is being undertaken is an attempt to dwell on how Surveillance has become a culture, the world over, and how it has aided capitalism, to take on one more of its ever new and evolving Avatars, in the guise of Surveillance Capitalism.

1. Towards a Kaleidoscopic Take on Surveillance

Surveillance is a multifaceted concept, usually connected to issues such as power and control, directed from societal authorities in order to control

citizens. Historical discussions have usually drawn upon the Benthamian concept of the panopticon, which was adapted and further developed in Michel Foucault's seminal work *Discipline and Punish*.[1] Foucault claimed that in modern society (18th century onwards), citizens have internalized the eye of the state (a theme also popularized in and through George Orwell's *Nineteen Eighty-Four*). Today, the possibilities to surveil people have been further refined with the implementation of digital communication, and the discussion has evolved from a unilateral focus on top-down surveillance to a broader understanding, where surveillance occurs between different actors and in different spheres of society – a development supported and enhanced by technological developments.[2] Instead of the Orwellian dystopia, in which the individual is monitored by the state, surveillance today permeates everyday life. This development was observed as early as the 1980s by Clarke 1988, when he introduced the concept of "dataveillance". He defined it as "the systematic use of personal data systems in the investigation or monitoring of the actions or communications of one or more persons."[3]

1.1. Baumanian Take on Surveillance

Zygmunt Bauman saw surveillance as the result of need for visibility in a liquid society, in the context of the transition from classical modernity to liquid modernity.[4] While modernity's maniacal need for order had given life to forms and techniques of coercion and regimentation, Bauman saw an epochal change driven mainly by the seductive power of consumerism. Since people's "conduct is made manageable, predictable and hence non-threatening, by a multiplication of needs rather than by a tightening of norms,"[5] the old repression was becoming superfluous or even counterproductive. Such self-monitoring, in particular linking it to the exclusionary policies and growing inequalities that have emerged with globalization. The old panoptical nightmare ("I am never on my own") now

1 Michel Foucault, *Discipline and Punish: The Birth of the Prison* (New York: Vintage Books, 1979).

2 Stefan Gelfgren, Coppelie Cocq, Lars Samuelsson and Jesper Enbom, "Introduction: The complex web of everyday surveillance," in *Everyday life in the Culture of Surveillance*, eds. Stefan Gelfgren, Coppelie Cocq, Lars Samuelsson and Jesper Enbom (Nordicom: University of Gothenburg):10, accessed October 3, 2023, https://doi.org/10.48335/9789188855732-i.

3 Clarke R., "Information Technology and Dataveillance," *Communications of the ACM,* 31, no. 5(1988):499, accessed October 3, 2023, https://doi.org/10.1145/42411.42413.

4 David Lyon, "Liquid Surveillance: The Contribution of Zygmunt Bauman to Surveillance Studies," *International Political Sociology* 4(2010): 325–338, accessed October 4, https://doi.org/10.1111/j.1749-5687.2010. 00109. x.

5 Zygmunt Bauman, *Legislators and Interpreters* (Cambridge: Polity Press 1987), 168.

recast into the hope of "never again being alone" (abandoned, ignored and neglected, blackballed and excluded), the fear of disclosure has been stifled by the joy of being noticed.[6]

After this brief kaleidoscopic of surveillance in general and its evolution, we come to the heart of our work, Surveillance Capitalism.

1.2. Slide into Surveillance Capitalism

For Shoshanna Zuboff, Big Data is the central component of a new logic of accumulation that she calls "surveillance capitalism."[7] Big Data arises from the four uses of the computer-mediated transactions: 'data extraction and analysis,' 'new contractual forms due to better monitoring,' 'personalization and customization,' and 'continuous experiments.' An examination of the nature and consequences of these uses sheds light on the implicit logic of surveillance capitalism and the global architecture of computer mediation upon which it depends. This architecture produces a distributed and largely uncontested new expression of power, which Zuboff calls the 'Big Other.'[8] It is constituted by unexpected and often illegible mechanisms of extraction, commodification, and control that effectively exile persons from their own behavior while producing new markets of behavioral prediction and modification. Surveillance capitalism challenges democratic norms and departs in key ways from the centuries-long evolution of market capitalism. Zuboff stresses how the use of Big Data by corporations and other organizations – in other words, dataveillance – should be seen not as an inevitable technology effect but as the intentional creation of the industry.[9]

1.3. Big Data and the Logic of Accumulation

'Big data,' according to Zuboff, is not a technology or an inevitable technology effect. It is not an autonomous process. It originates in the social, and it is there that we must find it and know it. 'Big data' is above all the foundational component in a deeply intentional and highly consequential new logic of accumulation that is surveillance capitalism. This new form of information capitalism aims to predict and modify human behaviour as a means to produce revenue and market control. Surveillance capitalism has

6 Zygmunt Bauman and David Lyon, *Liquid Surveillance: A Conversation* (Polity Press, 2013), 23.

7 Shoshanna Zuboff, "Big Other: Surveillance Capitalism and the Prospects of an Information Civilization," *Journal of Information Technology*, *30,* no. 1 (2015): 75–89, accessed September 8, 2023, https://doi.org/10.1057/jit.201.

8 Zuboff, "Big Other," 75.

9 Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (London: Profile Books Ltd, 2019), 15.

gradually constituted itself during the last decade, embodying a new social relations and politics that have not yet been well delineated or theorized. While 'big data' may be set to other uses, those do not erase its origins in an extractive project founded on formal indifference to the populations that comprise both its data sources and its ultimate targets.[10]

As a result of pervasive computer mediation, nearly every aspect of the world is rendered in a new symbolic dimension as events, objects, processes, and people become visible, knowable, and shareable in a new way. Now the enduring questions of authority and power must be addressed to the widest possible frame that is best defined as 'civilization' or more specifically – information civilization. Who learns from global data flows, what, and how? Who decides? What happens when authority fails? What logic of accumulation will shape the answers to these questions? Recognizing their civilizational scale lends these questions new force and urgency. Their answers will shape the character of information civilization in the century to come, just as the logic of industrial capitalism and its successors shaped the character of industrial civilization over the last two centuries.[11]

Against this background of the logic of accumulation of the Information Civilization, she defines surveillance capitalism as:

> i) A new economic order that claims human experience as free raw material for hidden commercial practices of extraction, prediction, and sales; ii) A parasitic economic logic in which the production of goods and services is subordinated to a new global architecture of behavioral modification; iii) A rogue mutation of capitalism marked by concentrations of wealth, knowledge, and power unprecedented in human history; iv) The foundational framework of a surveillance economy; v) As significant a threat to human nature in the twenty-first century as industrial capitalism was to the natural world in the nineteenth and twentieth; vi) The origin of a new instrumentarian power that asserts dominance over society and presents startling challenges to market democracy; vii) A movement that aims to impose a new collective order based on total certainty; vii) An expropriation of critical human rights that is best understood as a coup from above: an overthrow of the people's sovereignty.[12]

10 Zuboff, "Big Other," 75-76.

11 *Ibid*., 77.

12 Zuboff, *The Age of Surveillance Capitalism*, v.

1.4. The Everydayness of Surveillance Capitalism

Surveillance capitalism is no longer confined to the competitive dramas of the large internet companies, where behavioral futures markets were first aimed at online advertising. Its mechanisms and economic imperatives have become the default model for most internet-based businesses. Eventually, competitive pressure drove expansion into the offline world, where the same foundational mechanisms that expropriate your online browsing, likes, and clicks are trained on your run in the park, breakfast conversation, or hunt for a parking space.[13]

We are not surveillance capitalism's "customers." We are the sources of surveillance capitalism's crucial surplus: the objects of a technologically advanced and increasingly inescapable raw-material-extraction operation. Surveillance capitalism's actual customers are the enterprises that trade in its markets for future behavior.[14] This logic turns ordinary life into the daily renewal of a twenty-first-century Faustian compact. "Faustian" because it is nearly impossible to tear ourselves away, despite the fact that what we must give in return will destroy life as we have known it.[15]

In the next section, we shall see how this twenty-first-century Faustian Deal with the devil, operates as Negations, at different levels of our life in society, driven by the logic of instrumentarian power, which negates our personal identities and choices.

2. Surveillance Capitalism and Its Negations

Considering that the internet has become essential for social participation, the internet is now saturated with commerce, and that commerce is now subordinated to surveillance capitalism. Our dependency is at the heart of the commercial surveillance project, in which our felt needs for effective life vie against the inclination to resist its bold incursions. It disposes us to rationalize the situation in resigned cynicism, create excuses that operate like defence mechanisms ("I have nothing to hide"). In this way, surveillance capitalism imposes a fundamentally illegitimate choice that twenty-first-century individuals should not have to make, and its normalization leaves us singing in our chains,[16] leading to a series of negations.

13 *Ibid*., 10.

14 Zuboff, *The Age of Surveillance Capitalism,* 10.

15 *Ibid*., 11.

16 Roberto Mangabeira Unger, "The Dictatorship of No Alternatives," in *What Should the Left Propose?* (London: Verso, 2006): 1–11.

2.1. Asymmetries in Knowledge and Power

Surveillance capitalism operates through unprecedented asymmetries in knowledge and the power that accrues to knowledge. Surveillance capitalists know everything about us, whereas their operations are designed to be unknowable to us. They accumulate vast domains of new knowledge from us, but not for us. They predict our futures for the sake of others' gain, not ours. Just as industrial civilization flourished at the expense of nature and now threatens to cost us the Earth, an information civilization shaped by surveillance capitalism and its new instrumentarian power will thrive at the expense of human nature and will threaten to cost us our humanity.[17]

Surveillance capitalism's economic logic is directed through Big Other's vast capabilities to produce instrumentarian power, replacing the engineering of souls with the engineering of behaviour. Instrumentarian power cultivates an unusual "way of knowing" that combines the "formal indifference" of the neoliberal worldview with the observational perspective of radical behaviourism.[18] Instrumentarian society is an historical turning point comparable to the invention of the printing press or the internet. "Continuous streams of data about human behaviour" mean that everything from traffic, to energy use, to disease, to street crime will be accurately forecast, enabling a "world without war or financial crashes, in which infectious disease is quickly detected and stopped, in which energy, water, and other resources are no longer wasted, and in which governments are part of the solution rather than part of the problem."[19] This new "collective intelligence" operates to serve the greater good as we learn to act "in a coordinated manner" based on "social universals." The question "Whose greater good?" How is the greater 'good' determined when surveillance capitalism owns the machines and the means of behavioural modification? "Goodness" arrives already oriented toward the interests of the owners of the means of behavioural modification.[20]

The scheme of instrumentarian logic eschews clear of creative and often messy conflicts of politics, especially democratic politics, as a source of friction that threatens the rational efficiency of the community as a single, high-functioning "superorganism."[21] Individuality is a threat to instrumentarian society, troublesome friction that sucks energy from

17 Zuboff, *The Age of Surveillance Capitalism,* 8-9.

18 *Ibid*., 364-365.

19 Sandy Pentland, "Social Physics: How Good Ideas Spread," *Talks at Google*, 19, YouTube.com, March 7, 2014, accessed October 10, https://www.youtube.com/watch?v=HMB10ttu-Ow, 143.

20 Zuboff, 432.

21 *Ibid*., 433.

"collaboration," "harmony," and "integration." The surrender of the individual to manipulation by the planners clears the way for a safe and prosperous future built on the forfeit of freedom for knowledge.[22]

2.2. The Third Modernity of the Hive

Surveillance capitalism offers a new template for our future: the machine hive in which our freedom is forfeit to perfect knowledge administered for others' profit. Surveillance capitalists work hard to camouflage their purpose as they master the uses of instrumentarian power to shape our behaviour while evading our awareness. That is why Google conceals the operations that turn us into the objects of its search and Facebook distracts us from the fact that our beloved connections are essential to the profit and power that flow from its network ubiquity and totalistic knowledge.[23] Here the comparison with China is useful. An abnormal division of learning marks both China and the West. In China the state vies with its surveillance capitalists for control. In the US and Europe, the state works with and through the surveillance capitalists to accomplish its aims. It is the private companies who sit at the pinnacle of the division of learning, having amassed unprecedented and exclusive wealth, information, and expertise on the strength of their dispossession of our behaviour.[24]

3. Social Injustice in Surveillance Capitalism

The above Negations, which are embedded in Surveillance Capitalism, can be viewed through the lense of social justice, through the conceptual of framework of 'abnormal justice', as developed by Nancy Fraser.[25] Her suggestion is to submit all justice claims to the normative principle of Parity of Participation, based on the basic idea that "*Justice requires social arrangements that permit all to permit to participate as peers in social life.*"[26] Against this framework, personal data accumulation by surveillance capitalists is not only an economic injustice of maldistribution, but also the key mechanism by which two further, more opaque forms of injustice are made possible. The initial injustice of personal data maldistribution can lead to sociocultural misrecognition, which occurs when personal data are subject to algorithmic processing and classification, as well as political

22 *Ibid*., 439.

23 *Ibid*., 443.

24 *Ibid*., 444.

25 Nancy Fraser, "Abnormal Justice," in *Critical Enquiry,* 34, no. 3 (2008): 393-422.

26 Nancy Fraser, "Abnormal Justice," 405.

misrepresentation, which renders people voiceless to challenge any misuse of their personal data.[27]

3.1. Maldistribution and Personal Data: Asymmetrical Accumulation

Beyond the growing concerns about state surveillance or the loss of sensitive personal details in data breaches, the inability of data subjects to access their personal data creates significant injustices of maldistribution in which corporations are able to accumulate vast stockpiles of economically valuable personal data, enabled by strategies of dispossession which obfuscate the digital realm's materiality. Web companies have crafted the internet 'cloud' imaginary as a liminal space between the user and the website, a gap between the real and the virtual, which has encouraged both confusion and public apathy towards the accumulation of personal data by private companies.[28]

3.2. Misrecognition and Personal Data: Algorithmic Subjectivities

Injustices of misrecognition are manifest in sociocultural hierarchies that privilege some individuals and groups over others, preventing some from attaining full parity of participation in society. Maldistribution and misrecognition are thus closely entwined. Uneven access to economic and social goods contributes to the production of unjust status inequalities. In the case of personal data, the division of society into three 'data-classes'; those who create data, those who collect data, and those who can analyze data has serious implications that go beyond economic injustices of maldistribution. The data-class status hierarchy has more serious societal consequences than just the ability to access data and benefit from data analytics—relinquishing access to our data also gives up a degree of control over our lives as we are increasingly subject to classification and profiling,[29] leading to injustices of misrecognition.

3.3. Misrepresentation and Personal Data: Extraterritorial Accumulation

Apart from significant injustices of sociocultural misrecognition the initial injustice of maldistribution is, however, also leading to further injustices of misrepresentation, described by Fraser as political voicelessness. This voicelessness renders some individuals and groups unable to engage as peers in democratic society. When this occurs within a jurisdictional frame, this can

27 Jonathan Cinnamon, "Social Injustice in Surveillance Capitalism," in *Surveillance & Society*, December 2017: 611-612, accessed 11 October, 2023, DOI: 10.24908/ss.v15i5.6433.

28 David Lyon, "Surveillance, Snowden, and Big Data: Capacities, Consequences, Critique," in *Big Data & Society,* 1, no. 2 (2014): 1-13.

29 Cinnamon, "Social Injustice in Surveillance Capitalism," 615.

be referred to as 'ordinary-political misrepresentation'. In the case of data related injustices, this is exemplified by censuses and large-scale surveys that typically undercount minorities, the socioeconomically marginalized, and the homeless, leading to inadequate political representation for these groups and further marginalization. An important aspect of Fraser's third arm of her justice framework however, exposes meta-political injustices which "arise when a polity's boundaries are drawn in such a way as to wrongly exclude some people from the chance to participate at all in its authorized contests over justice"[30]

The ability to challenge forces that oppress you through formal institutionalized processes is a fundamental condition of democracy, however the separation of people from their data is eroding that possibility. The territoriality of personal data accumulation means that if you live outside of the data centre's political jurisdiction, it is likely that a foreign entity is controlling your data, restricting you from making a justice claim regarding access, the ways your data are used, or how it represents you.[31]

4. Interrogating Surveillance Capitalism and Its Negations

After having taken note of the Negations by Surveillance Capitalism, in this section, we engage ourselves in interrogating Surveillance Capitalism, in its Negations. Concretely, this is achieved by capturing the protests against the Instrumentarian negations by caused by Surveillance Capitalism and the attendant social injustices, indicated above. The section ends with a proposal towards a philosophical framework, centered on Kantian framework on Autonomy, to undergird any resistance against the incursions into privacy and other rights by Big Data.

4.1. Challenging the Obstacles to Parity of Participation

To challenge political misframing, Fraser's suggestion is to ascertain the appropriate frame within which to contest injustices by applying the *all-subjected principle,* the notion that "all those who are subject to a given governance structure have moral standing as subjects of justice in relation to it."[32] This principle provides the basis for undertaking a process of *reframing*, to ensure all those subjected to a governance structure are accorded equal consideration regardless of nationality or geographic location. In the case of the personal data injustices of surveillance capitalism, a recognizable governance structure is evident in the accumulation activities of the Web companies that

30 Nancy Fraser, "Who Counts? Dilemmas of Justice in a Post-Westphalian World," *Antipode,* 41 (2010.): 286.

31 Cinnamon, "Social Injustice in Surveillance Capitalism," 617.

32 Nancy Fraser, "Abnormal Justice," 411.

wield great power over their users, increasingly resembling sovereign nations in themselves.[33] A backlash against misframing concerning personal data is steadily growing, notably by Europeans contesting the misalignment between their subjection to the governance structures of Facebook, Google and others and their inability to access European data protection policies. Several recent rulings have made progress against this injustice, such as the 'right to be forgotten'[34] legislation brought against Google.[35]

4.2. Creating New Institutions to Resist Abnormal Injustice

As Fraser articulates in her discussion of the 'how' of abnormal justice, we must create "new institutions for staging and provisionally resolving such disputes democratically, in permanent dialogue with transnational civil society."[36] Potential model for a global agreement is illustrated by the reform of data protection laws underway in the European Union. The EU has developed an update to their 1995 Data Protection Directive, the recently adopted General Data Protection Regulation (GDPR) which is designed to produce a clear legal framework for governing the flow of personal data between EU members and outside EU boundaries, and, importantly, to hold foreign companies accountable to these regulations.

4.3. Dismantling Digital Enclosure to Address Social Lag

Global data regulation and localization initiatives are fundamentally about *preventing* data driven oppression and injustice; dismantling the digital enclosure of surveillance capitalism and placing control of personal data in the hands of data subjects could be a first step towards making further social justice advancements in the areas of socio-cultural recognition and political representation. Some form of ownership could empower people to reap the value of personal data, economic or otherwise, yet considerably more attention from the legal domain is needed, to address the "lag in social evolution", caused by the widespread phenomenon of legislations being co-opted by surveillance capitalists.[37]

33 Anupam Chander, "Facebookistan," *North Carolina Law Review* 90: 1807-1842.

34 Maria Romana Allegri, "The Right to be Forgotten in the Digital Age," in *What People Leave Behind: Marks, Traces, Footprints and their Relevance to Knowledge Society,* eds. Francesca Comunello, Fabrizio Martire and Lorenzo Sabetta (Gewerbestrasse: Springer, 2022), accessed October 4, 2023, 259-263.

35 European Commission, "Factsheet on the "Right to be Forgotten Ruling" - C-131/12 (Brussels: European Commission, 2014).

36 Nancy Fraser, "Abnormal Justice," 416.

37 Zuboff, "Big Other," 83.

4.4. Conferring Ownership Rights to Data Subjects

Conferring ownership rights to data subjects would require updating laws in such areas as informed consent, property, and copyright. Despite surveillance capitalism being powered by personal data, it is largely when individuals are aggregated to groups based on predicted risk and behaviour that economic value as well as injustice are produced, and so a key consideration will be to weigh the relative merits of legally defining personal data as the private property of the data subject, the collective property of a defined group based on the all-subjected principle, or a common property resource for broader public benefit. If some form of collective public ownership of personal data is deemed appropriate, efforts to achieve this outcome might be encouraged by recent academic attention to how privacy infringements in personal data analytics not only affect individuals, but also operate at the group level due to big data's aggregative instincts.[38]

Advocates for a legal right to 'group privacy' thus argue that we must protect not only 'their' privacy (meaning individuals) but also 'its' Privacy. Although privacy is an insufficient concept to characterize the diverse social harms under conditions of surveillance capitalism, privacy does have the advantage of being an existing, widely recognized right protected in most legal systems. Perhaps then a leap to 'group privacy' may not be too far off from legal recognition, however it would likely only serve to prevent harms rather than advance equality and social justice through data control or ownership by data subjects.[39] Together with the acquiring of control over personal or collective data, the advanced data analytics skills are also necessary to enable parity of participation. The question of agency, not just structures against oppression, should be central to any discussion about personal data, especially democratizing data mining, enabling the emergence of "knowing publics", beyond "known publics.[40]

4.5. Holding an Ethical Mirror to Ourselves and the Instrumentarian Capital[41]

Having lived from 1724 to 1804, Kant died 40 years before the invention of the telegraph and thus would have never dreamt of the current ethical challenges we face in the global and digital information age. Nevertheless,

38 Cinnamon, "Social Injustice in Surveillance Capitalism," 620.

39 *Ibid*.

40 *Ibid*.

41 This section draws inspiration from Casey Rentmeester, "Kant's ethics in the age of online surveillance: An Appeal to Autonomy," in *Everyday life in the Culture of Surveillance*, 187-204.

in *Groundwork for the Metaphysics of Morals*, Kant[42] engaged in "the search for and establishment of the supreme principle of morality", which he called "the categorical imperative" and argued that "from this one imperative all imperatives of duty can be derived."[43] Thus, any ethical question as to what obligations we have to one another as rational beings – our duties to each other who fall in the category of rational beings – can be approached from the perspective of Kant's categorical imperative, including questions of online surveillance.[44] McLuhan recognised that "once we have surrendered our senses and nervous systems to the private manipulation of those who would try to benefit from taking a lease on our eyes and ears and nerves, we don't really have any rights left."[45]

Regarding relationships between individual persons and governments, Kant's claim that "the touchstone of everything that can be concluded as a law for a people lies in the question whether the people could have imposed such a law on itself"[46] is highly relevant. He argued explicitly that the government should aspire "to treat men … in accordance with their dignity."[47]

Regarding the second relationship between individual persons and corporations, the most ethically problematic aspect from a Kantian perspective, is the potential of behavioural manipulation that Zuboff noted. We should be reminded of Orwell's warning of the power "in tearing human minds to pieces and putting them together again in new shapes of your own choosing."[48] In speaking of human beings, Kant once said that "we are dealing with beings that act freely, to whom, it is true, what they ought to do may be dictated in advance, but of whom it may not be predicted what they will do."[49]

5. Conclusion: Futuristic Trajectories

As we reach the end of our tryst with surveillance capitalism, few questions, which are also trajectories for the future, emerge. Surveillance capitalism looks irreversible, as much as the process of Globalization. Should we go along with this Irreversibility Dogma of surveillance Capitalism? Should we

42 Immanuel Kant, *Groundwork for the Metaphysics of Morals*, ed. and trans. A. W. Wood (New Haven: Yale University Press, 2002), 7–8.

43 Kant, *Groundwork for the Metaphysics of Morals*, 37.

44 Richad Herschel, R., & Virginia Miori, "Ethics & big data," *Technology in Society,* 49 (2017): 34, accessed 13 October, 2023, https://doi.org/10.1016/j.techsoc.2017.03.003.

45 Marshall McLuhan, *Understanding Media: The Extensions of Man* (New York: Mc-Graw Hill, 1964), 68.

46 Immanuel Kant, *On History*, ed. L. W. Beck, trans. L. W. Beck, R. E. Anchor and E. L. Fackenheim (London: Macmillan, 1963), 7.

47 Kant, *On History*, 10.

48 George Orwell, *1984* (New York: Signet Classics, 1950), 220.

49 Kant, *On History*, 141.

go along with the dictates of surveillance capitalism, foregoing our dignity, enjoying the 'bread-crumbs' falling from the table of phianthropocapitalist largesse.[50] Should we allow ourselves to be in the flow of digitalization, being datafied and commodified? Or would there be/should not be there spaces of resistance, relying on our agency, as subjects, respecting our autonomy and the autonomy of others, in the spirit of Kantian Categorical imperative, without allowing ourselves to becoming "the objects of a technologically advanced and increasingly inescapable raw-material-extraction operation."[51]

The first step, towards retaining a sense of human autonomy, begins with Herbert Marcuse's sense that "all liberation depends on the consciousness of servitude."[52] I like to end with the words of Zuboff, who exactly echo what is in my mind, as I end my reflections here: "Finally, and most important for all scholars and citizens, is the fact that we are at the very beginning of the narrative that will carry us toward new answers. The trajectory of this narrative depends in no small measure on the scholars drawn to this frontier project and the citizens who act in the knowledge that deception-induced ignorance is no social contract, and freedom from uncertainty is no freedom."[53]

50 Emanuela Susca, "Countering "Surveillance Capitalism: The Intertwining of Objective and Subjective Factors," in *What People Leave Behind*, 259-263, accessed October 4, 2023.

51 Zuboff, "Surveillance Capitalism," 10.

52 Herbert Marcuse, One-*dimensional Man: Studies in the Ideology of Advanced Industrial Society* (Boston: Beacon Press, 1964), 7.

53 Zuboff, "Big Other," 86.

17

Artificial Intelligence and Human Labour
Hope for Efficiency Verses Fear of Loss

John Peter Vallabadoss

Abstract

Creative potential of humans is manifested in the form of human labour for productivity, be it physical, mental or intellectual. Leisure and labour are part human activity. Artificial Intelligence poses a challenge to human labour as it is said to be enhancing the efficiency of productivity. Philosophical perspective on human labour in wake of rise of engulfing presence of AI in all walks of life views beyond AI efficiently handling the tools and machines invented, and how humans face the uncontrollable supra-human ability of AI. When it comes to human labour, it takes enormous creativity and authenticity, bringing imagination and originality, critical thinking to bring out a meaningful contribution to the world of work and design. Even AI is the creative potential of Human Labour. At the wake of AI influence in replacing much of human work force, it is a time to rethink and deepen our understanding of the challenges and opportunities the labour community face. We need to identify the innovations, practices, and solutions that these AI managed digital/virtual platforms may use to create decent and inclusive human labour. Positively moving forward in ensuring human welfare we ought to recommend the ways that

the AI revolution could contribute to governing changing labour practices by scaling solutions or alternatively replicate traditional forms of exclusion, fear of loss of human touch, livelihood issues of skilled labourers and unskilled work force. The paper surveys the challenges posed by AI induced technology that threatens human intervention, as it explores the possibilities of growing together with the AI, to be in pace with the technology humans have created, without losing sight of human dignity and welfare.

Keywords: *Human spontaneity, Efficiency, Replacement, Alienation, Mechanized-creativity, Productivity.*

Introduction

Human life and activities are measured and stored as data; All gadgets used become data centers; Every physical movement is calculated; Bodily functions are monitored; The reality today is that human is reduced to a 'bunch of data.' Data collection, storing and use is a serious concern and threat with anthropological, societal, economic and political implications.[1] Digital services replacing humanly offered services, and seemingly providing advantages, create obviously fears of increased dependency[2] and subjugation of humans to machines with Artificial Intelligence. Data is the driving energy, the fuel in all sectors of society, creating a technological revolution where a hype of wild exaggerated expectations is believed to solve major problems of humanity. As well, profound fears of the destructive potential of the new development, are loss of human personal interventions, control and surveillance, oppression of free speech and other human right issues. The high expectations and deep fears are implied with building trust and become aware of its challenges. The enormous speed of data-related developments in the current third decade of this century constantly pose new challenges, reducing human labour potential, stunning acceleration of efficient productivity, feeling suspended and side lined on the data driven globe.

1. Human Vulnerability, Freedom and Dignity

Human vulnerability has somehow is vividly exposed and increased at the fear of being victims of stealing and hacking data, violence and war due to abuse of data.[3] Internet from being a global platform has become more and

1 Christoph Stuckelberger, *Human-centered Data Ethics: How to become Better Humans?* (Geneva: Globethics Publications, 2023), 36.

2 See: "IMD - International Institute for Management Development, *Smart City Index,*" accessed September 30, 2023, https://www.imd.org/smart-city-observatory/home/.

3 Christoph Stuckelberger, *Human-centered Data Ethics*, 39.

more nationalized and controlled by national policies justified by security reasons. The fight about access, control and dominance over data becomes the key factor in the polarized world with few countries that patronize the giant data companies. Data is considered as soft power in geo-political situations. Data manipulation creates not only the ethical dilemmas, but deeper anthropological dilemmas of fundamental question of identity of being human. The hype on new data-related technologies with Artificial Intelligence raises fundamental questions about relation between 'man and machine', 'human ability to produce by labour,' as AI is more than 'mechanics', 'technologies' or 'engineering.' AI gets closer to biological and neurological phenomena, the distinction between human and AI is questioned. Human and non-human (animals and plants) distinction does not serve the purpose as AI is very close to humans, as 'co-human' storing and retrieving data, as human memory data.

Who are we as humans? If humans are being reduced as 'just' data and information, human specificity and identity are lost. If organized data can do many things better and faster, more precise and more powerful than we as humans, the question of who are we, is inevitable. Human functional brain, with connections based on social, spiritual and physical experiences, remains still a powerful tool, in thinking individuals and community and the environment, autonomy and interdependence. Basic roots of human existence with basic philosophical and ethical fundamentals are not totally sidelined. Human personality and reputation built on one's skills, knowledge, power, beauty, connections and performance, is said to be all in all the sum of one's bio-data. In the current narcissistic culture, the influence of social media, the definition of human identity is 'I am somebody because of my ability'; instead, human is somebody because one is loved, not because of one's skills, deeds, etc., but just because of being a human. Data based information and knowledge has a challenge of leading to a rich culture of wisdom and spiritual depth which only humans can give.[4]

Artificial Intelligence brings about profound changes in the lives of human beings, offering enormous potential in improving social coexistence, personal well-being, augmenting human capabilities and enabling or facilitating many tasks that can be carried out more efficiently and effectively.[5] While quantitative changes are underway, qualitative transformation takes place with technology and human relationship, respecting the inherent dignity of each. The objective has to be not to exclude anyone and to expand the areas of freedom that could be threatened by algorithmic conditioning.

4 Christoph Stuckelberger, *Human-centered Data Ethics*, 50.

5 Pontifical Academy for Life, "The Rome Call for AI Ethics and the Abrahamic Commitment," in *Data Ethics: Building Trust*, eds. Christoph Stuckelberger and Pavan Duggal (Geneva: Globethics Publications, 2022), 336.

Technological innovation with AI is a call to encouraging real commitment, guiding our actions and promoting technology to benefit humanity and the environment. However, improved, digital innovation behaving like rational actors but are in no way human, is in fact clear that technological progress affirms the brilliance of the human race. Fundamental condition of human freedom and dignity must be protected and guaranteed while developing and using AI systems. The time has come to question whether new forms of automation and algorithmic activity necessitate the development of stronger responsibility, act as tool for the good of humanity, and protect human rights in the digital era.[6]

As Artificial Intelligence and cyberspace transforms the way humans relate to each other, we are aware of the whole range of issues of justice and human dignity, option for the poor and vulnerable, with increased surveillance and censorship and the deliberate disinformation tactics.[7] The new moral governor, AI has no value in protecting human interest. Created by humans themselves, AI has the potentiality to create its own virtual value space where the masters creators of AI, could become submissive subjects to the world of virtual values. Anonymous, but very powerful governor, AI turns out to be where humans have no other way or help but only to follow AI dictates in choosing a particular life-style.

2. Protection of Labour and Digital Justice

Great advantages of cyber-based data is that there is no limit in time and space and robots can never get tired. The humans have lot of pressure in competing with AI powered robots; Humans need breaks for sleep, eating, and community sharing and refreshing energy. The existing and future potential of digital data world pose a great challenge with regard to the capacity to go on working without break. Except for unwanted break-downs of electricity or hacking-breaks, the accessibility to data, ability to be available 24 hours/7 days/365 days, the AI does not need breaks, weekly off, daily breaks and annual holidays.[8] Becoming better humans, with developing new ways of boundaries, breaks and limitations, with a data Sunday, a data holiday, a data fasting is recommended by 'well-being' masters. Abstain from round-the-clock availability; switch off the data collections devices for a day, to prevent data-burnout-syndromes and to offer data-fasting.

6 *Ibid.,* 340.

7 World Council of Churches Central Committee, "A Vision of Digital Justice," in *Data Ethics: Building Trust,* 334.

8 Christoph Stuckelberger, *Human-centered Data Ethics*, 52.

In the post globalization of the economy and liberalization of trade and industry, the rapid changes in technology and their consequences and ramifications, have effects not only on the nature and structure of industry, on the methods and places of production, but very much on the employment and the necessary skills to retain employability and mobility.[9] Ensuring minimum level of protection and welfare to labour, improving the effectiveness, evolving a healthy relations between ability and productivity, are all meant to increase concerns for protection and welfare of human labourers. Protection and welfare measures include the ability to meet the essential requirements of life and rights. Amid competitiveness in the international levels, the commitment of the workforce to quality and productivity, is dependent on the fair wages and fair share of profits and incentives. The link between productivity and efficiency with regard to improved quality by AI, cannot ignore the resources or potentials that humans have.

Constant updating and up-gradation of skills have to be paid special attention to the problems of the workforce with human limitations of physical age, work energy, mental ability and socio-economic difficulties. Ensuring the rights to freedom, right against slavery and servitude, right to rest, leisure period, holiday with pay and limitation on working hours, and right to standard living and wellbeing, both governments and industries have inviolable commitments. AI driven economy should not be violating or diluting these solemn commitments. Country like India, where population growth poses a major concern of developmental planning, has a steady increase in the proportion of population in the working age group of 15-59. It has resulted in a rapid growth in labour force over the years. Already widespread inequalities in income persisting and poverty, strikes over a quarter of the population. Introduction of AI managed labour market would further widen the gap between haves and have-not, with the considerable decline in workforce in agriculture and allied activities, and stagnancy in employment in many sectors.[10]

It is agreed that the most important factor for the improvement of productivity is the workforce and high productivity is necessary for the survival of the industry in a highly competitive global trade. However, at the cost of human labour, AI would further stand as a threat to already struggling workforce. Competitiveness and low cost of production are achieved through increased productivity, improved quality. Such moves affect the crucial role played by the workers' union in demanding human welfare, affecting thus

9 National Commission on Labour, *Conclusions and Recommendations* (New Delhi: Ministry of Labour, 2002), 4.

10 *Ibid.,* 8.

adversely the interests of the workforce while restructuring the industry with AI animated work.

While globalization and free trade has had an adverse impact on growing economy like India, highly competitive industry, even with IT related software businesses, employs only highly skilled work force, leaving behind huge percentage of population unemployed. The constraints on the modernization with technology, lay a heavy burden for already exploited workforce that are illiterate, socially backward, unskilled, unorganized, uninformed and poor. The technology has to be improved for high productivity, yet labour-based technologies are to be encouraged even with rapid industrialization requiring the use of more advanced technology and skilled personnel. Almost all unskilled and unorganized workforce would turn out to be victims of AI induced industries.

While trade unions representing workers are losing their hold on industry with the struggle for rights of workers, have to innovate new methods of protest and new tactics in the theatre of conflict. Genuine considerations for the welfare of the working class, with fall in employment generation, with policy of 'outsourcing' the production elsewhere, are only added to the AI replacing human labour. Frustration and indignation are bound to increase, with a strong impact on the development of the labour and under the persistent push for modernization. The underutilization of already excess human labour contributes to the problems of unemployment and related issues, even leading to major unrest and revolt among population.

Digital world transforms human life, offering us new ways to communicate, including advocating for human dignity and rights, and for multiple voices to be heard.[11] AI in digital world raises profound issues with power, justice, equity, participation, voices from the margins, as well as human dignity. We wrestle with the continuing and new opportunities and challenges of AI, relationality and vulnerability play a central role in our reflection on justice. The digital divides between those who have the access to new technologies and those who do not; such divide affects human beings in many levels of economic, geographic, cultural and technological accessibility. Access affects power relations and distribution of resources. Inequity is propagated with the control, use and analysis of data with few corporates and governments who have data control and manipulation.

3. AI: Terminators of Labour and Promoter of Alienation?

Humanoid, human-like robots are extensions of the dominant work force of the industrial robots in manufacturing sectors. These robots have replaced

11 WCC Central Committee, "A Vision of Digital Justice," in *Data Ethics: Building Trust,* 328.

millions of human work force and play a key role in production, with efficiency, speed and accuracy. The service robots behaving like humans are already in their upswing, with improved quality of being gentle and designed for direct contact with people. With AI and technical maturity these new batch of labourers are considered better than the industrial robots.[12] Major challenges of shortage of human labourers in special situations, like Pandemic and lesser young population, are addressed by the new arrivals. Early adopters in the market are getting well organized and better positioned to face the increasing demands of quality of care and service in everyday life. The lack of skilled workers in the industry is causing concerns for adaptation to newer machined labourers. The initial experiences of introducing the humanoids showed a great affinity and acceptance of the technology for better service. The automation and digital transformation through service robots show a new physicality of machines, with partially human-like appearance, however triggering philosophical discussions over replacement of the human labour. Increasing robot demand due to lack of skilled staff, is justified as a development simply to support humans with processes which could be automated, where human creativity and innovation are not involved. The routineness and repetitive labour could be easily replaced with automation. Lack of qualified personnel during and after Covid-19 pandemic, many leveraged digital networks to build opportunities.

While machination in industrial manufacturing firms treat human labours as and equalizing with machines, the corporate giants hail AI revolution for reducing the impact of striking labour unions, claiming for their rights and dignity. Labour laws influenced by the corporates, favour not the human persons contributing to the production and those consuming the products, but these regulations force the labourers to sign a contract to work like machines. Instead of machines at the service of humans, humans are made to work like machines, with speed and efficiency, where labour market look for only young and able, well-built work force generation to labour like machine to bring the expecting output in a short span of time. In Marxian terminology, the alienation takes place not only the relation between the owner and the labourer, but also the product and the producer – where the labourer is alienated from the fruit of his labour. Claim for share in the profit made out of the product, is what is contested in the post-Marxian draconian neo-capitalist market economy. AI is furthering the alienation and helps the capitalist-economist-policy makers to further manipulate the human work force. Humans are used only as machinated

12 Sylvia Stocker, "Robots for Social Services," in *Data Ethics: Building Trust,* 240.

work force, and as mere consumers of the product produced with or without human intervention.

Positively moving forward in ensuring human welfare we ought to recommend the ways that the AI revolution could contribute to governing changing labour practices by scaling solutions or alternatively replicate traditional forms of exclusion, fear of loss of human touch, livelihood issues of skilled labourers and unskilled work force. Questions of inclusion and representation of human labour in the digital labour market cannot be sidelined as the monstrous AI giant fulfilling the aspirations of market-profit driven commerce. Care responsibilities and flexibility offered in real human world, have hardly been seen in online platforms where access to and the nature of being trained and compelled to be complying with AI induced dictates. Social protection, benefits and insurance programs, are yet to be explored in the wake of AI dominated forms of work. Our approaches to labour collectives, has to offer an innovative approach bridging traditional work patterns to AI work force, including better organizing the platforms to share experiences and grievances, accessing information about human rights at work and communicating their conditions. When it comes to human labour, it takes enormous creativity to capture authentic perspectives, bringing imagination and originality, critical thinking to bring out a meaningful contribution to the world of work and design. At the wake of AI influence in replacing much of human work force, it is a time to rethink and deepen our understanding of the challenges and opportunities the labour community face. We need to identify the innovations, practices, and solutions that these AI managed digital/virtual platforms may use to create decent and inclusive human labour.

4. Digitalizing the Human and AI's Soft Labour and Smart Work

The development of AI changes human lives and the nature of work, as its power grows. It is opening possibilities to deal with humanity's contemporary challenge. Whether AI is beneficial or not, depends on how we develop it serving human creativity and respective human dignity.[13] Final consumer is the human – even in the digitized world. With AI, the labour force could be reduced considerably, without representation of human elements in the process of production and distribution. Possibility of shared vision on challenges and concerns of AI driven digitalized interactions, is much needed for pooling together the views of a diverse range of stakeholders from industry, development agencies, academics and workers to collectively brainstorm an approach to creating a better future

13 Pontifical Academy for Life, "The Rome Call for AI Ethics and the Abrahamic Commitment," in Data Ethics: Building Trust, 341.

of work.[14] The challenging time is this period of significant acceleration of the digital economy, where the lack of job-security, welfare and safety of many workers are being exposed. The potential impact for representation of human labour influences a change in the conversation between 'humanist' and 'technically savvy' profit interested CEOs of the corporates. Though the products are meant to serve humanity, AI driven production units have least concerns for the inclusive practices.

AI makes it easy and efficient the processes of governance of digitally driven economy, the power of networks in reshaping women's experience in the labour markets and laying out an ultimate vision for an inclusive and fair future of work.[15] A significant contribution to the future work through AI would be to have Smart Work and Soft Labour, beyond the masculine and feminine physical and gender-biased human labour. Deconstructing the gender-bias in labour market has benefits of change to lead an inclusive society and a more sustainable world. AI becomes thus a useful tool. Notions of work, fairness and inclusion of women's work, get changed from traditional occupations to newer vistas of world vista of work in relation to women. Nature of the labour market and the affordances of labour design encourage feminine participation in the economy of the world. Their contribution is much more recognizable than other 'feminine' duties undervalued as of home-making and domestic care work. The nature of women's work and its valuation in the market and in society have been changed, thanks to digitalization with AI.[16] Discourse of social justice has included the notion of fair work as the basis of faire trade. It emphasizes the need for economic actors to facilitate transparency around working conditions and redressal mechanisms, especially in the gig economy and digital platform work.[17] Education and sensitization of workers and employers on the evolving nature of workers' rights and responsibilities in the new digital economy, involve the re-imagination of all players, state and market, in ensuring safety and security of the workers, towards a concern not merely on productivity but also wellness and community. It is pertaining to the recognition of unorganized workers too.

14 Gillian Dowie, Preface, *Feminist Futures of Work: Reimagining Labour in the Digital Economy,* ed. Payal Arora, Usha Rama and Rene Konig (Amsterdam: Amsterdam University Press, 2023), 12.

15 *Ibid.,* 13.

16 Usha Rama, Payal Arora and Rene Konig, Introduction, *Feminist Futures of Work: Reimagining Labour in the Digital Economy* (Amsterdam: Amsterdam University Press, 2023), 18-19.

17 See: https://fair.work/en/fw/homepage, accessed September 22, 2023.

5. Human Flourishing – Our Goal

The productive labour is much counted in both traditional, industrial and today's digitalized world, whether the labour is done by hand (as in artisanal crafts) or by the heart (affective labour, physical and emotional care work), it is recognized and valued. Expanding the classical Marxist notions of use and exchange value of 'productive' labour within a capitalist system, labour by women that occurs within the home, creates the context and support for all other forms of work done outside the home. It has equal importance as it is also recognized as productive and worthy of valuing beyond economic terms. Smart work by AI creates the conditions to differential resistances towards chances of exploitation and precariousness. The liberative potential of AI induced works that values inclusive work, creates more favourable conditions encouraging women not only to enter the workforce but remain within it.[18]

Digitalization continues the historical narrative of technologically assisted human agents, with advanced digital systems which are intelligent and are evolving as self-generative spontaneous agents in their own right.[19] Automated thought and action in AI, are in practical and behavioral domains, with characters automatic, implicit, encoded in memory and heavily supervised by procedural routine and custom. As in human agent, the skill of a labour is obtained by routineness and repetitiveness, even though human labour begins in creative and speculative domains. In any ordinary human experience self-generative agency emerges as innovative, prior to repetitiveness, almost unsupervised. The difference in the artificial agents is, by contrast to human labour, conditioned by modelling through supervision. Process of meta-modelling in both artificial and human agents has implications for 'digitally augmented humanity' to integrate both types of agents and accommodate their different capabilities and potentialities.[20]

Self-generative capabilities and opportunities, throughout human history have been in most individuals and communities as empowering humanity, seeking to overcome deficits and providing opportunities to learn and develop, to expand well-being and the prospects of human flouring.[21] However powerful tool we humans create for utility, we seek human flourishing, welfare of the most vulnerable, enhancing of social relations and development of all, well-being of every individual, equitable economic and technological

18 Usha Rama, Payal Arora and Rene Konig, Introduction, *Feminist Futures of Work*, 19.

19 Refer, P. Norvig and S. Russell, *Artificial Intelligence: A Modern Approach,* 3rd ed. (Pearson, 2010).

20 Peter T. Bryant, *Augmented Humanity: Being and Remaining Agentic in a Digitalized World* (Cham, Switzerland: Palgrave Macmillan, 2021), 31.

21 Refer, Amarthya Sen, "Well-being, Agency and Freedom the Dewey Lectures 1984," in *Justice and The Capabilities Approach* (Routledge, 2017), 3-55.

development, common good of all.[22] As we are made to work in the digital space, the challenges are to enter there with the digital trust for continued and sustained engagement, advancing together with AI, for human welfare. "…we are at an important moment for digital transformation. The need and demand for trustworthy digital services is here and clear."[23] Trust becomes a key value and virtue in all human interactions. Besides careful application of AI with all its ethical concerns, growing erosion of trustworthiness is limiting the adoption of digital services even if they are considered beneficial, as in the case of Covid-19 related digital healthcare services. Security, data protection, reliability, fair use interaction and transparency – all such concerns in digital transformation require a holistic approach in the service of humanity.

For successful transformation where the potential of new AI technologies can be exploited, responsibility of offering trustworthy digital services is always insisted. Traditionally built through interpersonal relationship, trust might be a tricky value in the digital realm. Once lost, it is hard to regain and rebuild. AI driven future work demands that these services replacing human labour have to be trustworthy and those involving in development of AI workforce need to be trustworthy service providers. Still a question is whether AI by its self-generative mechanism would be forced to think in value-education and be accounted for ethical responsibility. The AI behavior would be hard to be determined, which poses an ongoing practical challenge for a solution. New possibilities and development open up new challenges and analytical approach for better interactions with AI virtual services. Creating better conditions for human-AI interactions for ushering in a new era of development and evolution. AI digital transformation offers previously unseen possibilities for rapid services. Omnipresence of digital services play an ever-important role in almost every aspect of human lives, and will do so even more in the future of humanity. For a sustainable and successful innovation, AI cannot be a hindrance but an enabler and a promoter of human flourishing.

Conclusion

As philosophers we are approaching human labour through lenses of social justice, anthropological concerns, inclusive economics, activism for human dignity, gender-bias freedom in labour market. AI challenges us for nurturing emergent identities and policies and principles of equity, flattening the world of work with digitalization, social and anthropological

22 Refer, "Rome Call for AI Ethics," Vatican City, 10 Jan, 2023.

23 Doris Leuthard, Foreword, "How to Build Digital Trust?: Swiss Digital Trust Whitepaper," by Nicholas Zahn and Niniane Paeffgen, in *Data Ethics: Building Trust,* eds. Christoph Stuckelberger and Pavan Duggal (Geneva: Globethics Publications, 2022), 546.

entrepreneurship that seeks to innovate new ways of relating with AI machines as equivalent or more efficient work force. As Marxist paradigm opposed the draconian capitalist exploitation of human labour, philosophers today are grounded with analyses and observations engaging the impact of AI on human labour and provide a ground for rethinking the future of work. The oppression of human labour leads to insensitive dominant group using AI through automation leading to deskilling and increasing precariousness among marginalized society. Even though profiting from the advantages of AI, we are skeptical about non-inclusive and potential oppression related to work and workers. Freed of care and concerns for community, AI induced future of work with technologically mediation, is to be carefully monitored for its shadows casting on the everyday lives of large junk of workers. The forces of digitalization, materially proximal and aspirationally distant for majority of population, especially in Global south, particularly in India. Presence of AI in the digital world is both as an opportunity to expand imaginations of creative human labour, and as a non-negotiable threat to suppress human work as creative for the living. AI design and framework pose challenges by digital interfaces in redefining human labour and concept of ownership of labour. We hope that AI usages in the future would serve to bridge the gender divide, to overcome oppressive practices to enable everyone to negotiate and to promote more equitable growth. AI influenced human labour would experience both liberation and oppression as humans navigate through digital world, application interface and develop creative workarounds.

18

Societies of Control

Victor Ferrao

Abstract

There has been always the fear the machine will become super intelligent and humans maybe enslaved or controlled by them. Here we are not raising this question. Here we are more interested to discern how the power elite in our societies will make use of this new digital power to increase their power over their societies.[1] *We are moving away from disciplinary societies identified by Michel Foucault*[1] *to societies of control of Gilles Deleuze.*[2] *This transition is accelerated by the rapid growth of artificial intelligence. AI seems to have led to the commodification of social control.*[3] *The*

1 See: https://pressbooks.pub/surveillancestudies/chapter/disciplinary-society/#:~:text=A%20disciplinary%20society%20is%20a,describe%20a%20condition%20of%20surveillance, accessed September 29, 2023. See also: Besley T., "Social Education and Mental Hygiene: Foucault, Disciplinary Technologies and the Moral Constitution of Youth," in *Educational Philosophy and Theory*, 34, no. 4 (2002), doi: 10.1080/00131850220000011835.

2 See: http://home.lu.lv/~ruben/Deleuze%20-%20Postscript%20On%20The%20Societies%20Of%20Control.pdf, accessed September 29, 2023.

3 Walter Pascarelli , "AI and the New Age of Social Control," accessed September 29, 2023, https://blog.worldsummit.ai/ai-the-new-age-of-social-control#:~:text=AI%20and%20the%20Commodification%20of,particularly%20through%20digital%20consumer%20technologies.accesed.

attention industry[4] *growing in the web produces data that Big Data Analytics is using to hack us.*[5] *The surveillance powered by AI has commodified social control. We are producing tons of data engaging the digital consumer technologies. Our digital life produced by connecting our smart phones and the Internet of Things is constantly providing data about our intimate life like never before. By feeding on this data, AI and predictive analytics, those willing to pay like big businesses can use the findings of data analytics to influence our behaviour. We have stepped into societies of control. This study is an attempt to understand societies of control. In the first part attempt to understand the impacts of Big Data Analytics, next we enter the zone of predictive analytics, then we enter the zone of data double, finally, we enter the world of the in-Dividual.*

Keywords: *Big data analytics, Data double, Hacking, Societies of control, Disciplinary societies, Predictive analytics, Dividual, In-dividual, Surveillant assemblage, Disassemblage, Panopticon, Banopticon, Synopticon, Attention industry.*

There has been always the fear the machine will become super intelligent and humans maybe enslaved or controlled by them. Here we are not raising this question. Here we are more interested to discern how the power elite in our societies will make use of this new digital power to increase their power over their societies. We are moving away from disciplinary societies identified by Michel Foucault[6] to societies of control of Gilles Deleuze.[7] This transition is accelerated by the rapid growth of artificial intelligence. AI seems to have led to the commodification of social control.[8] The attention

4 Yoric Pinto, "Understanding the Attention economy and Its Implications for the Advertising Industry," https://www.expresscomputer.in/guest-blogs/understanding-the-attention-economy-and-its-implications-for-the-advertising-industry/84174/.

5 Symbakali Gupta and Ayan Mukherjee, "Big Data in Hacking and Social Engineering," accessed September 29, 2023, https://www.degruyter.com/document/doi/10.1515/9783110606058-004/html?lang=en.

6 See: https://pressbooks.pub/surveillancestudies/chapter/disciplinary-society/#:~:text=A%20disciplinary%20society%20is%20a,describe%20a%20condition%20of%20surveillance, accessed September 29, 2023. Also see: Besley, T.

7 See: http://home.lu.lv/~ruben/Deleuze%20-20Postscript%20On%20The%20Societies%20Of%20Control.pdf, accessed September 29, 2023.

8 Walter Pascarelli , "AI and the new Age of Social Control," accessed September 29, 2023, on 29/09/2023.

industry[9] growing in the web produces data that Big Data Analytics is using to hack us.[10] The surveillance powered by AI has commodified social control. We are producing tons of data engaging the digital consumer technologies. Our digital life produced by connecting our smart phones and the Internet of Things[11] is constantly providing data about our intimate life like never before. By feeding on this data, AI and predictive analytics, those willing to pay like big businesses can use the findings of data analytics to influence our behaviour. We have stepped into societies of control. This study is an attempt to understand societies of control. In the first part attempt to understand the impacts of Big Data Analytics, next we enter the zone of predictive analytics, then we enter the zone of data double, finally, we enter the world of the in-Dividual.

Big Data Analytics

New analytics market is fast spreading among us and our intimate data is traded among the companies and Governments. Companies are interested to influence our individual purchasing behaviour while politicians are interested in influencing our individual and collective electoral behaviour. The commodification of social control, thus, is only on the rise. Cambridge Analytica and the Facebook has already shown how commercial data analytics can be used to build voter profiles in order to influence electoral campaigns.[12] Also, Palantir's predictive policing programme demonstrates how commercial entities are developing AI powered tools for the use of the state.[13] By collecting and mining personal data of citizens Palantir is constantly piling up data that is fully available without warrant enabling law enforcement agencies to predict who is about to commit a crime. China has been said to be in the front lane in the use of AI enabled social control of its citizens.[14] We are no longer citizens,

9 Yoric Pinto, "Understanding the Attention economy and Its Implications for the Advertising Industry," https://www.expresscomputer.in/guest-blogs/understanding-the-attention-economy-and-its-implications-for-the-advertising-industry/84174/.

10 Symbakali Gupta and Ayan Mukherjee, "Big Data in Hacking and Social Engineering," accessed 29 September, 2023, https://www.degruyter.com/document/doi/10.1515/9783110606058-004/html?lang=en.

11 See: https://www.oracle.com/in/internet-of-things/what-is-iot/, accessed on 29/09/23.

12 Nicholas Confessore, "Cambridge Analytica and Facebook: the Scandal and the Fall out so Far," accessed September 29, 2023, https://www.nytimes.com/2018/04/04/us/politics/cambridge-analytica-scandal-fallout.html.

13 See: https://www.axios.com/2018/04/20/peter-thiel-palantir-software-tracks-americans-silicon-valley, accessed September 29, 2023.

14 See: https://www.ispionline.it/en/publication/how-china-uses-artificial-intelligence-control-society-23244, accessed September 29, 2023.

perhaps we are fast becoming statizens[15] because we are all netizens today. Governments are keeping a digital eye on their citizens and are using AI to control their behaviour. The Government now can use drones enabled with face recognition AI to conduct surveillance upon problematic areas. Chinese company Biadu is said to be in the forefront in this venture.[16] Governments are increasingly putting up face recognizing AI in public spaces.

We have social credit system in China established in 2014 for instance, that is using AI and Data Analytics to reward and sanction individuals and companies for their social and economic behaviour.[17] This indicates that China is using AI to engineer social control. Earlier every employee as well as citizen was equal. The employee was given equal salaries for the work he/ she puts in. Today's corporations rate their workers based on their performance and reward or punish them and in the similar fashion a citizen is rated and rewarded or discredited by the Government. Its system black lists individuals. The number of discredited individual is 2020 alone reached 2.498.[18] This public labelling and shaming is an old technique enhanced by AI. More and more countries are likely to make use of these technologies of digital transformation of their societies. With the growth of digital technologies coupled with AI there is no hiding place under the sky and no citizen can cover himself or herself from the prying eyes of AI enabled digital technologies. Therefore, data driven social control is likely to be on the increase. Such a social credit system can be misused and put up oppressive regimes. This form of surveillance combines the panopticon of Michel Foucault[19] with what we may be called banopticon. This is close to the fate Kafka's Joseph K who constantly returns to the doors that are regularly closed to him by a malevolent overseer.[20] The discredited

15 See: https://thewire.in/rights/goodbye-citizenship-hello-statizenship, accessed September 29, 2023.

16 See: Jack Clark, "Import AI 196: Baidu wins City Surveillance Challenge; Covid Surveillance Drones; and a Data Set for building TLDR Engines," accessed September 29, 2023, https://jack-clark.net/2020/05/04/import-ai-196-baidu-wins-city-surveillance-challenge-covid-surveillance-drones-and-a-dataset-for-building-tldr-engines/.

17 Nicole Kobie, "Complicated Truth about China'a Social Credit System," accessed September 29, 2023, https://www.wired.co.uk/article/china-social-credit-system-explained.

18 Mo Chen and Grossklag, "Social Control in a Digital Control Society: Case Study of Chinese Social Credit System," accessed September 19, 2023, https://www.mdpi.com/2076-0760/11/6/229.

19 See: https://www.ucl.ac.uk/bentham-project/who-was-jeremy-bentham/panopticon#:~:text=The%20panopticon%20penitentiary%2C%20from%20the,supervision%20of%20unskilled%20workers%20by, accessed September 29, 2023.

20 See: https://www.gutenberg.org/files/7849/7849-h/7849-h.htm, accessed 29 September, 2023.

individual then is likely to be banned from state benefits. His or her personal accounts may be frozen and his or her travel plans may be also disrupted.

Our societies are today controlled without prohibitions. This means our societies are organized without walls. We are under the command to enjoy. This is mainly achieved by the application of information processing and predictive data analytics which make use of personal data of the consumers to lure them. Society of control controls our access. We need digital gate passes.[21] Our cars have toll tags and license-plate cameras at point of intersections. Google maps show the way as well as reveal our location. LinkedIn puts together employment histories. Amazon watches our reading habits. Facebook compiles our vacation stories. Tinder stores our romantic tastes. Several retailers catalogue our shopping histories. The data, thus, accumulated is collected by data brokers (Axiom and Live Ramp).[22] By using cross reference and co-relation, they link the common number of someone who regularly uses Uber with the same someone who spends a lot of time on Tinder. Similarly, they connect usernames, overlapping addresses and resembling faces. This information links aspects of professional, recreational, romantic, political etc., aspects of our life which are dispersed in the several platforms of the internet. These intersecting of names across data sets is called identity resolution[23] and it constantly builds our profile that remains known to us. This data of our intimate life is then sold by the data brokers to businesses. Thus, voracious social media platforms like LinkedIn and Tinder as well as online vendors like Amazon and Wayfair buy these data sets. Even in the YouTube the suggestions that we organically receive extend from what we are already watching making us trapped in our comfortable eco-chamber. Facebook constantly tweaks to keep people interested about the next round of selfie-illustrated gossips. Disciplinary societies observed and tormented people. Societies of control analyse and sell. We are put in the hands of profit seeking tycoons who use predictive analytics to enhance their economic opportunities. Unfortunately, most of us are not alarmed about our intimate data is being ogled by corporate surveillance especially when discounts and other benefits are attached to the products offered to us.

21 See: https://adda.io/blog/2019/07/7-reasons-why-you-shouldnt-use-gate-management-ledgers-anymore/, accessed September 29, 2023.

22 Bastian P. "Live Ramp (formerly Axiom) - selling You – Without Owning Your Data," accessed September 29, 2023, https://d3.harvard.edu/platform-digit/submission/liveramp-formerly-acxiom-selling-you-without-owning-your-data/.

23 https://www.techtarget.com/searchcustomerexperience/definition/identity-resolution#:~:text=Identity%20resolution%20is%20a%20data,and%20identifying%20non%2Dobvious%20relationships.

Predictive Analytics

Predictive analytics is an advanced form of data analytics that answers the question, 'what can happen next?'[24] As the name suggests, predictive data analytics forecasts the future outcomes. The process employs data analysis, machine learning, artificial intelligence, and statistical models to find pattern to predict future behavior with a great deal of precision Today with the rise of Big data, huge data sets are available for data mining and predictive insights are offered for business, policing, administration, medical care etc. The advancement of Big data machine learning has advanced the capabilities of predictive analytics. Data scientists study selected sets to identify correlations between elements in the given data sets and predict future trends and performances. The use of predictive analysis enables us to make sound decisions, increases profit making in businesses, and reduces risks by identifying dangers at the early stage. Predictive analytics finds its applications in online retail, health care, risk reduction, education, fraud detection, improvised market campaigning, weather forecasting, social media analysis, cyber security, recommendation and search engines, administrative needs and other Government needs etc.[25]

There are about Eight Steps Employed by Data Scientists to Perform Data Analytics:[26]

a) Define problem statement - The data scientist defines the project outcome, sets up the scope the efforts, determine the objectives and identify the data sets to be used.
b) Data collection - It involves the gathering of raw data that is necessary to perform predictive analysis from authorized sources.
c) Data Cleaning - The data scientists refine data sets. This process involves removing of un-necessary, erroneous, duplicate and redundant data from the data sets that are set apart to perform predictive data analytics.
d) Data analysis - The data scientists explore the data to find patterns or outcomes. Data analysis offer useful information that manifests patterns or trends.
e) Build predictive models - Data scientists make use of algorithms to build predictive models based on the observed patterns. Here AI becomes very useful.

24 See: https://cloud.google.com/learn/what-is-predictive-analytics, accessed September 29, 2023.

25 See: https://www.turing.com/kb/predictive-analytics-with-ml, accessed September 29, 2023.

26 *Ibid*.

f) Validation - The data scientists check the efficiency of the predictive model by performing several tests. The model is checked for its accuracy by testing it on some samples. Model is studied in simulated conditions.
g) Deployment - The scientists them employ the predictive models in real conditions. This means that a model is put to real use.
h) Model monitoring - The data scientists regularly monitor the predictive model to check its performance and to ensure that it gives proper results.

There are Two Models of Predictive Analytics:[27]

a) Classification Model - Data scientists put data objects like consumers or potential outcomes into one category or the other. Thus, for instance, if the retailer has a huge data sets, it can be used to categorize the kind of his consumers and their specific needs. One friend of mine who is involved in data analytics told me that he had advised the owner of a mall in Japan to keep beers and nappies together. In Japan the rate of divorces were high and the babies were with the fathers. This is why after a long stressful day, the men would come and buy beer and nappies. It was found that the mall that kept beer and nappies in the closest selves exhibited optimal sales.
b) Regression Model - Data scientists attempt to predict continuous outcomes such as how much revenue a customer would generate during his or her relationship with the company.

There are Three Techniques of Predictive Data Analytics:[28]

a) Regression analysis - Regression is a statistical analysis technique that estimates the relation between variables. It is often used to determine how a change in one or more variables affects another variable. Thus for instance, Data scientists use regression analysis for instance to determine how increase in price will affect the sale of a product.
b) Decision trees - They are classification models that place data into different categories based on distinct variables. Data scientists make use of this method to understand individual decisions. The model appears like a tree with each branch standing for a potential decision and the leaf projects the possible result of that decision.
c) Neural networks - Data scientists make use of neural networks or machine learning methods to do predictive analysis of complex data sets. Essentially, they powerhouse pattern recognition engines. Neural networks are used to determine non-linear relationships in data sets,

27 Clay Halton, "Predictive Analytics: Definition, Model Types and Uses," accessed September 29, 2023, https://www.investopedia.com/terms/p/predictive-analytics.asp#:~:text=People%20often%20confuse%20predictive%20analytics,%2C%20regression%2C%20and%20neural%20networks.

28 *Ibid*.

especially when no mathematical formula exists to analyse the data set. Neural networks can be employed to validate the results of decision trees and regression analysis.

The central issue in the context of predictive analytics is access to data. Data being knowledge. It gives power to those that have access to it.[29] Data protections laws often do not apply to the Governments. Most humans are objects of data. Therefore, immense control is given to Government and other power elites. Deleuze points out that when the individual human being becomes data, he or she becomes a dividual.[30] It divides the human person in accordance to the categories that they use to analyse data about him or her. The big data world has led to a new form of human body, a form of becoming that transcends human corporeality and reduces flesh to mere information.[31] This new body is our data double which involves multiplication of the individual (dividual) and constitution of an additional self.[32] This additional self stands in the future and is constituted by Big Data Analytics. Societies of control are open societies but what is closed in such societies is access to information and therefore to power.[33] Disciplinary societies were closed societies that disciplined individual without forcing them.[34] Societies of control are based on the command to enjoy. But what an individual does is archived and used again to influence the future becoming of the same individual. An individuals who apparently feels free is actually controlled through predictive data analytics. The same is true of collectives or groups.

Data Double

When the internet first came into being, we saw it as a platform of freedom and unregulated activity. But over time it got regulated for what is deemed as the good of our society. But with the advent of AI, there is real fear that those who write the code that regulates the web might well have more power and influence over the rest of us. This inequality of power is important to us as each of us has a data double, digital duplicate of our lives captured in

29 Michel Foucault Power Knowledge relations. See: https://www.powercube.net/other-forms-of-power/foucault-power-is-everywhere/, accessed September 29, 2023.

30 Micaela Ott, "Dividual Subjectivations in the Society of Control," accessed September 29, 2023, https://ul.qucosa.de/api/qucosa%3A71584/attachment/ATT-0/.

31 See: Isa Seow, "Deleuze and Data Syestems: the Concept 'Access' in the Control Society,'" accessed September 29, 2023, https://medium.com/@seow.isa/deleuze-and-data-systems-the-concept-of-access-in-the-control-society-26cb8c1cf75d.

32 *Ibid*.

33 *Ibid*.

34 Sovereign Societies used force in public on the public.

data and spread across assemblages of information systems.[35] The systems of surveillance abstract human bodies from their territorial settings and separate them into series of discrete flows which are then reassembled into distinct data doubles which can be studied and put to targeted individuals for vested interest or their good as in the case of medicine.[36] This means bodies are de-corporealized and rendered data double of pure virtuality.

As we surf through the internet or move with our smart phone, it is difficult to know what is being captured of our life. It is even more difficult to know to what ends that our data self are made to serve. The services that we seek on the web, discounts that we avail, opportunities that we enjoy or are shut out from are increasingly defining our data doubles and the algorithms in which they are analysed render us vulnerable to be exploited for commerce or politics. Students today for instance, are subject to data capture, mining and analysis using learning analytics and methods. Using nudging techniques and predictive statistics, a student's data double is made a subject of critical analysis that influences admissions decisions, financial aid packages, motivational messaging, strategic advising etc. We may therefore, modify the famous dictum of Rousseau that says 'man is born free and is everywhere in chain' to be 'humans are born free and are immediately electronically monitored.'[37]

Prof. Kevin Warwick of Reading University was the first self-proclaimed cyborg.[38] He had implanted silicon chip transponder into his forearm. The surveillance potential of this technology has been used to monitor pets and may be even used to monitor the sick and the old. It could be used to monitor those who hold gun licences. If they enter a school for instance, where they are not welcome, the computer of the school would sound alarm bells alerting everyone inside. While there are good and bad sides of cyborgized surveillance, the new surveillant assemblage does not approach the body in the first instance as a single entity to be moulded, punished or controlled.[39] What it does is capture the body as it navigates different flows. Thus, for instance, photography captures the body in its flow of light waves. This means body itself is an assemblage whose parts and processes are broken down for

35 Kyle M. L Jones, "What is a Data Double," accessed September 29, 2023, https://datadoubles.org/2018/ 05/01/what-is-a-data-double/.

36 *Ibid*.

37 Kevin D. Haggerty and Richard V. Ericson, "The Surveillant Assemblage," accessed September 29, 2023, https://www.researchgate.net/publication/12188129_The_Surveillant_Assemblage.

38 *Ibid*.

39 *Ibid*.

observation. A lie detection test for instance, aligns and compares assorted flows of respiration, pulse and electricity.

Thus, the surveillant assemblage standardizes the flesh/information flows of the body. This renders the body into pure information that can be analysed and put to use for governance, commerce and control. This indicates that surveillant assemblage in operation today transcends the flesh and reduces the body into pure information.[40] This new body becomes a data double. It then circulates in host of different calculation centres which then enable access as well as its denial to resources, services and powers in ways that are not known to its referent. Thus, for instance, our biometric IDs are gate passes to access airports, banks etc., so also our PAN card is watch keeping tool on use or abuse of the banking system. This means our data self or data double is the object to which governmental and marketing practices are directed.

The surveillant assemblage in operation today is not one of soul formation through self-monitoring achieved through the panopticon as demonstrated by Michel Foucault. Surveillant assemblage of today is one of seduction to be consumer in a market economy or that of becoming an obedient uncritical citizen. It domesticates us for the market and reigning mode of politics.[41] That is why they construct several profiles of a single individual and attempt to influence him or her to achieve their vested interest. The sale of data doubles is to which we have the challenge to be awaken. Otherwise, our society of control will fast devolve into a burnout society where we are over fad and fatigued on the same. Data analytics bundles us along with what they deem as individual of similar tastes. Foucault's panopticism dissolves into synopticism.[42]

A large number of individuals are targeted with the same product or service when it comes to commerce and fear and bias when it comes to politics. We are more and more moving into psychopolitcs.[43] This is why I opine that our society becomes transformed into a society of banality of evil.[44] Evil no longer shocks us. It loses its monstrosity and we can then kill invoke the name of God. We are now into a synopticon of the society of control. In the olden days people living in a cities enjoyed anonymity which gifted them freedom and privacy. The individual in the city, therefore, could

40 *Ibid*.

41 *Ibid*.

42 Kevin D. Haggerty and Richard V. Ericson, "The Surveillant Assemblage."

43 See: Byung Chul Hans, *Psychopolitics of Power*: *New Liberalism and New Technologies of Power* (London: Verso, 2017).

44 Victor Ferrao, "When Speech Turns Silent," accessed September 29, 2023, https://www.heraldgoa.in/Edit/Opinions/When-speech-turns-silent-/208289.

disappear into invisibility. Today with the rise of surveillant assemblage this disappearance has disappeared. Everyone one is visible to the surveillant assemble.[45] Everyone is vulnerable and can be seduced and exploited for gains in commerce or politics. Everyone is now under the gaze even those who do not have voluntary digital profiles are not free. If one wants to go away from the gaze, one will have to be content with bare life.[46] This means one cannot access banking system, work and use the internet. Access to services of both market and the government requires us to embrace to be the objects of the gaze of surveillant assemblage. This is why we have the challenge to become wakefully aware of the challenges of society of control and critically find emancipative ways of resistance.

The In-Dividual

Our data-ized self is a schizo. It is atomizing and fragmenting ourselves. The self has become a dividual in a society of control. We do have the challenge to come to face to face with our reality. An individual is not really an isolated singularity. An individual is always a dividual. An individual is not One but is in-One.[47] This in-One is not all-One. The individual does not have to die into the all-One. Finding one's rightful place in-One may be an important means of emancipatively responding to the society of control. The individual is an in-dividual. This means our behaviour online influences the services that we are enjoying. Thus, for instance, Amazon keeps a tab on what I buy. Let us say I buy important books of Philosophy and read them on my Kindle as well as take notes about them on it. This data that is mechanically readable can in it turn lead to the production of books. The publishers are interested in the reading tastes of the readers. Hence, it will influence the production or publication of books. With the coming of the digital books and e-readers, bookstores has fast been closing down. The coming of OTT platforms has also harmed the prospects of cinema theatres as well as types of films that are produced. This is why our resolute participation in the digital world can change the world itself. This potential agency that we have has to be nurtured and developed in order to respond and resist the lures and entanglement of the society of control. We as participants are not simply at the mercy of the society of control. We can respond critically and emancipatively and control the society of control. This means the society of control is not Sisyphean

45 Kevin D. Haggerty and Richard V. Ericson, "The Surveillant Assemblage."

46 Georgio Agamben, *Homo Sacer*, *Sovereign Power and Bare Life* (Stanford: Meridian, 1998).

47 See: Roco Gangle, *Fracois Laruelle's Philosophies of Difference*: *a Critical Introduction and Guide* (Edinburg: Edinburg University Press, 2013).

world[48] that seals our fate and render us helpless. We can participate creatively and intelligently to let the data analytics work for our good and the good of our society.

Our participation in the digital world cannot lose sight of that fact that we as individuals are in-One relation with the world and others. We cannot escape our being-in-the-world. We as in-Dividual have to work with other in-Dividuals. The internet already allows crowd sourcing of information and action. We have the challenge to be proactive right in the middle of the world of Big Data Analytics. We can therefore, work to get our wired data-ized bodies to work for our advantage. This way we may be able to resist surveillant assemblage from becoming simply means of commercial or political ends or propaganda tools. This is why we have to become the wakefully aware of the seductions that invites us to self-stage ourselves to serve the attention industry. Perhaps, we have the challenge to become militant intellectuals that understand the allurements of the attention industry and wage a collective war against the same. We cannot choose to be individuals in isolation. That position will imprison us into our echo-chambers and we shall miss the diversity of the experience that the digital world provides as well as not be able to effectively respond to the lures and entanglements of world of Big Data bubble. We have the challenge to invent new forms of disobedience or new forms of disassemblages that will produce new forms of resistance.[49] This forms of resistance has to equip us to give up what Deleuze and Guattari calls molar identities and accept molecular multiplicities as a way of life.[50] The challenge is to resonate with the society of control to sing a dissonate note. This means we cannot allow our agency be weaken or robbed.

Spell of the societies of control is indeed strong. But it is possible to expel the bewitching spell. To rise up to an emancipative response to the societies of control, we have the challenge to give up the Latin etymology of the word Individuum where the prefix in is the negation of dividuum. The notion of the individum drives from its Greek roots and is stressing the transcendence of the individual.[51] It is forgetting the immanent or molecularising aspect of the individual. This means we do not look upon the individual as a self-contained atomised entity but the individual as a being-in-the-world is in dynamic relation with the world. That is why we

48 Myth of Sisyphus of Albert Camus.

49 See: https://www.youtube.com/watch?v=44b14rknSFc, accessed September 29, 2023.

50 John Brandy, "Deleuze's Becoming Molecular: An Introduction," accessed September 29, 2023, https://medium.com/@jn.bradi/deleuzes-becoming-molecular-b5c636732296.

51 Gerald Raunig, *Dividuum*: *Machinic Capitalism and Molecular Revolution*, Vol. 1. (South Pasadena: Semiotext, 2016).

think of the individual as in-Dividual. This means that the individual is in relations and therefore dividual and it is that very aspect of being a dividuum can be a source of resistance to society of control. The becoming of the divisibility of the individual can be the line of resistance.[52] We therefore, have to give up the myth of domination of one over many and have to reject all in-One metaphysics and embrace a metaphysics of in-One. We then have to focus on the 'in' of the in-One relation and attempt to transform it in emancipative ways. The 'in' in the in-One relation indicates the interrelation and transversality[53] of the dividual. It is only in riding the chaosmosis[54] of the societies of control that we will be enabled to bring order or cosmos into it. Being a dividual is a disjointed experience. The dividual is always in the gaps. The dividual is a disjointed multiplicity. It will never be joined completely in One. It is in-One. It is a disjointed-disobedient disassemblage. To find our freedom in the society of control, we have the challenge to burst its self-amplifying bubble that tempts our ego. This does not mean that we have to move from the One to all-One (die into Community/ nation). It is challenge for us to remain in-One. The time is certainly out of joint.[55] But we do not have set it right by joining it into One. We have the challenge to remain in a tensive in-One to continuously set it right. We have the challenge therefore, to give up the fallacy of jointedness, which is myth of world without rupture.

Conclusion

Our study reveals that we have stepped into a society of control and each of us have a data double that is being archived by the data power barons. This data that is mined is used to control us to suit big businesses or political vested interests. But we are not up against the wall. In the middle of this vulnerabilities, we have possibilities of resisting and move the world of Big Data Analytics towards emancipative ends. To this we have become creative disobedient in-Dividuals who creatively navigate collectively and turn the assemblage of Surveillance into Disassemblage of resistance for our good and the good of our society.

52 Like the line of flights of Deleuze.

53 Border crossing ability.

54 Chaotic conditions of our society that can rise to a higher order. See: Felix Guattari, *Chaosmosis*: *An Ethico-aesthetic Paradigm* (Indianapolis: Indiana University Press, 1995).

55 See: http://www.shakespeare-online.com/quickquotes/spiteright.html#:~:text=Hamlet%20Glossary%20%2D%20The%20time%20is%20out%20of%20joint&text=out%20of%20joint%2C%20utterly%20disordered,that%20of%20setting%20a%20clock, accessed September 29, 2023.

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Social Media and Artificial Intelligence

Biju Rockey Pallippadan

Abstract

Today, AI can create social media posts for you. It can draft and target social ads. It can automate monitoring. And it powers most of what you see on any given social network; "AI in social media" market is projected to grow from $633 million in 2018 to more than $2.1 billion by 2023, according to estimates. Our smartphone uses AI to power voice assistants and real-time navigation. Our favourite companies, like Amazon and Netflix, use AI to recommend products and content. And email clients like Gmail even use AI to automatically write parts of emails for you. AI's most impressive capabilities are powered by machine learning. Machine learning is a subset of AI that allows machines to make accurate predictions based on large sets of data. The smartest AI tools then actually improve the accuracy of their predictions over time. This ability to get smarter, often without direct human involvement, makes AI very powerful. Currently, many machine predictions help with decisions, but humans still make the final call. In the future, governments might let machines make simple decisions. But what if a decision made by a machine is wrong or unethical? Who's responsible? Is it the AI system, the one who made the AI, or the person who used its

data? These are some of the tough questions that the world is going to face.

Keywords: *AI, Social media, Facebook, Instagram, Tiktok, Snapchat.*

Introduction

The foundation of Artificial Intelligence is said to have been laid in the early 19th century when Charles Babbage, an English mechanical engineer, and polymath, began his work on the concept of a programmable computer. Considered the "father of the computer," he invented the first computer, a fully automated calculating machine designed to perform mathematical calculations. The input was to be provided to this mechanical devise machine via punched cards. For output, the machine would have a printer, a curve plotter, and a bell ("Computer"). Engineers had to wait another four decades before the introduction of the first operational electronic digital computer. It was the Colossus, built in 1943 by British engineer Tommy Flowers to help decrypt coded messages sent by the German military. Colossus may be regarded as the first artefact that may be classified as an instance of AI.

1. Historical Development of AI

It is important to notice that during these early days, the focus of computer engineering was on technical matters, i.e., making electronic machines for scientific and military applications. The goal was to increase speed and efficiency in the data processing. Interestingly enough, engineers could design and build an "intelligent" system, without understanding the nature of intelligence at all.

Interest in theoretical questions surrounding AI began with Alan Turing, regarded as the founding father of artificial intelligence and of modern cognitive science. He was a leading early exponent of the hypothesis that the human brain is in large part a digital computing machine. He theorized that the cortex at birth is an "unorganized machine" that through "training" becomes organized "into a universal machine or something like it." (Aron)[1]

2. AI in Modern World

In a ground-breaking paper published in *Mind* in 1950, Turing raised the question: "Can a machine think?" He contended that this question should be replaced with the question, Can a machine respond to queries in the way humans do, so that the interrogator is unable to judge whether the answer

1 Copeland B., "Alan Turing," *Encyclopaedia Britannica*, accessed August 17, 2023, https://www.britannica.com/biography/Alan-Turing.

came from a computer or a person?" Turing proposed what subsequently became known as the Turing Test (TT) as a criterion for whether an artificial computer is thinking.

Turing's paper marked the beginning of a field of research that would later develop into what we know today as Artificial Intelligence. The field comprises technical matters and theoretical discussions. It got recognized as an independent field of study during the now-famous, summer conference in 1956 at Dartmouth College in Hanover, New Hampshire. The conference is memorable for coining the term 'AI' and establishing it as an independent field of research (Artificial Intelligence 2018). *It was and still continues to be defined in terms* of TT, as is seen in Nilsson's (1998) textbook for the field. In this book, he defines AI as the field devoted to building an artefact able to pass the TT.[2]

Work on AI has mushroomed since the Dartmouth conference in 1956. Research is extensive and diverse, covering all of the aspects of artificial intelligence. Hans Moravec (1999), in his *Robot: Mere Machine to Transcendent Mind*, informs us that in the speed of computing computer hardware doubles every 18 months. "Fourth generation" robots, he predicts, will soon enough exceed humans in all respects, from running companies to writing novels. These robots, he forecasts, will evolve to such lofty cognitive heights that we will stand to them as single-cell organisms stand to us today.[3] Keeping pace with technological advancement are debates on theoretical issues. Today, many theoretical issues related to AI such as the nature of intelligence, moral responsibility, soul, subjectivity, etc. are widely discussed. This is evident from the abundance of material available on the internet, and the reader has only to do an internet search to find up-to-the-minute discussion on the topic.[4]

Despite the stunning growth in content and usage, interestingly, scholars don't agree on what exactly defines intelligence and how the term can be applied to artefacts. Clarity on these questions goes a long way in making our study focused and clear. This paper aims to bring clarity and precision to what AI is.

3. Artificial Intelligence: A Modern Approach

As Russell and Norvig (2009) tell us, the main unifying theme is the idea of an artificially intelligent agent. The "artificial" part seems self-explanatory:

2 Stuart J. Russell and Peter Norvig, *Artificial Intelligence a Modern Approach* (New Jersey: Pearson Publishers, 2003).

3 Stephen Hawking, "AI Robots will Take Over Humans Completely," accessed September 25, 2023, https://www.indiatoday.in/education-today/gk-current-affairs/story/stephen-hawking-ai-robots-taking-over-humans-1076444-2017-11-03.

4 *Ibid*.

entities are called artificial if they are manufactured by humans out of pre-existing materials. But what about "intelligence"? In traditional philosophy, intelligence is associated with the mental capacity of human beings (or other living beings) who can perceive, predict, manipulate, and understand the world around us.[5] Analogously, the word can also be used to denote the mental capacity of other living beings that manifest similar characteristics. But what about the so-called "intelligence" when the term is applied to artificial systems? Optimistic researchers in AI point out that today machines outperform humans in many fields and claim that before long artefacts will outperform humans in all fields that we traditionally associated with the mind – perceive, predict understand and manipulate the world around us.[6]

Demis Hassabis CEO of AI company DeepMind, calls artificial intelligence the "science of making machines smart."[7] That means we teach machines to mimic human intelligence. We give them the ability to see, hear, speak, move, and write. We see these abilities on display in the smart hardware and software you use every day. Our smartphone uses AI to power voice assistants and real-time navigation. Our favourite companies, like Amazon and Netflix, use AI to recommend products and content. And email clients like Gmail even use AI to automatically write parts of emails for you. AI's most impressive capabilities are powered by machine learning. Machine learning is a subset of AI that allows machines to make accurate predictions based on large sets of data. The smartest AI tools then actually improve the accuracy of their predictions over time. This ability to get smarter, often without direct human involvement, is why AI is so powerful.[8]

Today, AI technology uses data to make increasingly relevant and accurate predictions about everything from what product you want to buy next to which ad campaign to run to which content topics to cover on your blog. AI can read and write using natural language generation and natural language processing. It can detect and mimic tone of voice using sentiment analysis. It can detect images, video, and faces using image recognition and computer vision capabilities. AI can even predict performance and recommend actions.

4. Various Social Media Platforms

In the computer age AI in social media usage is at its peak on the following platforms such as Facebook, Twitter and so on.

5 Stuart J. Russell and Peter Norvig, *Artificial Intelligence a Modern Approach,* 87.

6 Terry Dartnall: *Artificial Intelligence and Creativity* (Dordrechta-Netherlands: Kluwer Academic Publishers, 1994), 108.

7 Terry Dartnall: *Artificial Intelligence and Creativity,* 108.

8 *Ibid*.

4.1. Facebook

Facebook was perhaps one of the premier platforms that introduced Artificial Intelligence usage on social media platforms. They have actively implemented multiple AI-powered tools to improve the overall user experience. Facebook was also one of the first platforms that used Artificial Intelligence tools to create an engaging user browsing experience. Facebook suggests people and content to users based on their choices, web preferences, and browning pattern.AI helps Facebook serve its users better. Facebook marketing is the new sensation in online marketing, and most of its strategies are based on Artificial Intelligence tools.[9]

4.2. LinkedIn

LinkedIn is a purely professional space that uses Artificial Intelligence and machine learning extensively. LinkedIn uses various techniques of machine learning and artificial intelligence techniques to sort out candidates based on the company's specific needs. AI in social media platforms like LinkedIn helps its members to connect with suitable candidates for their businesses, while those looking for jobs locate suitable employers as per their qualifications. For example, AI-powered tools used on the platform suggest the candidate's names in a priority hierarchy according to their job search status. In addition, it helps to create a pleasant and positive communicative environment between the job seeker and employer. LinkedIn also uses AI extensively for rating candidates' performance. The entire information processing and sorting of candidate profiles on LinkedIn are done through complex AI algorithms.

4.3. Pinterest

If you have visited Pinterest to find ideas for your new room decor or ideas for organizing your bookshelf, you just know how precise Pinterest is regarding suggestions. Pinterest will recommend and display only relevant results according to the searches, and this is done by none other than Artificial Intelligence tools. Pinterest analyses user sentiments and employs a specific neural networking tool to analyse user preferences and boost pins. This ensures you are not overburdened with irrelevant information when using Pinterest. Using AI in social media platforms like Pinterest is a perfect example of a streamlined search using AI.[10]

9 "Importance of Artificial Intelligence in Social Media," accessed September 20, 2023, https://ccomsys.net/million-dollar-idea/importance-of-artificial-intelligence-in-social-media/.

10 Nilam Yaraghi, "How should social media platform combat misinformation and hate speech," accessed Sept 20, 2023, https://ccomsys.net/million-dollar-idea/importance-of-artificial-intelligence-in-social-media/.

4.4. Twitter[11]

Twitter is a platform known for using face detection, mitigating the use and spread of inappropriate images, and filtering inappropriate comments and posts. Who do you think does that? Twitter is a powerful social communication platform used globally to spread important messages. Therefore, filtering and monitoring Twitter content is of great importance. AI in social media platforms like Twitter helps detect fraud propaganda and do away with hateful accounts. Twitter also uses various AI-powered tools for face detection – a unique Twitter feature.[12] You would love to use Twitter because of the numerous engaging features like reply suggestions while commenting on posts or answering a tweet. However, none other than Artificial Intelligence tools help Twitter to do so and provide an excellent user experience. Various other social media platforms are using Artificial Intelligence in their daily functioning. These are:

4.5. Instagram

Instagram, is a photo and video sharing social networking service owned by American company Meta Platforms. The application allows users to upload media that can be edited with filters, be organized by hashtags, and be associated with a location — via geographical tagging. Instagram is a free social media platform for sharing photos and videos with your followers. It's become an especially popular way to connect with brands, celebrities, and friends. Instagram has an array of features, from short-form videos to live streaming and private messaging.[13]

4.6. Tik Tok

TikTok is a popular social media app that allows users to create, watch, and share 15-second videos shot on mobile devices or webcams. With its personalized feeds of quirky short videos set to music and sound effects, the app is notable for its addictive quality and high levels of engagement.[14] TikTok allows users to watch, create, and share short videos online. The app

11 The erstwhile Twitter is changed into "X", when Elaon Musk acquired it in October 2022, for a deal of $44 Billion; the website Twitter.com is changed into "X.com".

12 Nilam Yaraghi, *How Should Social Media Platform Combat Misinformation and Hate Speech*, accessed September 20, 2023, https://ccomsys.net/million-dollar-idea/importance-of-artificial-intelligence-in-social-media/.

13 See: https://www.businessinsider.com/guides/tech/what-is-instagram-how-to-use-guide?IR=T, accessed Sept. 20, 2023.

14 Mahendra Dutta, *Unleashing the Power of AI: Game-Changing Facebook Advertising Features for Performance Marketing Success,* accessed August 15, 2023, https://www.linkedin.com/pulse/unleashing-power-ai-game-changing-facebook-features-marketing-dutta.

was launched in 2016 by the Chinese technology company ByteDance. Now available in more than 150 different markets, TikTok has offices in Beijing; Los Angeles; Moscow; Mumbai, India; Seoul, South Korea; and Tokyo. The app saw about 1 billion global daily users by early 2022. Despite political pressure on operating in certain places such as the United States, it continues to gain in popularity.

4.7. Snapchat

Not only social media platforms, even our mobile phone uses Artificial Intelligence to recognize our voice commands and write based on what we speak with voice typing features. The core goal of Artificial Intelligence is to mimic human intelligence. Even the movies and series that Netflix recommends to you for binge-watching are all guided by AI-powered tools based on what genres you usually like to watch. Isn't that incredible?

5. Benefits of Using AI in Social Media

After having learnt so much about how popular social media platforms use Artificial Intelligence and AI-powered tools, let us study the key benefits of using AI in social media platforms:

5.1. To Recognize Images

Did you know that the world's leading social media platform, Facebook (Meta), uses Artificial Intelligence to recognize your face in photos and suggest who you want to tag? Platforms like Snapchat use AI to study your facial specs and advise filters accordingly. The impact of using AI in social media is enormous. Facebook also uses various other tools to analyze inappropriate image content like nudity and prevent posting them on the platform. AI can, for example, predict and even recommend actions through image recognition and computer vision capabilities.[15] Artificial intelligence has dramatically helped to prevent data and identity thefts. Apart from it, users can use platforms better when they know their data is secure.

5.2. AI-Powered Chatbots

AI-powered customer service chatbots have escalated the automaticity of customer communication to a large extent. As a result, AI can help brands cater to their customer needs and queries better. AI in social media marketing is potent for answering customer queries precisely. It can simultaneously handle a large volume of data and offer tailored customer service. Businesses

15 Simplilearn, *Social Media: Advantages and Disadvantages*, accessed October 2, 2023, https:// www.simplilearn.com/real-impact-social-media-article.

can now bid farewell to mismatched time-zones and inefficient time loss in the case of human customer representatives. In this way, users can get instant problems to solutions, and their grievances are addressed more readily. AI-powered chatbots help provide optimum customer support and improve business quality.

5.3. Analysing Sentiments

AI is a highly beneficial tool for discovering brand new audiences based on recent trends and sentiment analysis. AI usage in social media marketing focuses on improving customer targeting and user customization. Artificial Intelligence can detect sentiments by listening to the voice of the user. AI uses the customers' past browsing history, choices, and purchase history to improve the user-centricity of the websites and e-commerce sites. AI can also analyze users' comments to predict how the user feels about the service. These are immensely useful for business modification. Making customer journeys hassle-free, tailor-fit, and smooth will improve brand loyalty and help create a solid customer base who shall keep returning to the brand. In a nutshell, Artificial Intelligence helps in refined content targeting.

5.4. Increased Security

Artificial Intelligence helps social media platforms protect user data and enhance customer privacy. For example, AI checks, monitors, and improves security chains for the information that users post online. AI has introduced various techniques, such as pattern detection, fraud prevention, and user authentication, to enhance the security facets of social media. You can even use AI-powered tools to protect and lock various private information and content on your personal social media accounts.[16]

6. Social Media Marketing

Artificial Intelligence helps social media marketers establish close bonds with customers and analyze their preferences. Data thus obtained are valuable indicators of brand performance and often shape product lines. Apart from the primary benefits of AI in social media mentioned above, there are certain secondary usages of Artificial Intelligence in social media marketing. These are:

6.1. Revenue Acceleration

Artificial Intelligence uses the data obtained from social media to help customize a brand-appropriate strategy and thus gain maximum benefits. AI

16 Simplilearn, *Social Media: Advantages and Disadvantages,* accessed August 10, 2023, https://www.simplilearn.com/real-impact-social-media-article.

also helps to learn about various words, images, or tags that can generate maximum revenue.[17]

6.2. Reduce Operational Costs

Artificial Intelligence helps to reduce operating costs for social media campaigning. First, through automation of various intermediate tasks involved in social media campaigning of a brand, and second, by promoting effective marketing, thereby reducing unnecessary costs. Artificial Intelligence helps in customizing the browsing experience. Thus, artificial intelligence is a powerful tool to improve customer satisfaction and make interfaces user-friendly. Artificial intelligence uses complex algorithms to analyze user actions and recommend ads and products accordingly.[18]

Posting and engaging on social media is a key piece of most companies' digital marketing strategy - and AI can help you do it better than before. In social media, AI can both accelerate revenue and reduce costs across a range of use cases. Accelerate Revenue. Because AI learns from data, it can use data from social media audiences to accelerate revenue in a number of ways, such as: Learn which headlines, words, and images in posts lead to the most engagement, discover new audiences and trends based on sentiment analysis, predict who will take action and buy more, then target them with social messages.[19]

Using AI in social media offers several advantages social media marketers can leverage to achieve better client results. Here are some of the key benefits: Increased Efficiency and Accuracy, Better Audience Targeting, Improved Customer Service, Cost Reduction, Crash Course: Generative AI For Agencies, Increased Efficiency And Accuracy. By using AI, social media agencies can increase their efficiency and accuracy in delivering content and analysing data.

7. Improved Customer Service

AI-powered chatbots can provide 24/7 customer service, reducing response times and improving the overall customer experience. With NLP, chatbots can provide personalised and meaningful interactions that mimic a human's tone and style, making it easier for customers to get the help they need.

17 *Ibid.*

18 *Ibid.*

19 *Ibid*.

8. Cost Reduction

Businesses can reduce their social media management and overall marketing costs by automating tasks and improving efficiency. AI-powered tools can analyse data and provide insights that would take human analysts hours to produce (such as keyword research and idea generation), reducing the need for manual labour and the associated costs.

9. Philosophical Issues

Philosophically, the main AI question is "Can there be such?" or, "Can a machine think?" What makes this a philosophical and not just a scientific and technical question is the scientific recalcitrance of the concept of intelligence or thought and its moral, religious, and legal significance. Can a machine act intelligently? Can it solve any problem that a person would solve by thinking?[20] Are human intelligence and machine intelligence the same? ... Can a machine have a mind, mental states, and consciousness in the same sense that a human being can?

10. The Ethical and Social Implications of AI

When decisions that were traditionally made by humans are handed over to algorithms and AI, there's a risk that the capacity for moral reasoning could be compromised. This implies that relying solely on AI might diminish the human ability to engage in thoughtful ethical thinking.[21]

11. Can a Machine Display General Intelligence?

Is it possible to create a machine that can solve all the problems humans solve using their intelligence? This question defines the scope of what machines could do in the future and guides the direction of AI research. It only concerns the behavior of machines and ignores the issues of interest to psychologists, cognitive scientists and philosophers, evoking the question: does it matter whether a machine is really thinking, as a person thinks, rather than just producing outcomes that appear to result from thinking.

The basic position of most AI researchers is summed up in this statement, which appeared in the proposal for the Dartmouth workshop of 1956: Every aspect of learning or any other feature of intelligence can be so precisely described that a machine can be made to simulate it. Arguments against the basic premise must show that building a working AI system is impossible because there is some practical limit to the abilities of computers or that there is some special quality of the human mind that

20 See: ep.utm.edu/artificial-intelligence/#:~:text=Philosophically%2C%20the%20main%20AI% 20question, %2C%20religious%2C%20, accessed September 15, 2023.

21 *Ibid*.

is necessary for intelligent behavior and yet cannot be duplicated by a machine (or by the methods of current AI research). Arguments in favor of the basic premise must show that such a system is possible.[22] AI is the ability of a computer, or a robot controlled by a computer to do tasks that are usually done by humans because they require human intelligence and discernment. Although there is no AI that can perform the wide variety of tasks an ordinary human can do, some AI can match humans in specific tasks.

12. Ethical Challenges of AI

Immanuel Kant's ethical philosophy underscores three key principles: autonomy (the ability to make one's own decisions), rationality (using logic and reason to make choices), and moral duty (following ethical obligations). Application to AI in Governance: The act of delegating decision-making processes to AI systems carries the risk of eroding the capacity for nuanced moral reasoning. Letting machines decide instead of humans might weaken the important ideas of Kantian ethics.[23]

12.1. Job Displacement and Socioeconomic Impact

Automation powered by AI can lead to job displacement in certain industries. The resulting socioeconomic impact, including unemployment and income inequality, poses ethical questions about the responsibilities of governments and organisations in addressing these consequences.[24]

12.2. Threat to Moral Reasoning

When decisions that were traditionally made by humans are handed over to algorithms and AI, there's a risk that the capacity for moral reasoning could be compromised. This implies that relying solely on AI might diminish the human ability to engage in thoughtful ethical thinking.

12.3. Challenges of Codifying Ethics

Attempting to translate ethics into explicit rules for robots or AI-driven governmental decisions is highlighted as a challenging task. Human morals are very complex, and it's tough to make these complicated ideas fit into computer instructions.

22 John-Stewart Gordon, *Ethics-of-artificial-intelligence,* accessed August 29, 2023, https://iep.utm.edu/ethics-of-artificial-intelligence/.

23 "Exploring the Ethical Implications of AI," The Hindu, accessed October 3, 2023, https://www.drishtiias.com/daily-updates/daily-news-editorials/exploring-the-ethical-implications-of-ai.

24 *Ibid*.

12.4. Lack of Accountability & Transparency

It can be difficult to assign responsibility when something goes wrong with an AI system, especially when it involves complex algorithms and decision-making processes.

12.5. Informed Consent

AI systems can be used to collect and analyse personal data without the knowledge or consent of the individuals involved. This raises concerns about informed consent and the right to privacy. Some research says that machines can, in a way, be responsible for their actions in ethical terms. James Moore (Professor at Dartmouth College) classified machine agents related to ethics into four groups.[25]

12.6. Ethical Impact Agents

These machines, like robot jockeys, don't make ethical choices themselves, but their actions have ethical effects. For example, they could change how a sport works. These machines can make and explain ethical judgments. Adults and advanced AI with good ethical understanding fall into this category.[26]

Conclusion and Critical Appraisal

Overall, using AI in social media can help businesses achieve their goals faster, more accurately, and at a lower cost. By automating routine tasks and providing valuable insights, marketers can focus on creating high-quality content and engaging with their audience, ultimately improving their presence on social media. Increasingly, machines and Artificial Intelligence (AI) are assisting humans in decision-making, particularly in governance. Consequently, several countries are introducing AI regulations. Government agencies and policymakers are leveraging AI-powered tools to analyse complex patterns, forecast future scenarios, and provide more informed recommendations. However, the use of AI in decision-making comes with challenges. AI can have built-in biases from the data it learns from or the viewpoints of its creators. This can result in unfair outcomes, posing a significant obstacle to effectively incorporating AI into governance. Currently, many machine predictions help with decisions, but humans still make the final call. In the future, governments might let machines make simple decisions. But what if a decision made by a machine is wrong or unethical? Who's responsible? Is it the AI system, the one who made the AI, or the person who used its data? These are some of the tough questions that the world is going to face. Putting ethics into machines is tough, and everyone needs to be careful moving forward.

25 *Ibid*.

26 *Ibid*.

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Artificial Intelligence, Self-Respect and Democracy A Rawlsian Analysis

Jeff Shawn Jose

Abstract

The significant progress in AI, exemplified by Large Language Models (LLM) such as ChatGPT and Gemini, has broadened public exposure to AI. This increased public engagement with AI complicates discussions about its potential and challenges. While the use of AI in democratic processes like elections and governance has sparked reasoned debates about its potential and challenges, self-respect is an often-overlooked aspect in these discussions. This paper analyses AI's influence on democracy from the point of view of John Rawls' conception of self-respect. John Rawls is an influential political philosopher who reinvigorated twentieth-century political thought. For Rawls, the conception of self-respect is an essential aspect of a just society. Through this paper, I attempt to answer the following question: What is the impact of AI on democracy when viewed through the lens of Rawls's conception of self-respect?

Keywords: *Artificial intelligence, John Rawls, Democracy, Self-respect, Surveillance.*

Introduction

Even though discussions and developments in Artificial Intelligence research gained momentum in the second half of the twentieth century, the first quarter of the twenty-first century is witnessing both the benefits and burdens of its practical applications. The recent emergence of LLMs such as ChatGPT and Gemini further complicates the landscape by bringing AI into the mainstream. In comparison to Netflix, which took three and a half years to get one million subscribers, Chat GPT garnered 100 million monthly users after a month of its release.[1] The practical application of AI in democratic processes, such as elections and governance, has sparked reasoned debates about its possibilities and challenges. AI has the potential to enhance the election process by assisting both citizens and politicians and increasing the efficiency of governance. On one hand, AI can help citizens better understand the policies and promises of candidates and express their opinions more effectively. On the other hand, AI also carries the risk of manipulating citizens' choices through disinformation and misinformation.[2] For instance, the Cambridge Analytica scandal underscored the potential for manipulating citizens' choices using their personal data on social media without consent.[3] The increasing pervasiveness of AI in the realm of democracy necessitates a critical analysis of the current state of affairs.

This paper critically analyses AI and its impact on democracy through the lens of John Rawls's conception of self-respect. John Rawls is an influential political philosopher who revitalised twentieth-century political thought. Katrina Forrester states: "The impact of Rawls's theory on the landscape and language of political philosophy was immense."[4] For Rawls, the conception of self-respect is an essential aspect of a just society. Even though there is a considerable number of papers on Democracy and AI and the Approach of Technology from Rawls's perspective, the conception of self-respect has received little attention. This paper seeks to bridge this gap by examining, from Rawls's perspective, the impact of AI on self-respect and Democracy. Towards this end, I will begin by sketching a general overview of AI and its impact on democracy, followed by an outline of

1 Sarah Kreps and Doug Kriner, "How AI Threatens Democracy," *Journal of Democracy,* 34, no. 4 (2023): 122.

2 Adam Michael, "Artificial Intelligence, Democracy and Elections" (European Parliamentary Research Service, 2023), https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI (2023), 751478.

3 Mathias Risse, "Artificial Intelligence and the Past, Present, and Future of Democracy," Carr Center Discussion Paper Series, 2021.

4 Katrina Forrester, *In the Shadow of Justice: Postwar Liberalism and the Remaking of Political Philosophy* (New Jersey: Princeton University Press, 2019), x.

Rawls's understanding of self-respect. Finally, I will critically analyse the impact of AI on democracy through the prism of self-respect.

Artificial Intelligence and Democracy

The influence of AI extends across different levels of human interactions, encompassing the individual, societal, and institutional domains. To analyse the impact of AI, particularly on democracy, it is essential to outline its fundamental characteristics, including recent developments. Even though there is no consensus on the definition of Artificial intelligence, the following statement highlights its general feature: "The field of artificial intelligence (AI) can be characterised as the investigation of computational systems that exhibit intelligent behaviour (including algorithms and models used in these systems)."[5] These AI-based machines "are potentially capable of imitating or even exceeding human cognitive capacities, including sensing, language interaction, reasoning and analysis, problem solving, and even creativity."[6] This capability is enabled by neural networks, which mimic the functioning of neurons in the human brain, and is grounded in machine learning or deep learning, which operates on the principle of learning from data. As a result, machines can exhibit intelligence akin to humans through self-adjustment and self-correction.[7] In general, technologies using AI can be divided into Artificial General Intelligence (AGI) (Strong AI) and Artificial Narrow Intelligence (ANI) (Weak AI). AGI refers to the prospect of creating a type of AI that could be intelligent like human beings or be like superhumans and can perform a wide variety of tasks. In contrast, ANI refers to the type of AI which are specialised in doing tasks particular to a domain. The LLMs such as Chat GPT and Gemini are examples of ANI. Andreas Jungherr observes: "…in examining the impact of AI on democracy, it is important not to get sidetracked by imaginary AGI and instead focus on specific instances of narrow AI…."[8]

In addition to existing AI machines, the release of Chat GPT and the subsequent arrival of LLMs such as Gemini has popularised AI, thereby opening the possibility of impacting democracy. LLMs are called generative AI because of their ability to synthetically generate unique content, such as

5 Ron Sun, "Artificial Intelligence," in *A Companion to Cognitive Science*, ed. William Bechtel and George Graham (Oxford: Blackwell Publishers, 1999), 341, https://doi.org/10.1002/9781405164535.ch26.

6 World Commission on the Ethics of Scientific Knowledge and Technology, "Preliminary Study on the Ethics of Artificial Intelligence" (UNESCO, 2019), 3, https://unesdoc.unesco.org/ark:/48223/pf0000367823.

7 World Commission on the Ethics of Scientific Knowledge and Technology, 3.

8 Andreas Jungherr, "Artificial Intelligence and Democracy: A Conceptual Framework," *Social Media + Society* 9, no. 3 (July 1, 2023):2, https://doi.org/10.1177/20563051231186353.

text, images and music, based on user prompts. It is estimated that "[b]y 2026, 90% of online content may be synthetically generated."[9] Notwithstanding the advantages, the challenges posed by synthetically generated output to democracy are immense. The potential of disinformation and misinformation by AI "can lead to deception on a large scale and damage public trust in democratic institutions."[10] Even though the manipulation of information is not new to the political landscape, the arrival of AI has increased the pace and quantity of misinformation produced and blurred the lines between discerning fact and fiction. This can result in "citizen disengagement with the political landscape if they cannot discern fact from fiction."[11] This disengagement can escalate if AI becomes tools to negatively impact society, such as by perpetuating social inequalities by reinforcing racial and gender bias, engaging in surveillance to exercise control over citizens and aiding in attaining power and suppressing the rights of the citizens. This, consequently, erodes trust and spirit in democracy.[12] Hence, there is a need to critically analyse the impact of AI on democracy, and this is done in this paper through Rawls's conception of self-respect.

Rawls, Democracy and Self-Respect

Rawls's understanding of a just society is democratic. He states in the preface of *A Theory of Justice* that his conception of justice (justice as fairness) "constitutes the most appropriate basis for a democratic society."[13] Furthermore, he adds in the preface of its revised edition that "the central ideas and aims [of justice as fairness are]… as those of a philosophical conception for a constitutional democracy."[14] However, Rawls doesn't continue to refer to the term democracy in his works explicitly. Amy Gutmann observes that in Rawls' two key works, namely *A Theory of Justice* and *Political Liberalism*, only the former has an entry on democracy. However, she argues that Rawls implicitly identifies "…democracy with the broad

9 Michael, "Artificial Intelligence, Democracy and Elections," 1.

10 *Ibid*., 3.

11 Sarah Kreps, R. Miles McCain, and Miles Brundage, "All the News That's Fit to Fabricate: AI-Generated Text as a Tool of Media Misinformation," *Journal of Experimental Political Science* 9, no. 1 (2020): 18, https://doi.org/10.1017/XPS.2020.37.

12 Fernando Filgueiras, "The Politics of AI: Democracy and Authoritarianism in Developing Countries," *Journal of Information Technology & Politics* 19, no. 4 (October 2, 2022): 450, https://doi.org/10.1080/19331681.2021.2016543.

13 John Rawls, *A Theory of Justice: Revised Edition* (Cambridge: The Belknap Press of Harvard University Press, 1999), xviii.

14 *Ibid*., xi.

ideal of free and equal citizens in a fair system of social cooperation…"[15] Jeffrey Bercuson says that the starting point of democracy for Rawls is from the individual and their need for certain primary goods. Furthermore, the social values and the good of the community are extracted from the good of individuals.[16] Notably, self-respect is one of the primary goods for Rawls, which will be explored further in this paper.

Rawls accords a pride of place to self-respect among the primary goods. The prominence of self-respect is clear when Rawls states, "… perhaps the most important primary good is that of self-respect."[17] Primary goods are the basic goods free and equal persons need in a democratic society to pursue their higher order interests. Higher order interests are basic, regulative and effective interests to exercise and advance two moral powers of citizens. There are three higher order interests: a capacity for a sense of justice, a capacity for a conception of the good (two moral powers) and protecting and advancing a citizen's conception of the good.[18] A sense of justice refers to a capacity to understand, apply and act according to the principles of justice. On the other hand, a capacity for a conception of the good is the capacity to form, revise and pursue one's understanding of the good.[19] The primary goods to advance the higher order interests include basic rights and liberties, freedom of movement, free choice of occupation, income and wealth, and the social bases of self-respect.[20] The primary goods are distributed by the society's basic structure, which Rawls considers the modern constitutional democracy consisting of major social, political and economic institutions. These major institutions distribute it based on the two principles of justice.[21]

15 Amy Gutmann, "Rawls on Relationship between Liberalism and Democracy," in *The Cambridge Companion to Rawls*, ed. Samuel Freeman (New York: Cambridge University Press, 2003), 170.

16 Henry S. Richardson, "Democracy," in *The Cambridge Rawls Lexicon*, ed. Jon Mandle and David A. Reidy (Cambridge: Cambridge University Press, 2015), 193.

17 Rawls, *A Theory of Justice,* 386.

18 John Rawls, *Political Liberalism* (New York: Columbia University Press, 1996), 74.

19 Rawls, *Political Liberalism*, 19.

20 *Ibid*.,76

21 These principles are: "(a) Each person has the same indefeasible claim to a fully adequate scheme of equal basic liberties, which scheme is compatible with the same scheme of liberties for all; and (b) Social and economic inequalities are to satisfy two conditions: first, they are to be attached to offices and positions open to all under the conditions of fair equality of opportunity; and second, they are to be to the greatest benefit of the least-advantaged members of the society (difference principle)." John Rawls, *Justice as Fairness: A Restatement*, ed. Erin Kelly (New Delhi: Universal Law Publishing Co. Pvt. Ltd., 2011), 42–43.

From my perspective, Rawls emphasises three different dimensions of self-respect: individual, societal, and institutional. Rawls identifies self-respect with self-esteem in *A Theory of Justice*. Here, Rawls highlights two aspects of self-respect, which Cynthia A. Stark calls the "sense of one's value aspect" and the "self-confidence aspect."[22] An individual gets a sense of his own value from the conviction that his idea of good or his plan of life is worth pursuing. The ability to fulfil one's plan gives him confidence. Rawls argues that when we are not confident about the value of our plan of life, it becomes a burden for us to execute it. Furthermore, there will be fear of failure and self-doubt clouding our self. Hence, Rawls states, "It is clear then why self-respect is a primary good."[23]

One can identify individual dimensions of self-respect in the sense of one's value aspect. The situation that leads to attaining self-respect from an individual perspective is about having a rational plan of life that satisfies the Aristotelian principle. Aristotelean principle states that "other things equal, human beings enjoy the exercise of their realised capacities (their innate or trained abilities), and this enjoyment increases the more the capacity is realised, or the greater its complexity."[24] If the plan of life does not create interest in an individual, thereby not satisfying the Aristotelian principle, it results in "…no feeling of competence or a sense that they are worth doing."[25] In *Political Liberalism*, Rawls responds to the fact of reasonable pluralism and doesn't point to the Aristotelian principle but to the political conception that is free from any comprehensive doctrine (philosophical or religious). Attaining self-respect from the perspective of a citizen now involves not the worthiness of the conception of the good but involves "our self-confidence as a fully cooperating member of society rooted in the development and exercise of two moral powers."[26]

The societal dimension of self-respect is emphasised when Rawls indicates "the companion effect of the Aristotelian principle" where "others confirm and take pleasure in what we do." The associative ties of an individual offer situations where others affirm one's rational plan of life, thereby experiencing the worth of pursuing one's plan of life and becoming self-confident. Furthermore, the associative ties also become a safety net during failures and self-doubt.[27] Hence, Rawls states, "…our

22 Cynthia A. Stark, "Rawlsian Self-Respect," in *Oxford Studies in Normative Ethics*, ed. Mark Timmons, Vol. 2 (Oxford: Oxford University Press, 2012), 240.

23 Rawls, *A Theory of Justice: Revised Edition*, 386.

24 *Ibid*., 374.

25 *Ibid*., 387.

26 Rawls, *Political Liberalism*, 319.

27 Rawls, *A Theory of Justice: Revised Edition*, 387.

self-respect normally depends upon the respect of others", and it is from this mutual respect that the citizens would be willing to explain the grounds of their actions when the claims of others are overruled.[28] In *Political Liberalism*, this mutual respect is reflected in public reason connected to the criterion of reciprocity: "Our exercise of political power is proper only when we sincerely believe that the reasons we offer for our political action may reasonably be accepted by other citizens as a justification of those actions."[29] The content of public reason is drawn from the public political culture of the democratic society, which consists of the fundamental idea of citizens as free and equal and society as a fair system of cooperation.

The institutional dimension of self-respect is affirmed when Rawls underlines the importance of basic institutions in a democratic society to function based on the two principles of justice. The representatives in the original position would choose the principles of justice over utilitarianism or other political principles because of the emphasis on self-respect. For instance, Rawls argues that adopting Utilitarian principles as the basis for institutions "entails some loss of esteem"[30] but "the contract conception of justice supports the self-esteem of citizens more firmly than other political principles."[31] Furthermore, the public affirmation of the status of equal citizenship without discrimination secures self-respect in a well-ordered society.[32] In Rawls's view, the priority given to liberty in his theory secures self-respect of citizens. It is because equal liberty allows for the existence of "many free communities of interest" that support and encourage the plan of life of citizens. Another aspect is that it guarantees equality of all citizens such that "…everyone has a similar and secure status when they meet to conduct the common affairs of the wider society."[33] Finally, Rawls underlines the institutional dimension of self-respect in *Justice as Fairness: A Restatement* when he clarifies that self-respect is not an attitude but the social-bases of self-respect. He states, "These social bases are things like the institutional fact that citizens have equal basic rights, and the public recognition of the fact that everyone endorses the difference principle, itself a form of reciprocity."[34]

28 *Ibid*, 154.

29 Rawls, *Political Liberalism*, xlvi.

30 Rawls, *A Theory of Justice: Revised Edition*, 158.

31 *Ibid*., 470.

32 *Ibid*., 478.

33 *Ibid*., 477.

34 Rawls, *Justice as Fairness: A Restatement*, 60.

Self-Respect and AI

AI tools can impact self-respect of a citizen. As Anna Lauren Hoffman rightly observers, "Importantly, our self-respect is not won or lost only in our interactions with others; it is also shaped by our interaction with non-human dimensions of the world – like technological artefacts, information systems, and the built environment – that codify and reproduce self-respect's social bases."[35] It can be argued that using AI tools could place barriers to citizens acquiring self-respect in a constitutional democracy. Based on the above discussion, these obstacles are analysed from the individual, societal and institutional dimensions of self-respect.

AI and Individual Dimension of Self-Respect

Acquiring self-respect from an individual point of view includes living a life of self-esteem with a sense of one's value when one is convinced that one's plan of life or good is worth pursuing. In a constitutional democracy, where the principles of justice distribute benefits and burdens, an individual is guaranteed equality and freedom and can develop and exercise their moral powers, namely, the capacity for a sense of justice and the capacity for the conception of the good. However, AI tools can interfere with the autonomy of the individual to make informed decisions in a democracy.

AI tools have the potential to undermine the self-respect of citizens in a democracy by facilitating manipulation and perpetuating bias. Mark Coeckelbergh observes that AI can manipulate human experiences and thoughts. It directly affects democracy because "AI and other digital technologies can nudge voters towards supporting a particular politician or party through personalised advertisements, and so on."[36] Furthermore, voters could be influenced by AI chatbots that could apply manipulating tactics in real-time by customised interactions with the voters. AI also can generate content to manipulate the voter by simulating human emotions. There is potential for manipulation because "…users may not be able to distinguish between human and AI generated content."[37] Andreas Jungherr point out that even though the claims of Cambridge Analytica that they could influence the behaviour of voters have been proved false, it is a case in point to understand the general perception to accept such an exaggerated claim about digital

35 Anna Lauren Hoffmann, "Rawls, Information Technology, and the Sociotechnical Bases of Self-Respect," in *The Oxford Handbook of Philosophy of Technology*, ed. Shannon Vallor (New York: Oxford University Press, 2022), 236, https://doi.org/10.1093/oxfordhb/9780190851187.013.15.

36 Mark Coeckelbergh, *The Political Philosophy of AI: An Introduction* (Cambridge, UK: Polity Press, 2022), 80.

37 Michael, "Artificial Intelligence, Democracy and Elections," 4.

manipulation uncritically. He adds that transformer models such as Chat GPT can generate automated text or images. It is not concerned about the truth but only about the plausibility of arguments. Furthermore, mass production of false or misleading content can make it impossible for citizens to trust and identify correct information.[38] Hence, instead of self-esteem, the citizen is pushed to self-doubt and his autonomy is compromised by questioning his self-worth. This could drive him away from actively participating in democratic processes such as elections.

AI and Societal Dimension of Self-Respect

An individual attains self-respect from his associative ties in society when fellow beings affirm their life plan. Furthermore, they act as a support in times of failure and self-doubt. This dimension also involves a criterion of reciprocity, where citizens offer reasons for political actions that could reasonably be accepted by other citizens. AI has the potential to disrupt the societal connection between citizens and weaken self-confidence and self-respect.

The use of AI can lead to citizens being locked in information bubbles and echo chambers. Algorithms, fuelled by personal user data, present users with information that reinforces their existing beliefs and shields them from opposing viewpoints. Coeckelbergh argues that this has a detrimental impact on democracy. It hinders the development of an empathetic form of politics where citizens can understand and consider opposing viewpoints. This, in turn, fuels increasing political polarisation, where divides widen and compromise becomes difficult. Social media platforms like Facebook and search engines, with their AI-powered algorithms, can exacerbate this problem by feeding us information based solely on our past choices, effectively trapping us in filter bubbles and echo chambers. He cites El-Bermway to indicate a shift from a global village to "digital islands of isolation that are drifting apart each day."[39] Associative ties would be lost, leading to a loss of self-respect. Furthermore, the insular nature of echo chambers and information bubbles can make citizens unresponsive to the need for public justification of their political views. Trapped in chambers of unfounded certainty, they may feel no pressure to engage in deliberation or mutual understanding with those with differing viewpoints. This discourages the very aspects of democratic discourse that are essential for finding common ground and building a cohesive society.

38 Jungherr, "Artificial Intelligence and Democracy," 5–6.

39 Coeckelbergh, *The Political Philosophy of AI*, 76–77.

AI and Institutional Dimension of Self-Respect

In John Rawls' theory of justice, the institutional dimension of self-respect is secured when a society's political, social, and economic structures function according to his two principles of justice. These principles ensure equal basic liberties for all citizens and allow for social and economic inequalities only if they ultimately improve the lives of the least advantaged members. This framework fosters self-respect by guaranteeing fair treatment and the potential for upward mobility within the system.

AI weakens democratic institutions by fostering inequality and injustice through surveillance. LLMs respond to prompts based on the data sets on which it has trained. Mostly, it cannot trace the source of the response we receive from LLMs such as Chat GPT. There is opaqueness in its functioning. Furthermore, it can give wrong, misleading and biased information. Jungherr and Schroeder state: "Many studies have shown that inherent gender, racial, and other biases lead AI to reproduce societal inequalities and injustices."[40] The temptation for surveillance also weakens the credibility of democratic institutions. The multi-national companies, called "the Big Nine", namely, Google, Microsoft, Facebook, IBM and Apple in the US, and Tencent, Alibaba and Baidu in China, collect and organise large amounts of user data. In China, this data is served to the hands of the government and in the US, they resort to surveillance capitalism.[41] According to Shoshana Zuboff, surveillance capitalism is the puppet master that imposes its will on the people through digital apparatus as the Big Other. She defines the Big Other as "… the sensate, computational, connected puppet that renders monitors, computers, and modifies human behaviour."[42] She states: "Surveillance capitalists quickly learned to exploit the gathering momentum aimed at hollowing out democracy's meaning and muscle."[43] This directly impacts the self-respect of citizens as they are treated as objects constantly under surveillance by multinationals and democratic institutions.

Conclusion

Globally, democracy is at a crossroads for many reasons, and AI is one of the most prominent contributing factors. As I have argued in this paper, the detrimental impact of AI on self-respect, from the individual, societal

40 Andreas Jungherr and Ralph Schroeder, "Artificial Intelligence and the Public Arena," *Communication Theory* 33, no. 2–3 (August 1, 2023): 170, https://doi.org/10.1093/ct/qtad006.

41 Risse, "Artificial Intelligence and the Past, Present, and Future of Democracy," 11.

42 Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (London: Profile books, 2019), 376.

43 *Ibid*., 518.

and institutional dimensions, can erode democracy. From an individual perspective, self-respect includes living a life of self-esteem with a sense of one's own value, equipping one to pursue the rational good. However, AI algorithms can isolate citizens in echo chambers and information bubbles, shutting them off from reality and discouraging them from engaging with diverse voices. When confronted with opposing viewpoints, they may lose confidence, leading to self-doubt and, eventually, a lack of self-respect. The institutional dimension of self-respect points to the fair distribution of basic rights and liberties. However, AI tools raise serious ethical questions for citizens concerning their self-respect. When political institutions use surveillance methods and the inequality fostered by AI, trust in these institutions is lost. The citizens lose self-respect as they are reduced to mere data points in the hands of multinationals and democratic institutions.

The unrestricted deployment of AI could inflict significant harm on individual self-respect and democratic institutions. Therefore, ethical considerations and legal frameworks are crucial for effective AI regulation. Governments and NGOs must actively educate citizens about AI's potential for manipulation and exploitation. Additionally, promoting online and offline platforms that break echo chambers and information bubbles is vital to foster open dialogue and rebuild trust. Democratic institutions must ensure transparency in data gathering. Both governments and corporations should provide clear visibility into how data is processed and used. In conclusion, to preserve the dignity of citizens in a democratic society, AI should remain a tool controlled by humans. We must prevent citizens from becoming mere pawns manipulated by AI.

Statement of the ACPI

We, the Association of Christian Philosophers of India (ACPI), met at Jnana Niketan, Institute of Philosophy and Religion, Little Flower Seminary, Aluva, Kerala, for the 46th Annual Research Seminar from 21 to 23 October, 2023 in which 23 research papers were presented and discussed. There were 90 participants at the seminar.

Key insights and Salient points emerged from the Discussions and Deliberations

1. We become increasingly aware of the exponential growth and development of technologies using Artificial intelligence (AI) in every possible domain of human existence. The engulfing presence of AI is even equated to the revolutionary events like Big-Bang and the Origin of Life. However, the unsettling connections between AI and human existence raise several profound questions about our existence, human identity and dignity, making and implementing of policies, safeguarding privacy and asserting the autonomy. The contemporary technologies obviously enhance not only our comforts and communication abilities but also shape our attitude towards one another, Nature and God.
2. AI's evolution intersects science, philosophy, anthropology, ethics and theology, prompting multi-disciplinary debates including discourses on bio-conservatism and transhumanism. This convergence offers an opportunity to have different perspectives where one can see AI as the realisation of our potentialities enabling us to have a positive outlook at the interfacing of AI and humanity.
3. It appears to be a legitimate fear that not only machines would enslave human beings but also the huge data collected in and through AI may be used to manipulate to suit the interests of the societal elites, to the alarming disadvantage of the poor and the

socially oppressed citizens. AI powered surveillance capitalism puts the privacy and the autonomy of individuals in a greater risk.

4. Though AI deals with the unimaginably huge and varieties of data and processes it with enormous speed, yet *human judgement* has to be involved not only in creating the algorithms for AI technology but also in interpreting and applying the outcomes of such technologies.
5. Along with the considerations of consciousness issues, ethical dilemmas, societal responsibility, and our commitment towards nature and future generations, we need to pay attention to the linguistic difficulties with regard to the languages used by AI and for AI; because philosophical analysis of languages makes it clear that languages always involve crucial ambiguities, not only in ordinary languages but also in mathematical/scientific languages. Similarly, AI languages might also lead to serious misunderstanding and communication gap.
6. AI is certainly capable of improving and enhancing the functional capabilities of human beings. However, it is a serious question whether it can replace human agencies with all their emotional, interpersonal, cultural, and spiritual aspects.
7. While it is obvious that AI manifests a certain level of consciousness and the neural networking systems mimic the neural activities of the human brain, both the machine consciousness (intelligence) and human consciousness (intelligence) are categorically different with nuances. It is highlighted that Indian philosophical insights can enhance the understanding of the themes related to AI, namely, human nature, creativity, rationality, corporeality, consciousness, intelligence, intuition, aesthetics, morality, mind and cognition.
8. The contributions of Indians to the AI, space and bio-technologies at large is amazingly appreciable. However, the benefits of the technologies and the scientific temperament have not reached the last person on the periphery of Indian societal strata.
9. The rise of AI challenges our understanding of human labour and its value. AI has the potential to replace many forms of human labour, but it also creates new opportunities for creative and meaningful labour. We need to find ways to ensure that AI is used to enhance human dignity and welfare, rather than to replace or devalue human labour.

In the light of such deliberations, we the members of ACPI sincerely commit ourselves to:

1. Evolve an educational curriculum to create awareness of the serious need for updating the ethical principles as the technologies evolve, ensuring that human dignity and identity are not compromised at any cost.
2. Use the AI tools and technologies starting from mobile phones to magnificent quantum computers, keeping in mind ethical implications, societal commitments and safeguarding the rights of the marginalised in society.
3. Keep away the AI technologies from interfering with the social fabric of the nations, distorting democratic practices, and legitimising the discriminations that polarise citizens on different accounts.
4. Revisit the understanding of traditional concepts such as free will, consciousness and transcendence in the context of new developments of AI-related technologies and reaffirm the uniqueness of humanity.
5. Be ready to avail the healthy contributions of AI technology of human enhancement provided they do not tamper or meddle with the fundamental identity and dignity of human beings.
6. Take the maximum benefits of AI technologies to uplift every section of humanity by reducing all inhuman discriminations prevailing in the society, preserving the intrinsic value of Nature.

Thus, we are convinced of the necessity that all those who are directly or indirectly involved in AI technologies need to be aware of the responsibility to '*humanize robots'* (to the extent possible) rather than *'robotize humans'*, acknowledging the potential dangers and irreversible consequences for humanity and Nature.

23 October, 2023

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